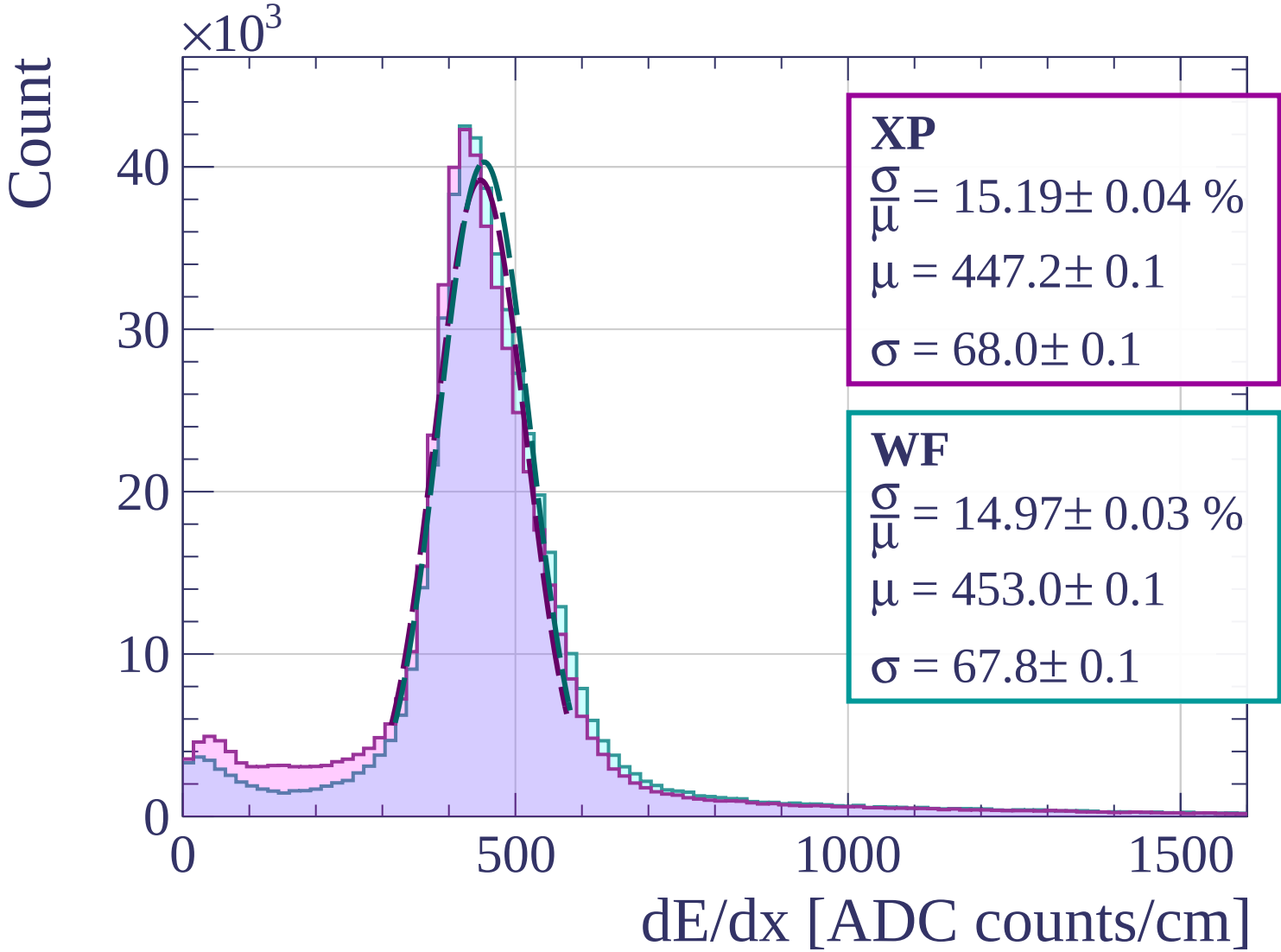
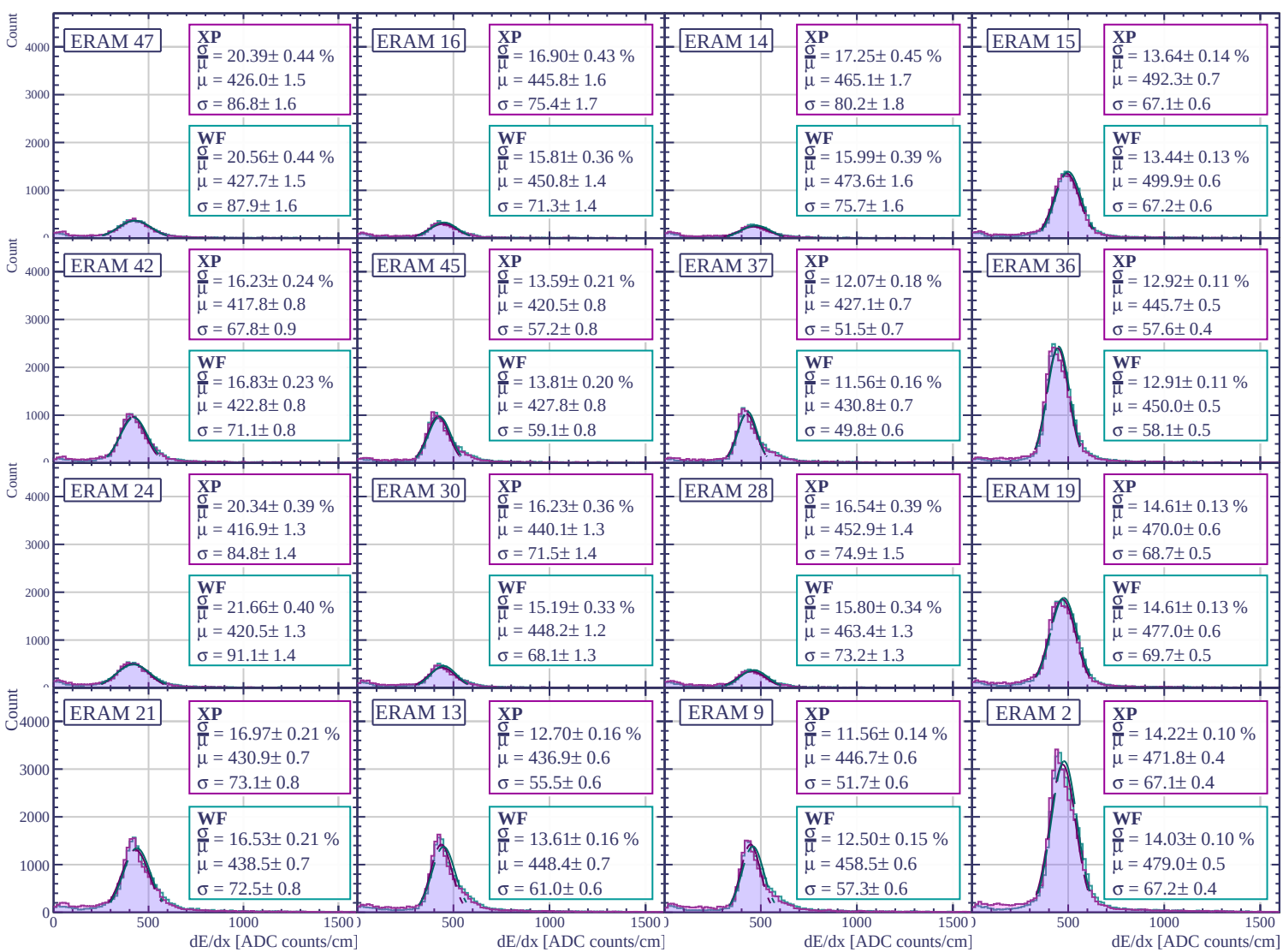
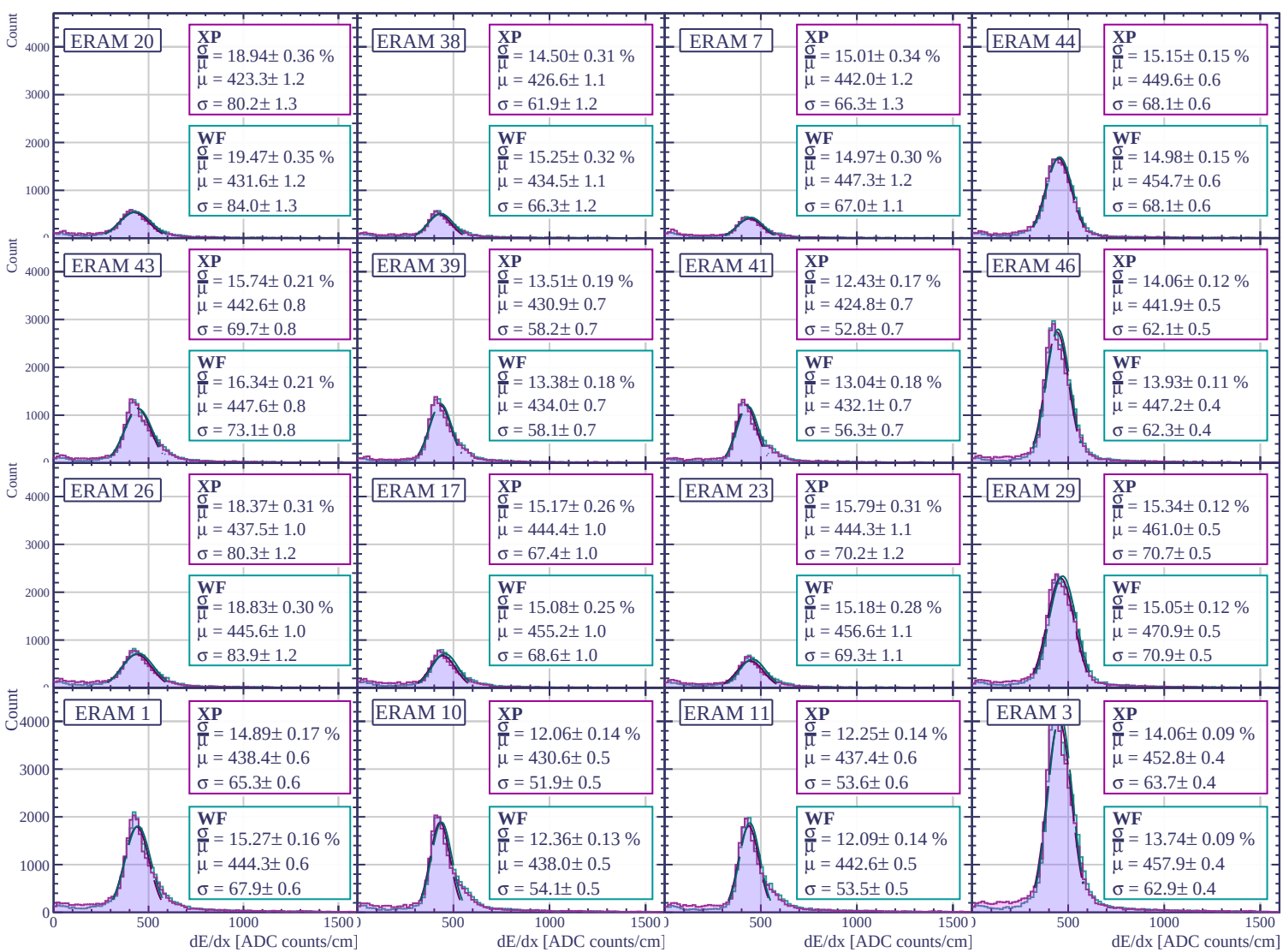
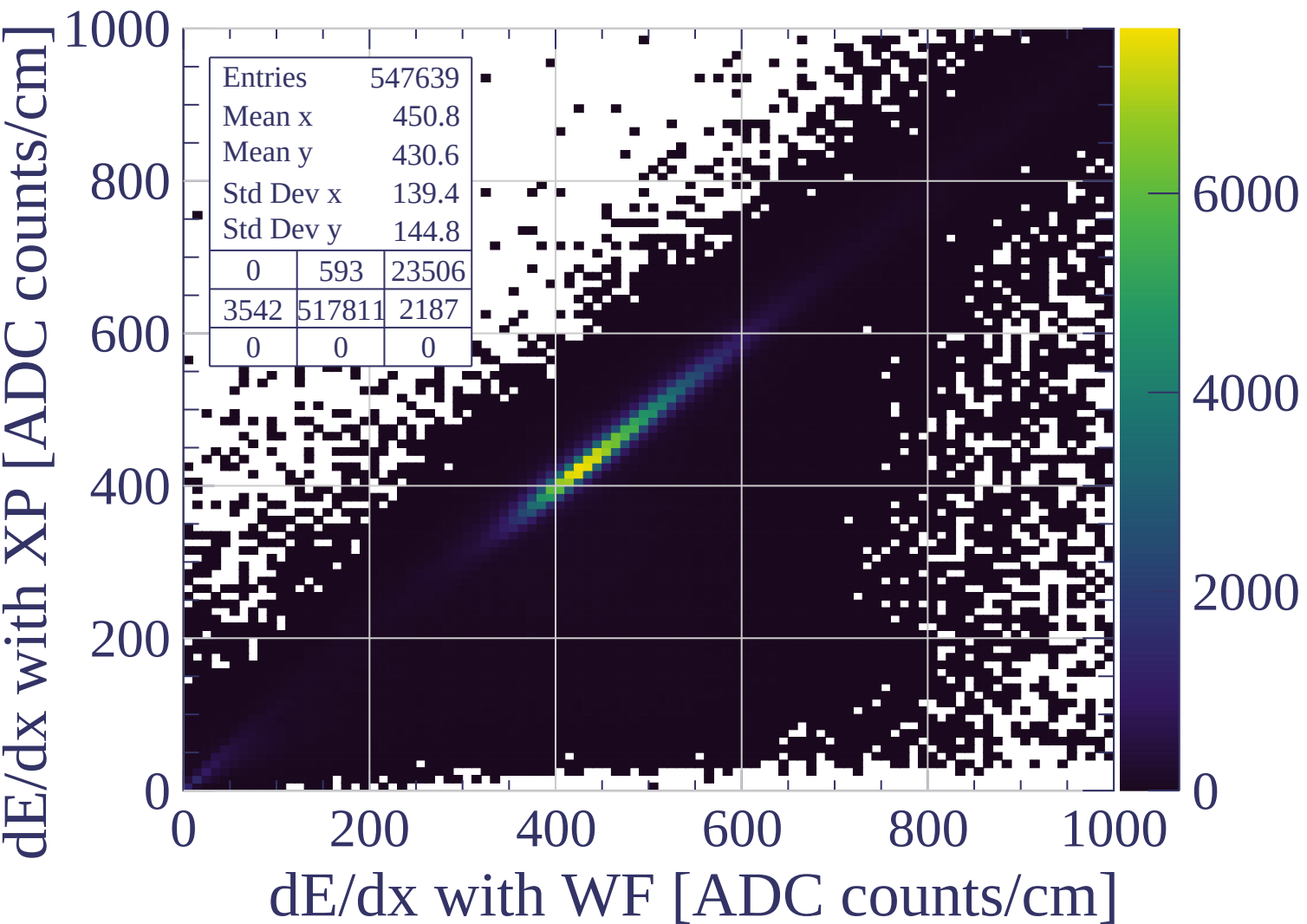
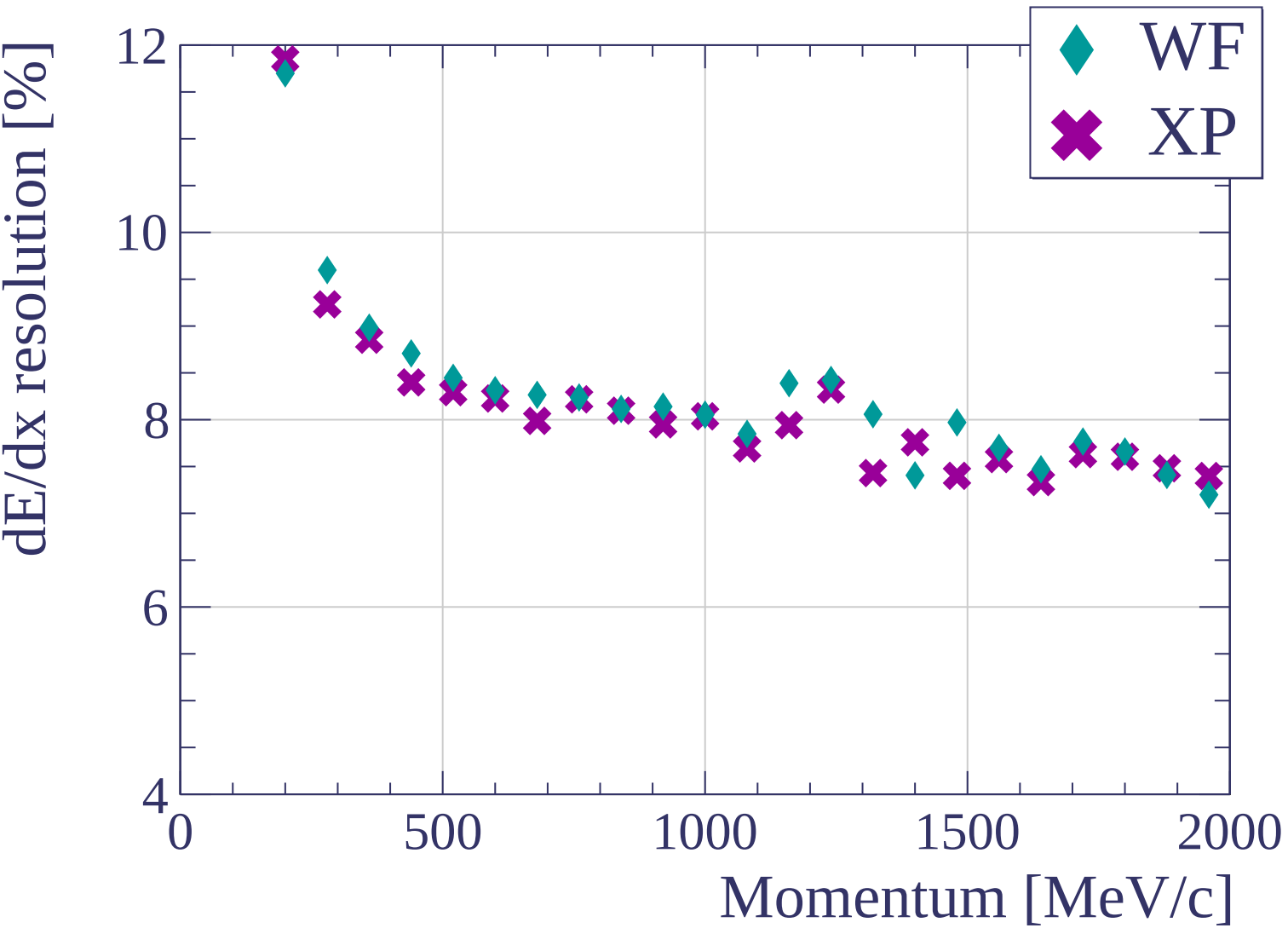


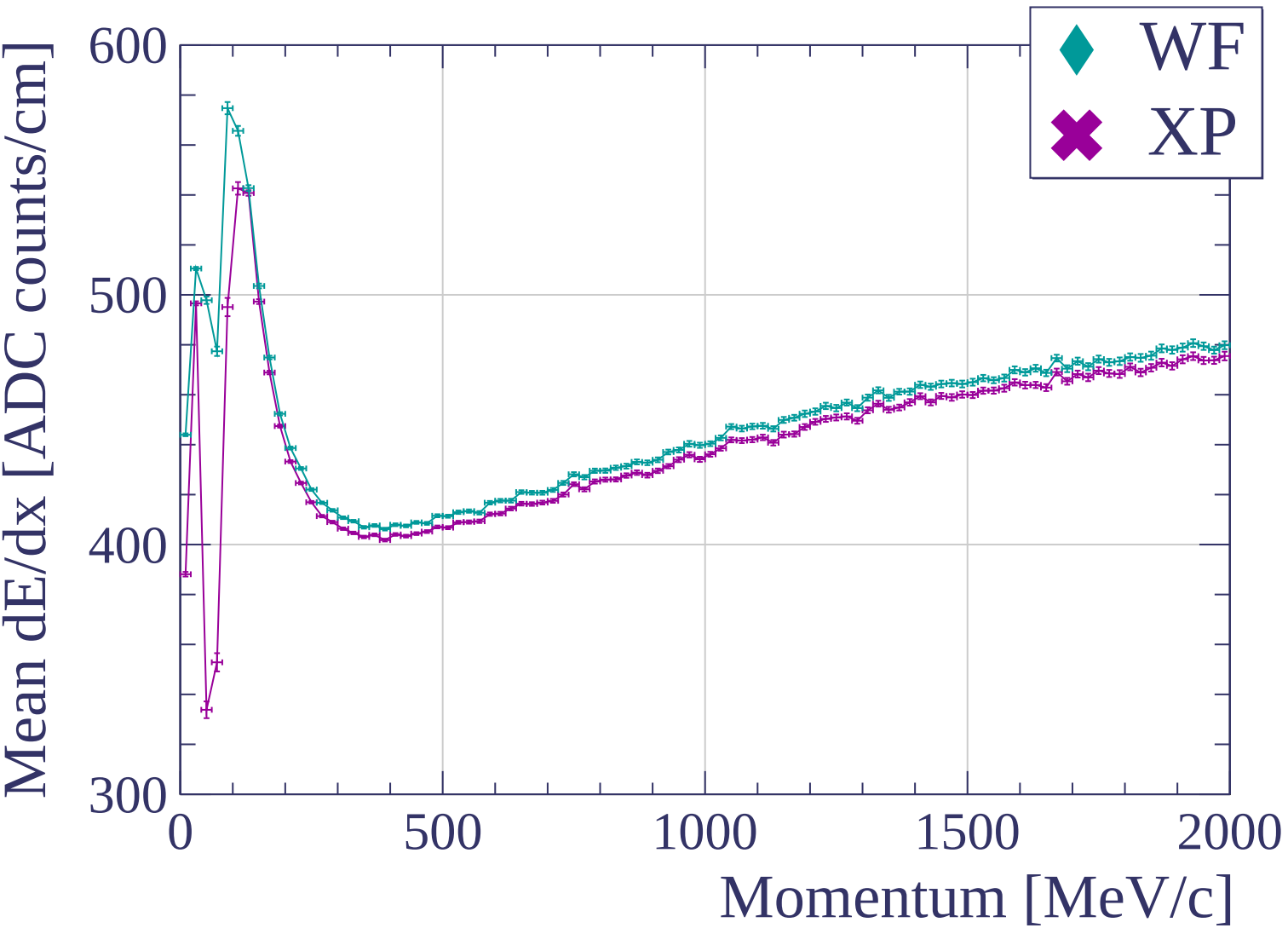


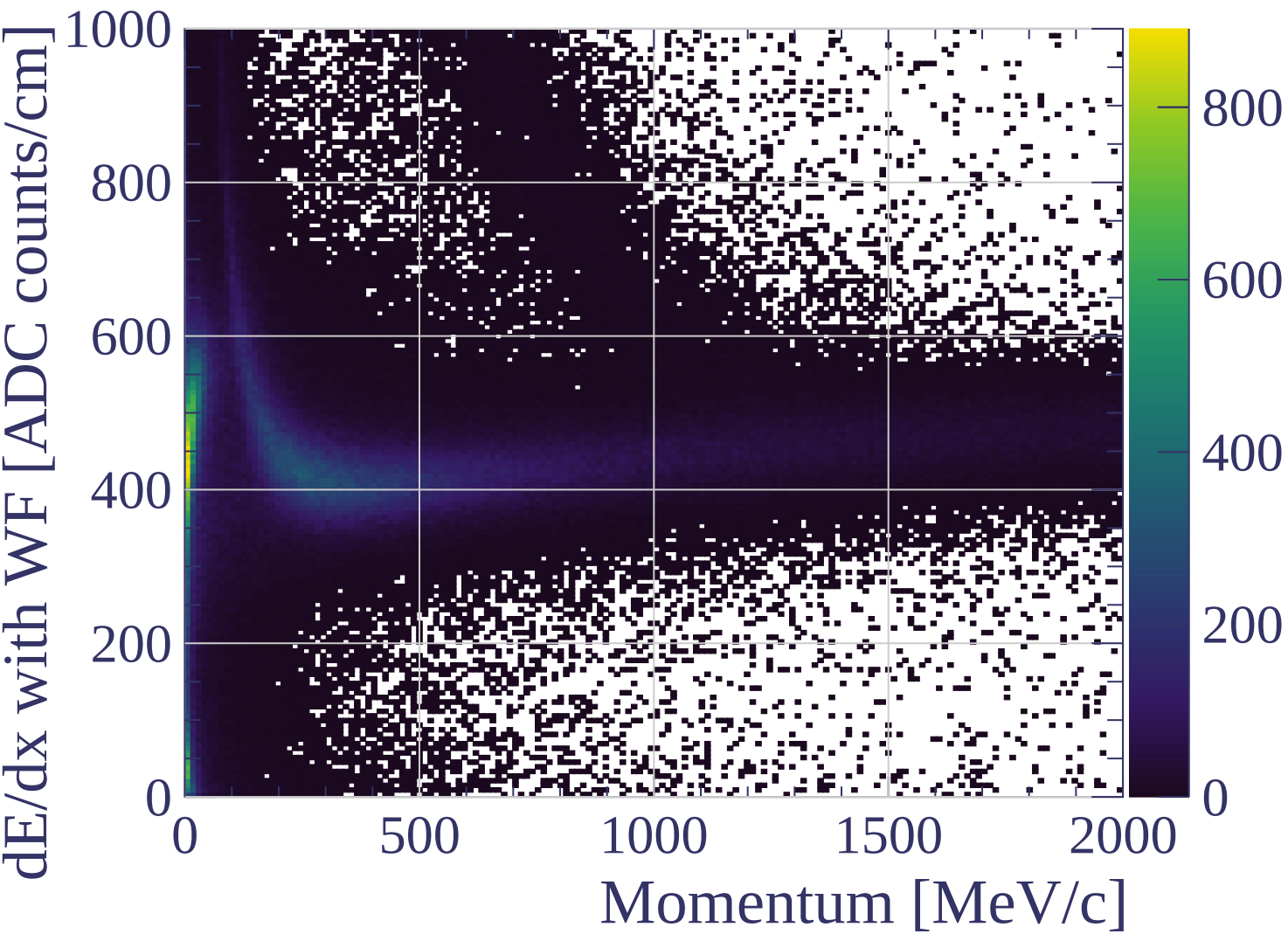


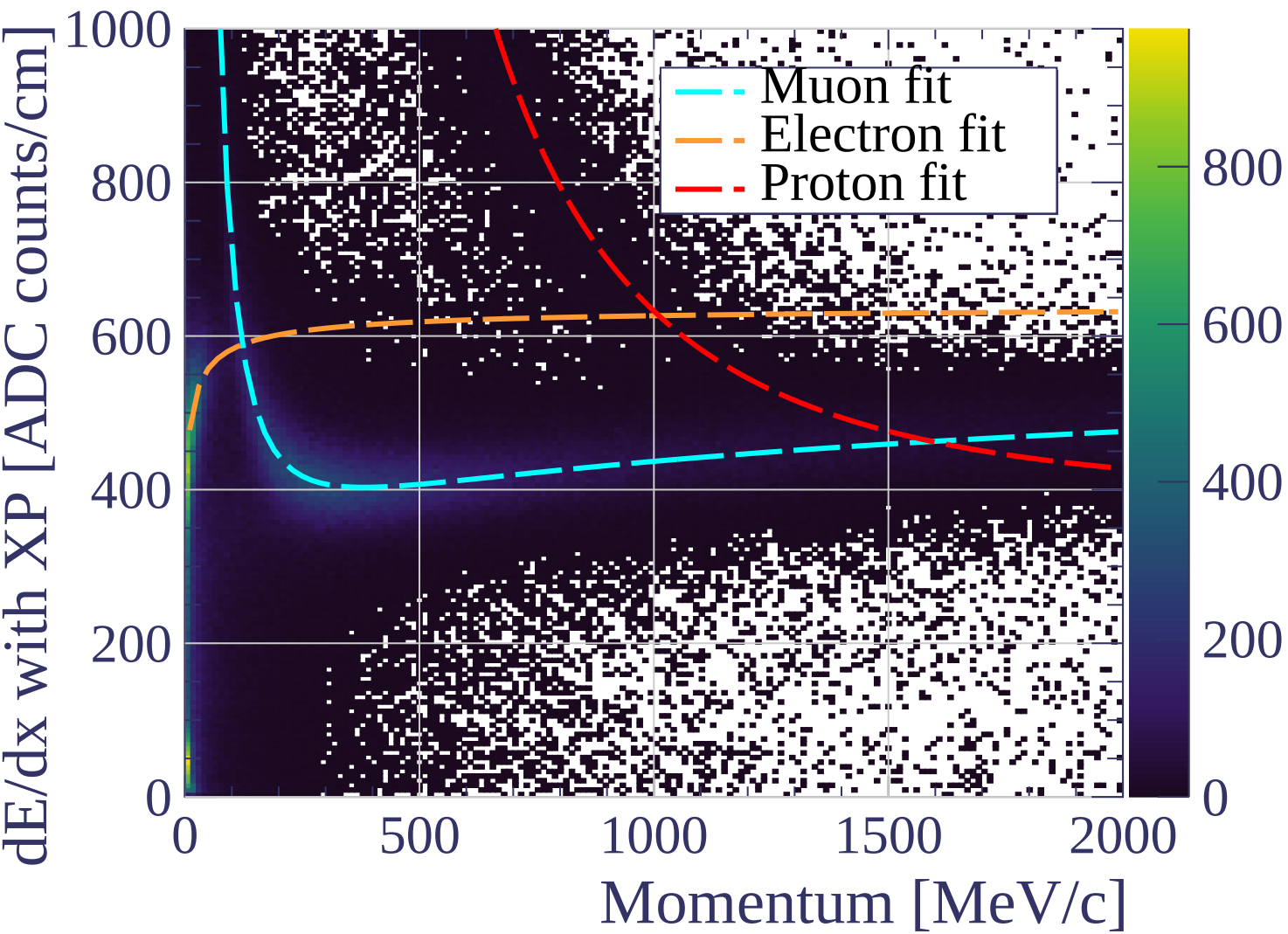


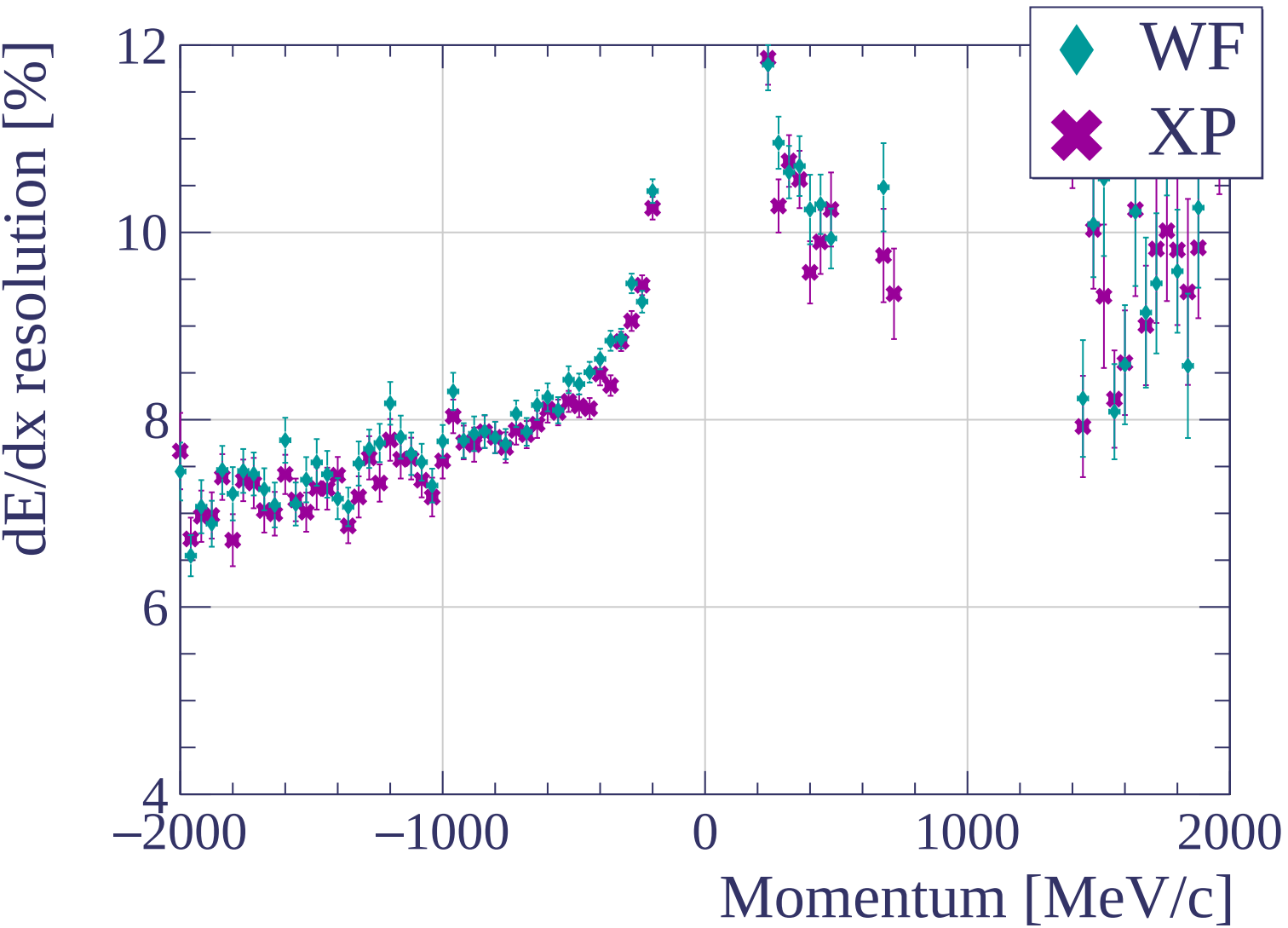


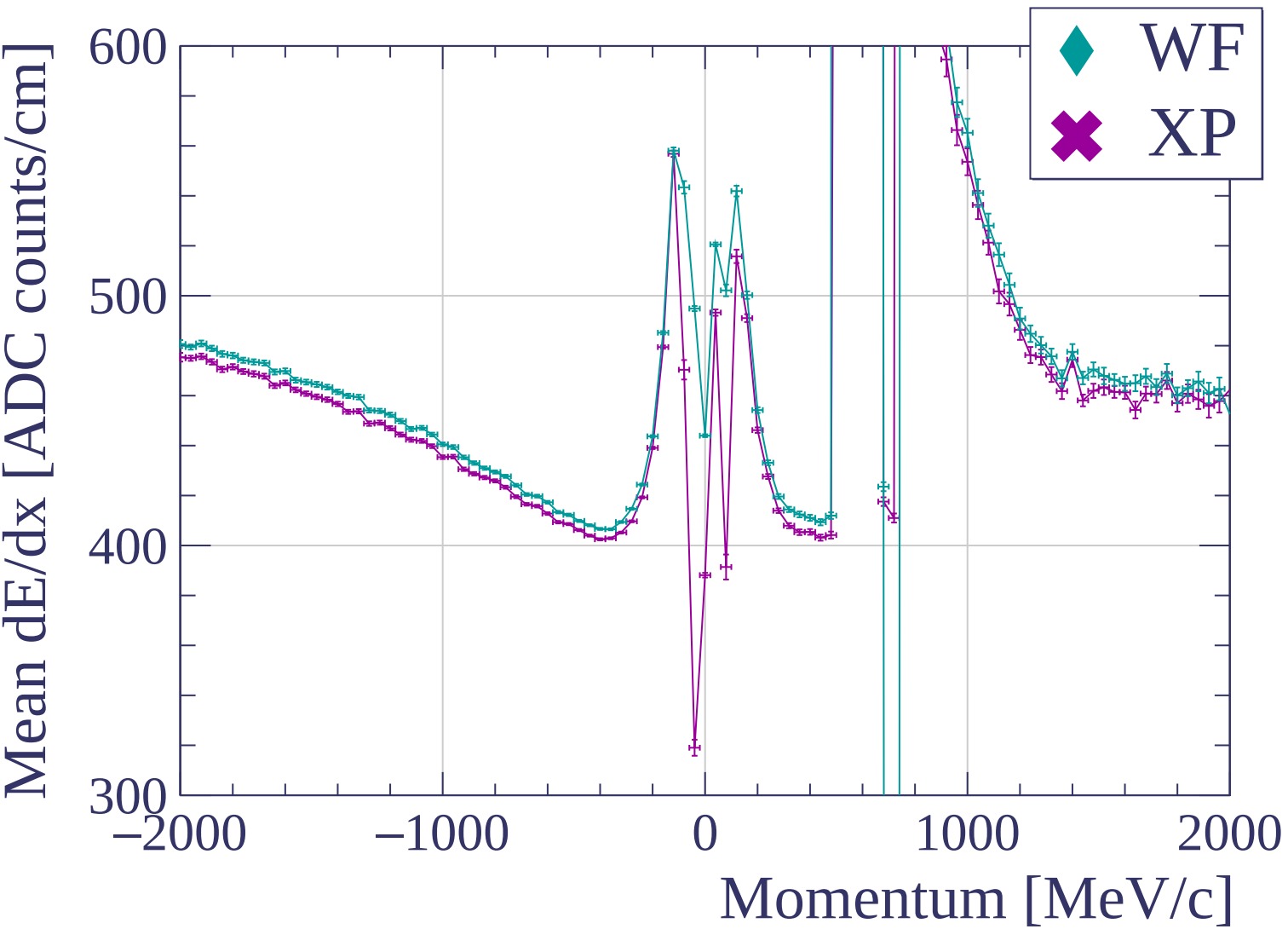


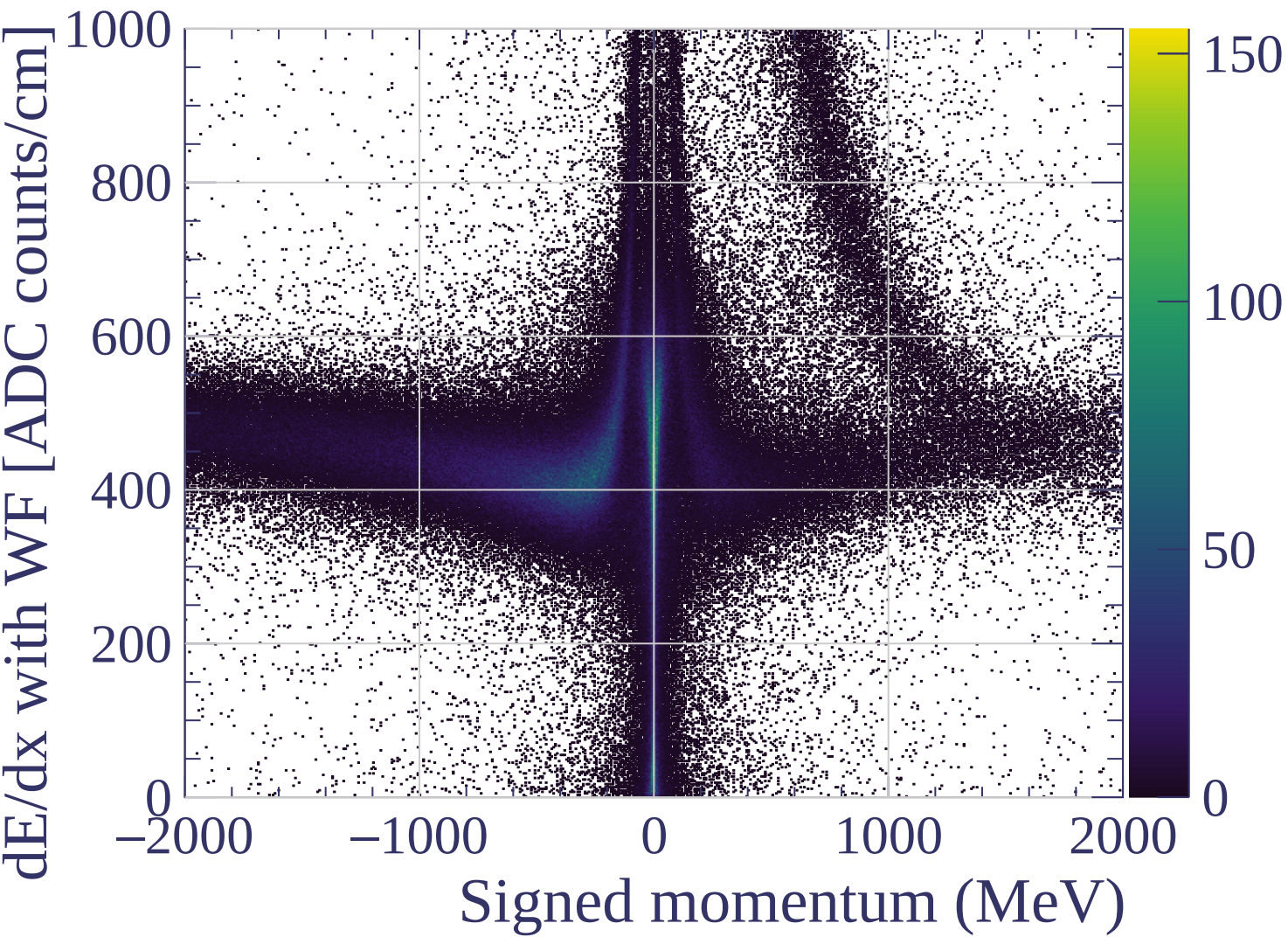


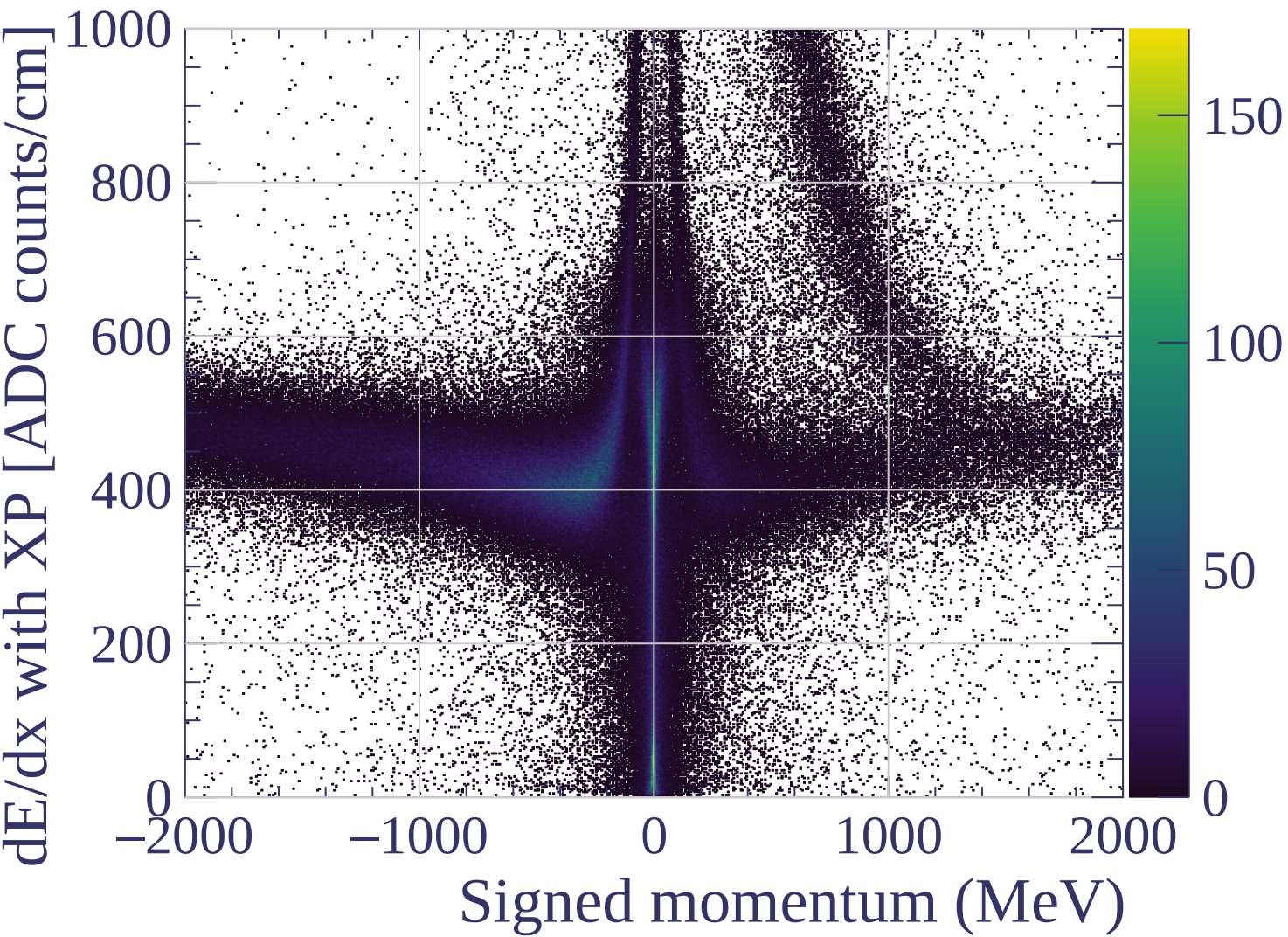


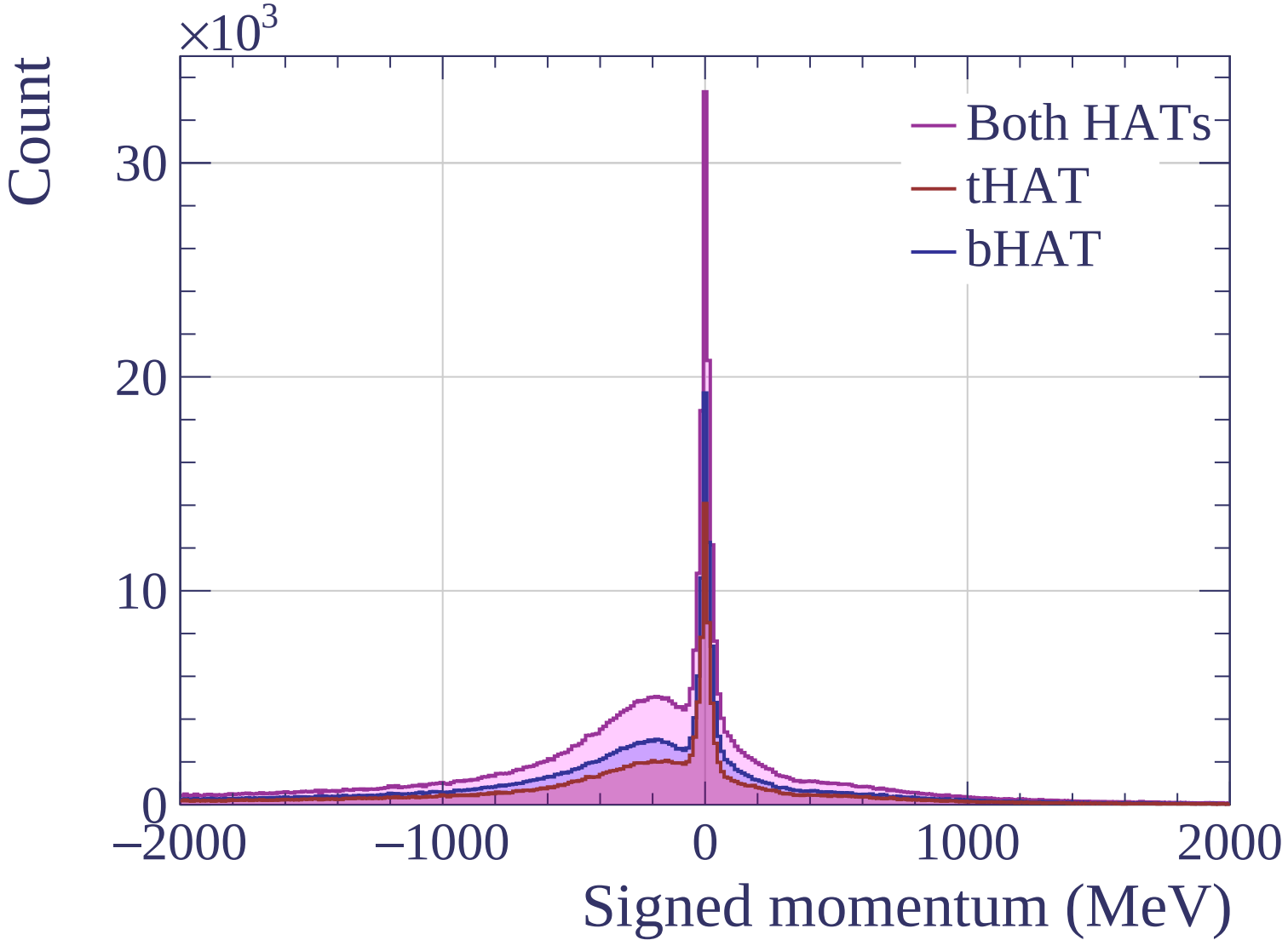


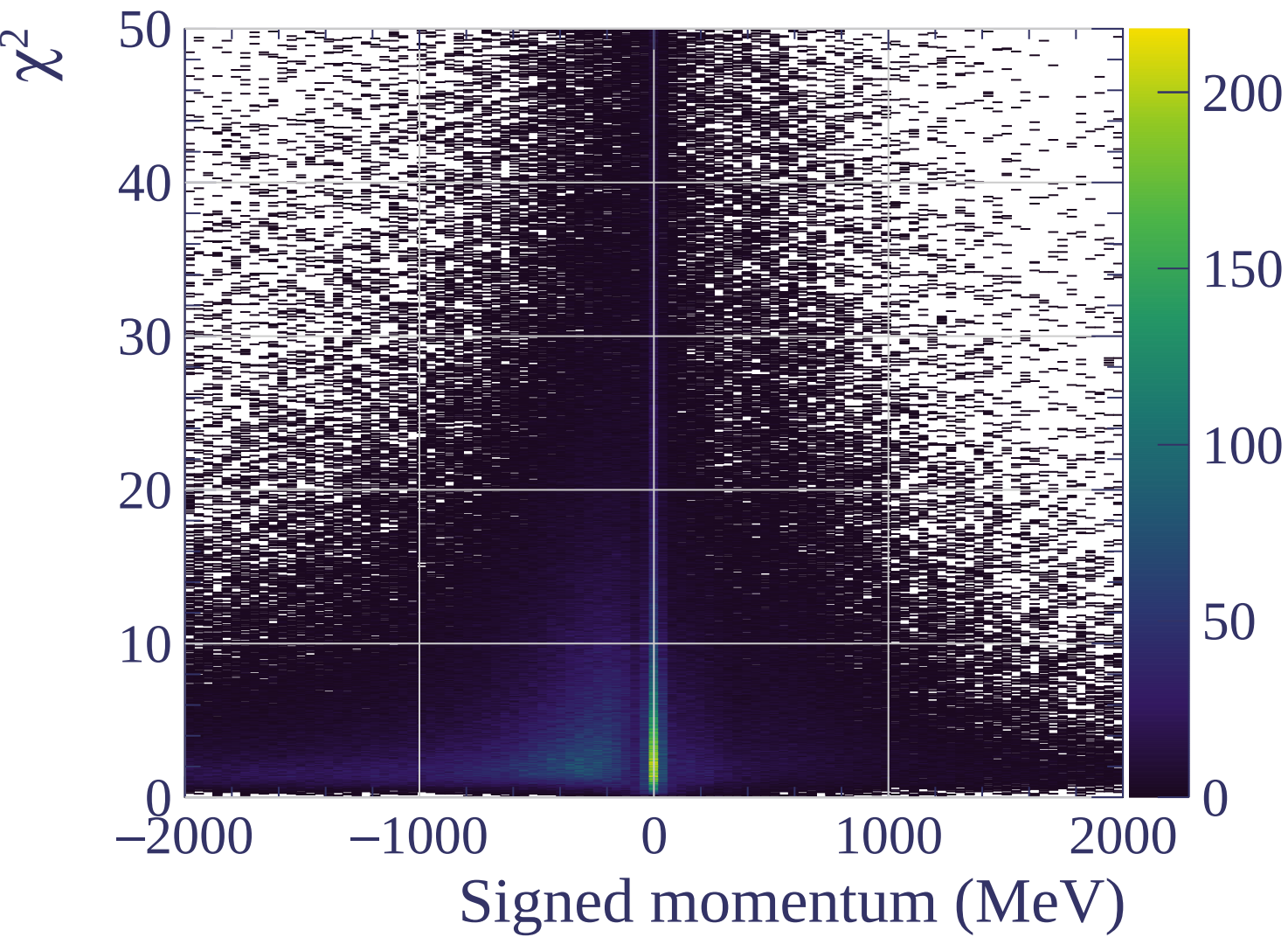


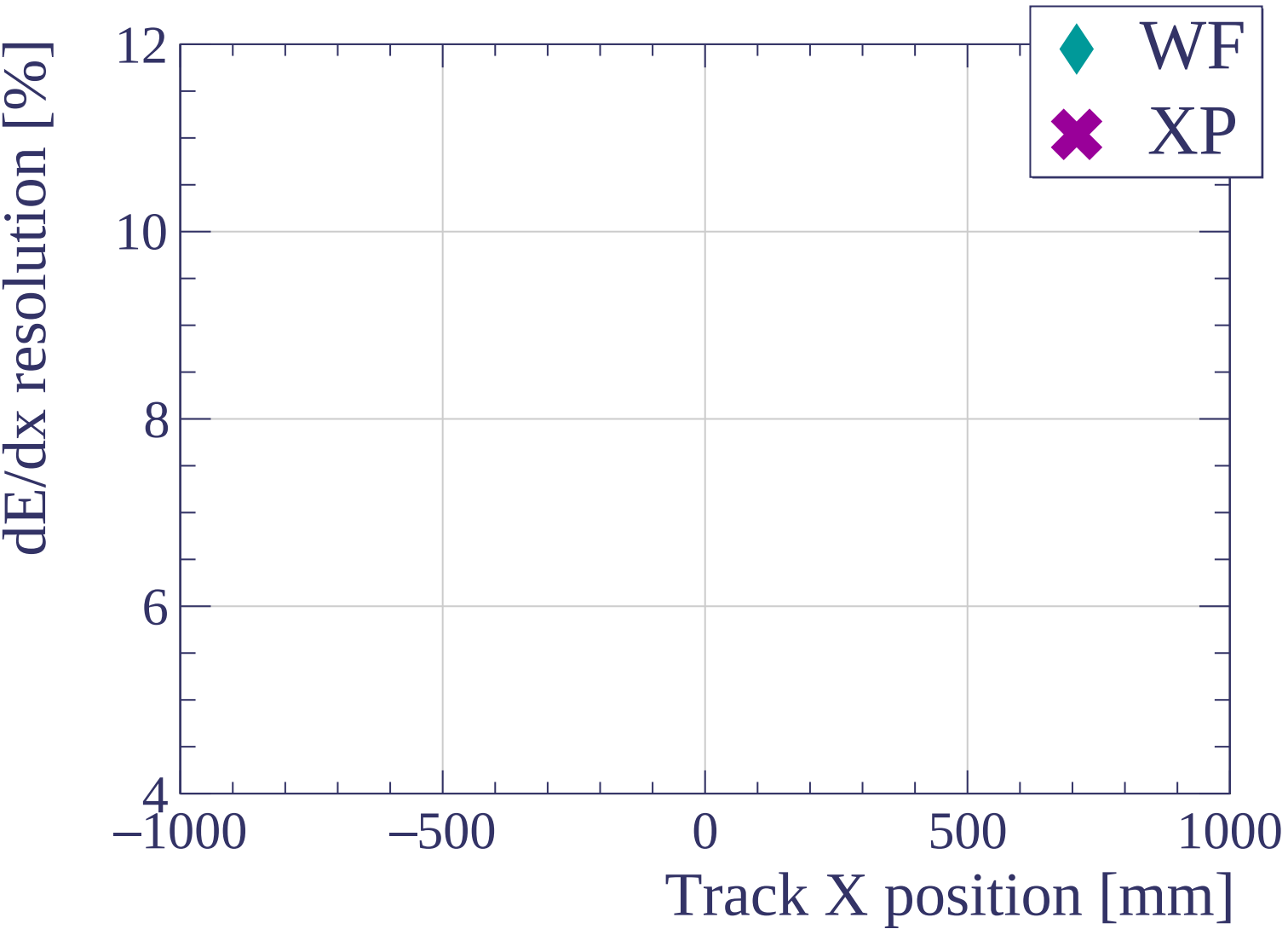


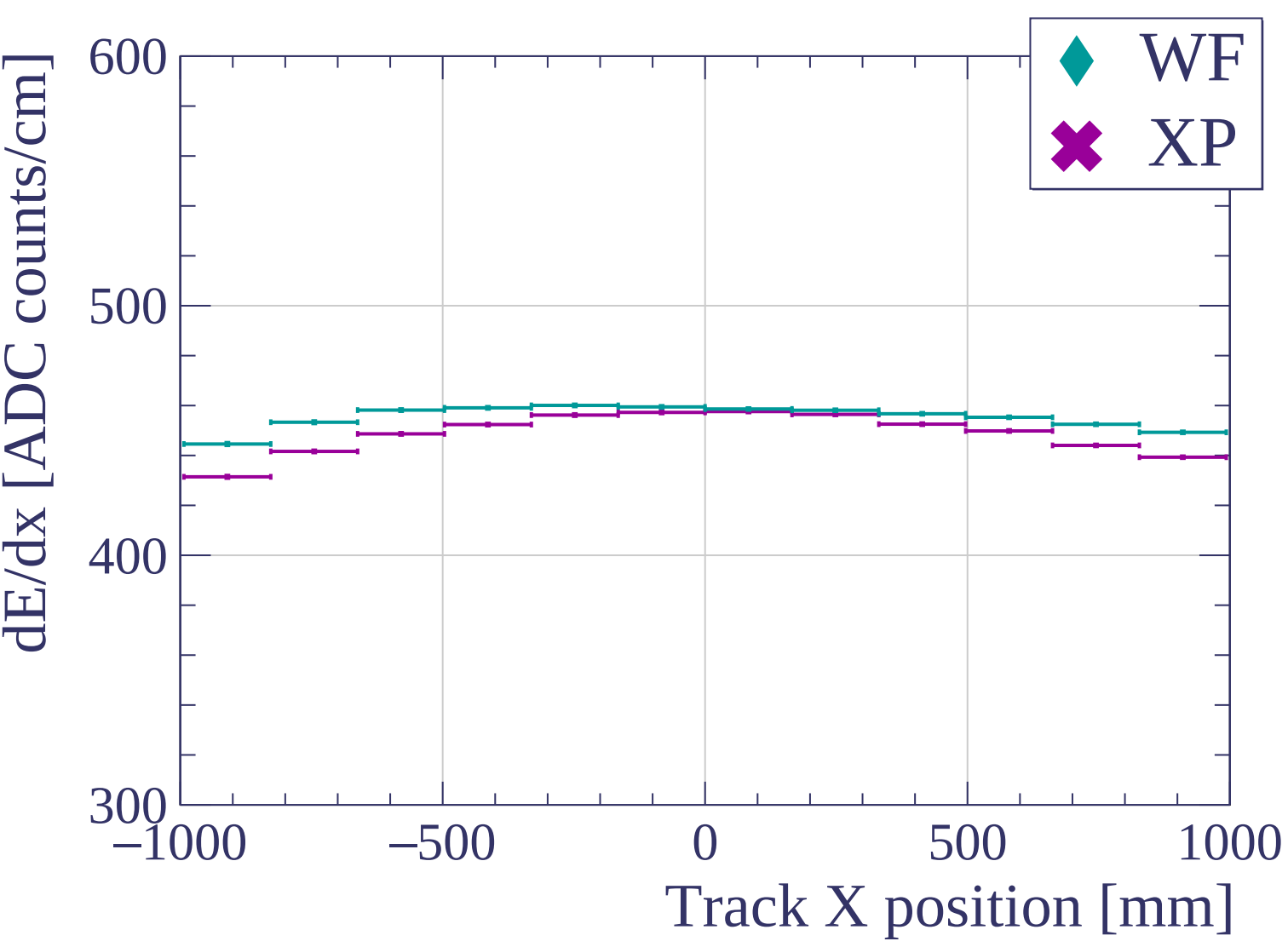


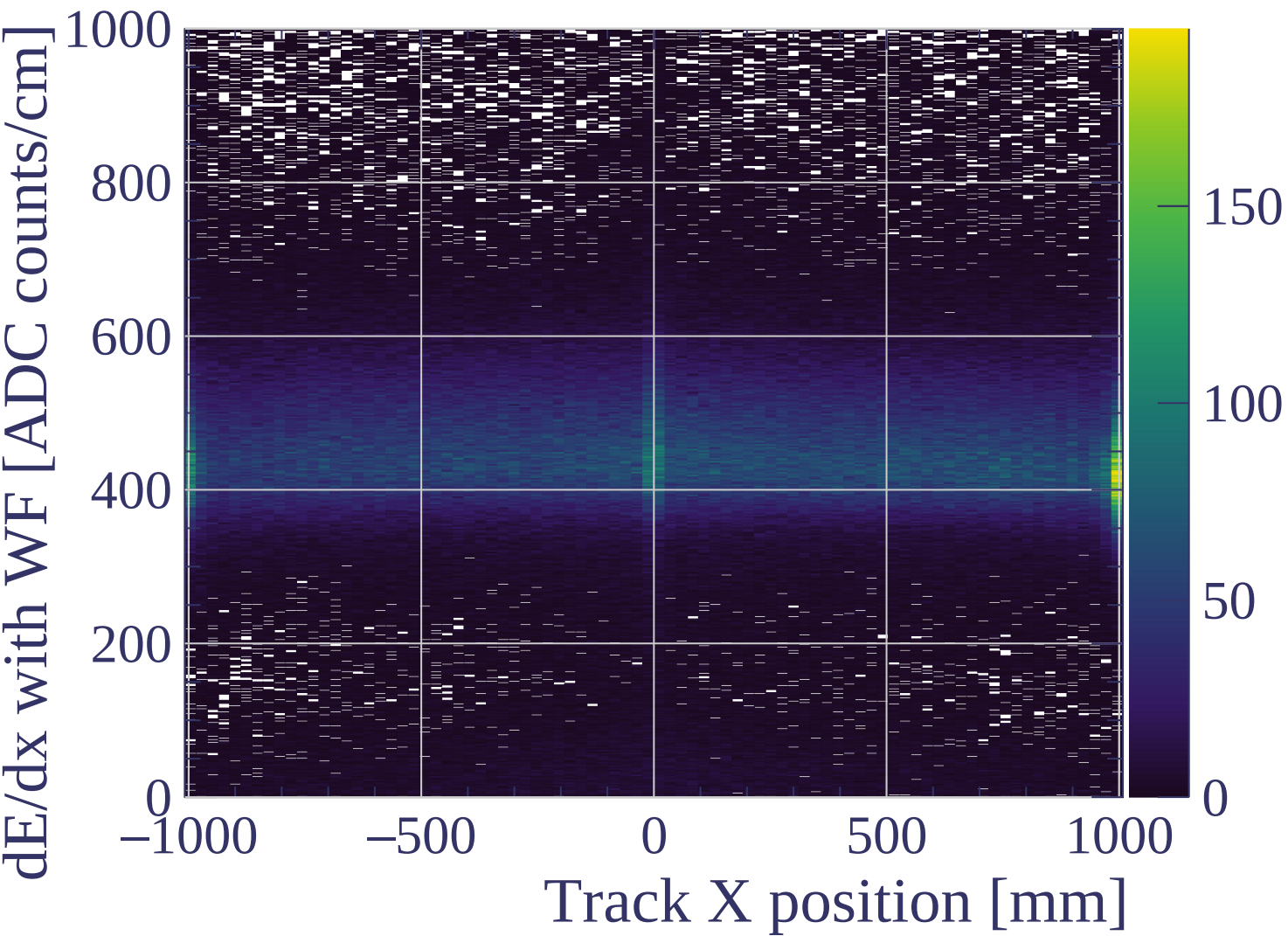


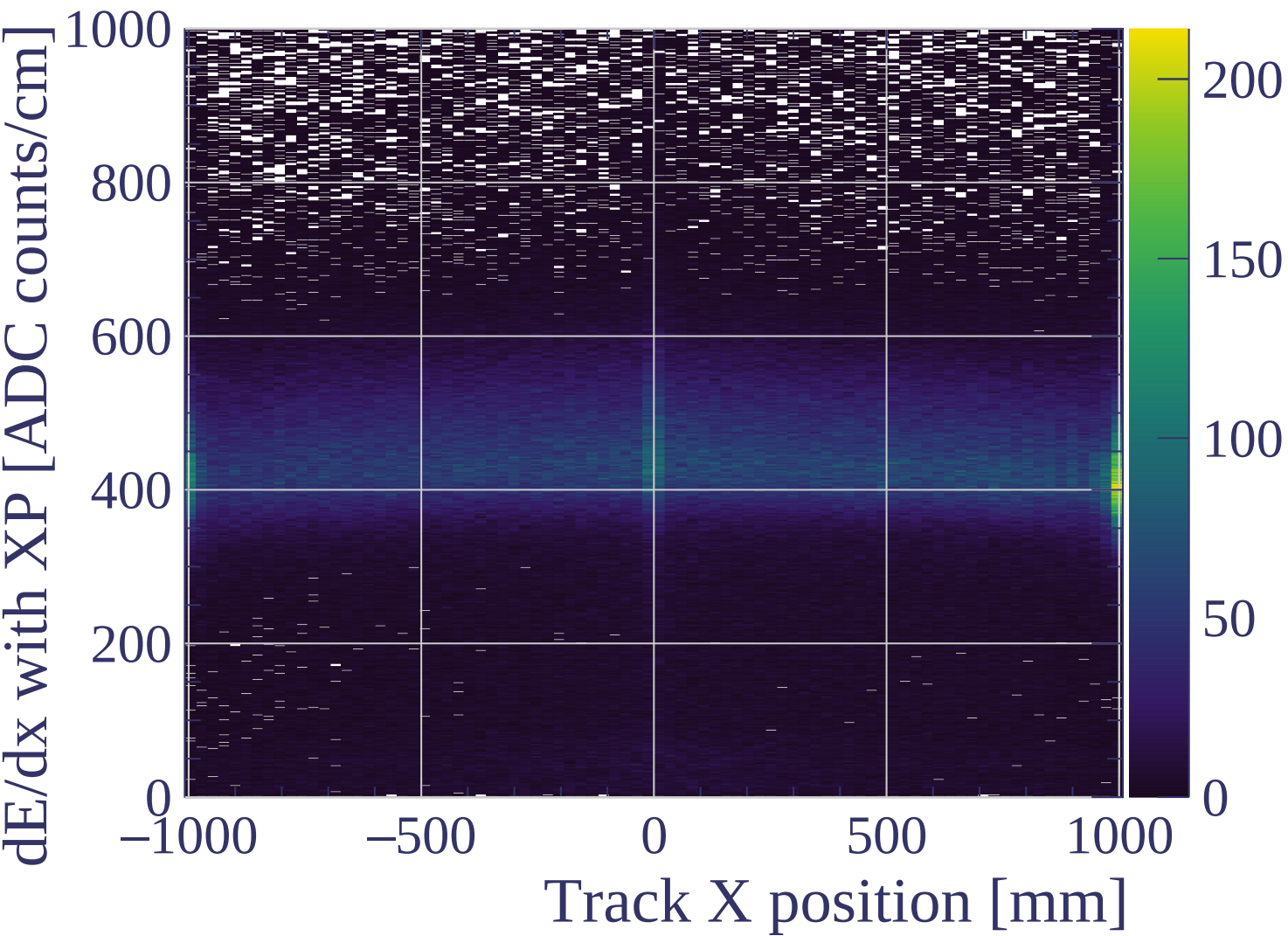


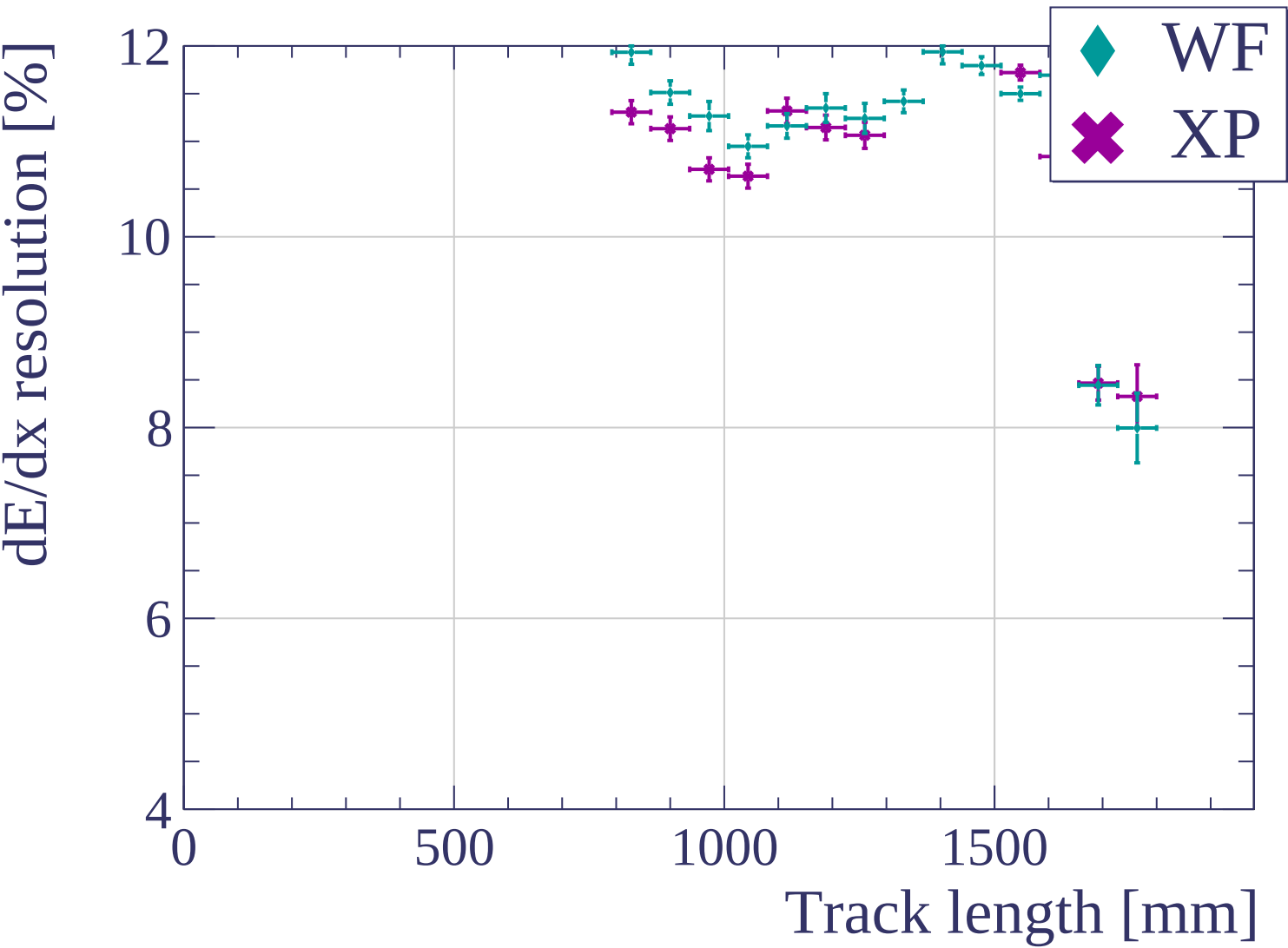


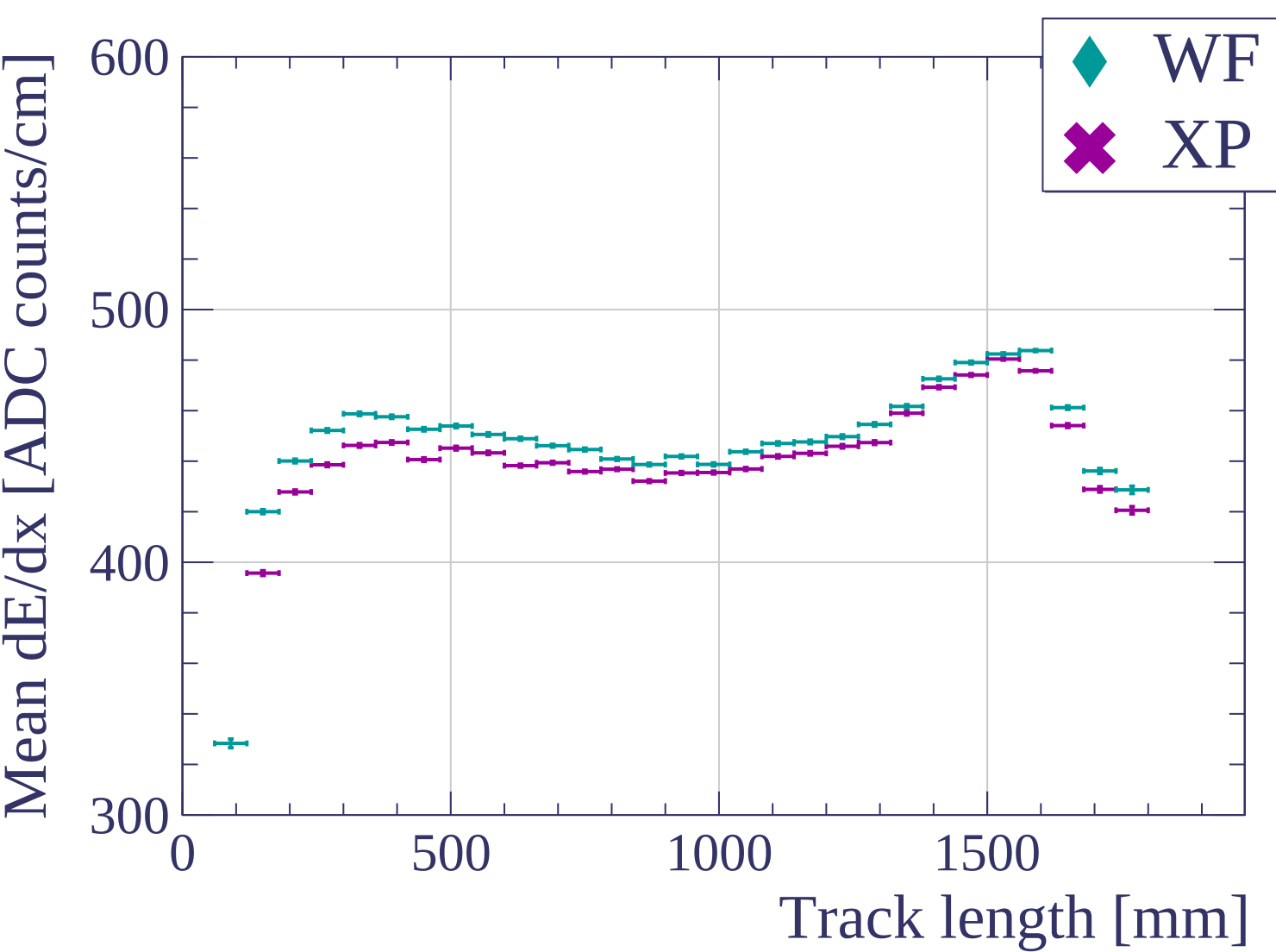


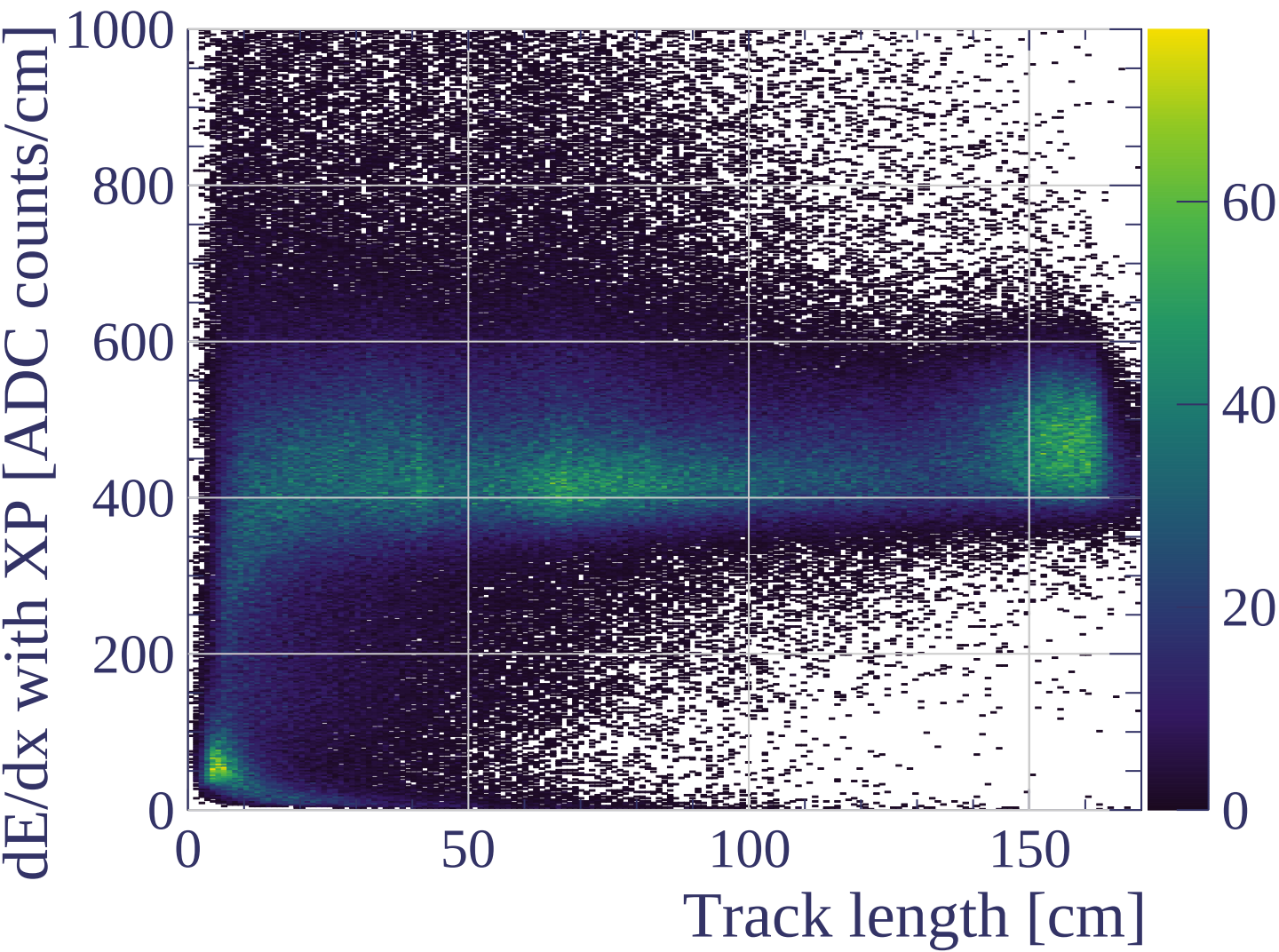


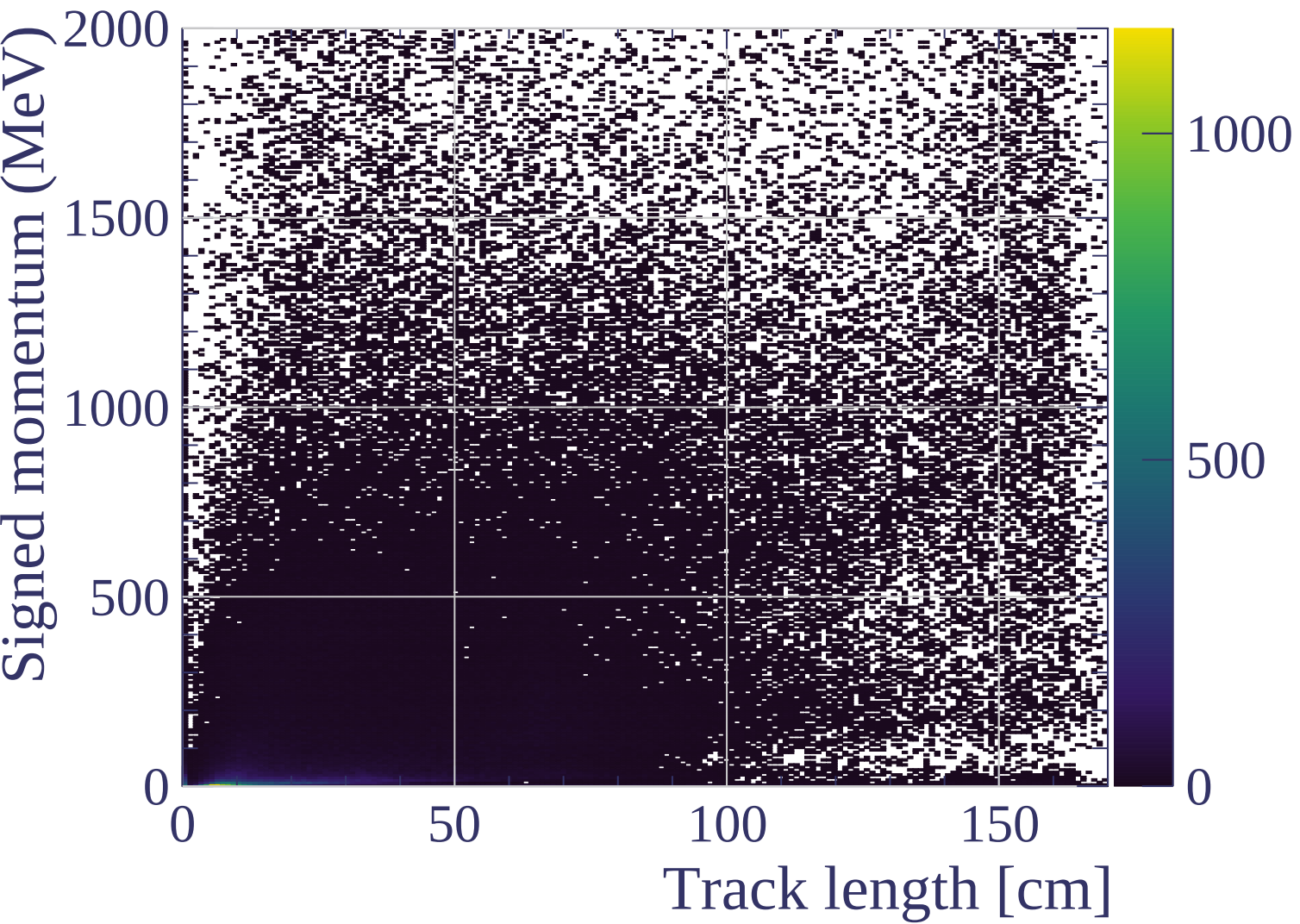


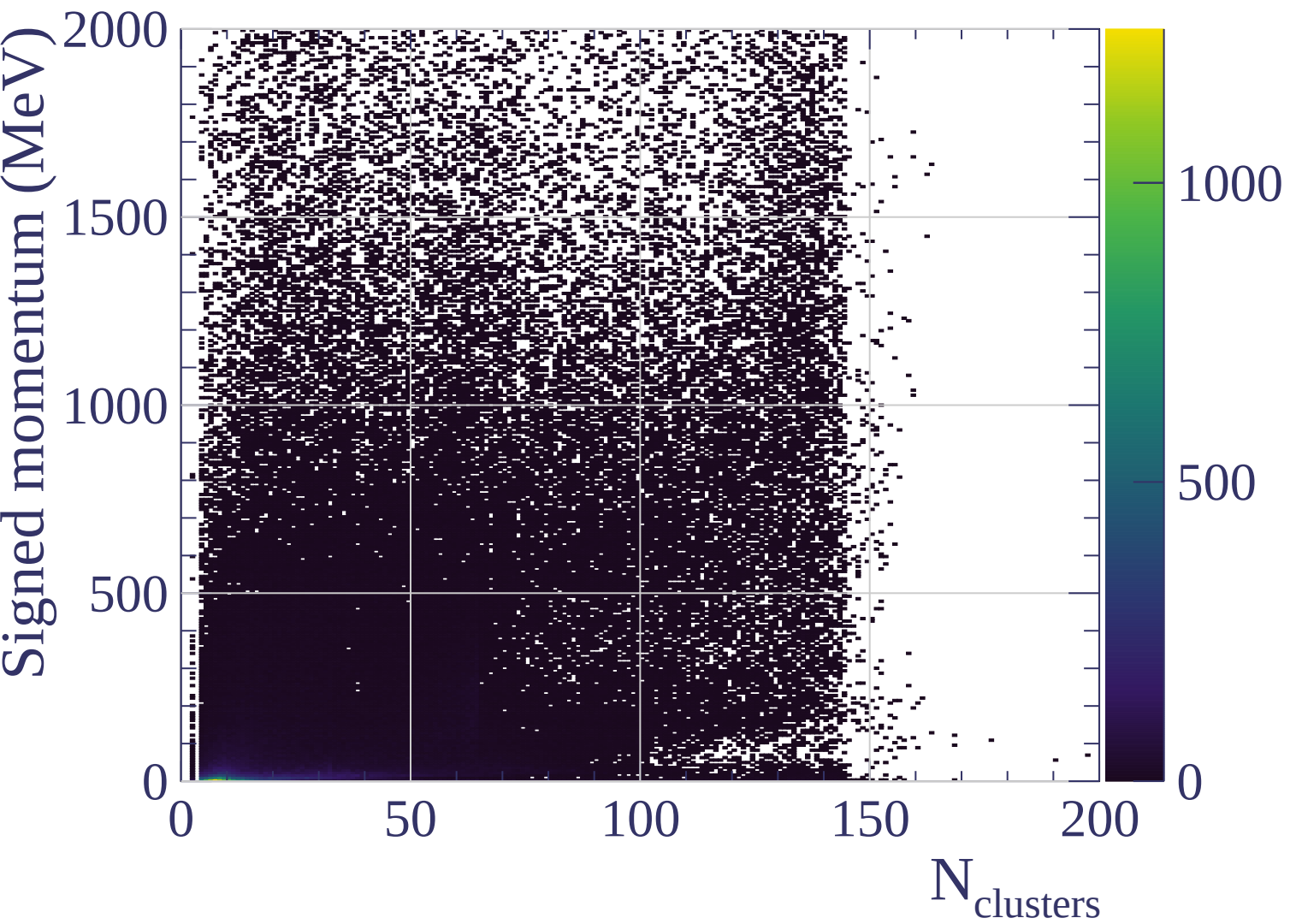


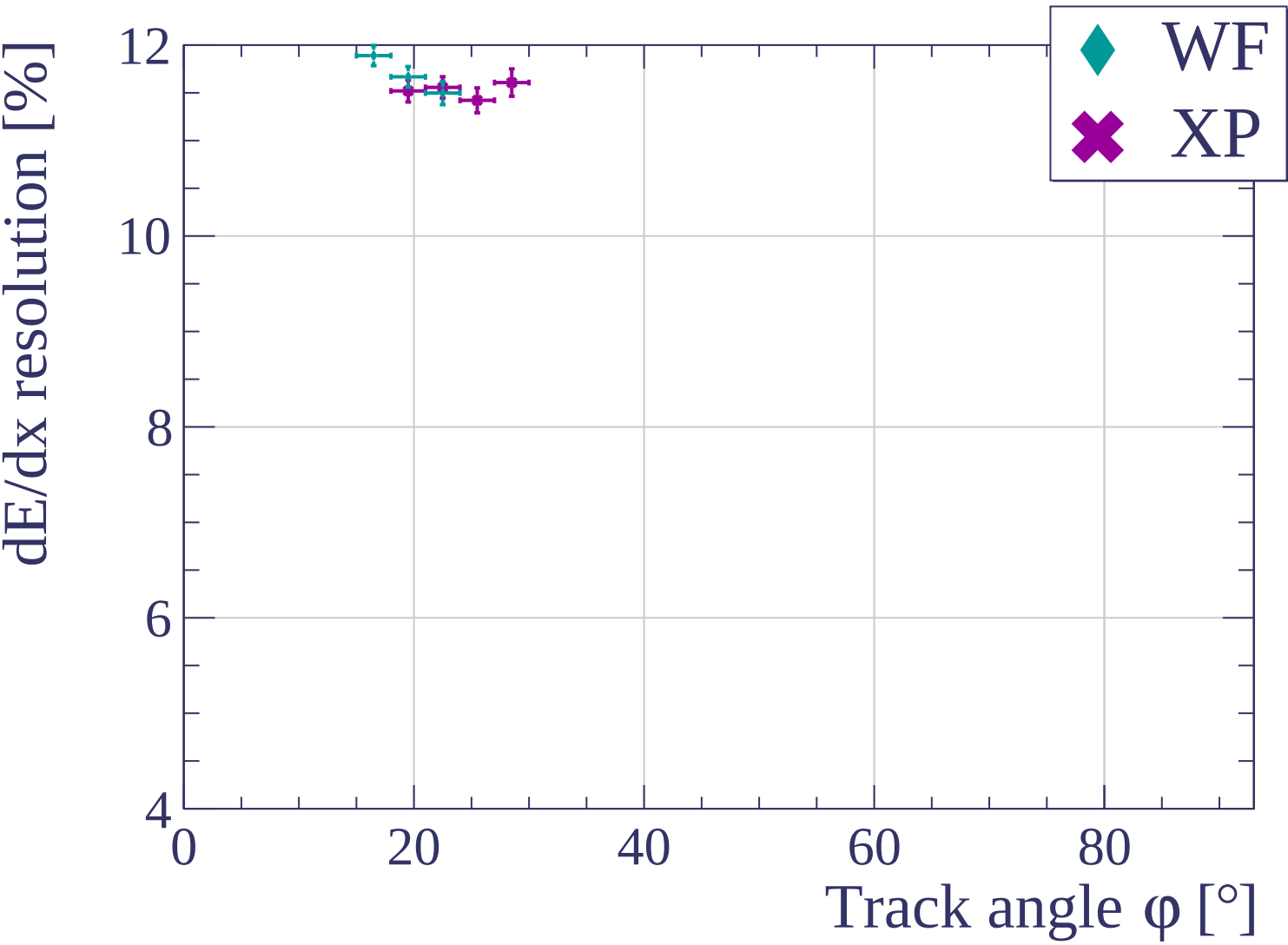


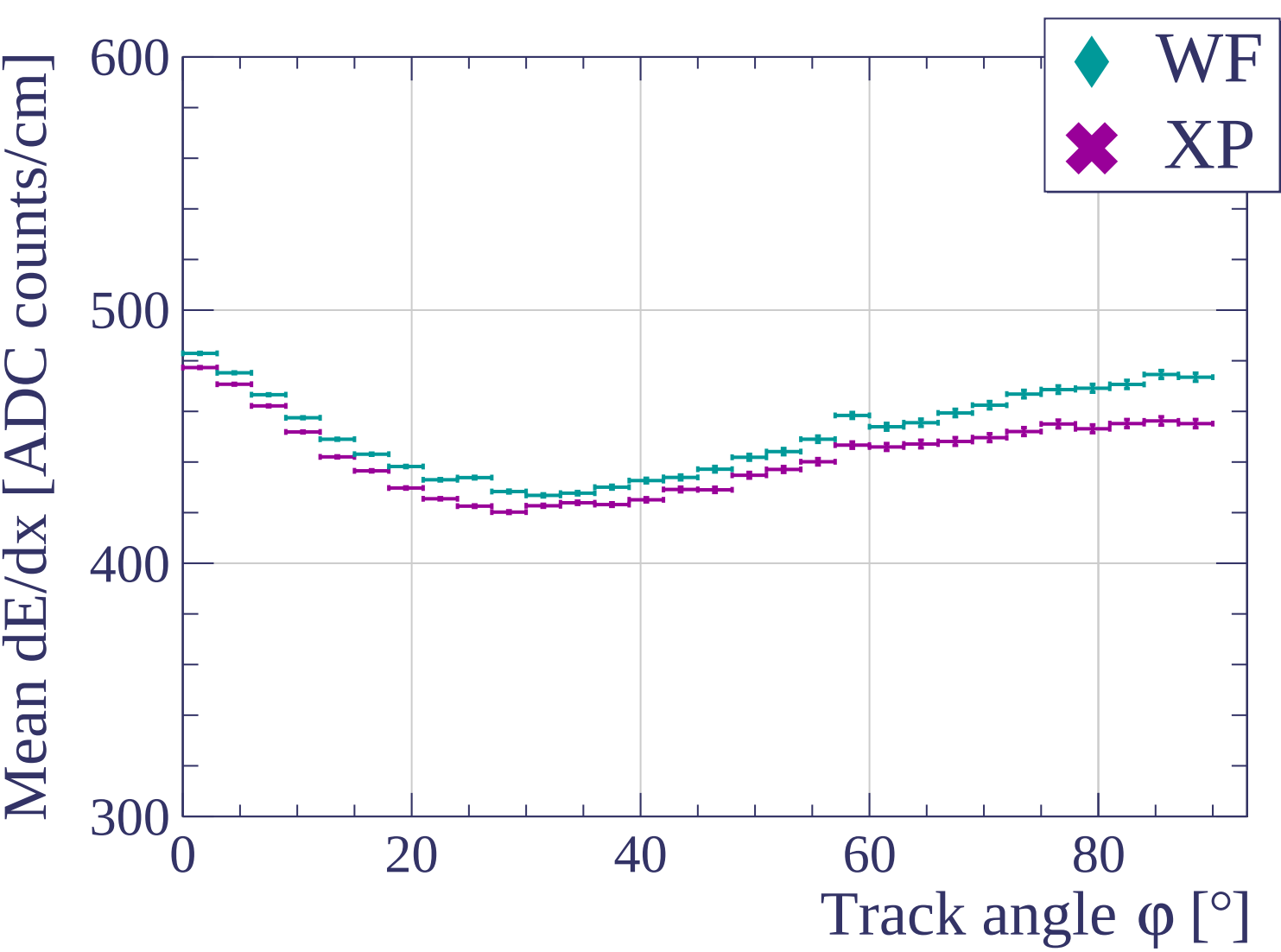


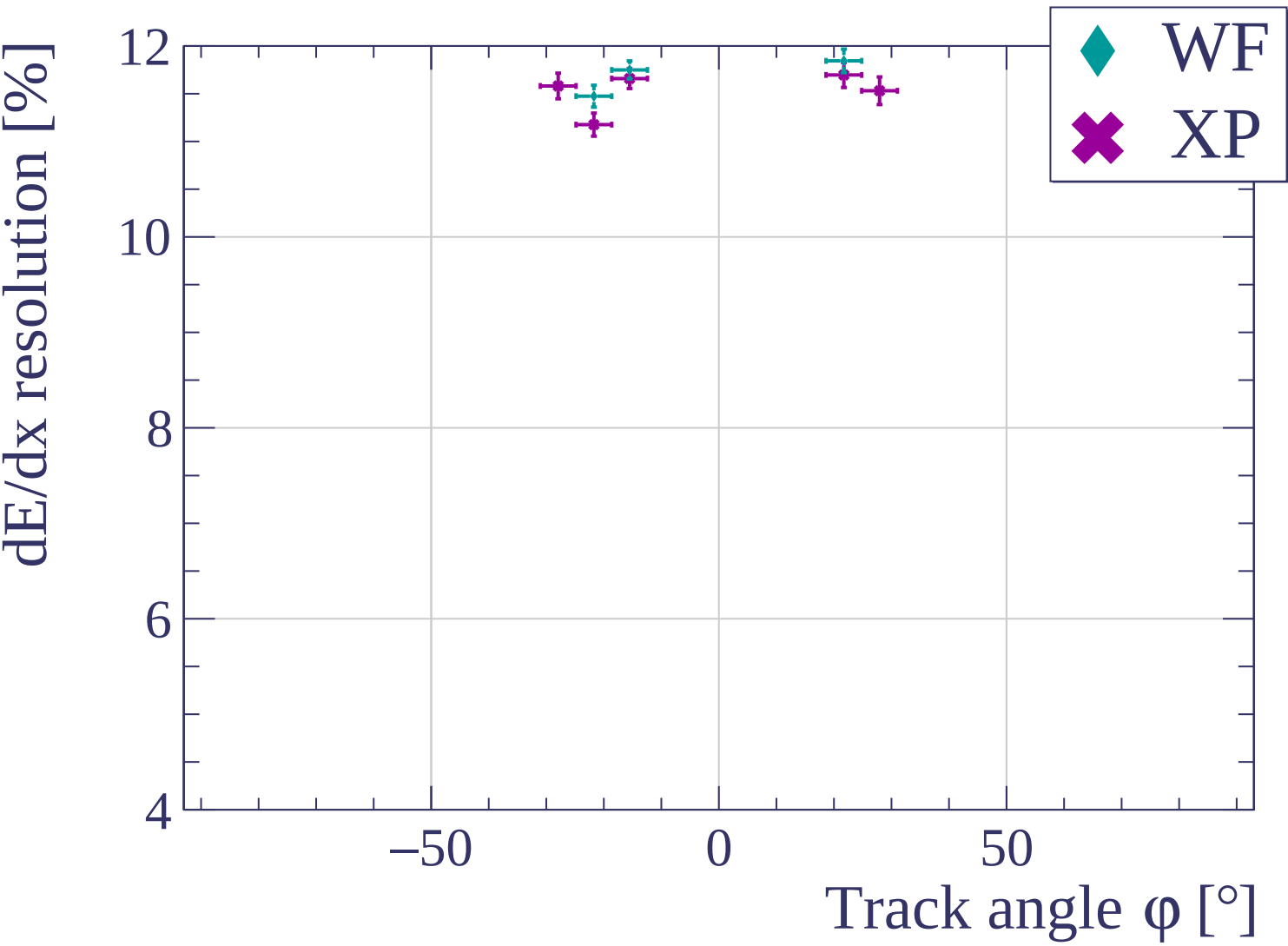


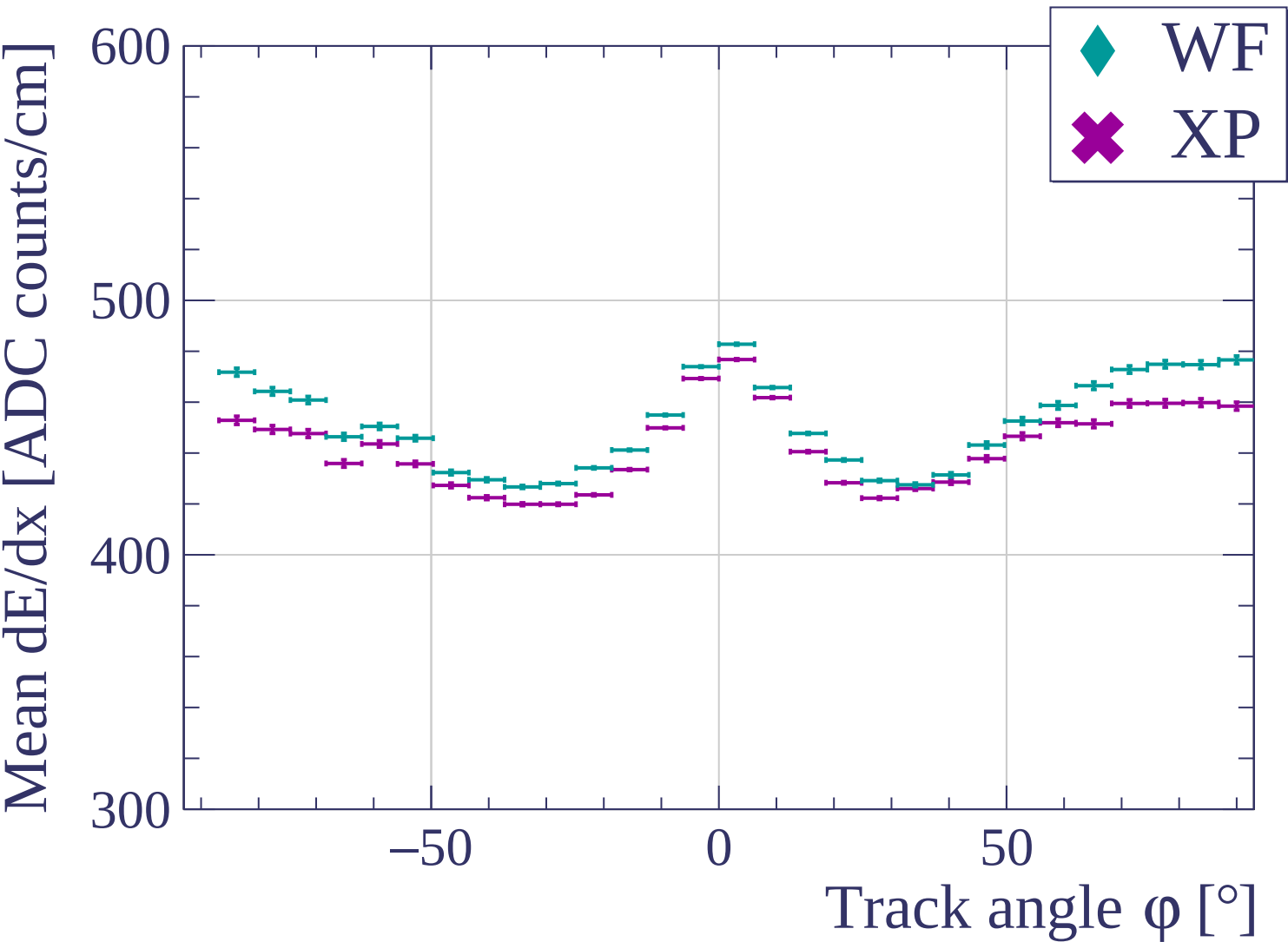


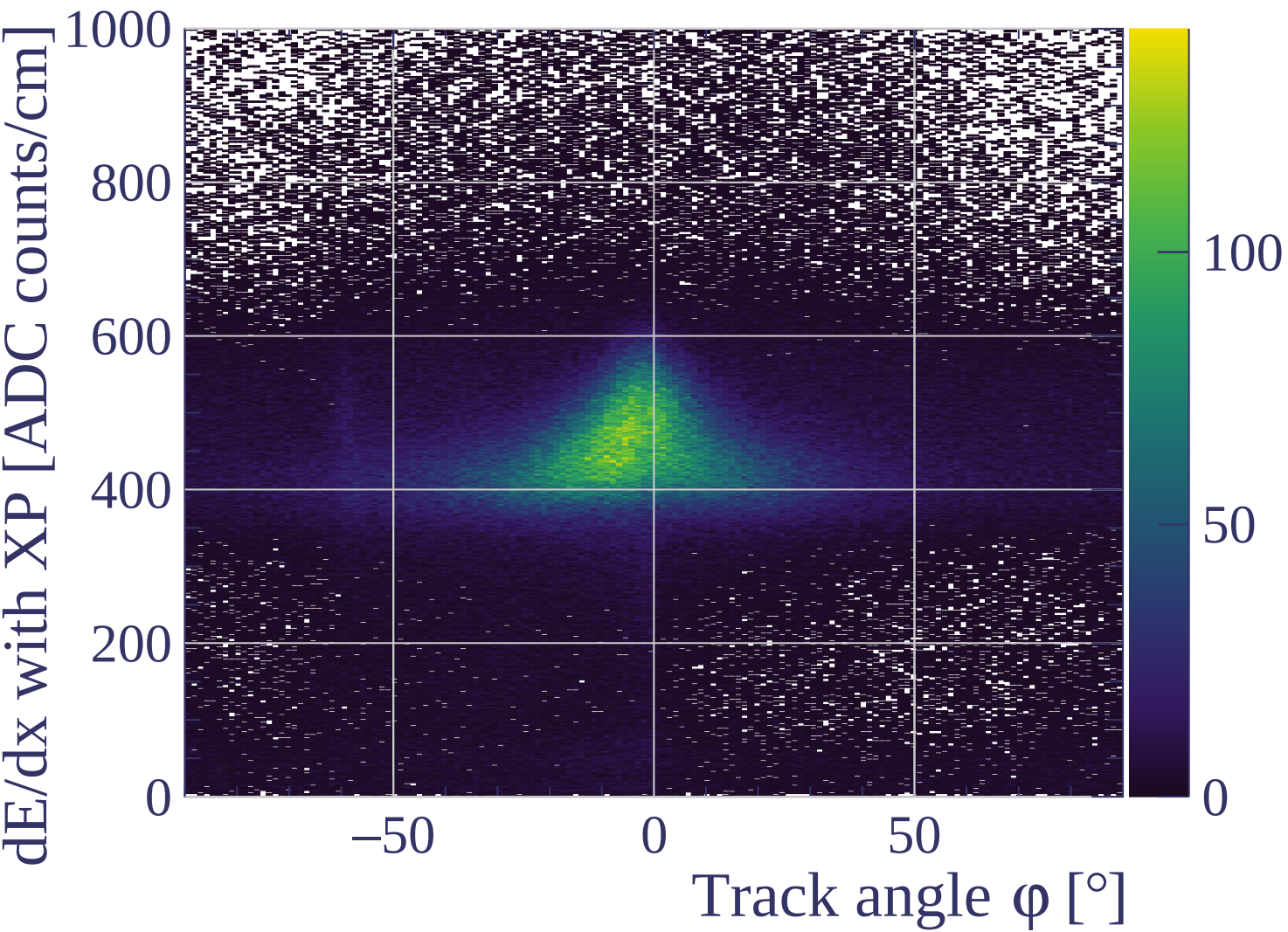




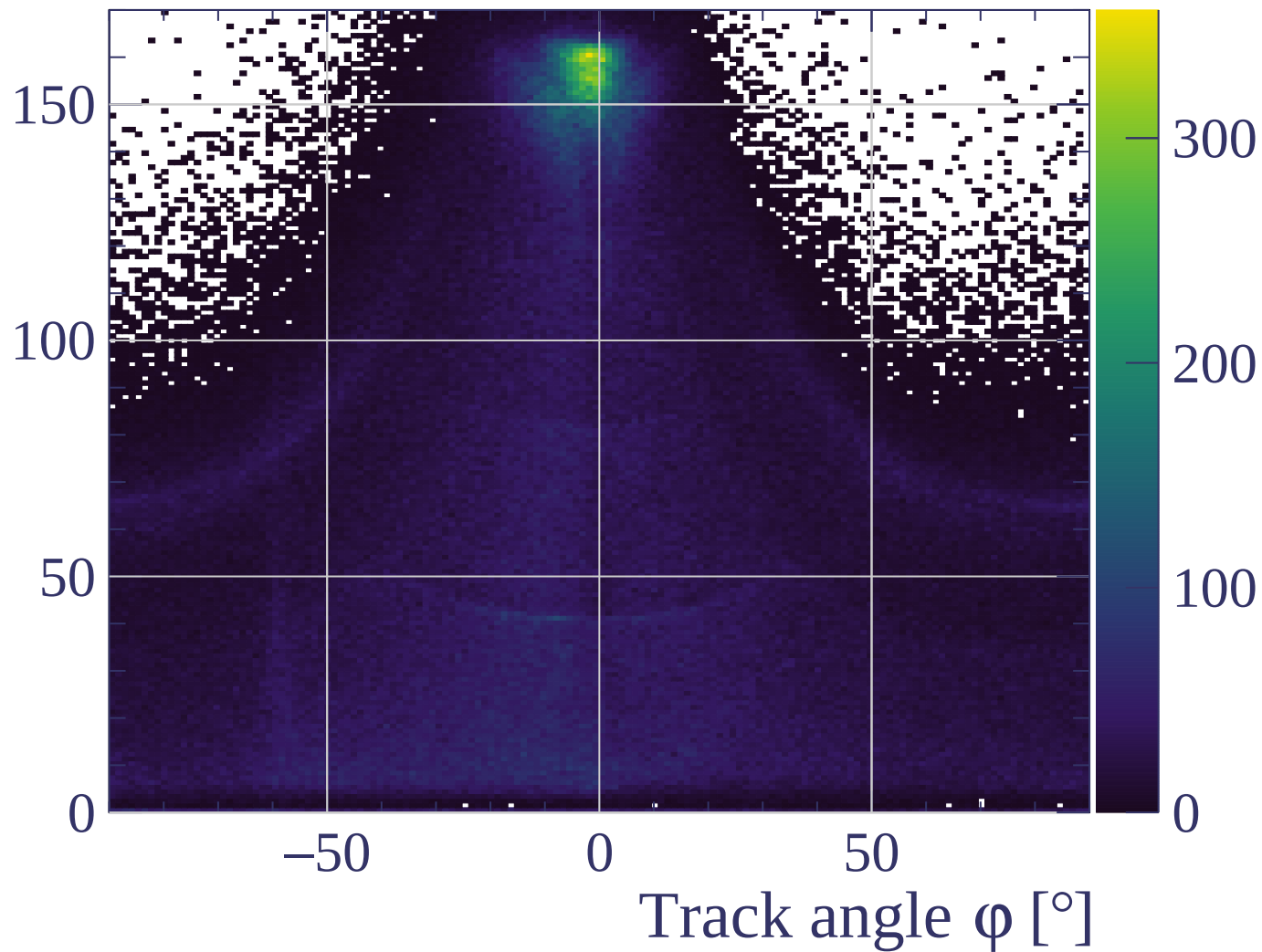


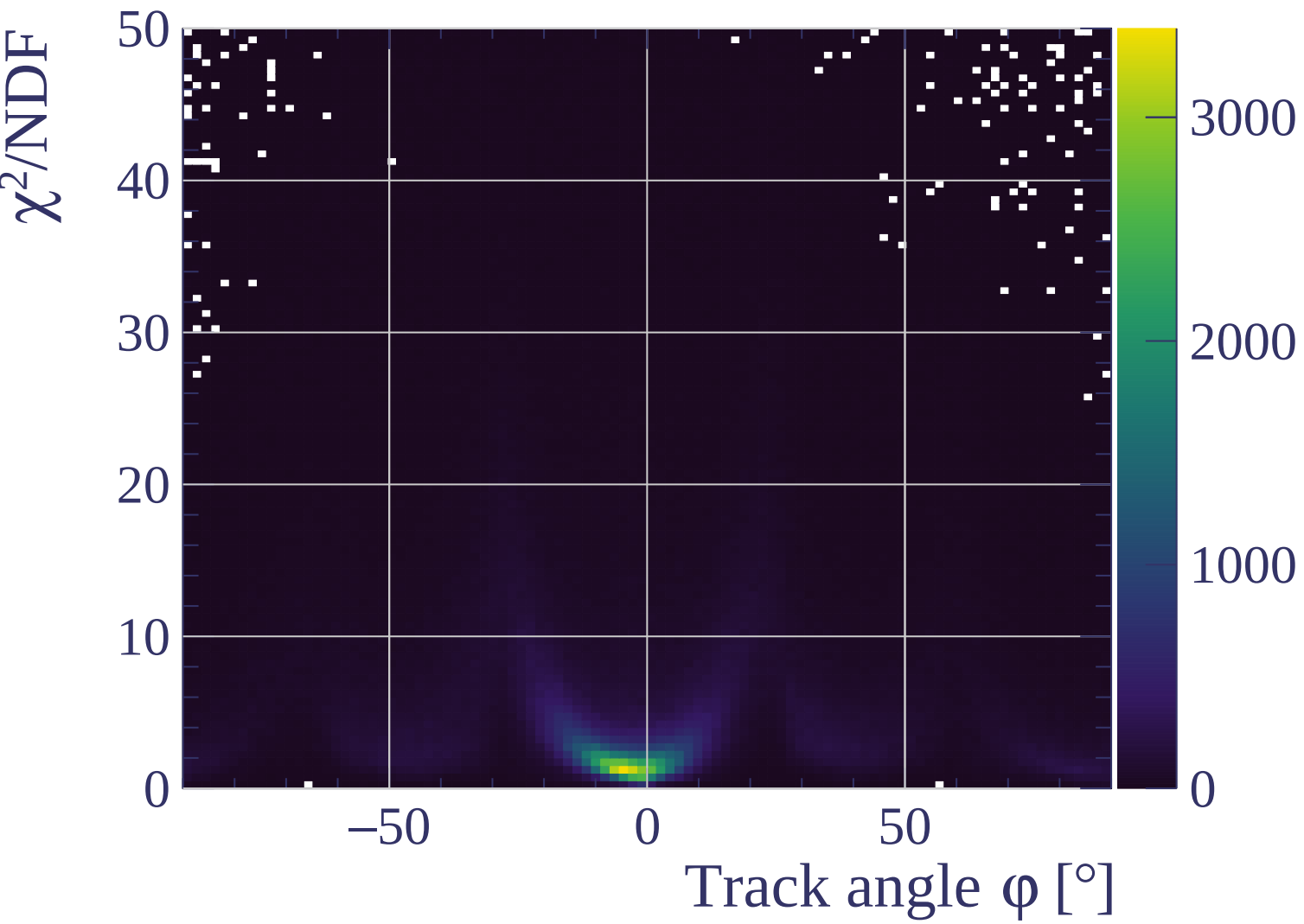


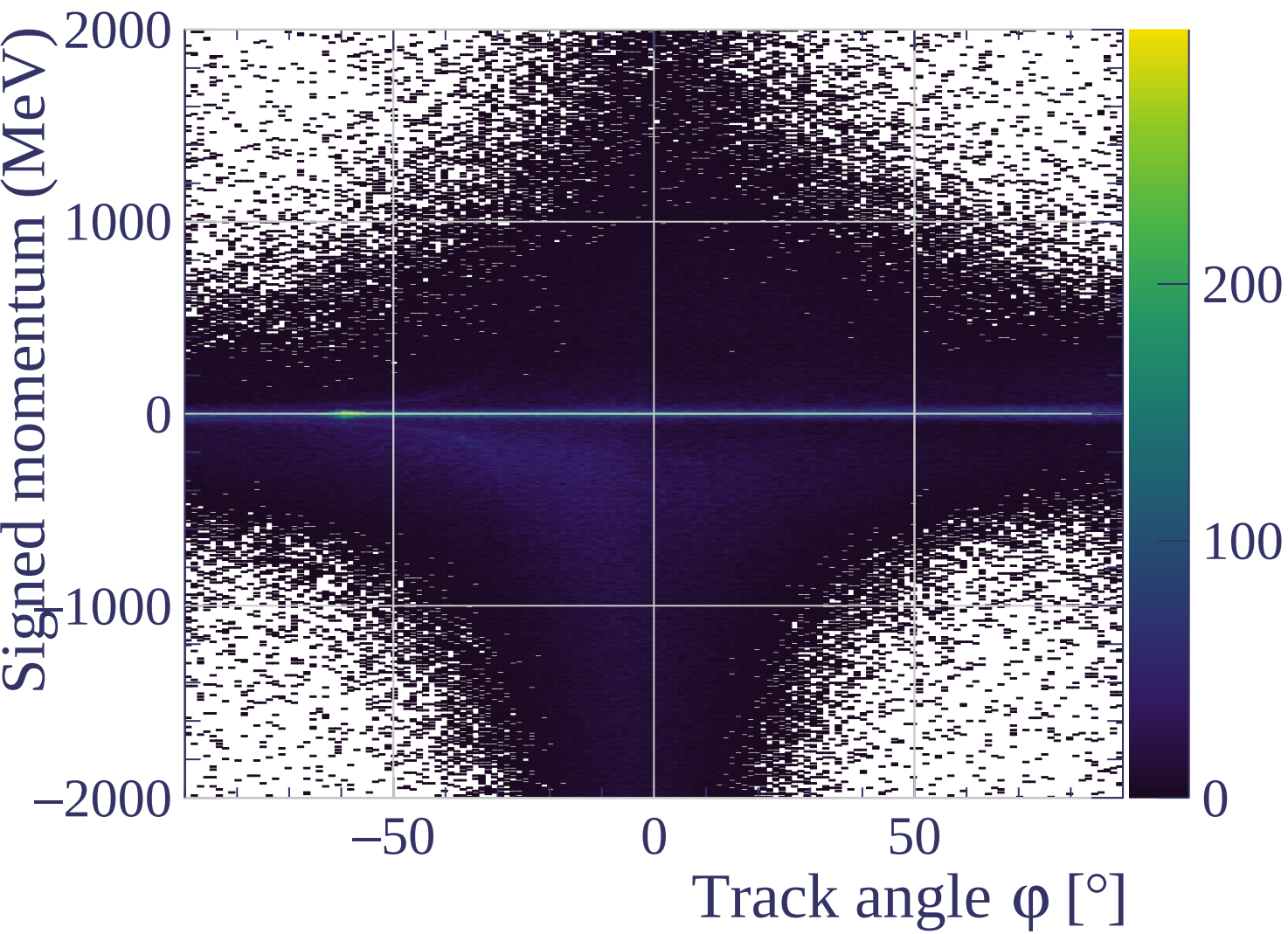


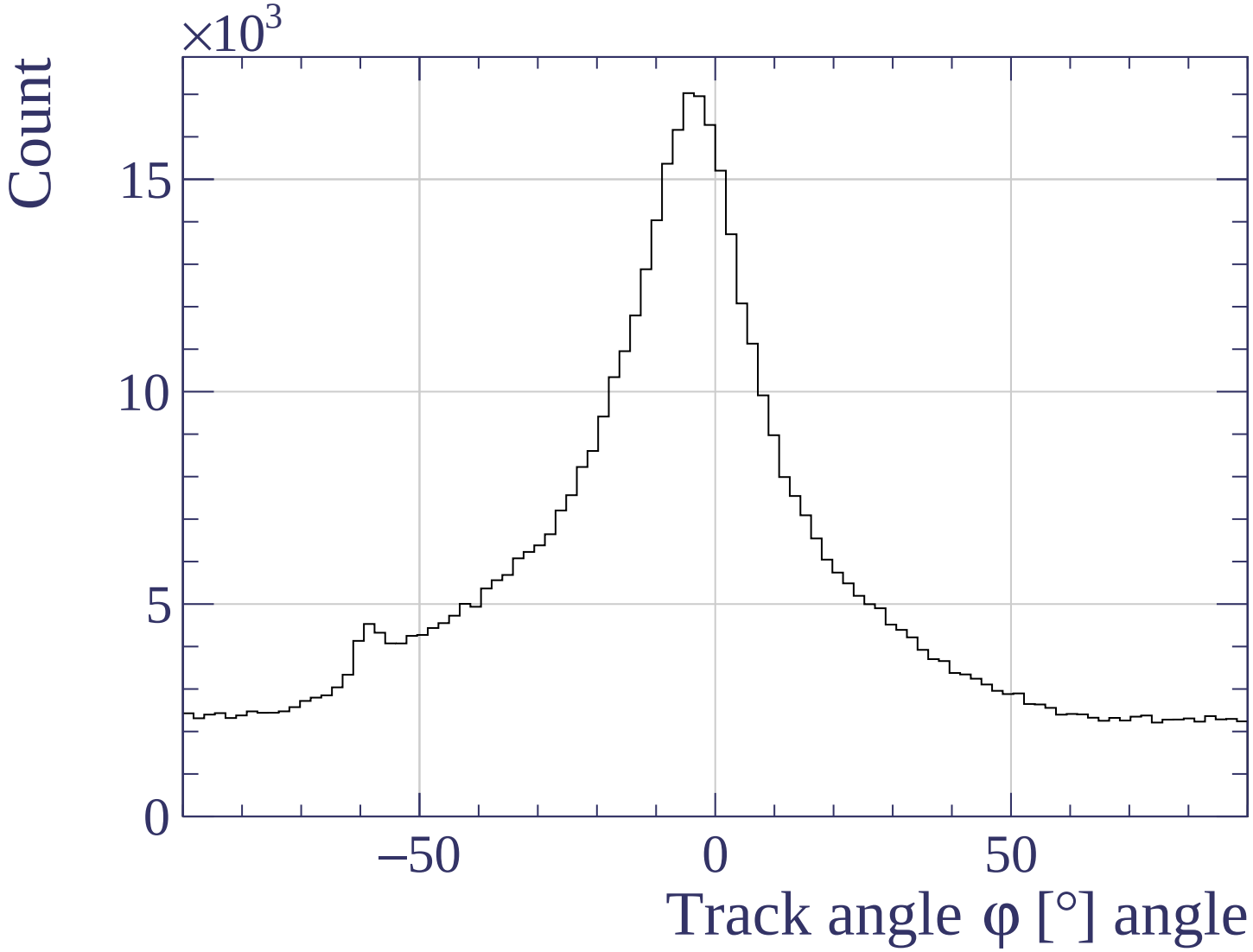


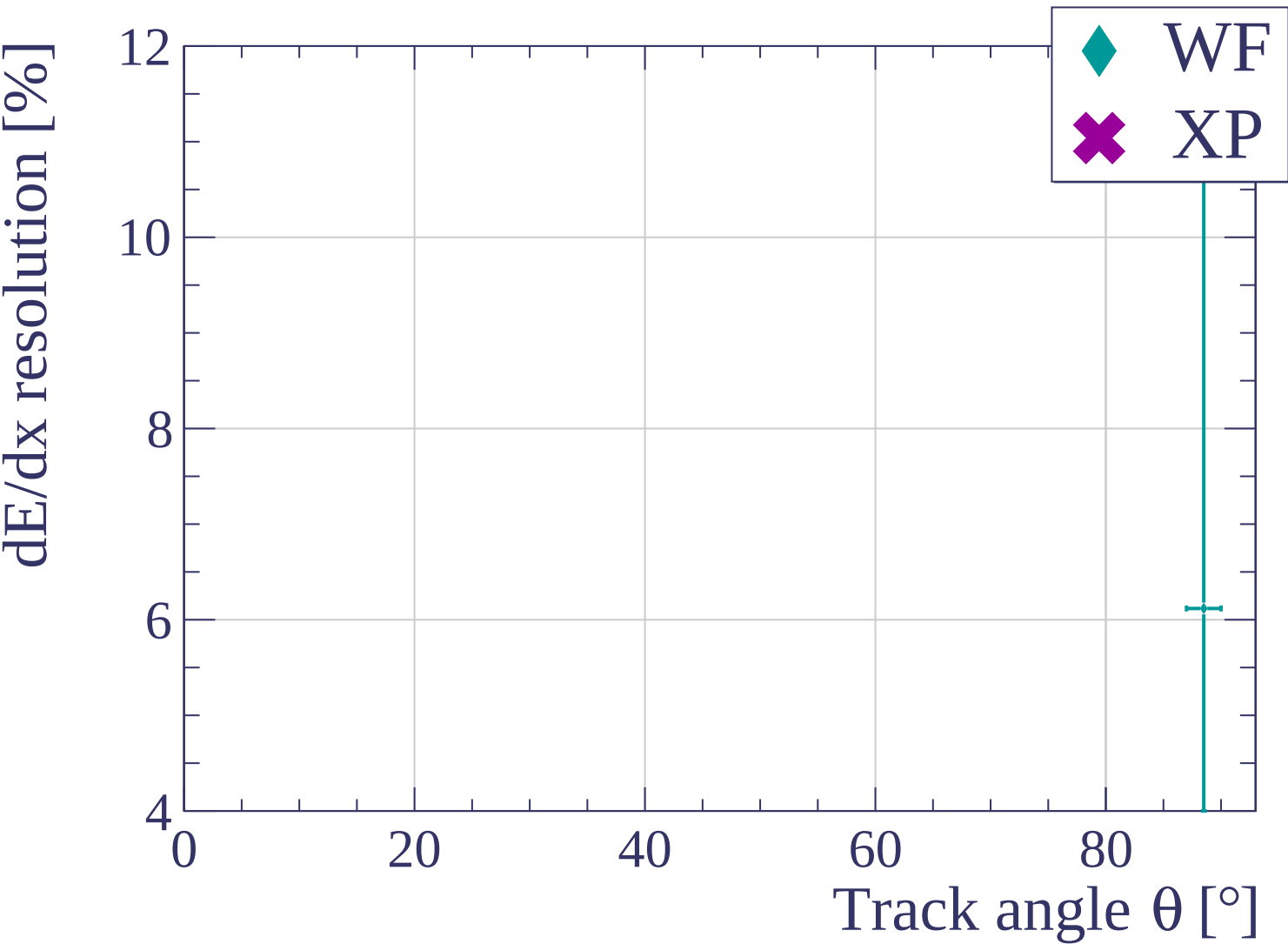
Track length [cm]

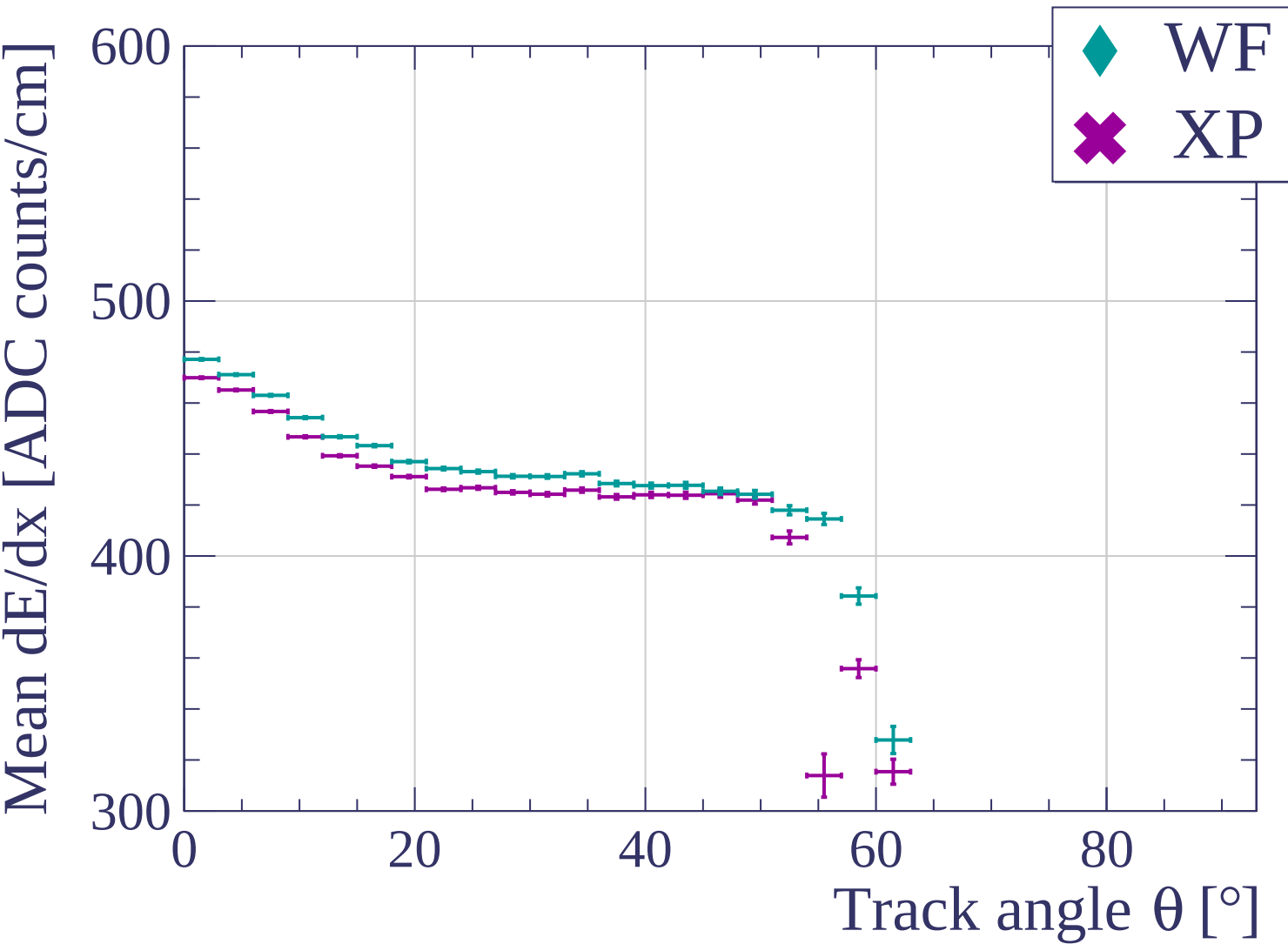


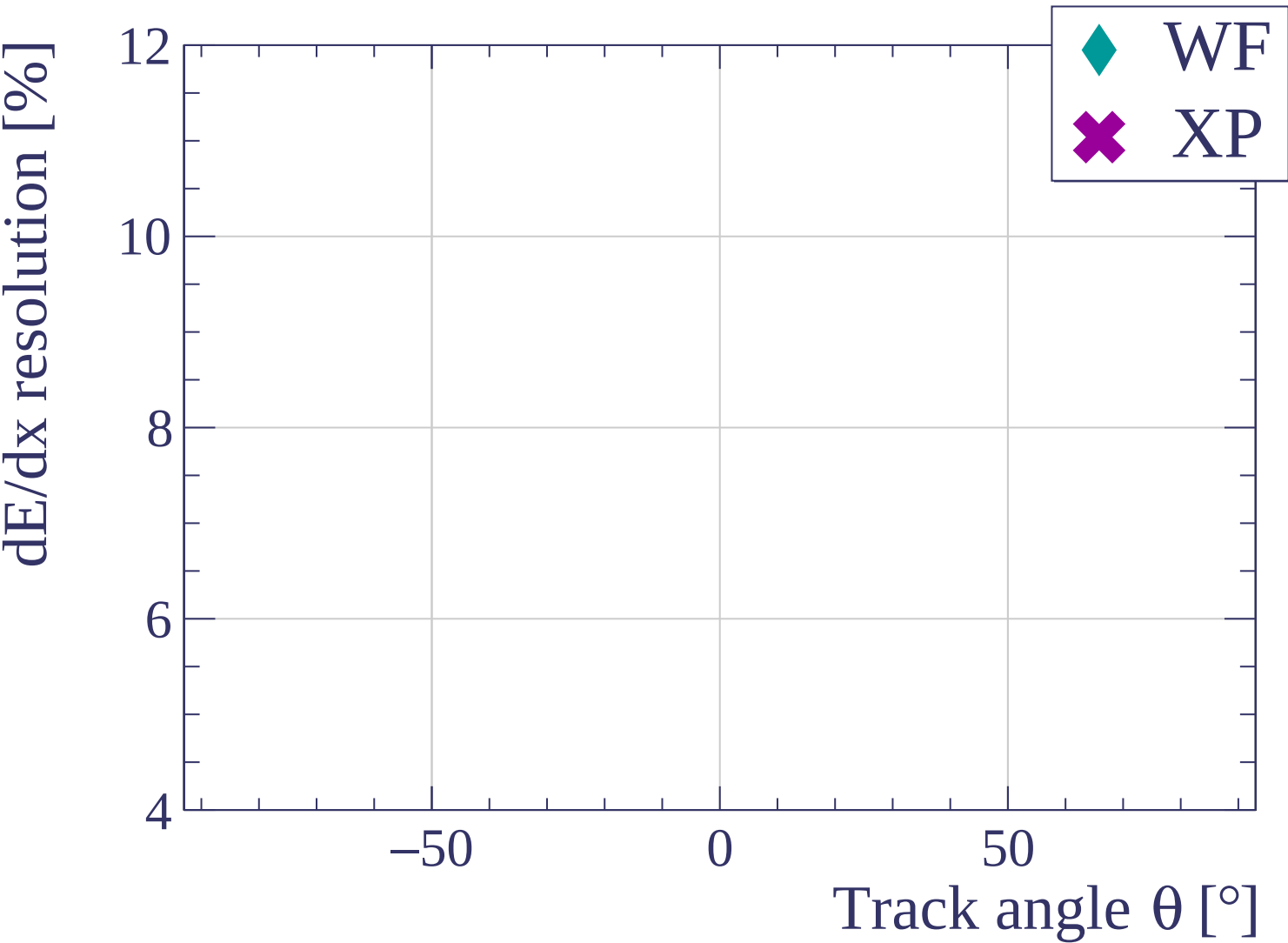


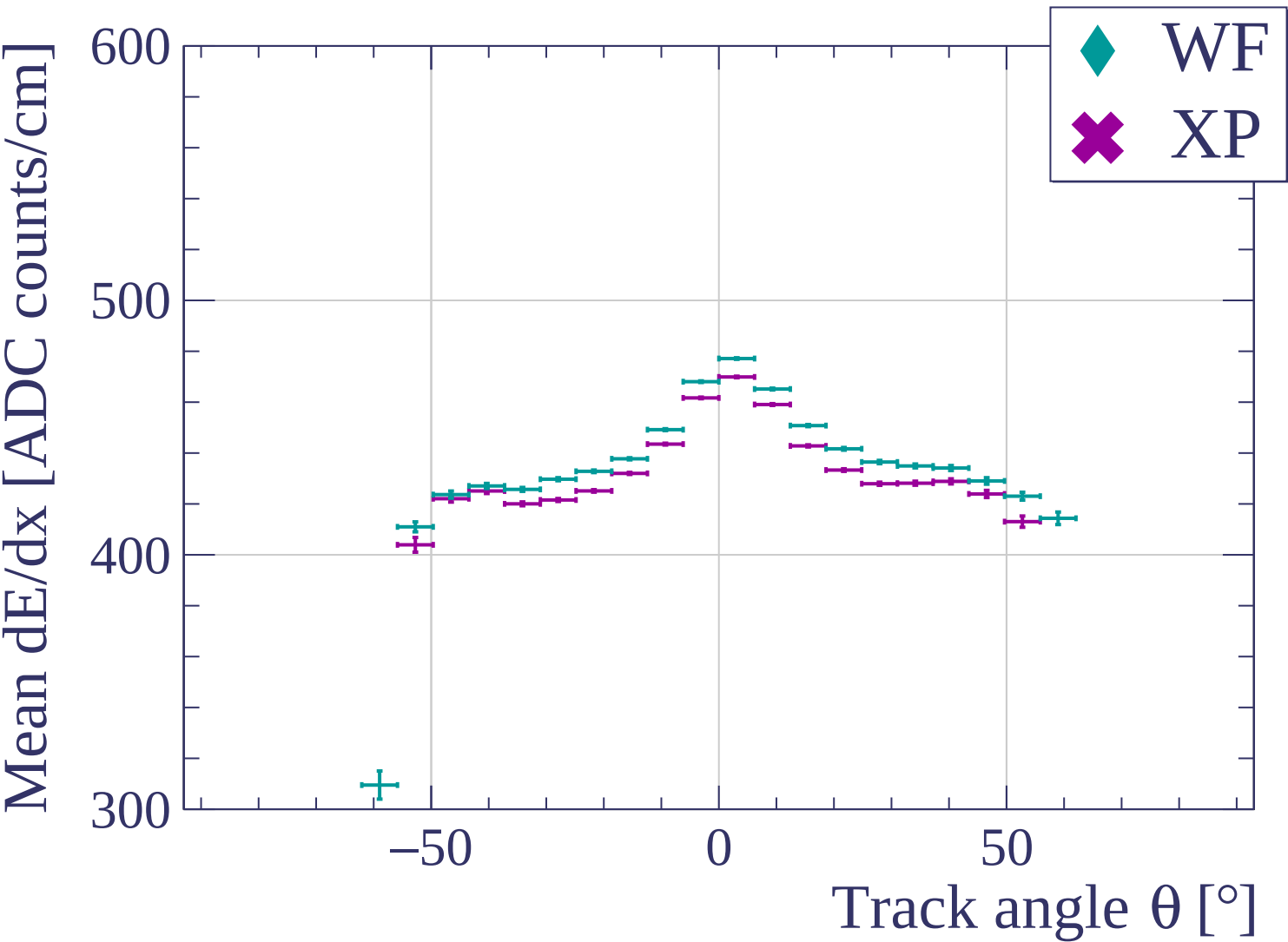


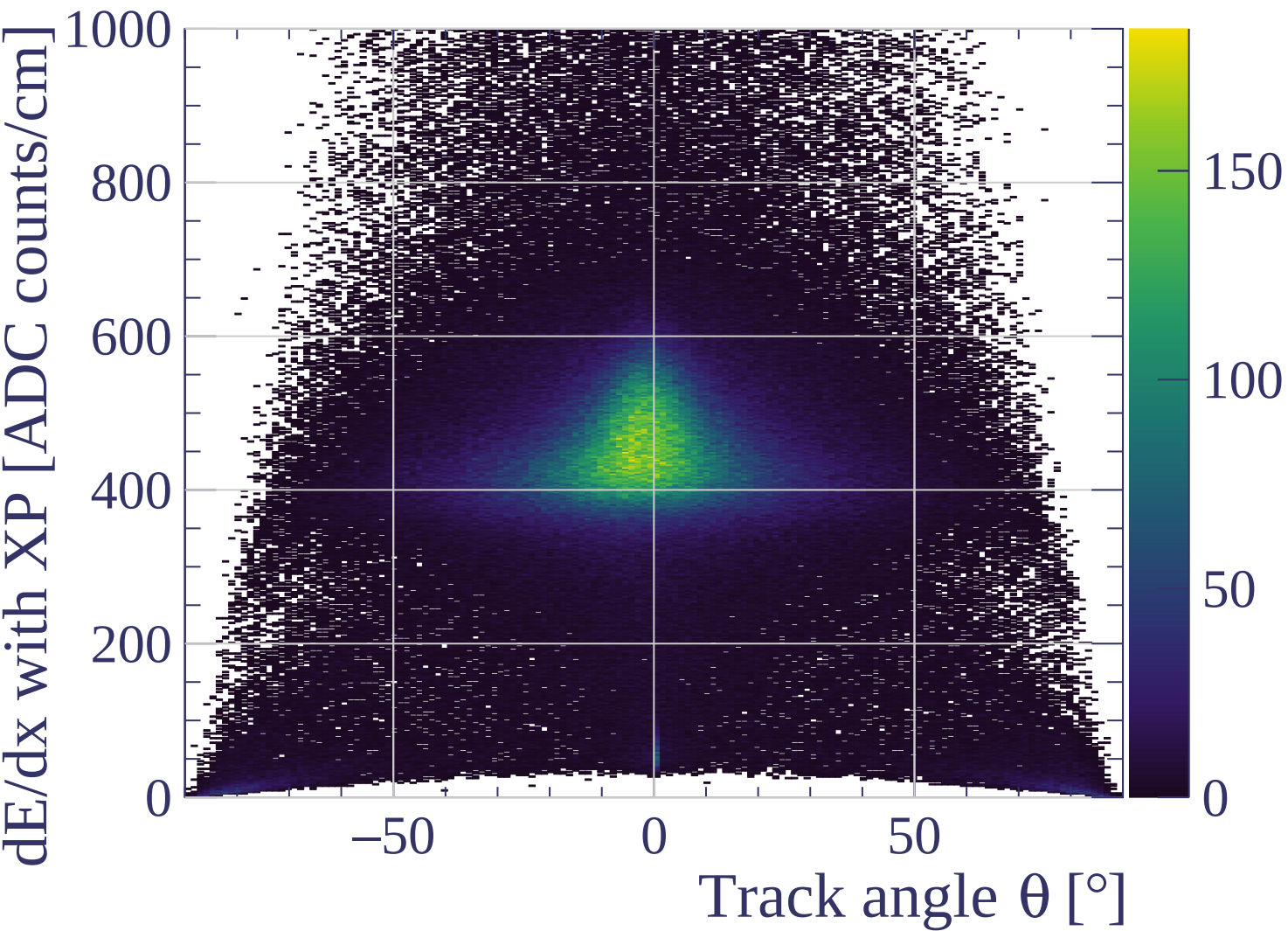




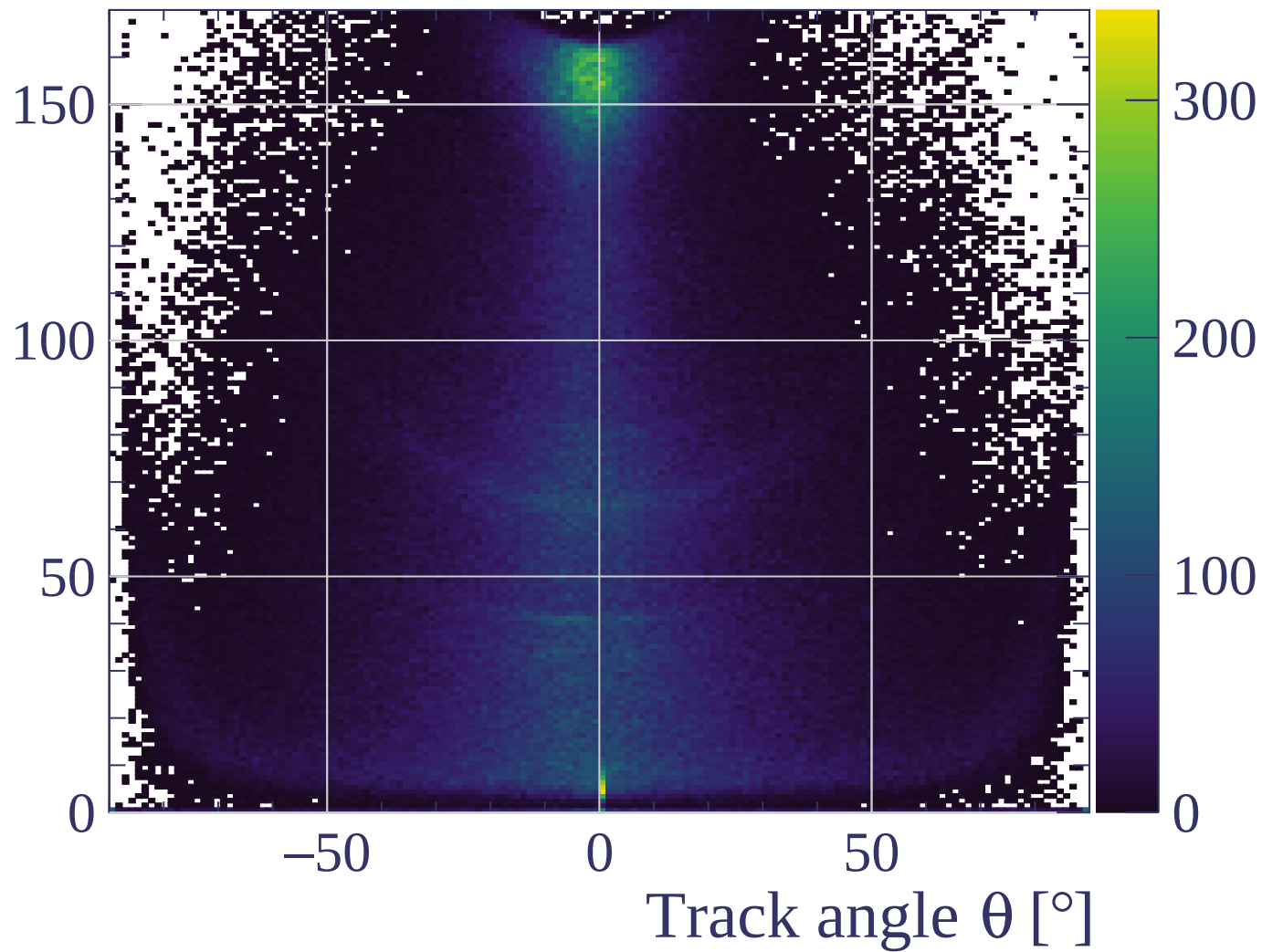


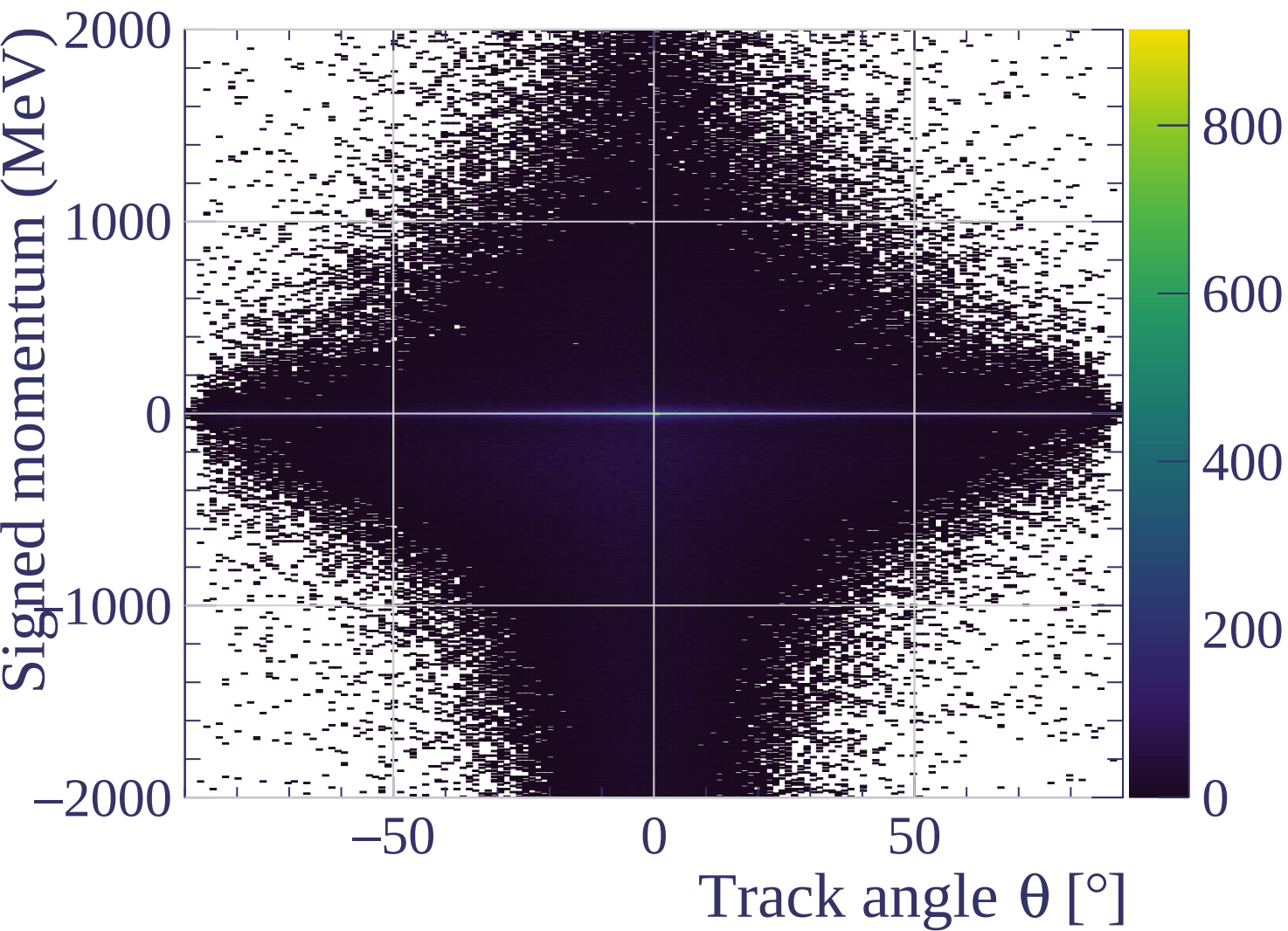


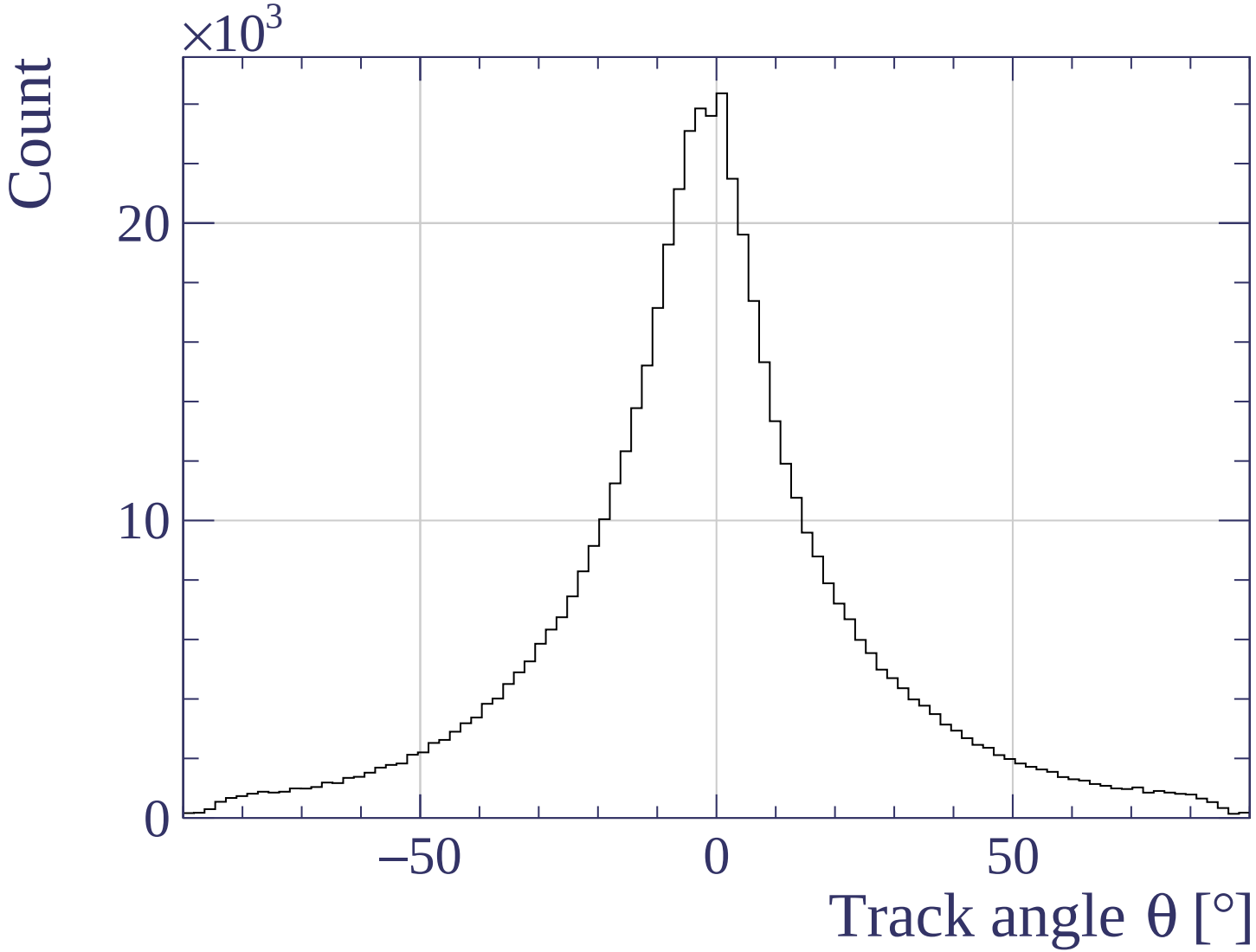




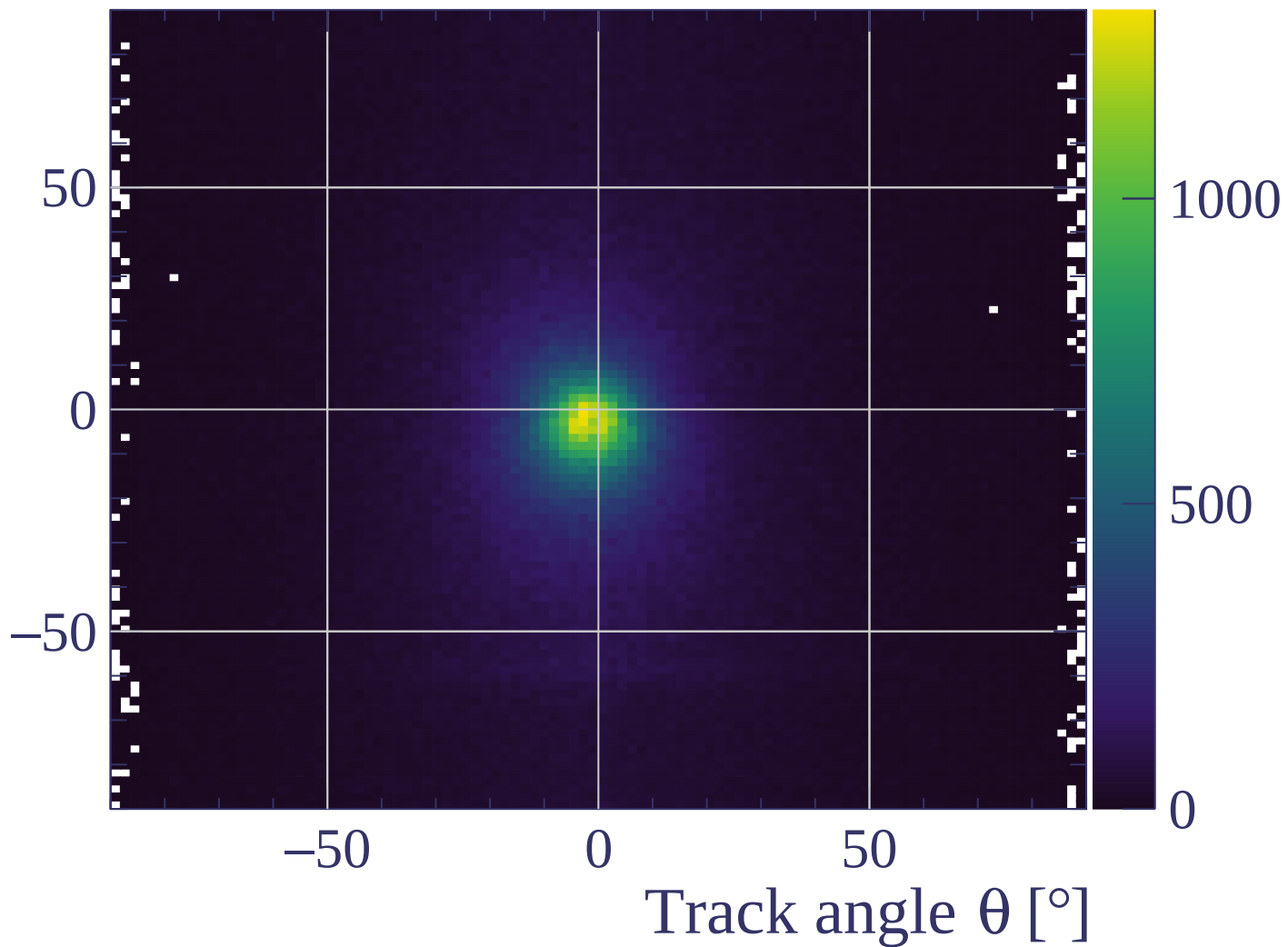
Track length [cm]



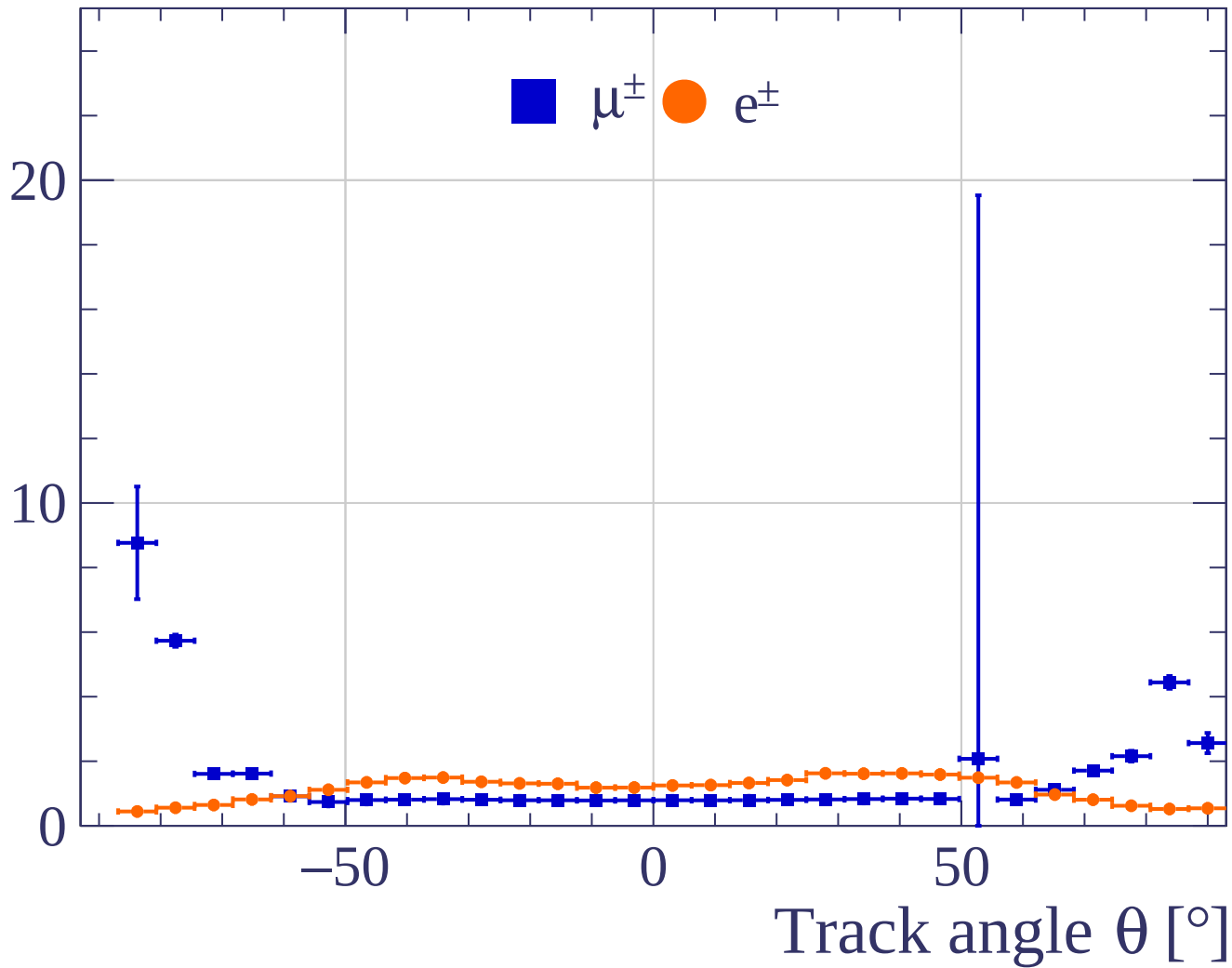




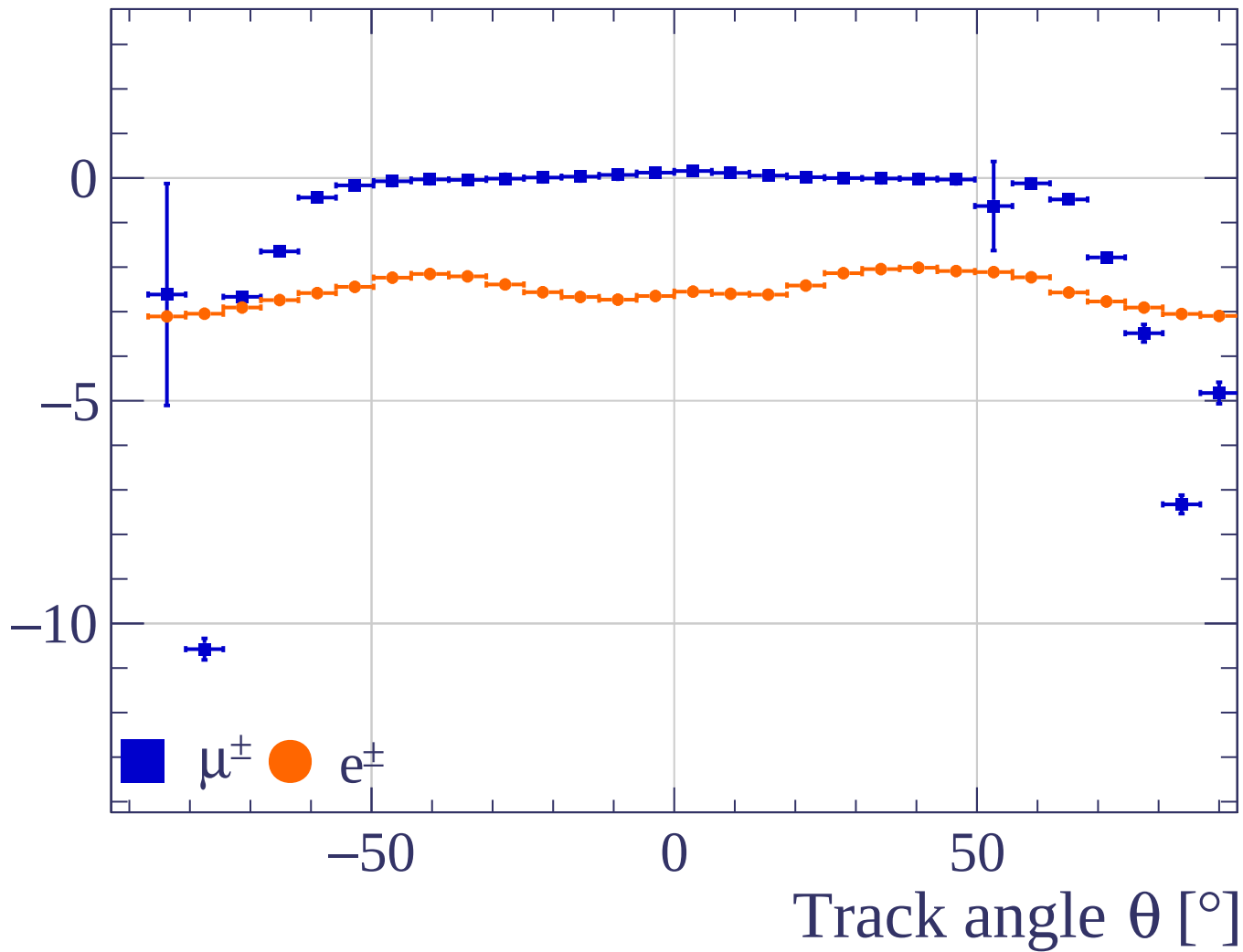
Track angle ϕ [°]



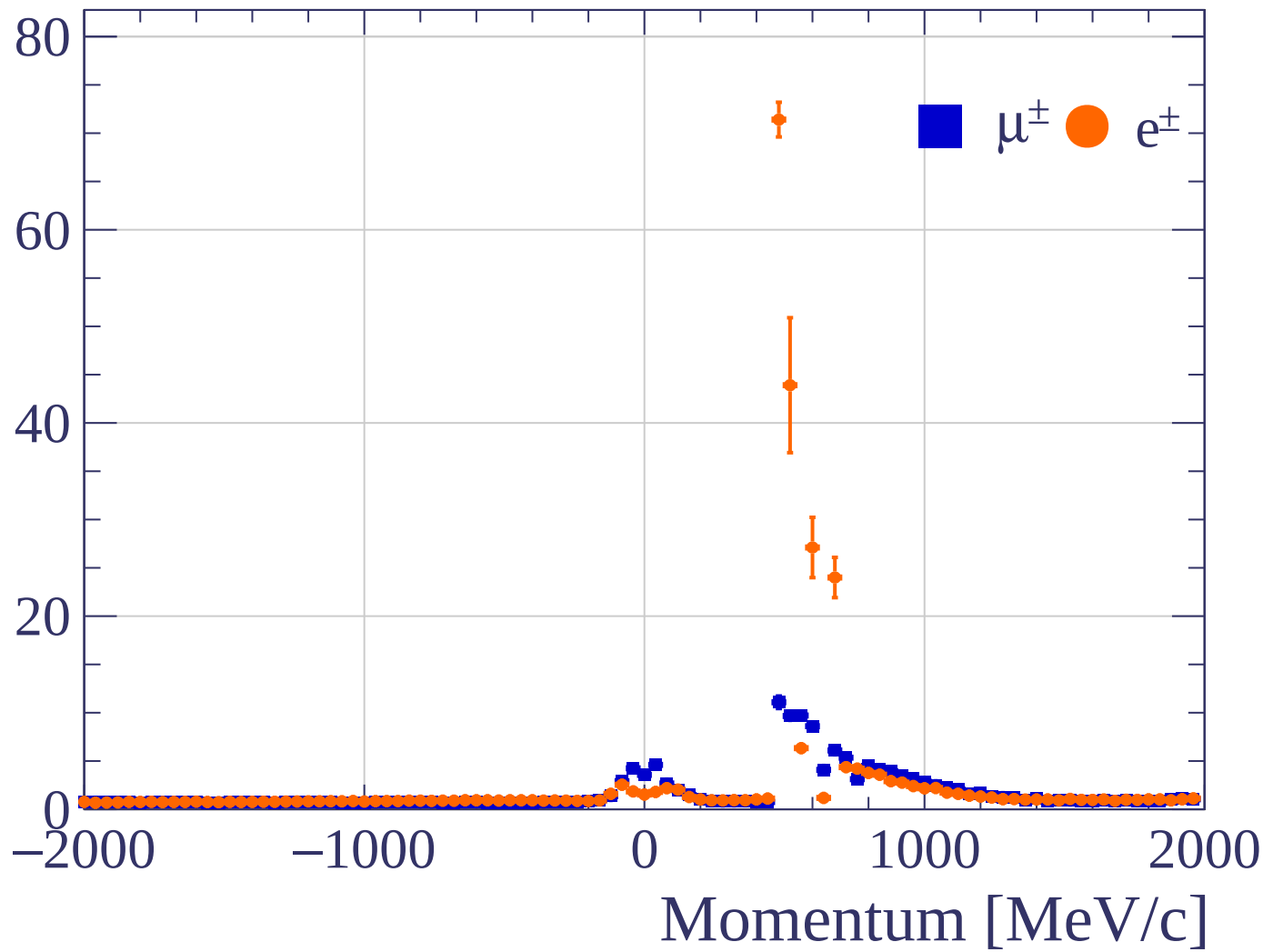
Pulls standard deviation



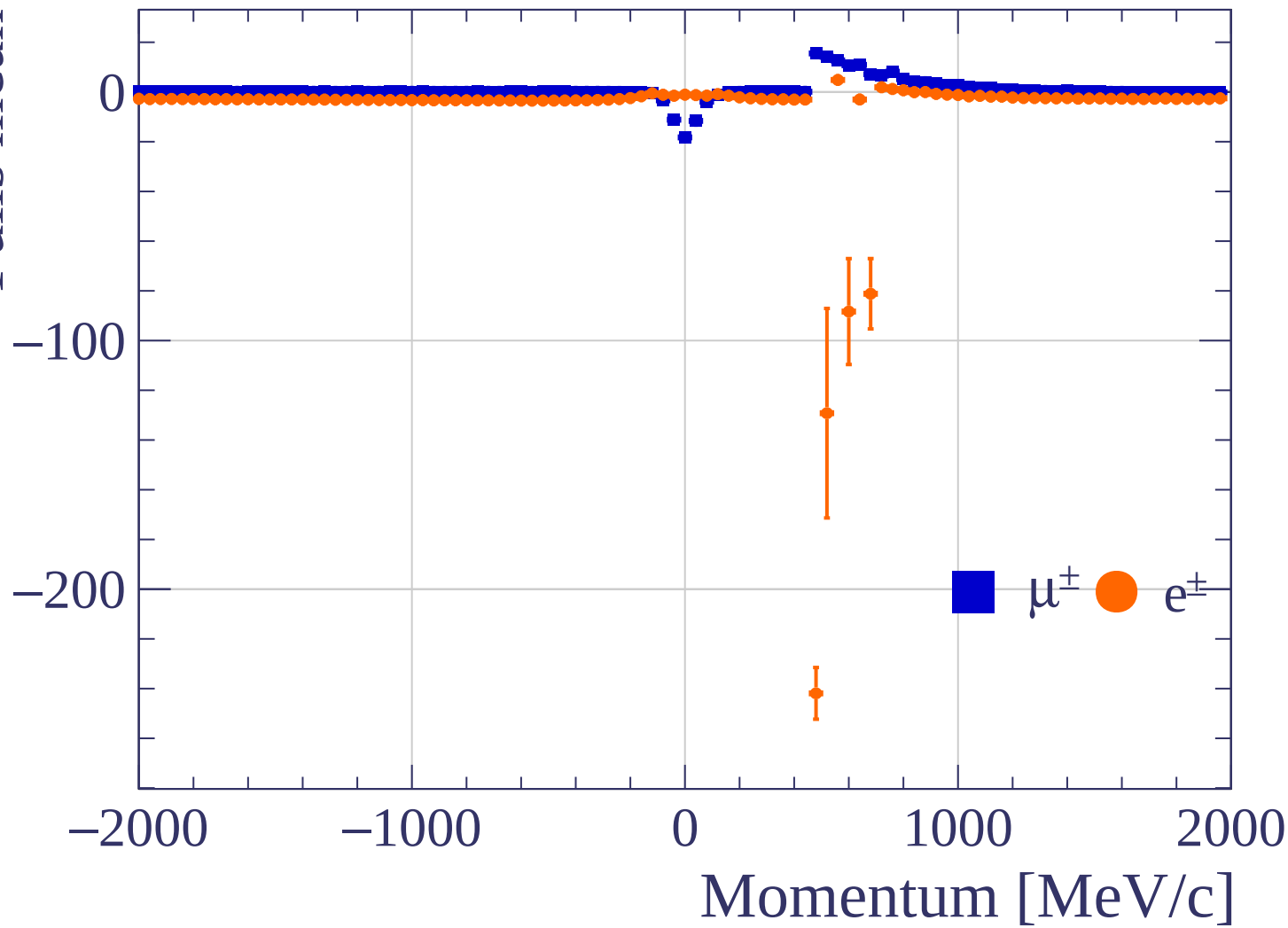
Pulls mean

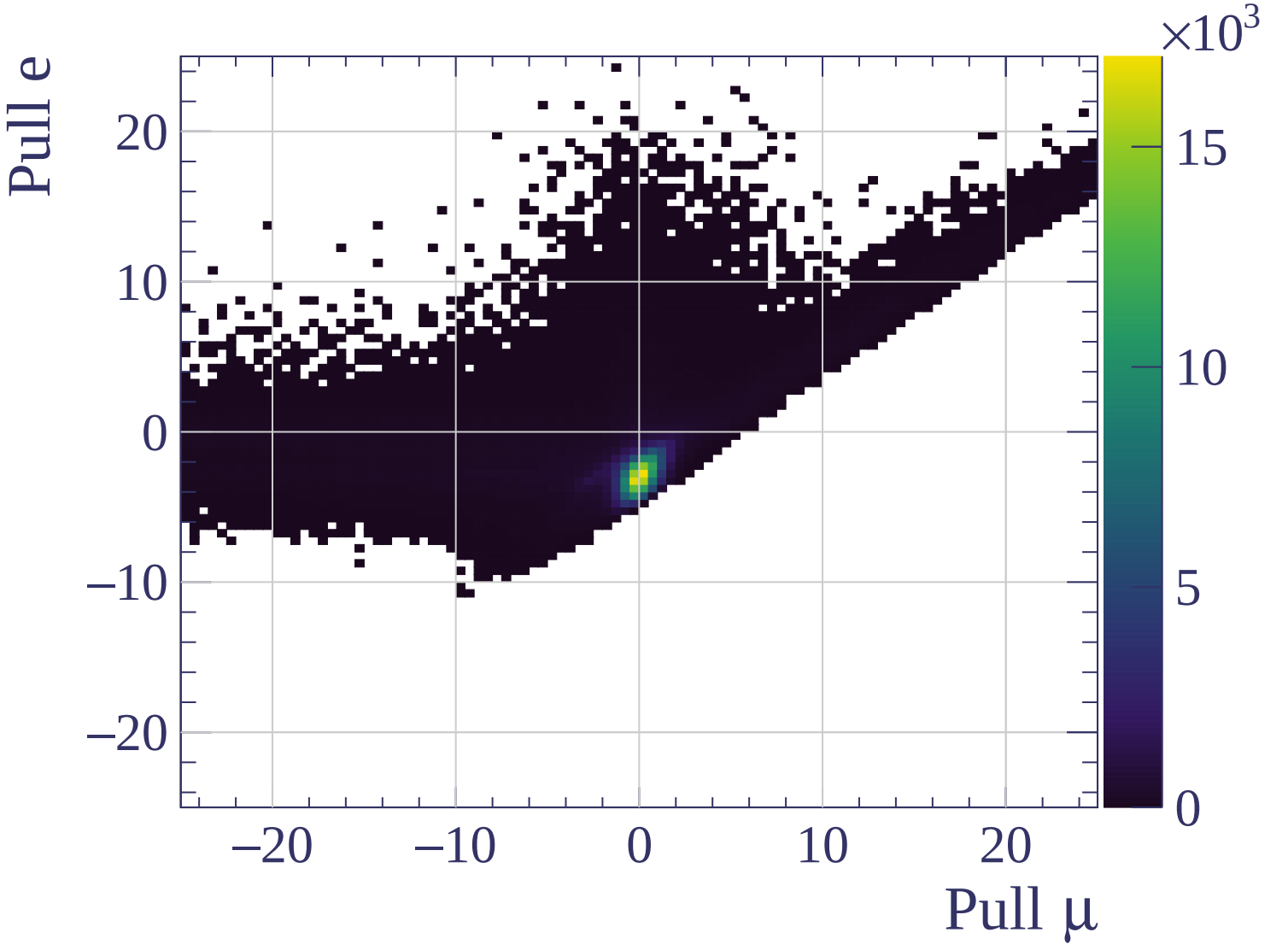


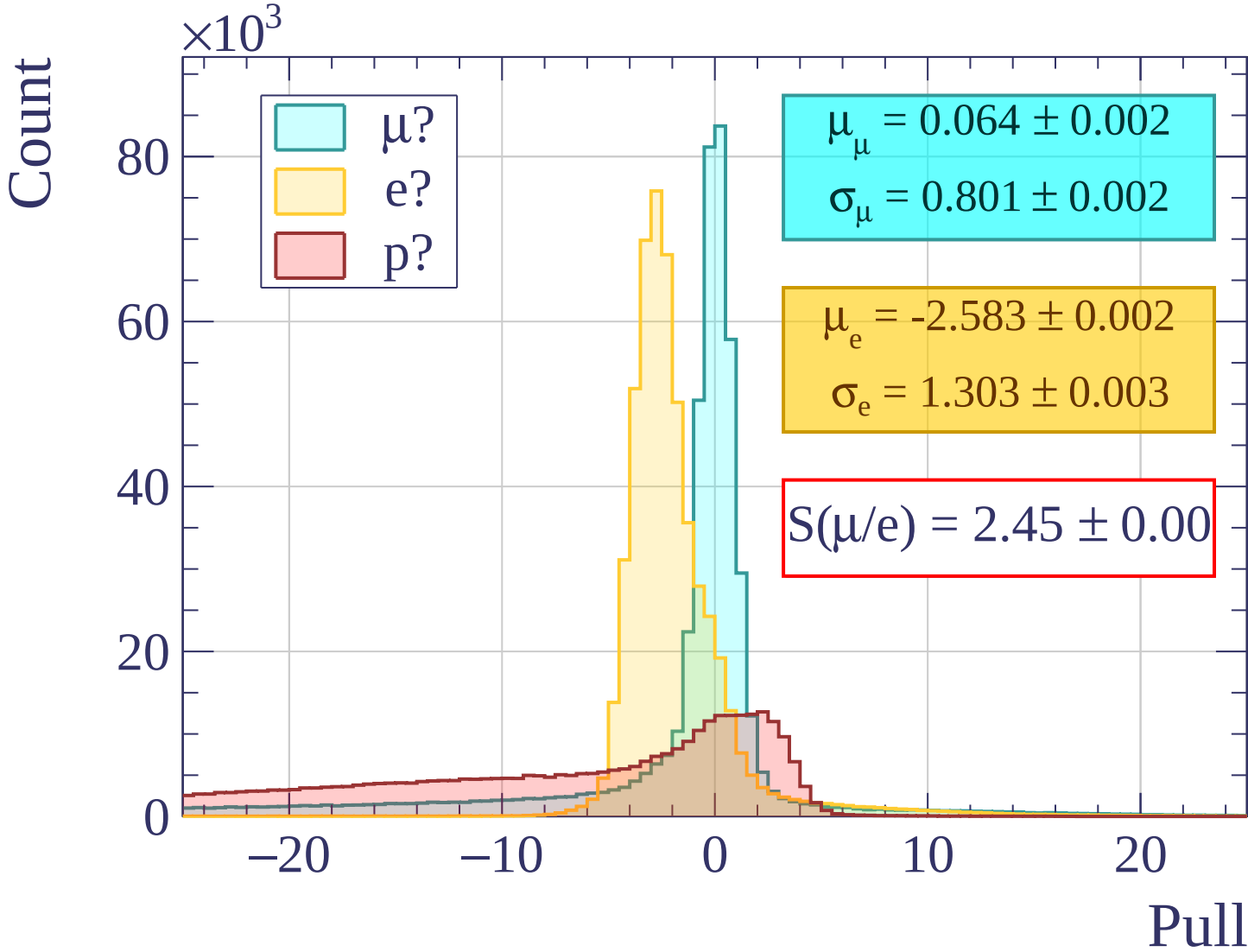
Pulls standard deviation

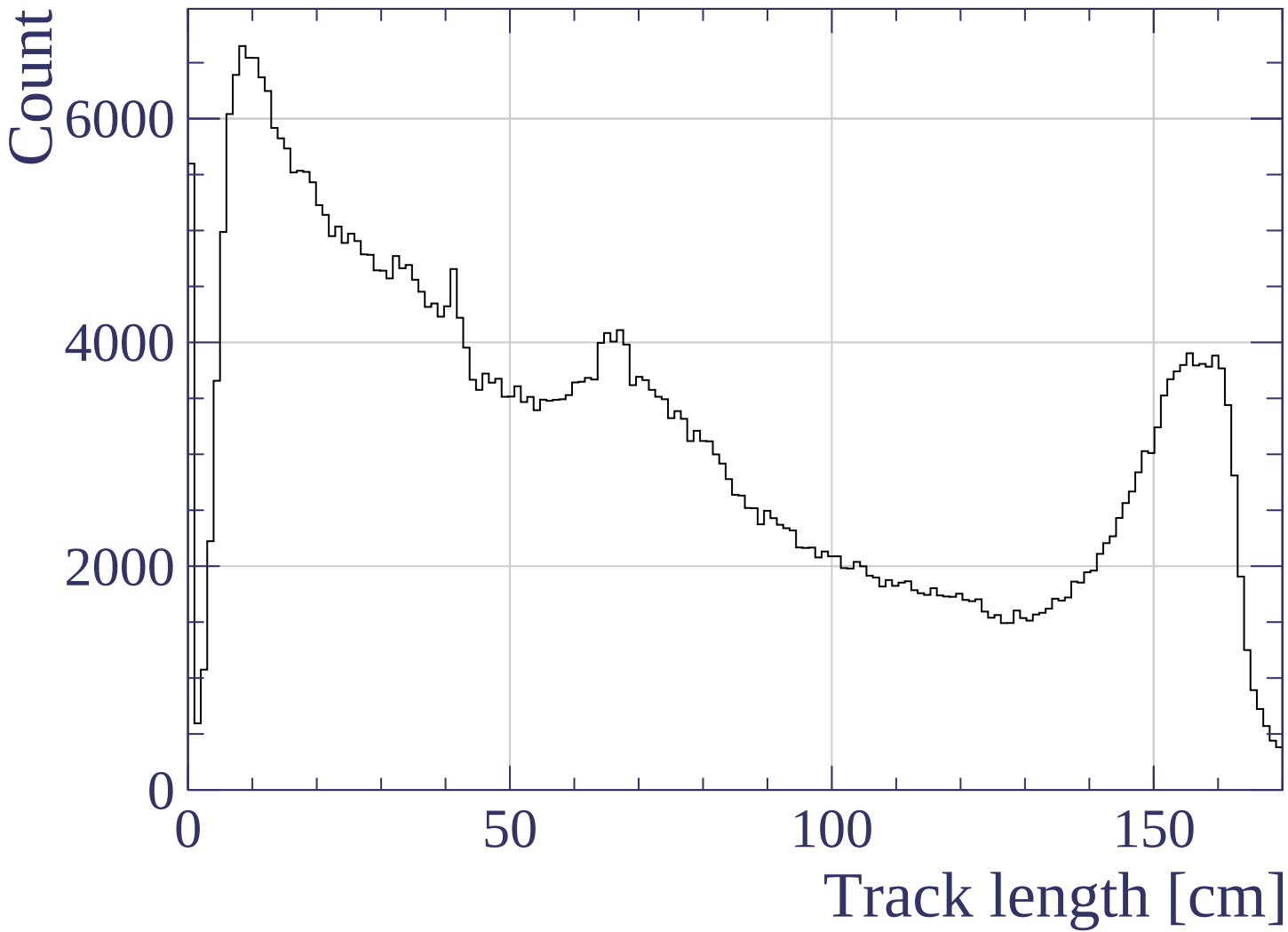


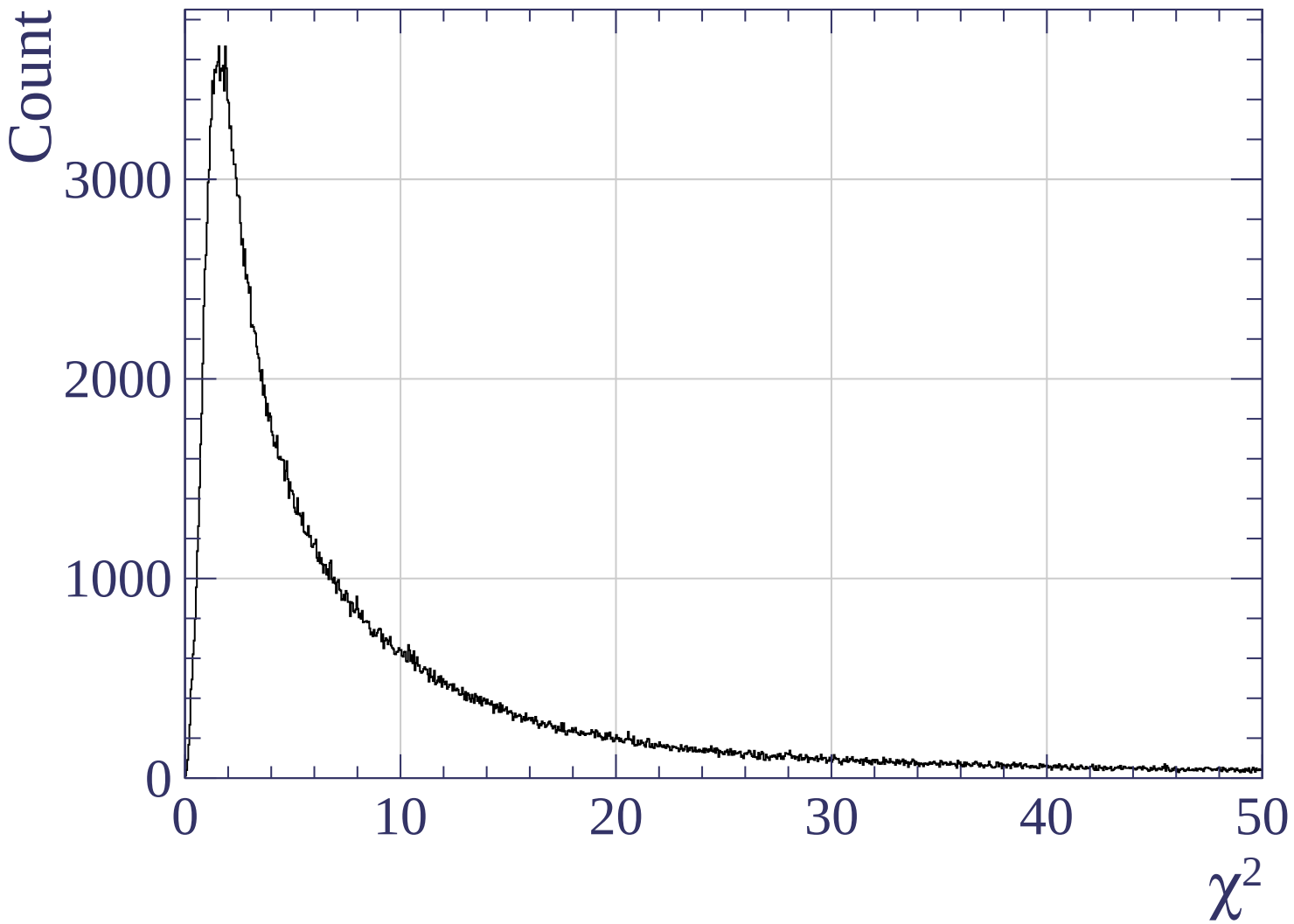
Pulls mean

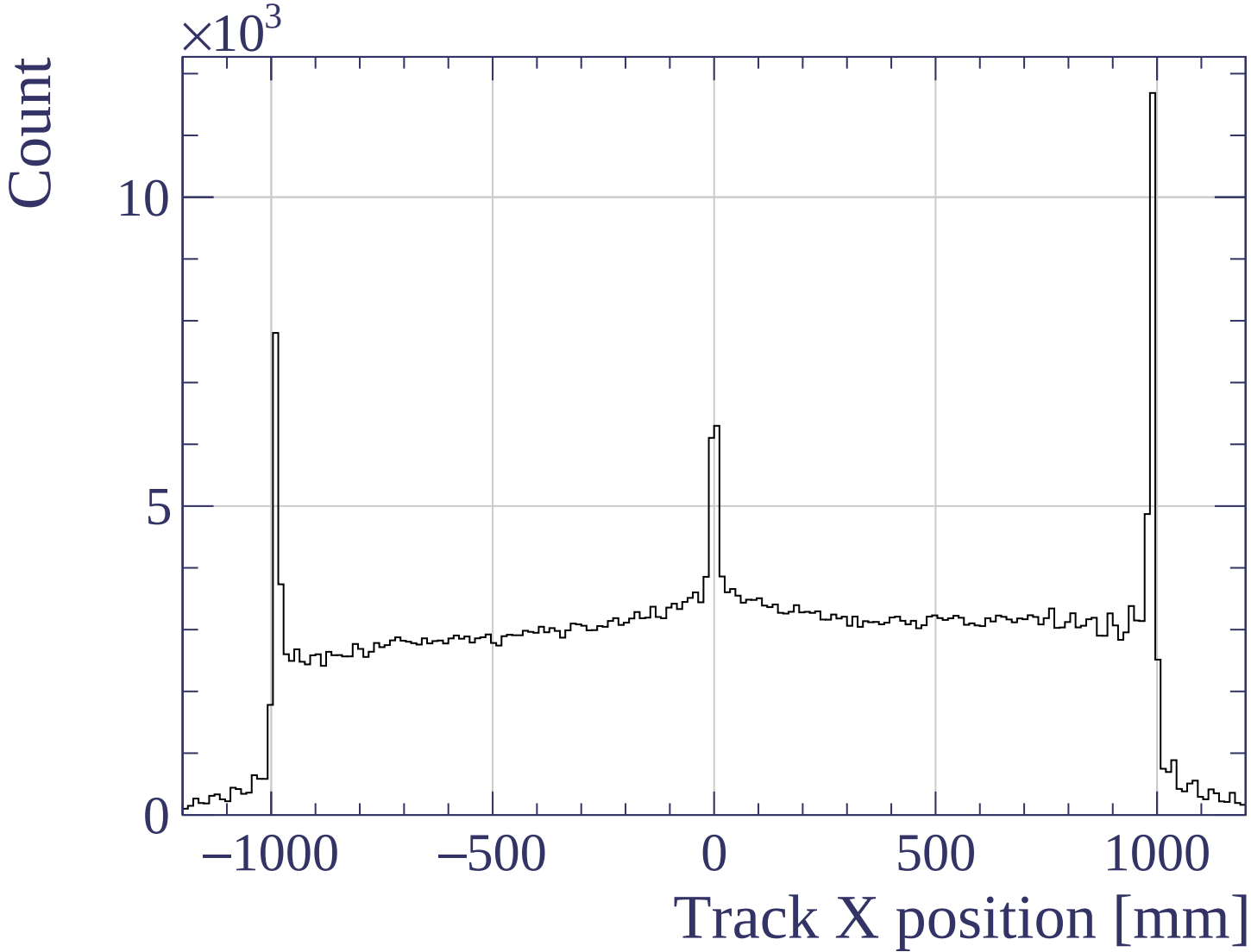


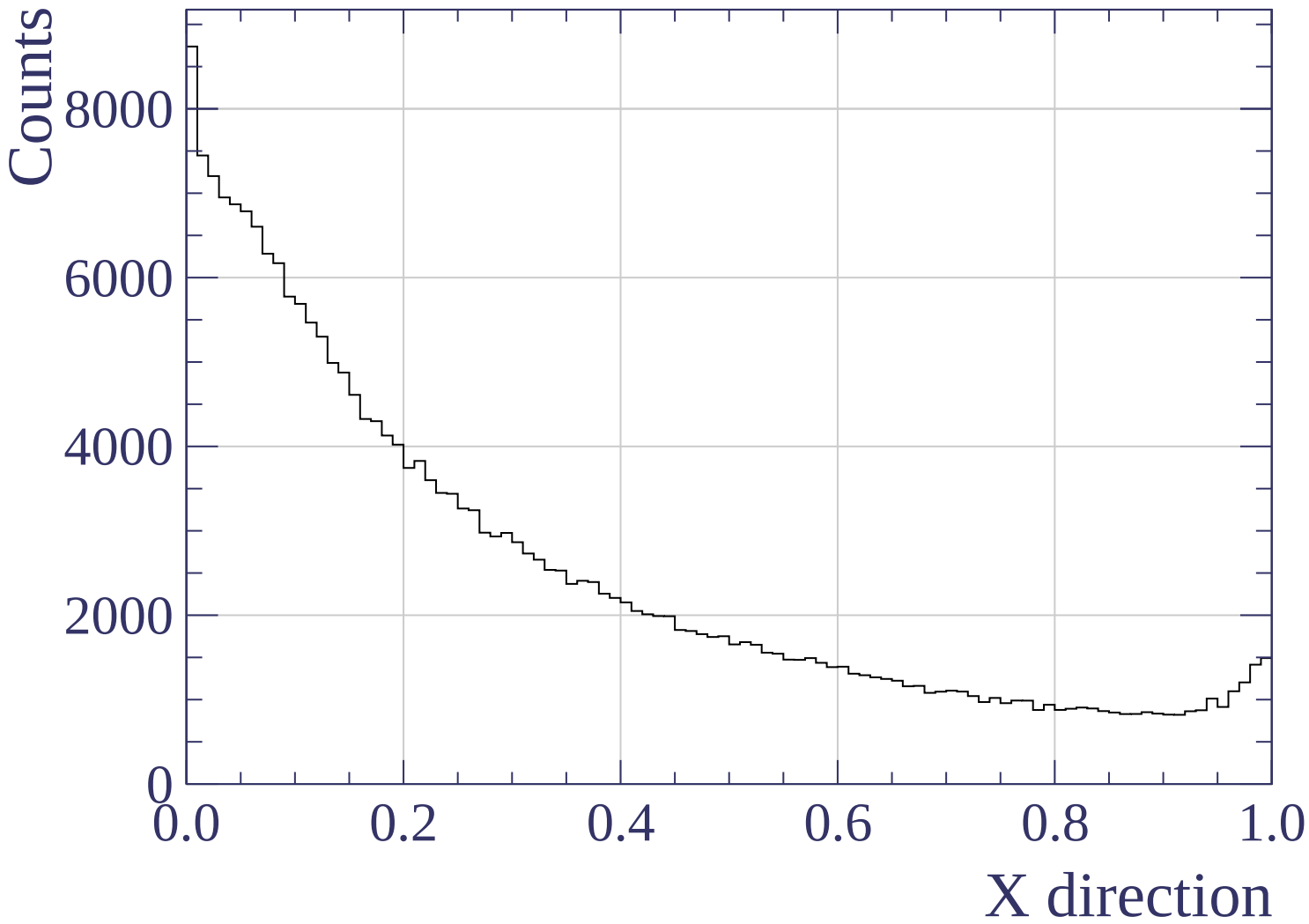


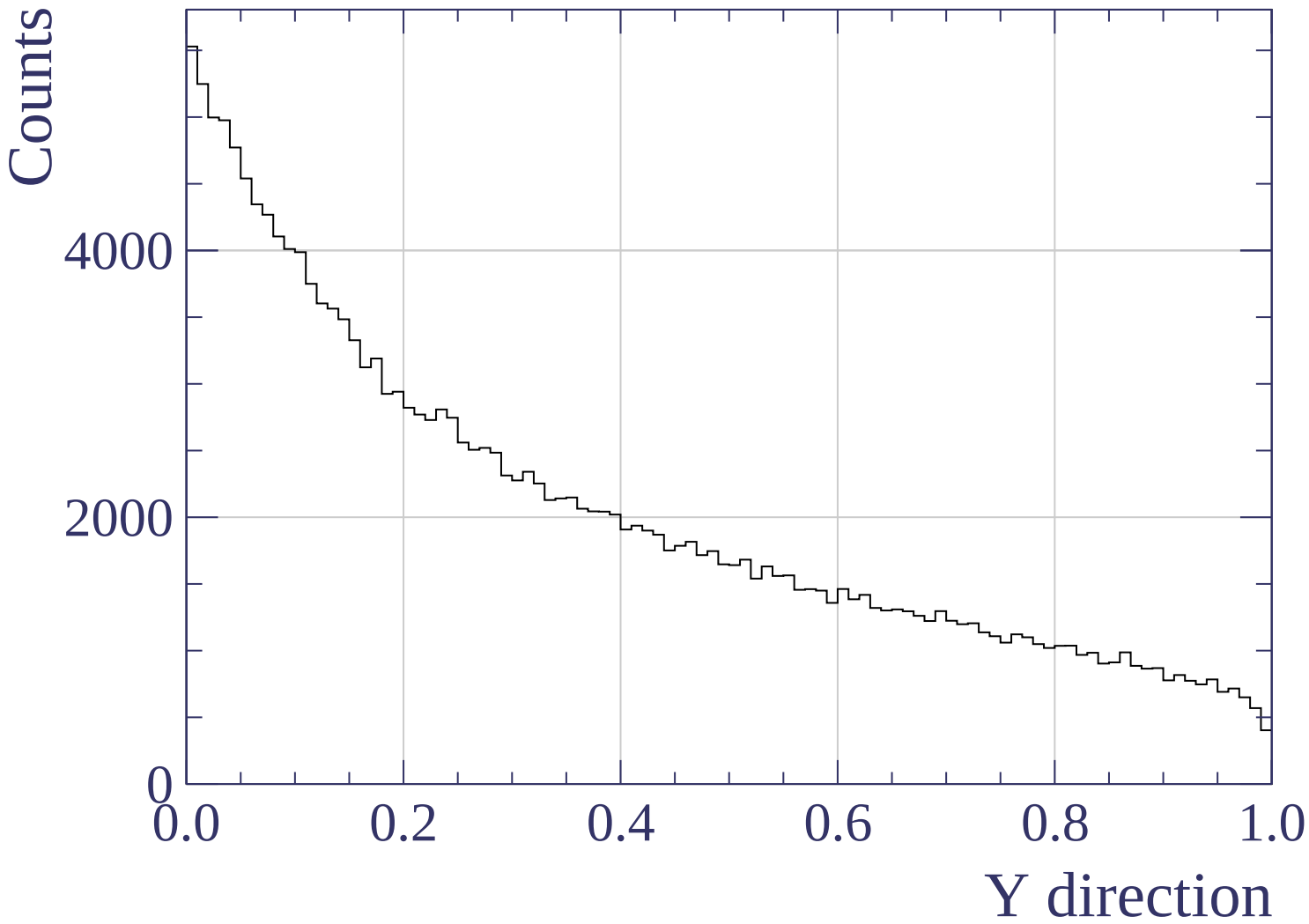


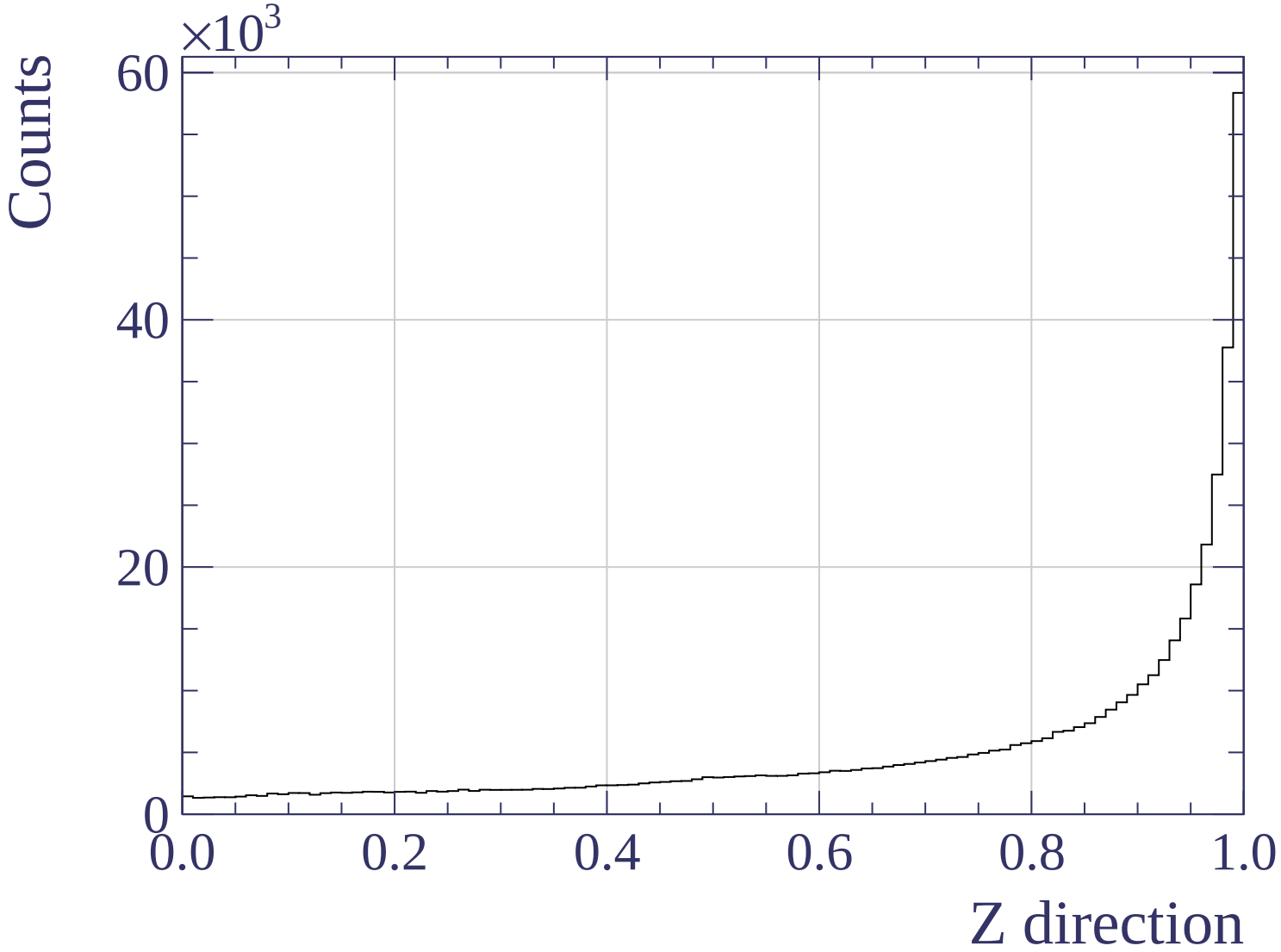


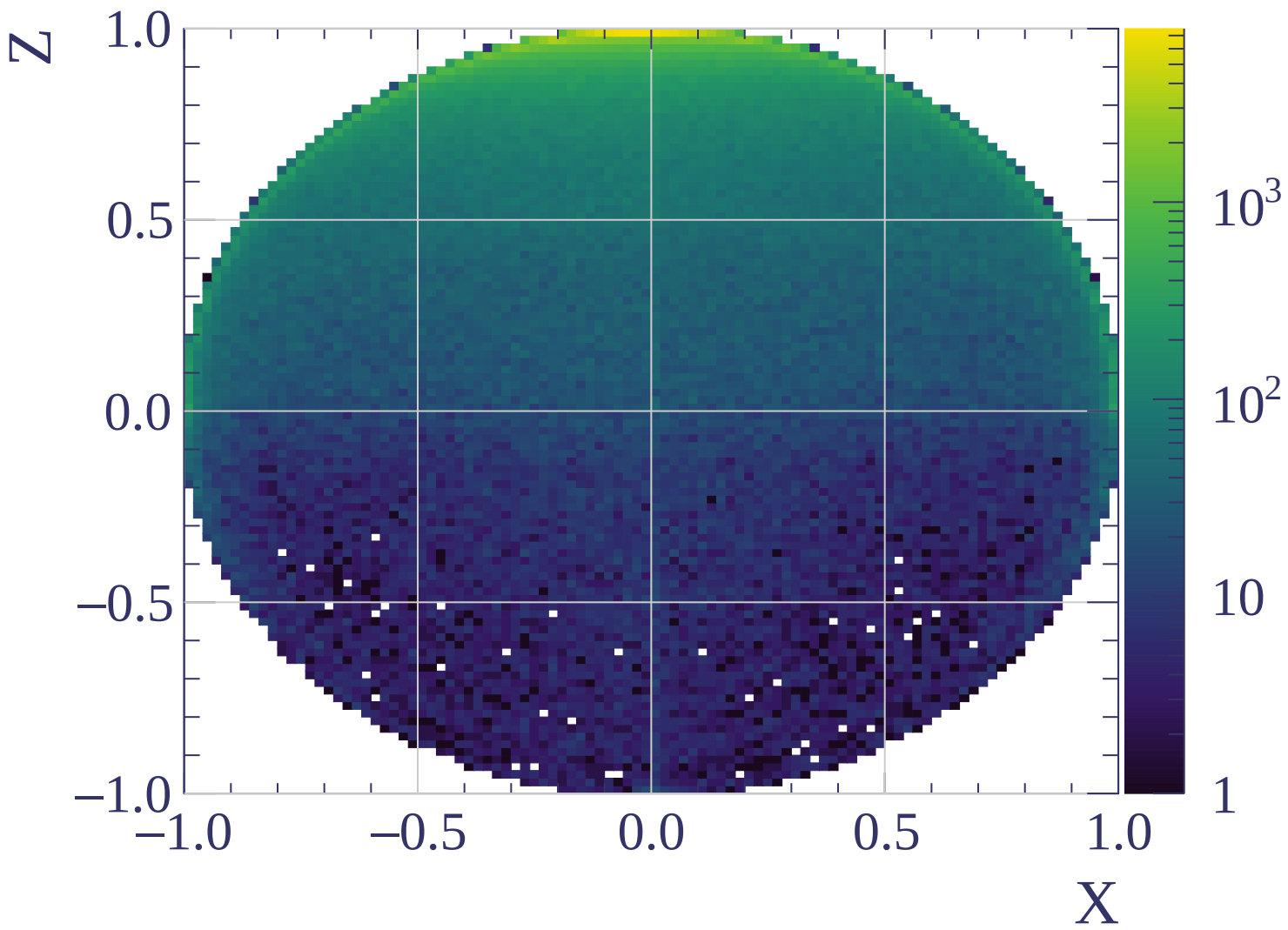


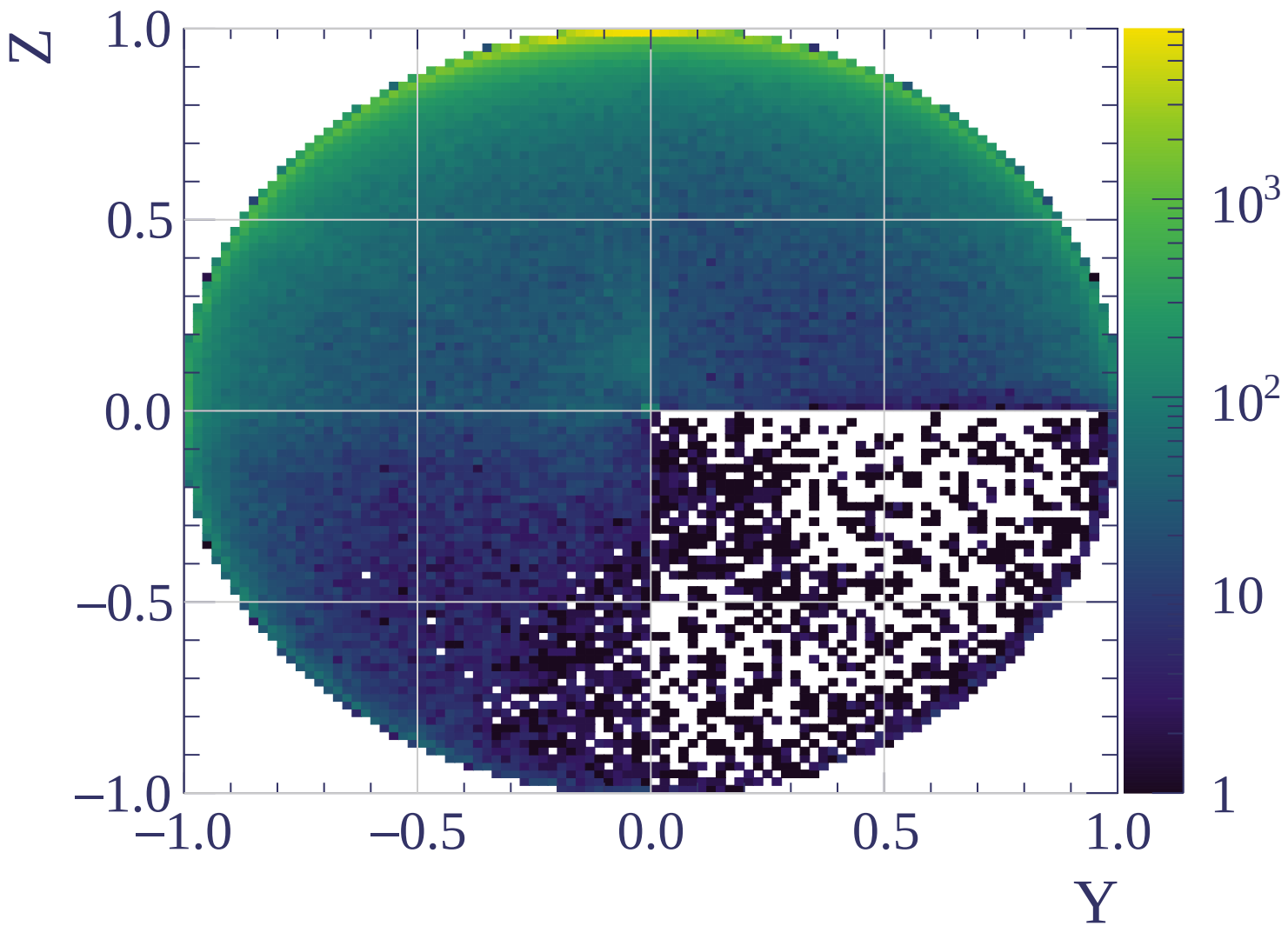


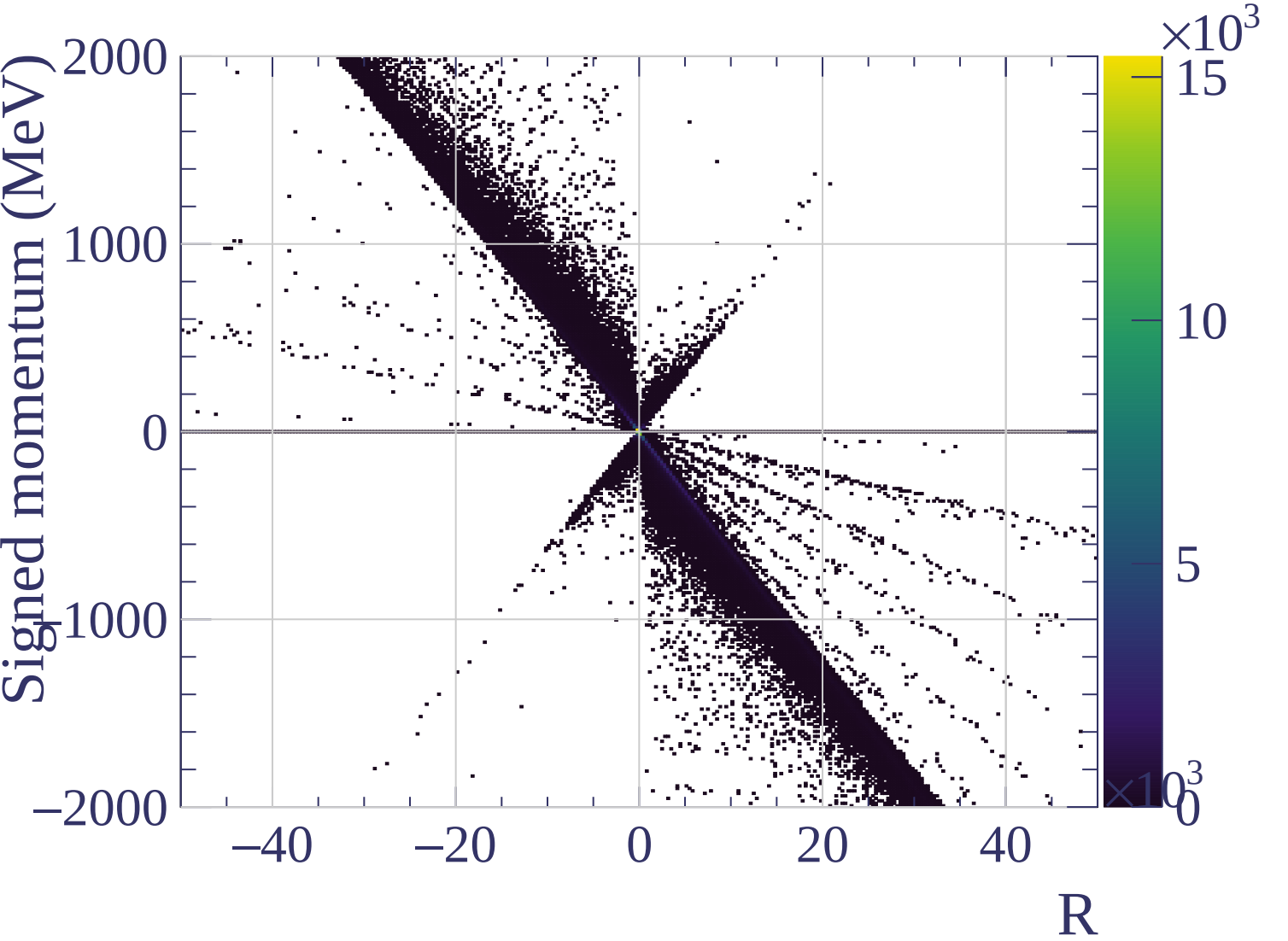


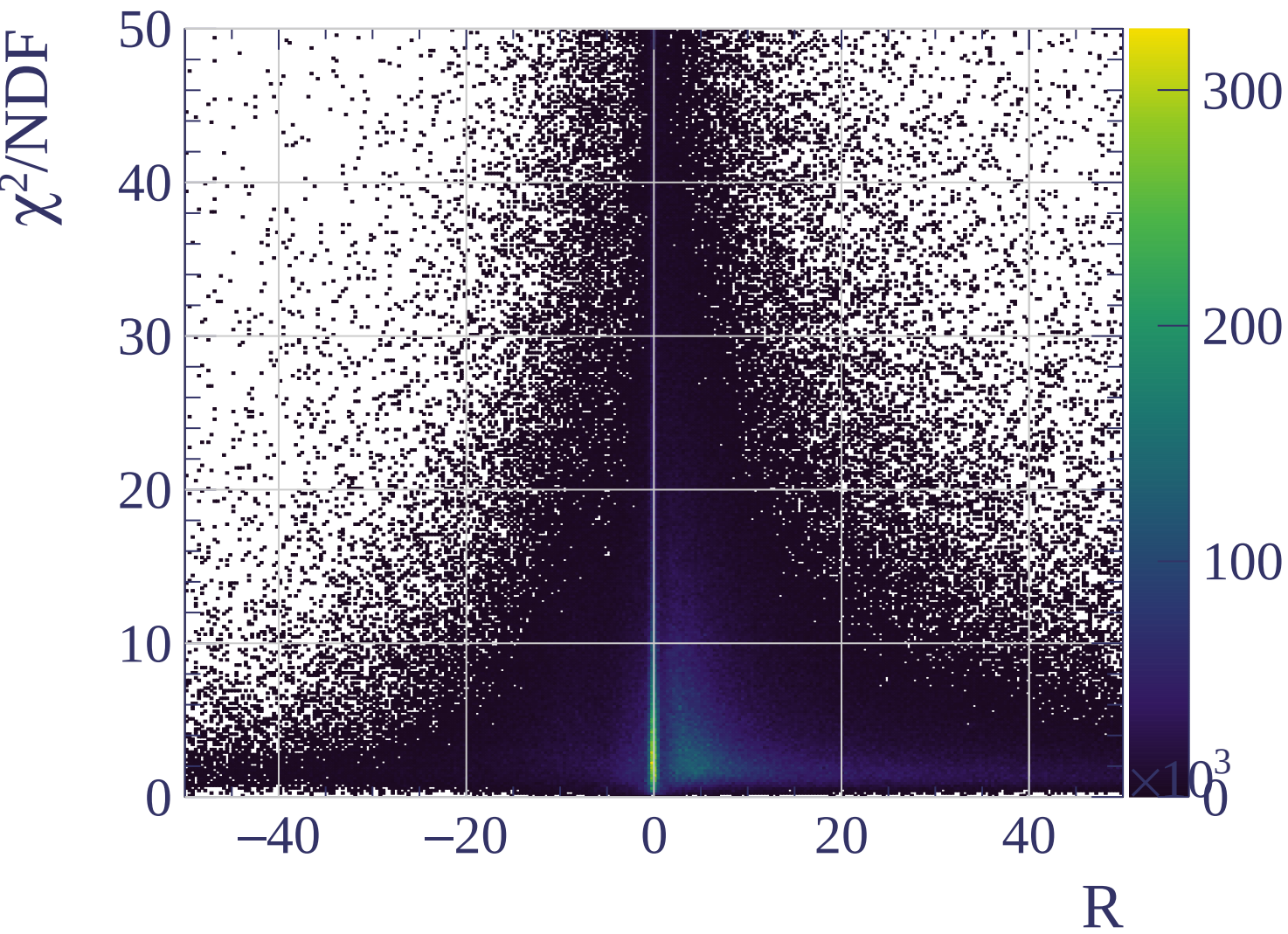




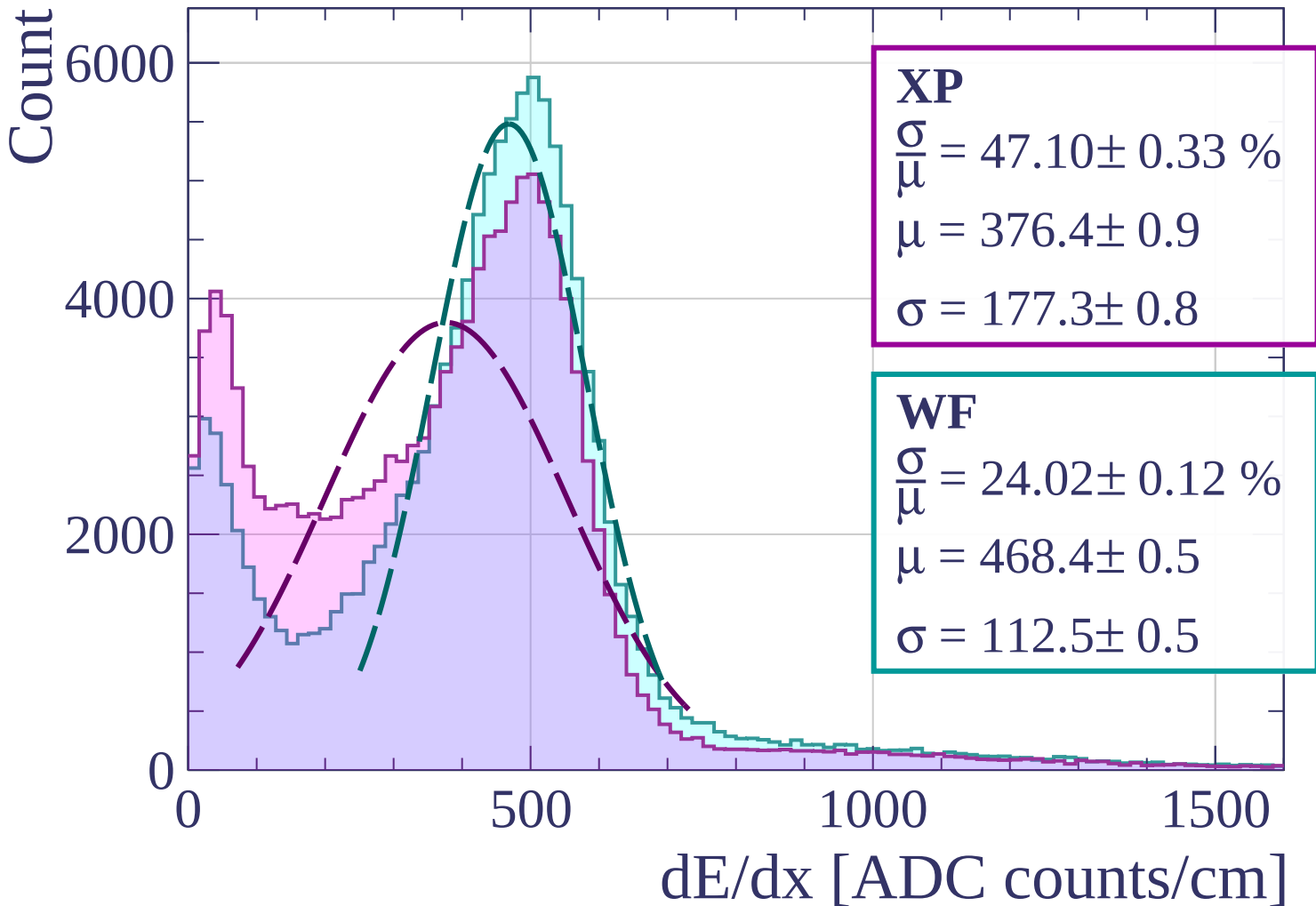








Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$

Count

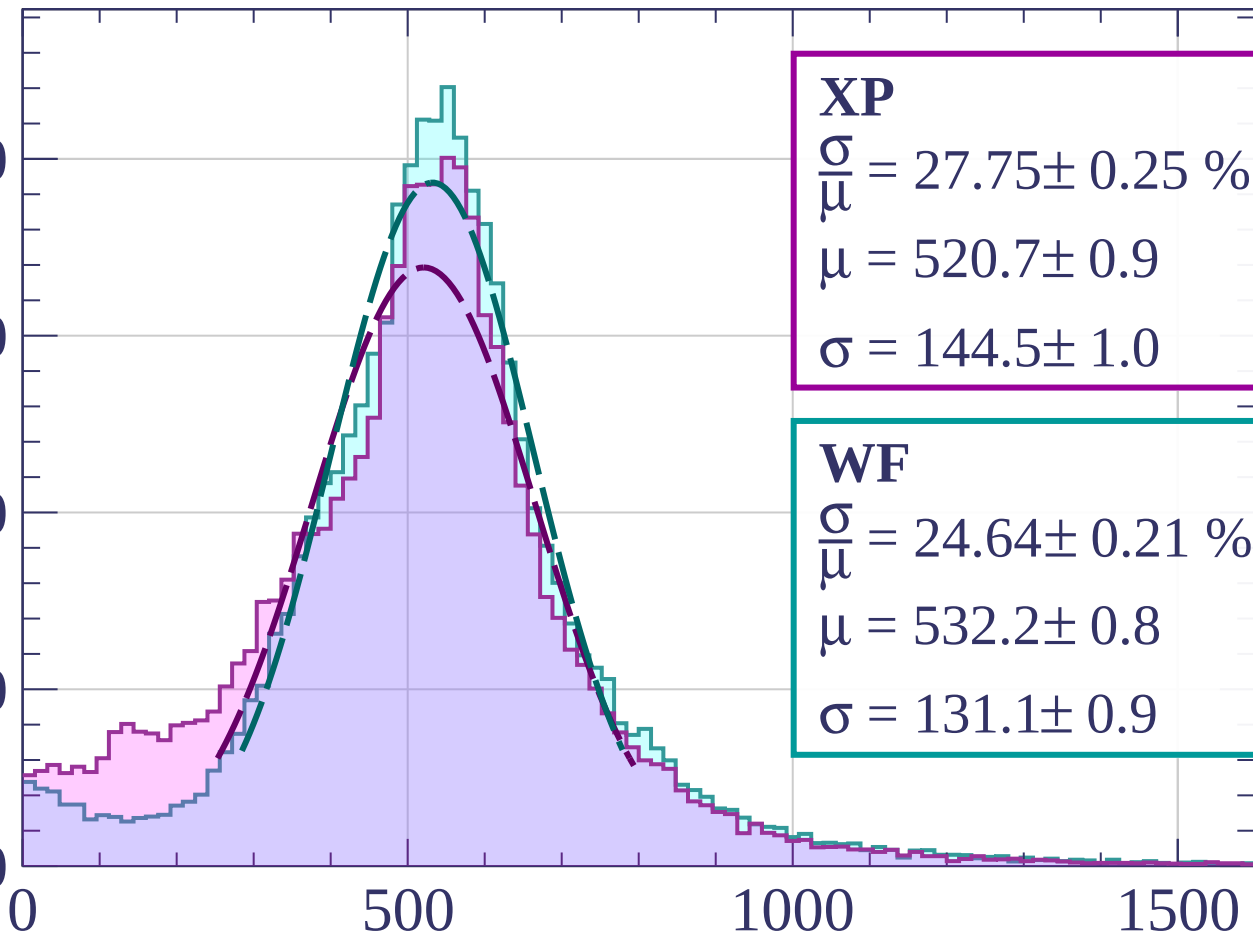
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 27.75 \pm 0.25 \%$$

$$\mu = 520.7 \pm 0.9$$

$$\sigma = 144.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 24.64 \pm 0.21 \%$$

$$\mu = 532.2 \pm 0.8$$

$$\sigma = 131.1 \pm 0.9$$

dE/dx [ADC counts/cm]

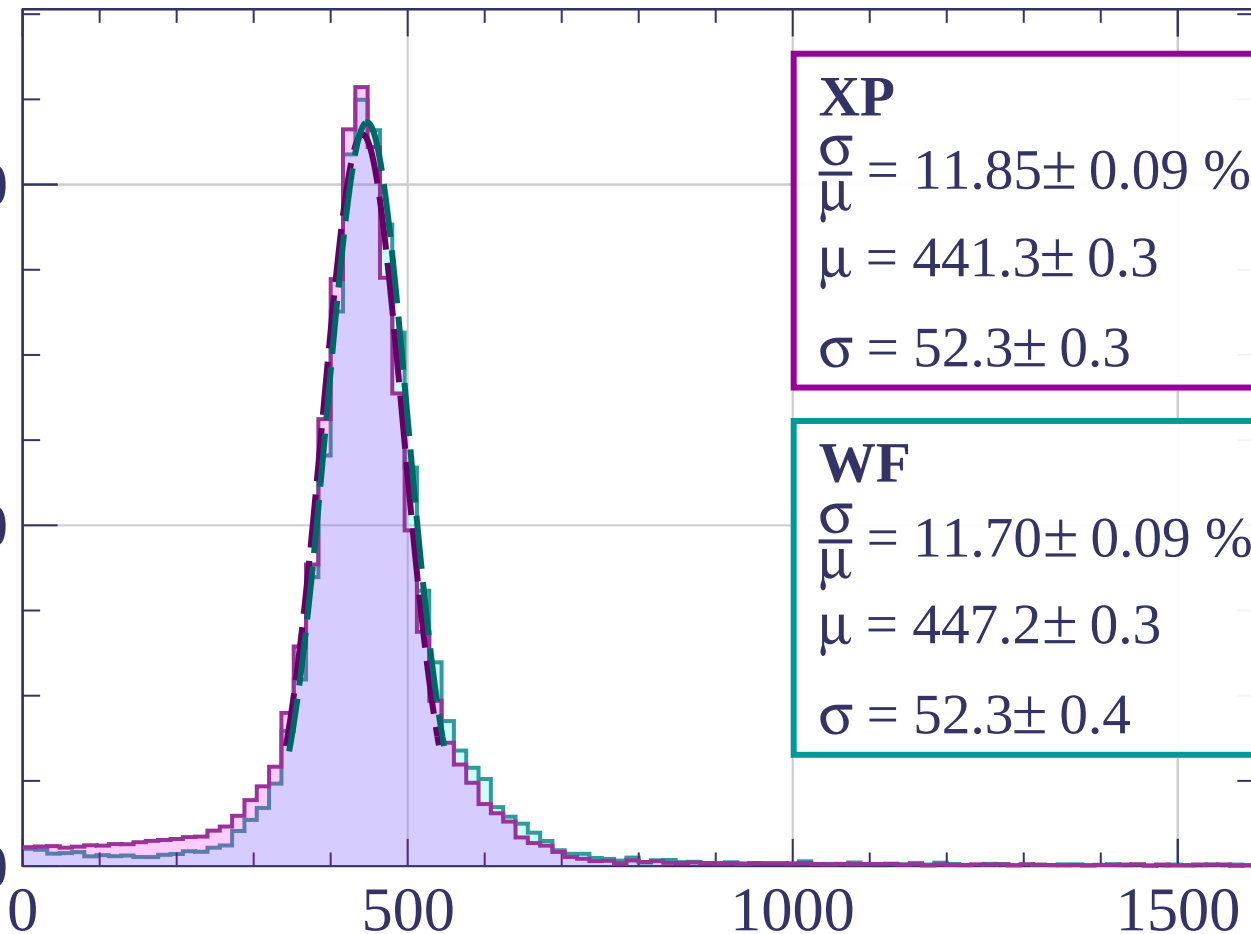
Energy loss | $160 < p < 240$

Count

4000

2000

0



XP

$$\frac{\sigma}{\mu} = 11.85 \pm 0.09 \%$$

$$\mu = 441.3 \pm 0.3$$

$$\sigma = 52.3 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 11.70 \pm 0.09 \%$$

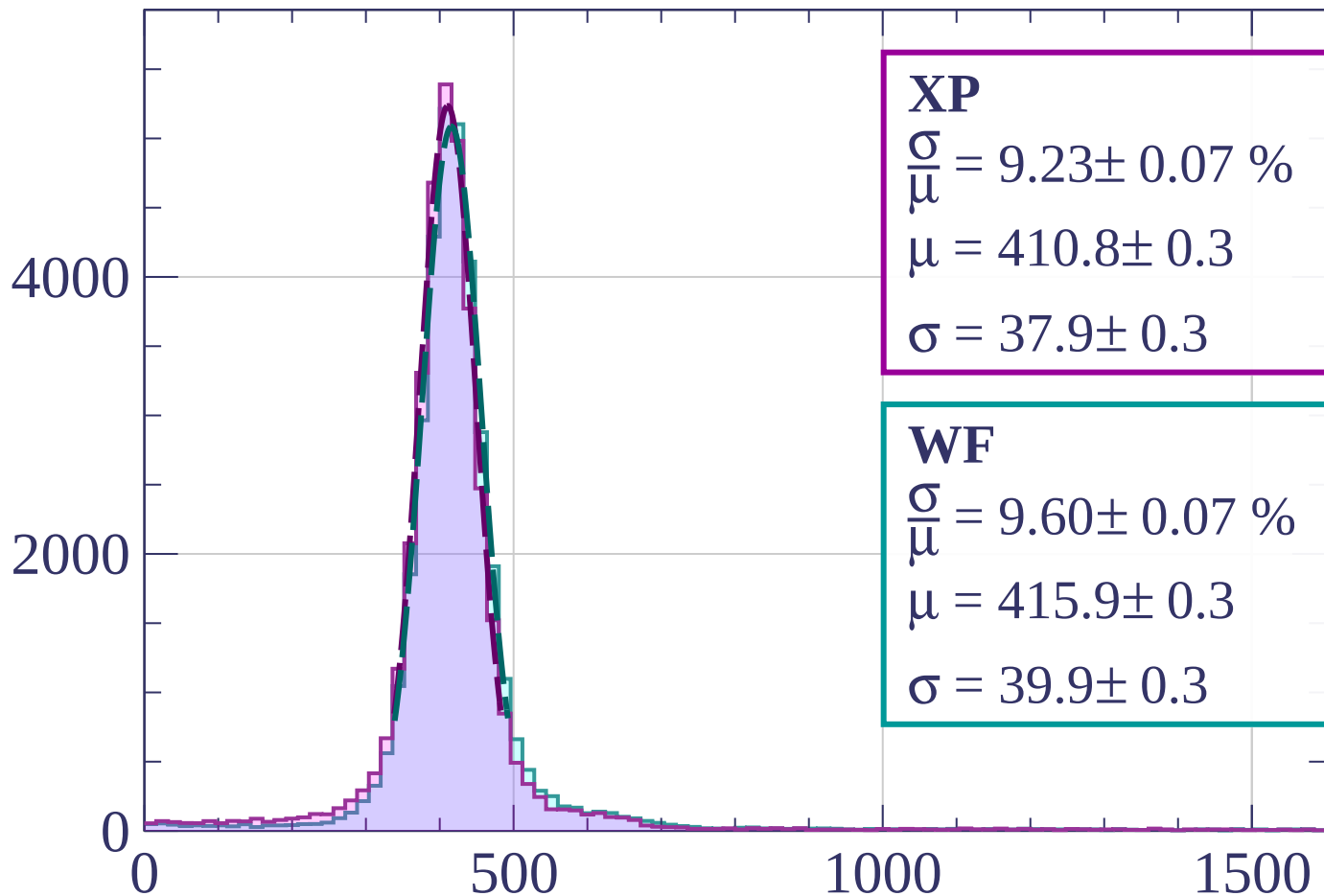
$$\mu = 447.2 \pm 0.3$$

$$\sigma = 52.3 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.07 \%$$

$$\mu = 410.8 \pm 0.3$$

$$\sigma = 37.9 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.07 \%$$

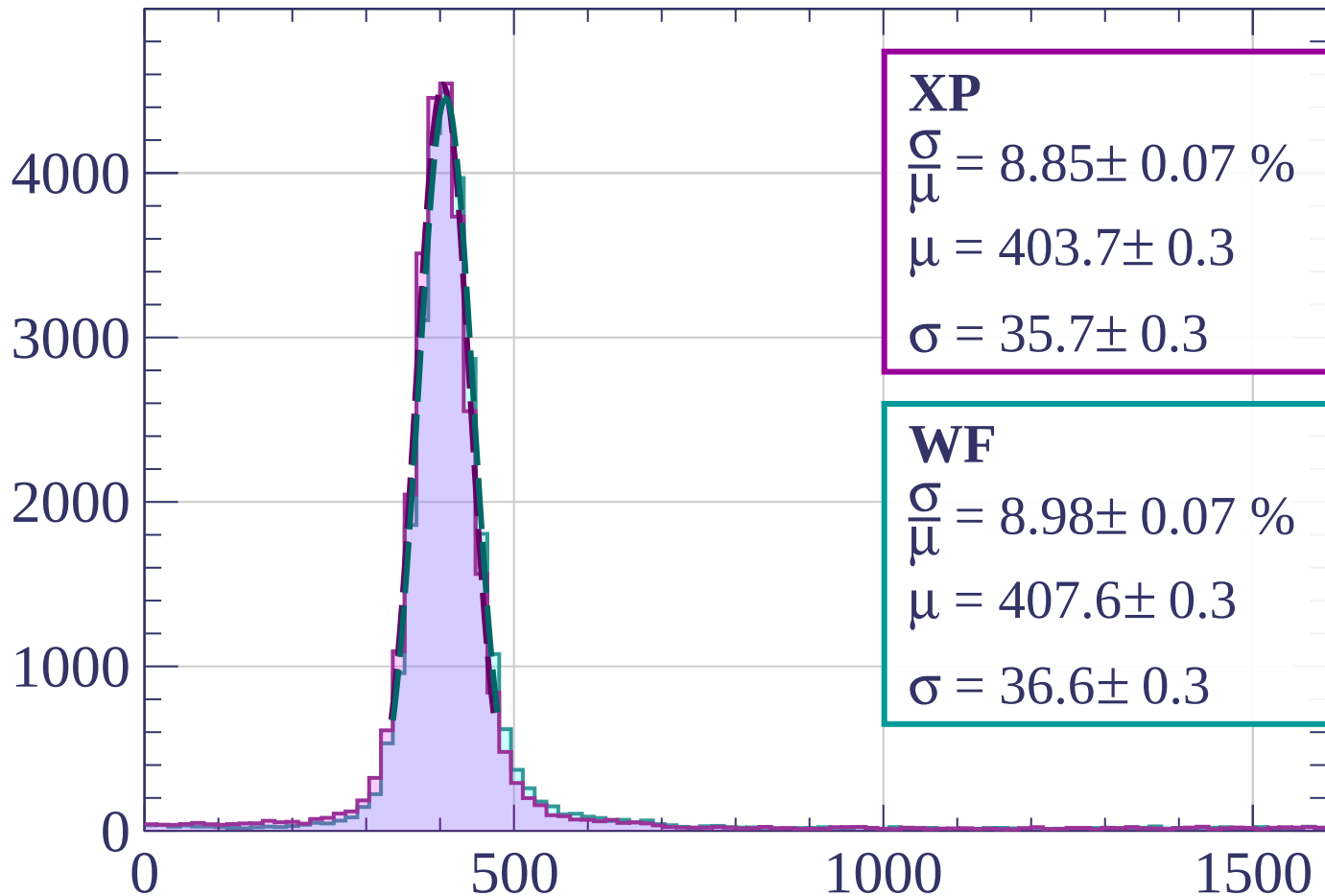
$$\mu = 415.9 \pm 0.3$$

$$\sigma = 39.9 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 8.85 \pm 0.07 \%$$

$$\mu = 403.7 \pm 0.3$$

$$\sigma = 35.7 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.07 \%$$

$$\mu = 407.6 \pm 0.3$$

$$\sigma = 36.6 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count

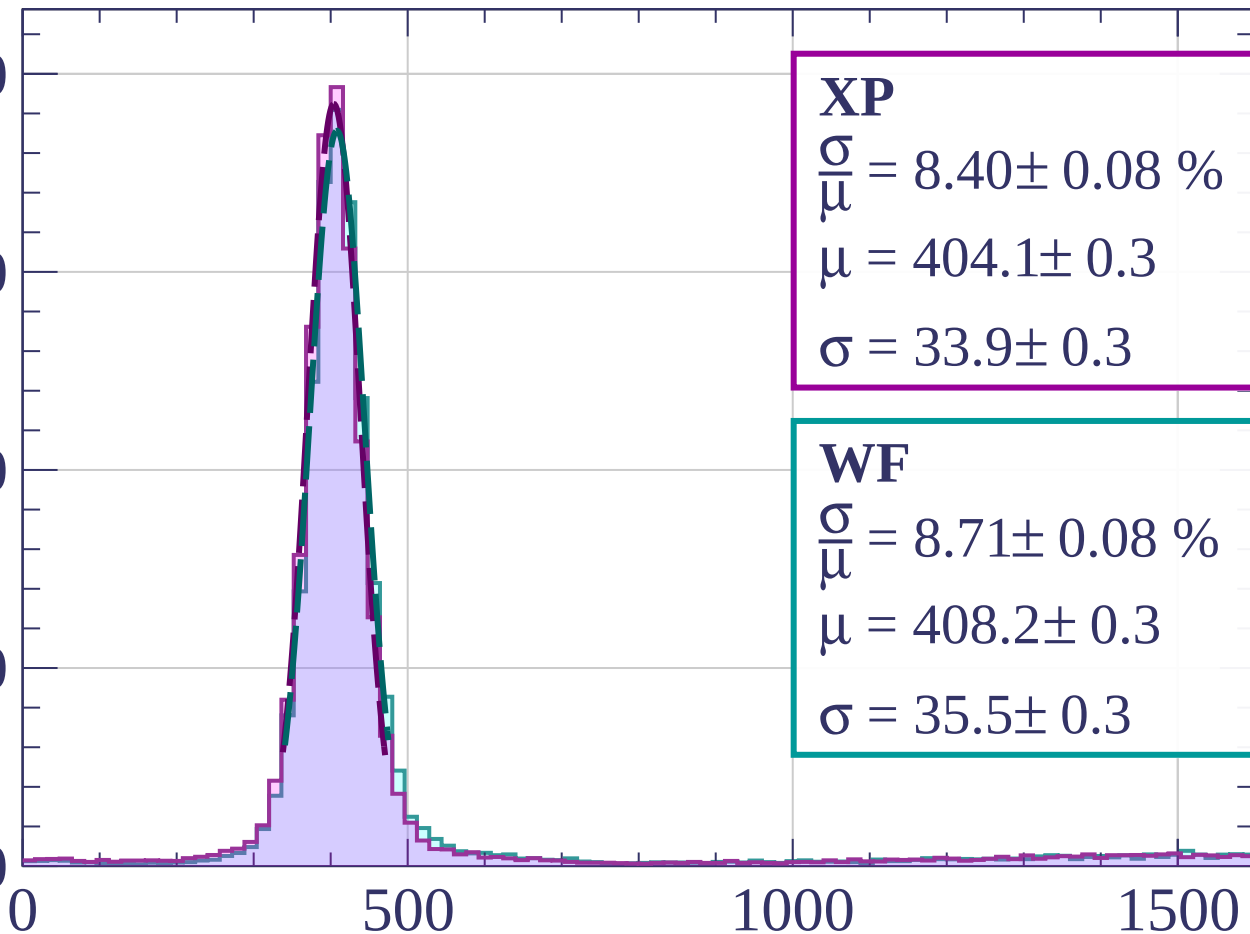
4000

3000

2000

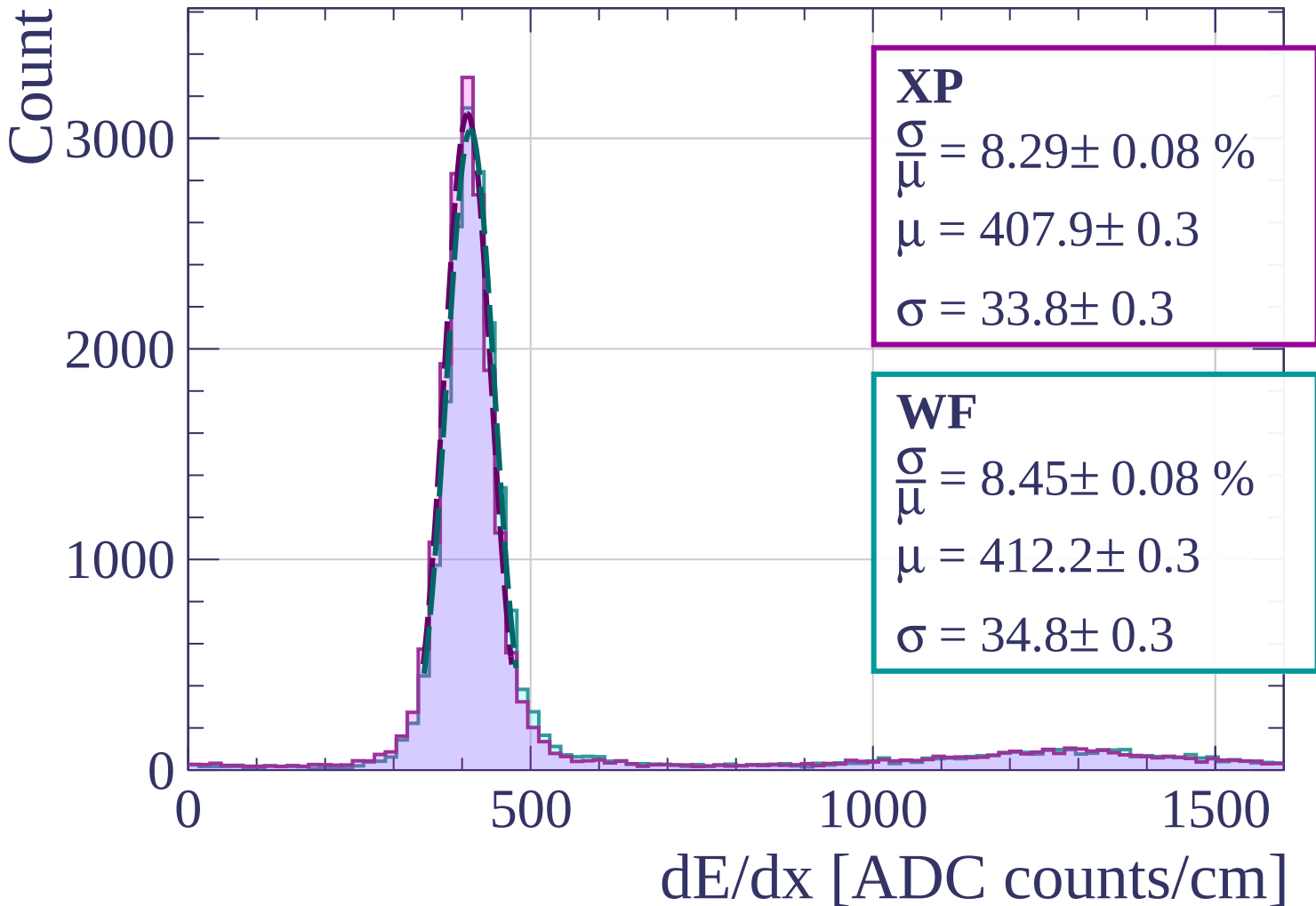
1000

0



dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$



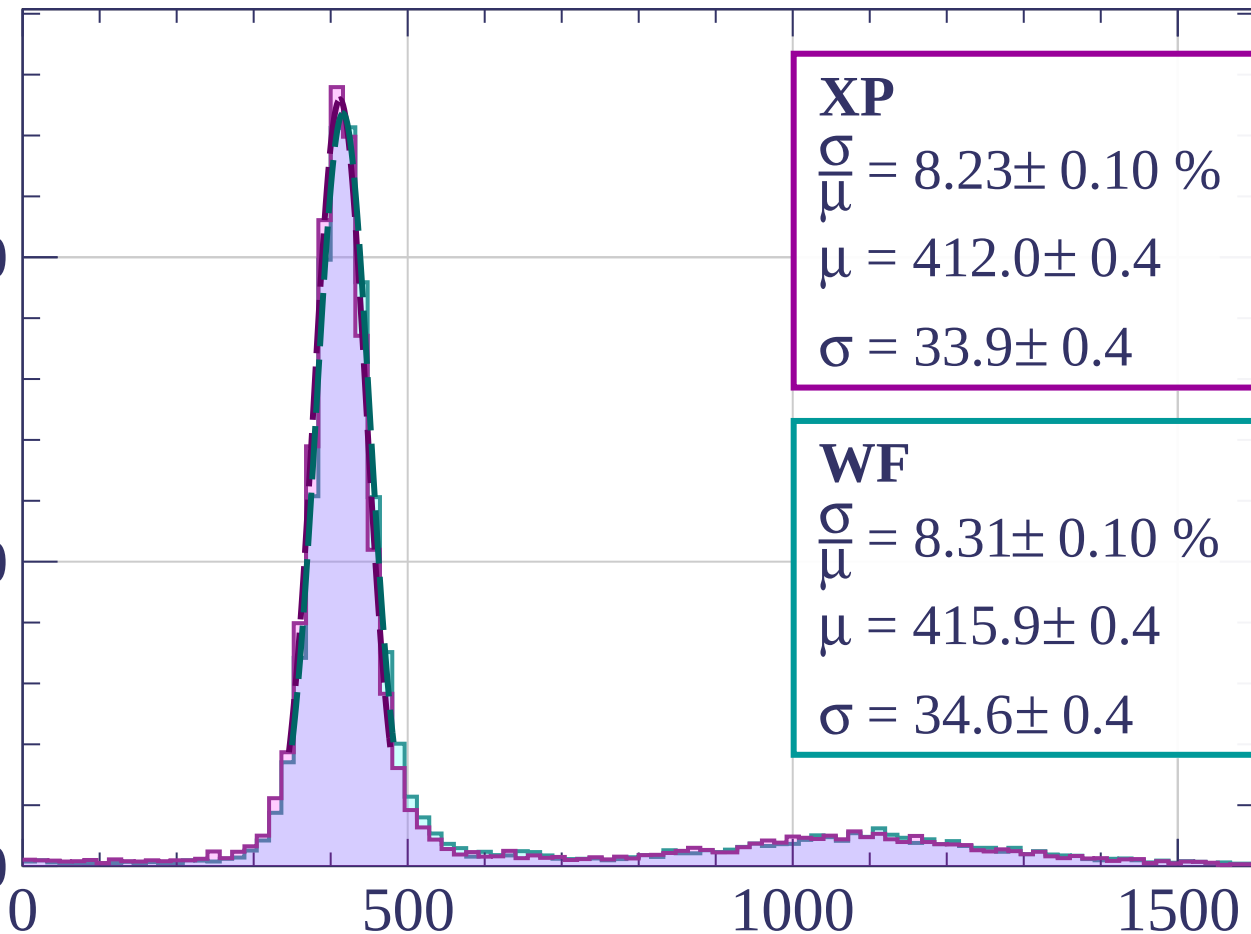
Energy loss | $560 < p < 640$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.23 \pm 0.10 \%$$

$$\mu = 412.0 \pm 0.4$$

$$\sigma = 33.9 \pm 0.4$$

WF

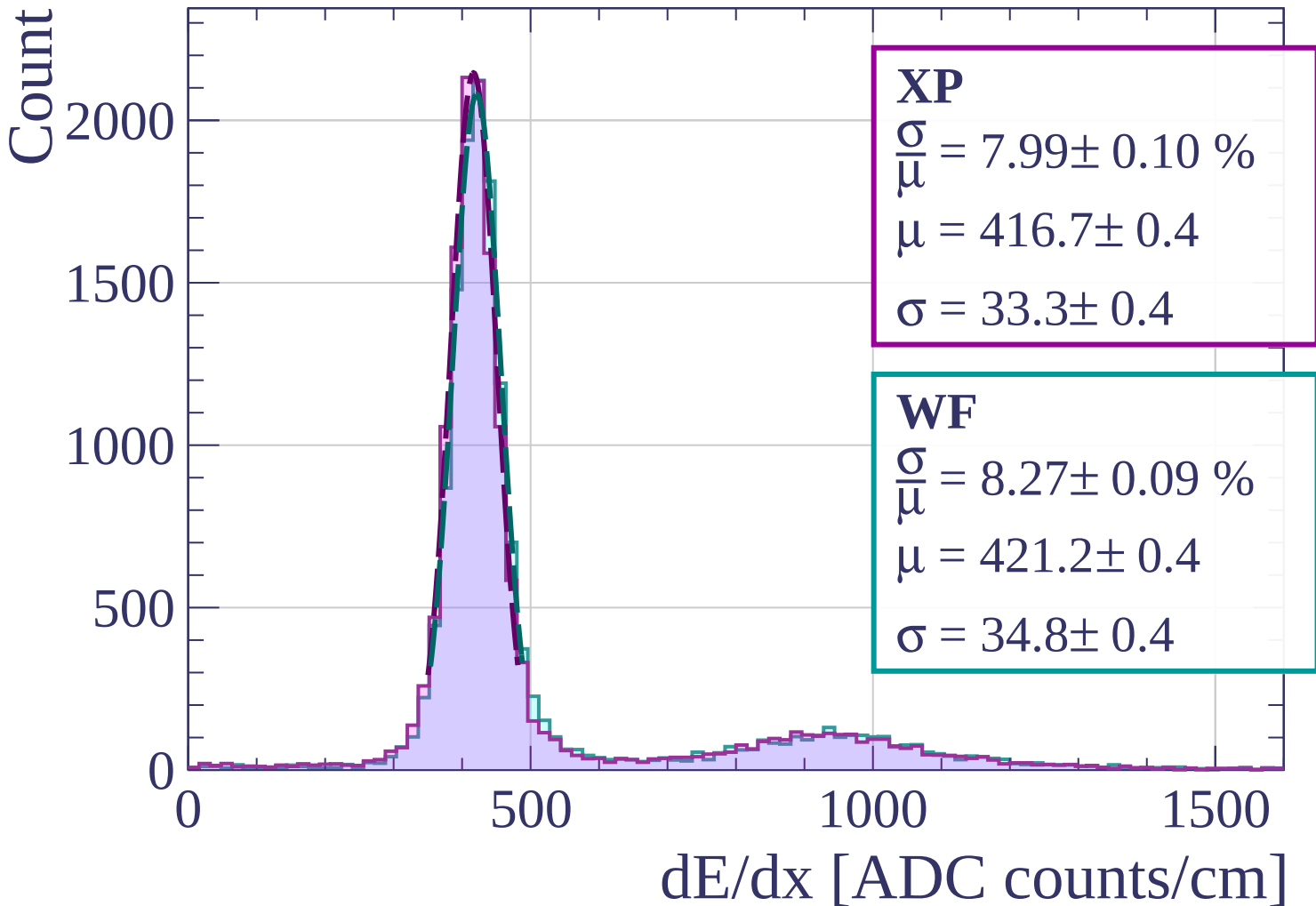
$$\frac{\sigma}{\mu} = 8.31 \pm 0.10 \%$$

$$\mu = 415.9 \pm 0.4$$

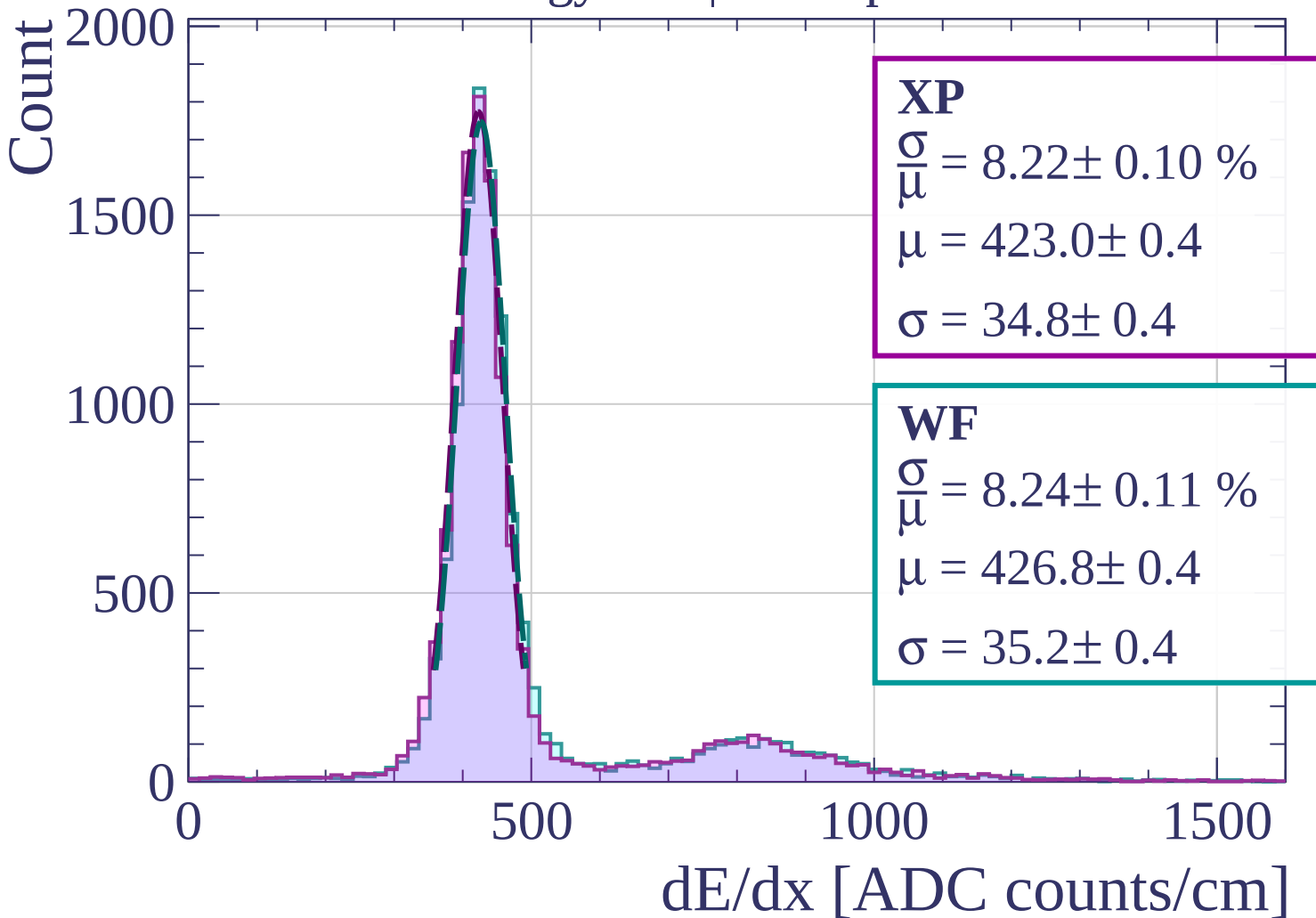
$$\sigma = 34.6 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

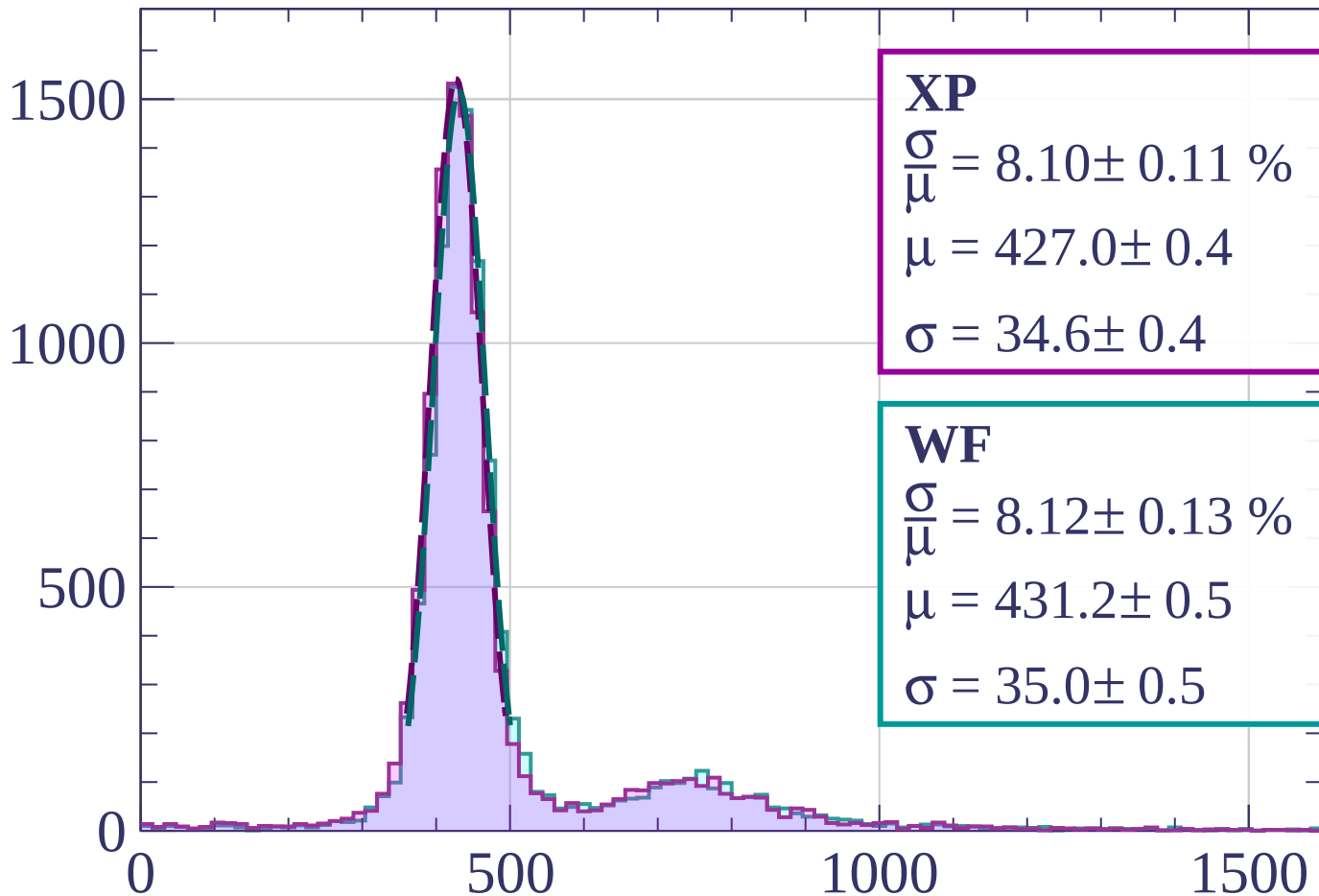


Energy loss | $720 < p < 800$



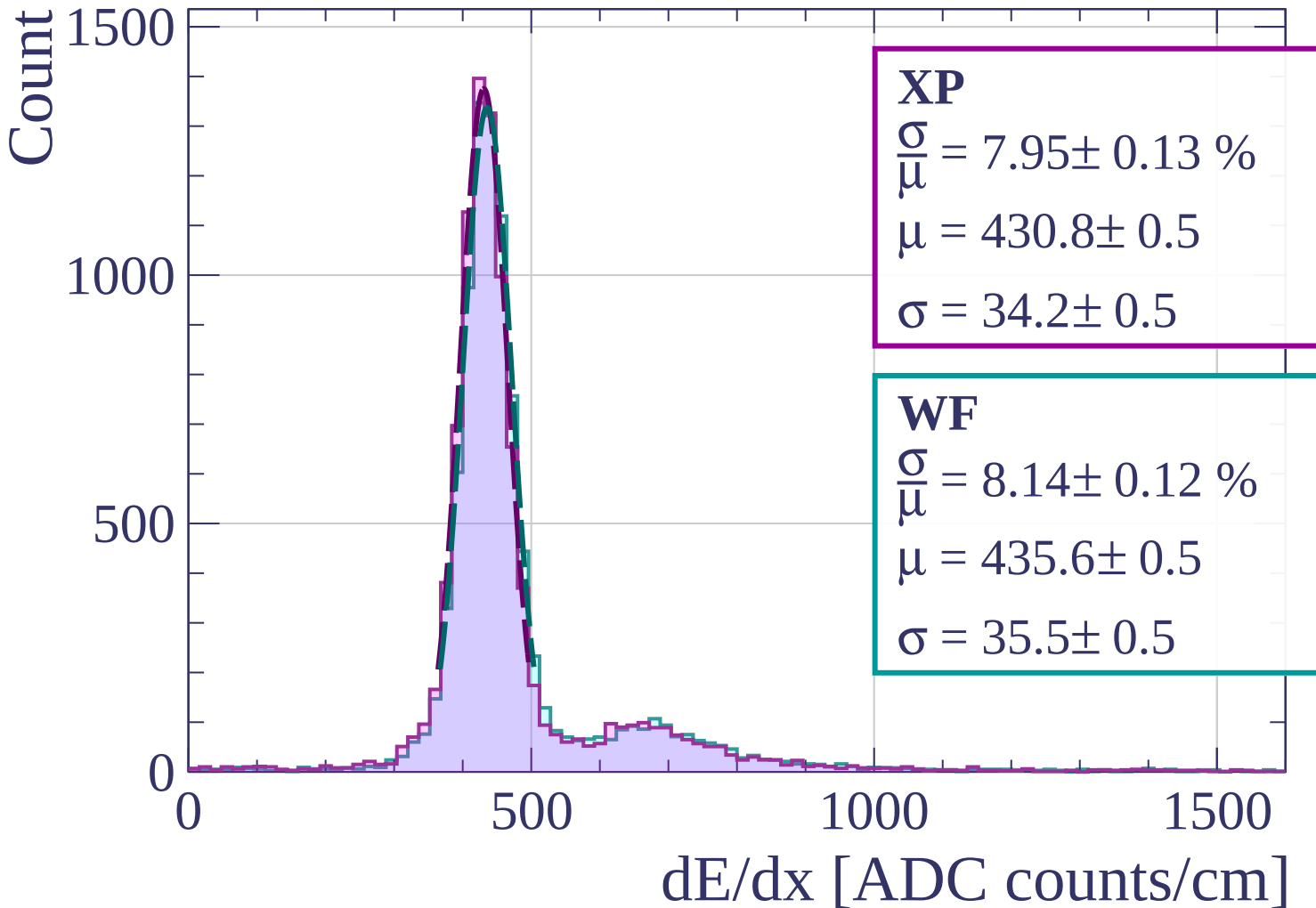
Energy loss | $800 < p < 880$

Count



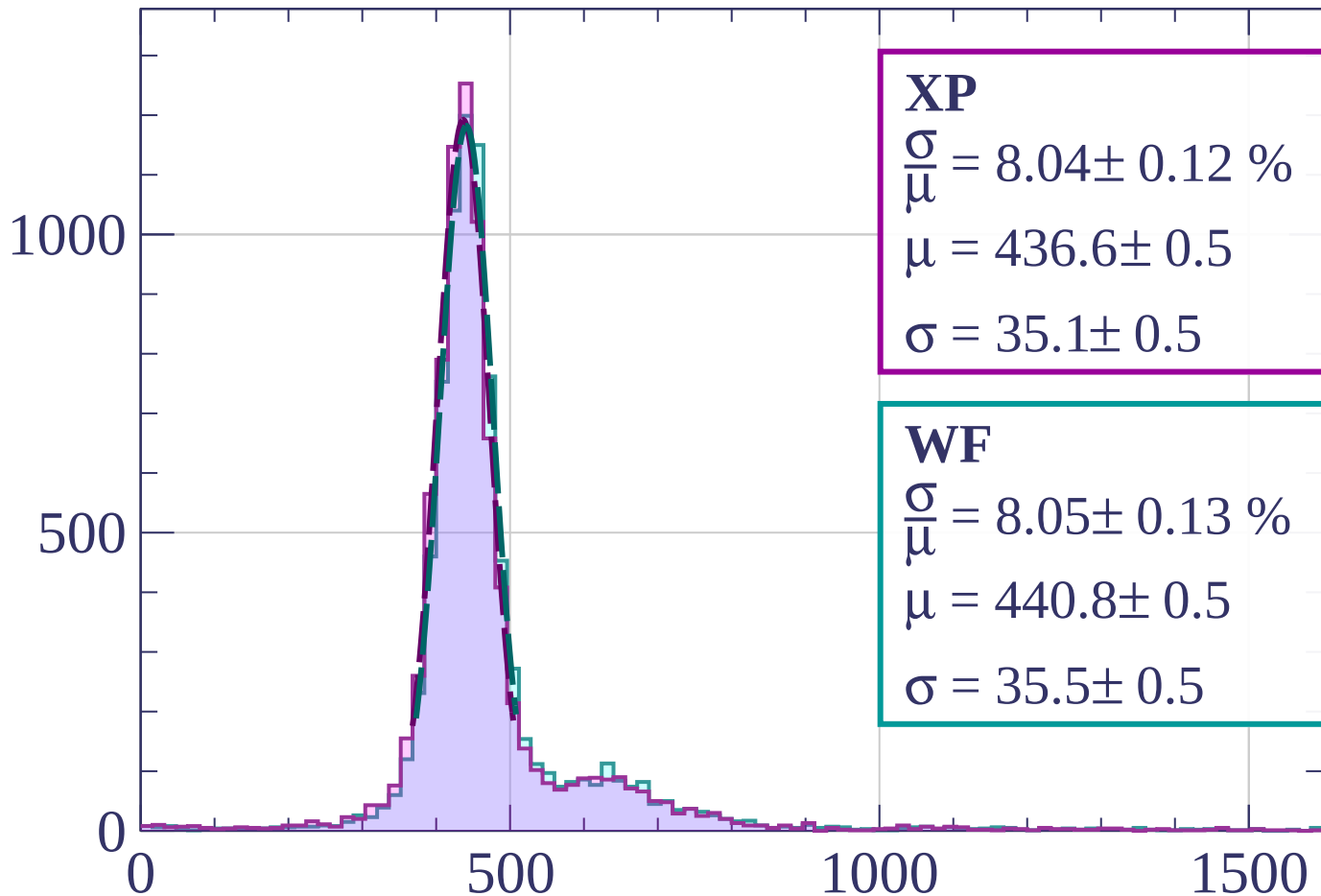
dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.04 \pm 0.12 \%$$

$$\mu = 436.6 \pm 0.5$$

$$\sigma = 35.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.05 \pm 0.13 \%$$

$$\mu = 440.8 \pm 0.5$$

$$\sigma = 35.5 \pm 0.5$$

dE/dx [ADC counts/cm]

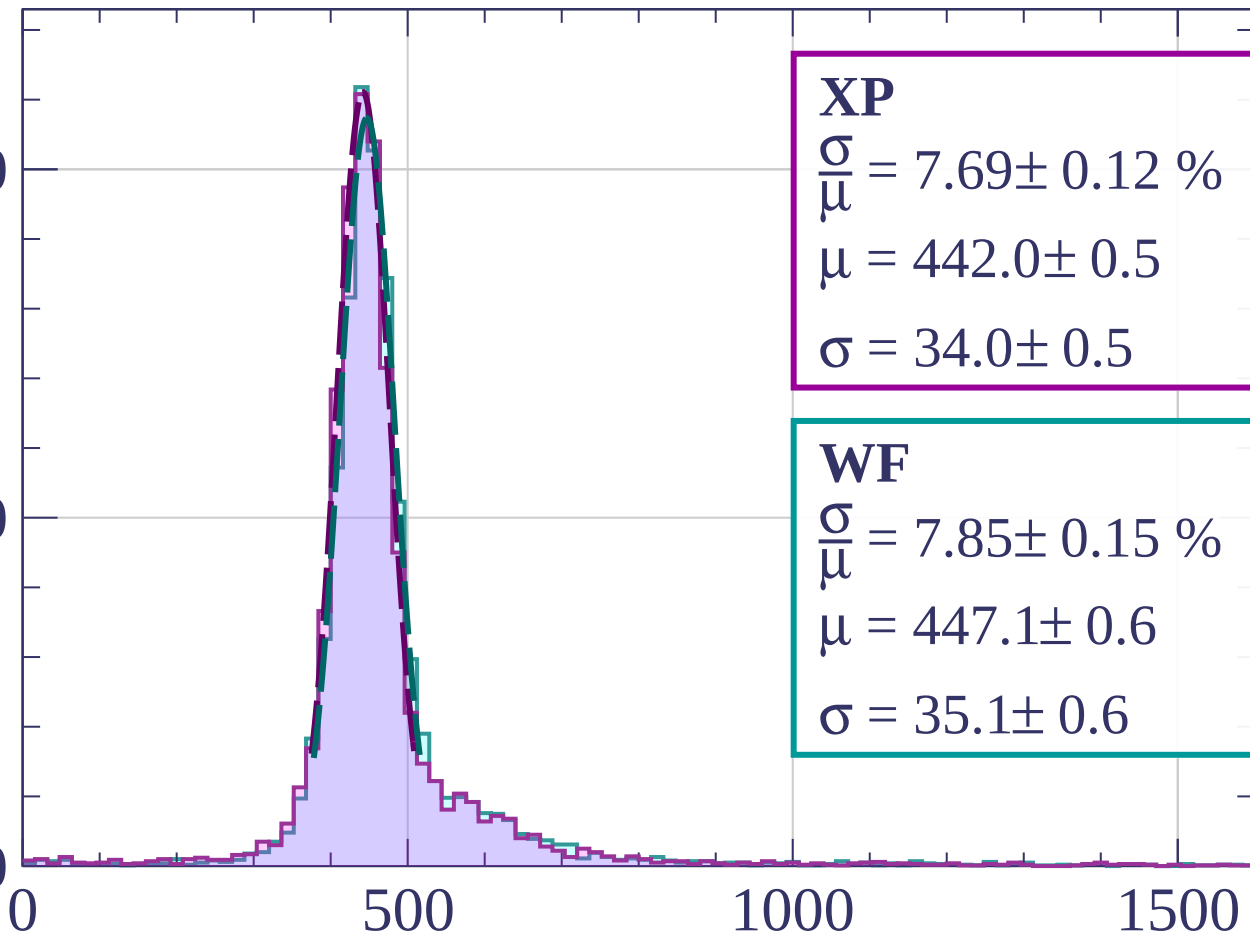
Energy loss | $1040 < p < 1120$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 7.69 \pm 0.12 \%$$

$$\mu = 442.0 \pm 0.5$$

$$\sigma = 34.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.15 \%$$

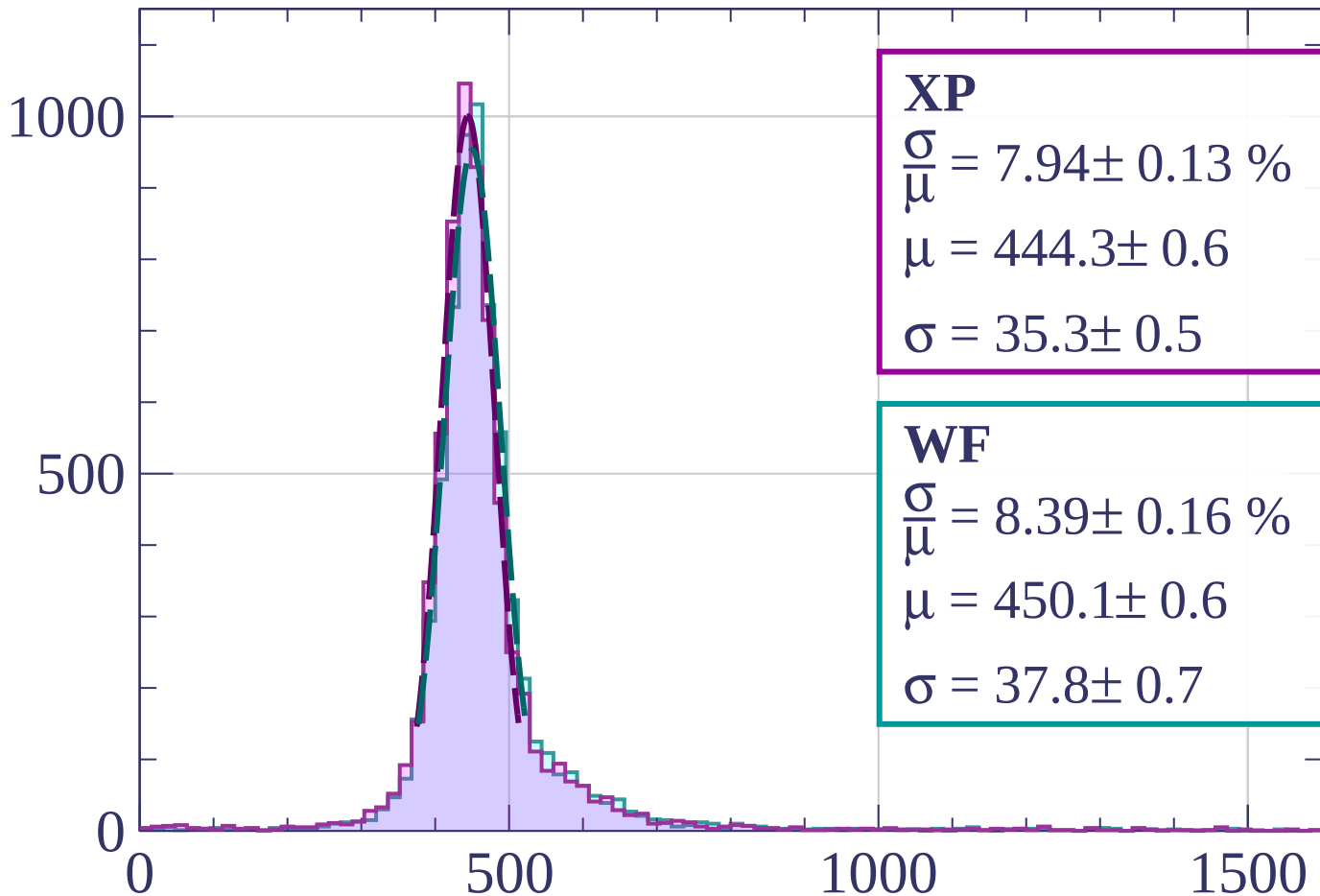
$$\mu = 447.1 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.94 \pm 0.13 \%$$

$$\mu = 444.3 \pm 0.6$$

$$\sigma = 35.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.16 \%$$

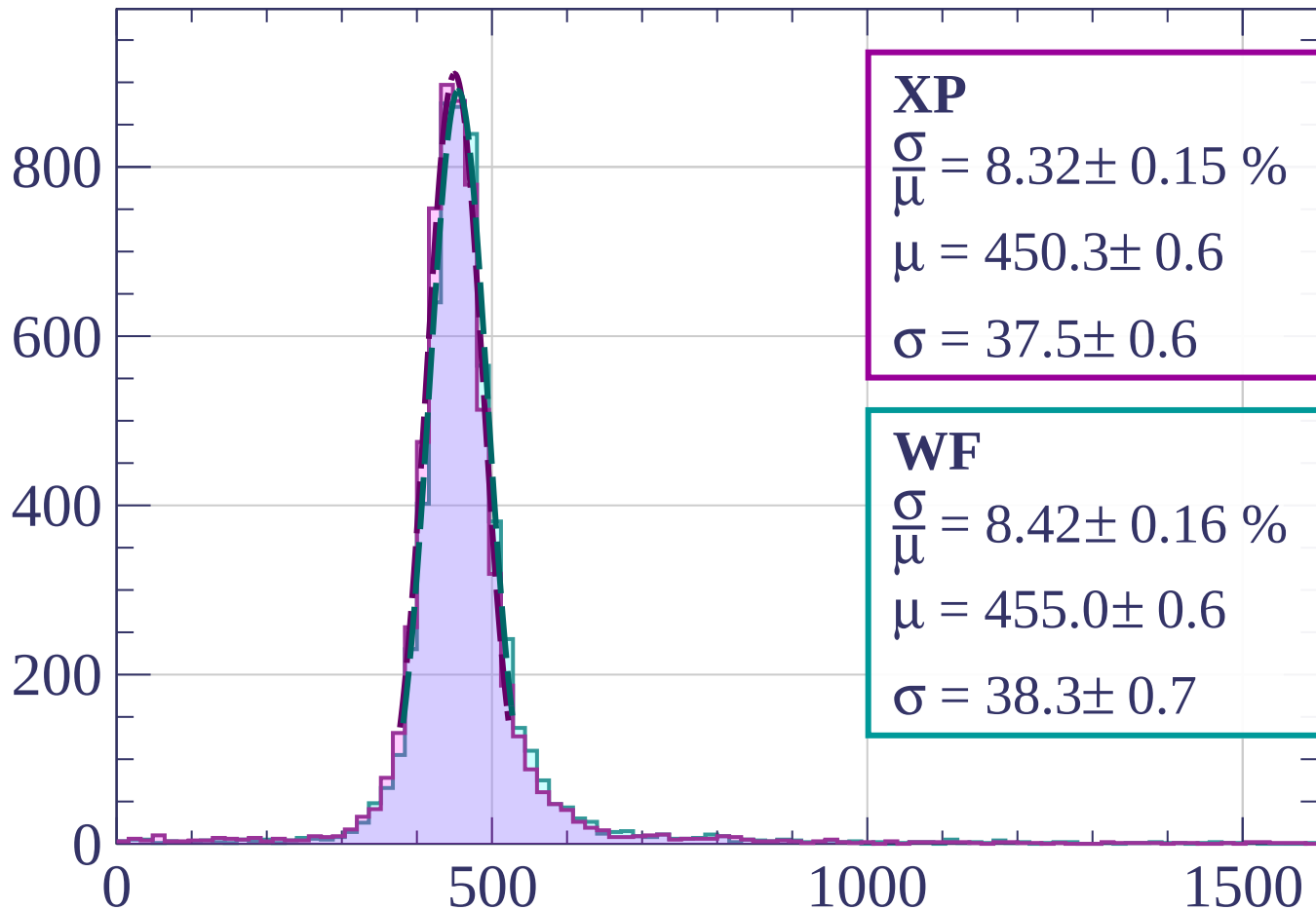
$$\mu = 450.1 \pm 0.6$$

$$\sigma = 37.8 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

$$\mu = 450.3 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.16 \%$$

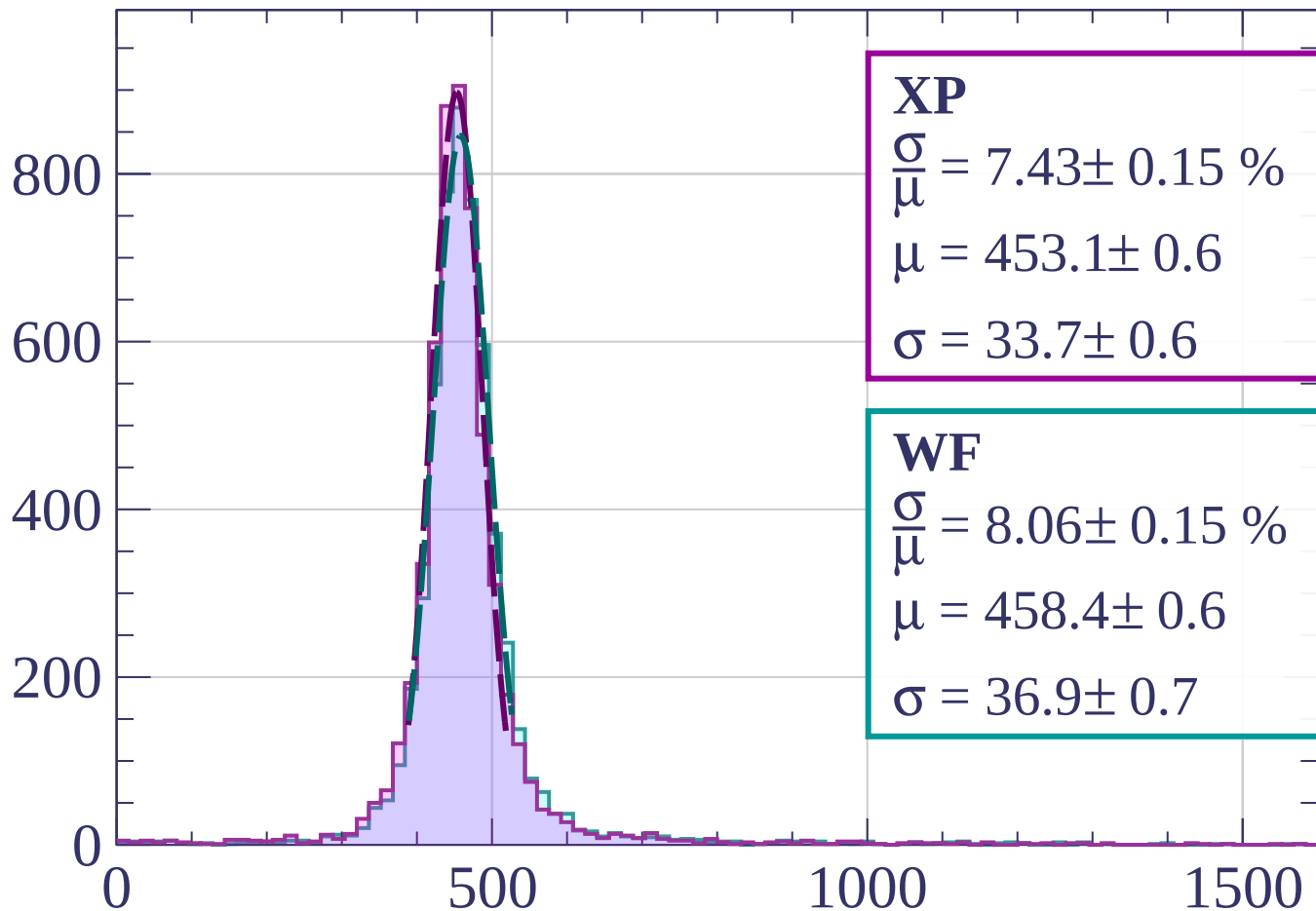
$$\mu = 455.0 \pm 0.6$$

$$\sigma = 38.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.15 \%$$

$$\mu = 453.1 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.15 \%$$

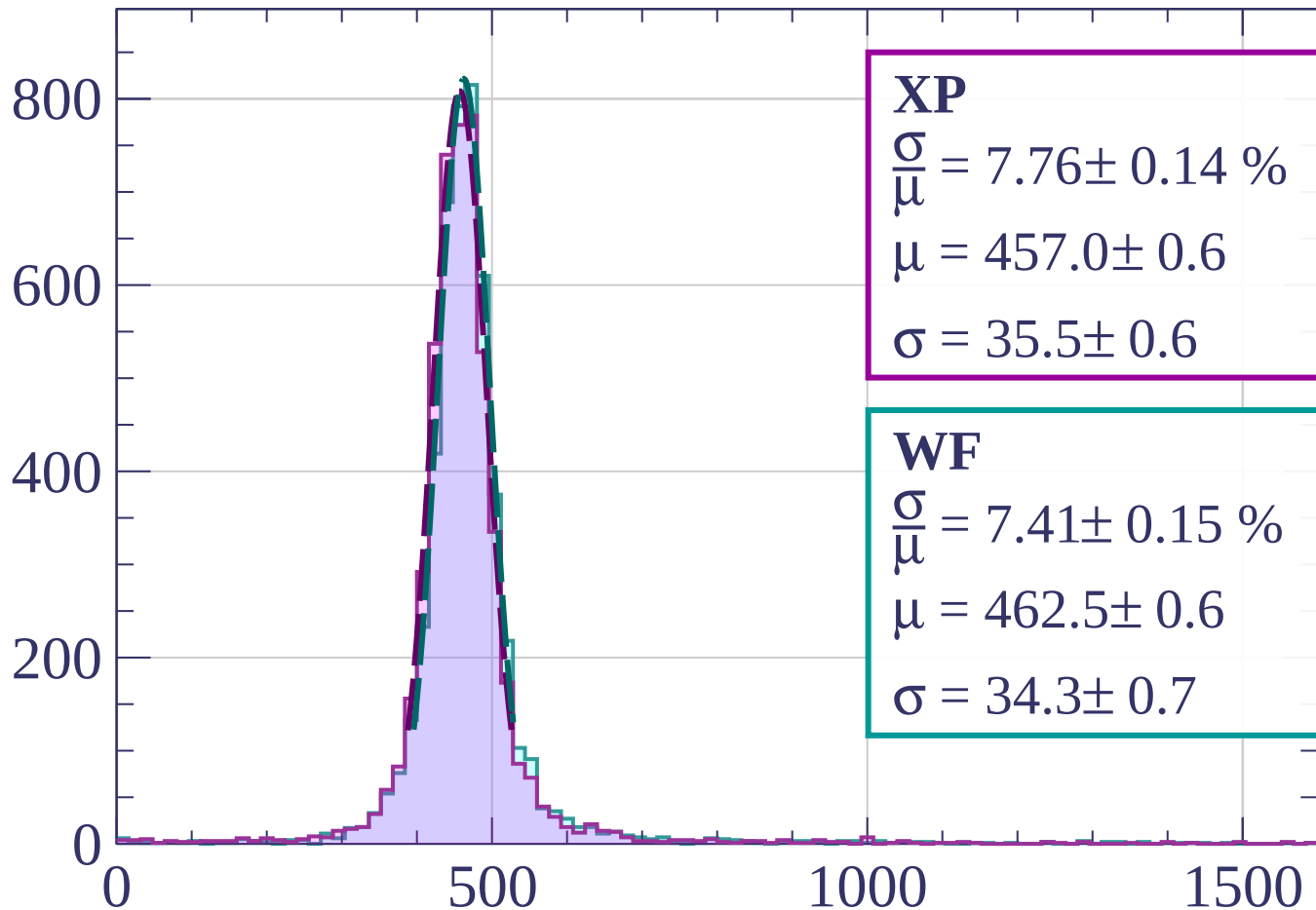
$$\mu = 458.4 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.14 \%$$

$$\mu = 457.0 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.15 \%$$

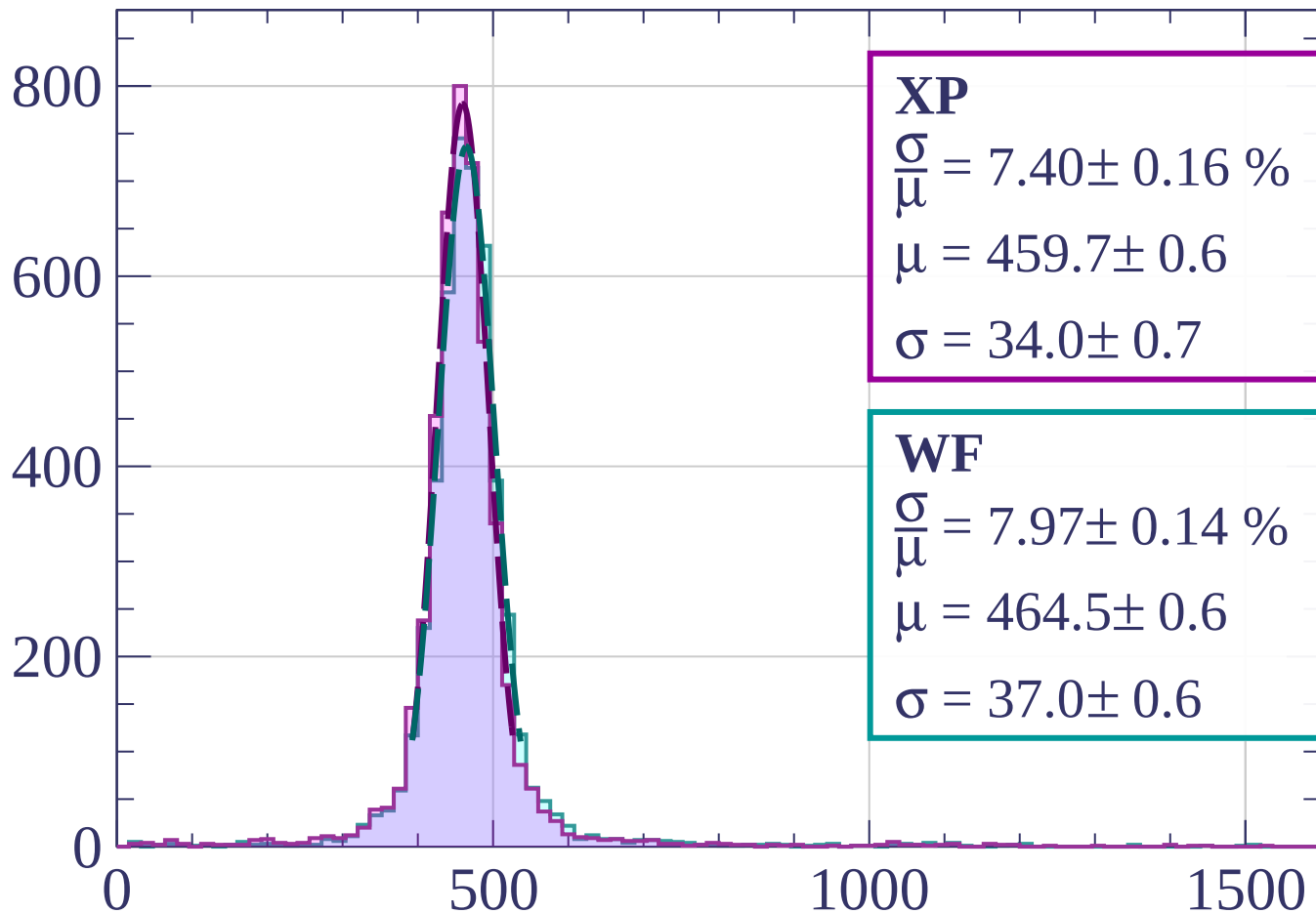
$$\mu = 462.5 \pm 0.6$$

$$\sigma = 34.3 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.16 \%$$

$$\mu = 459.7 \pm 0.6$$

$$\sigma = 34.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.14 \%$$

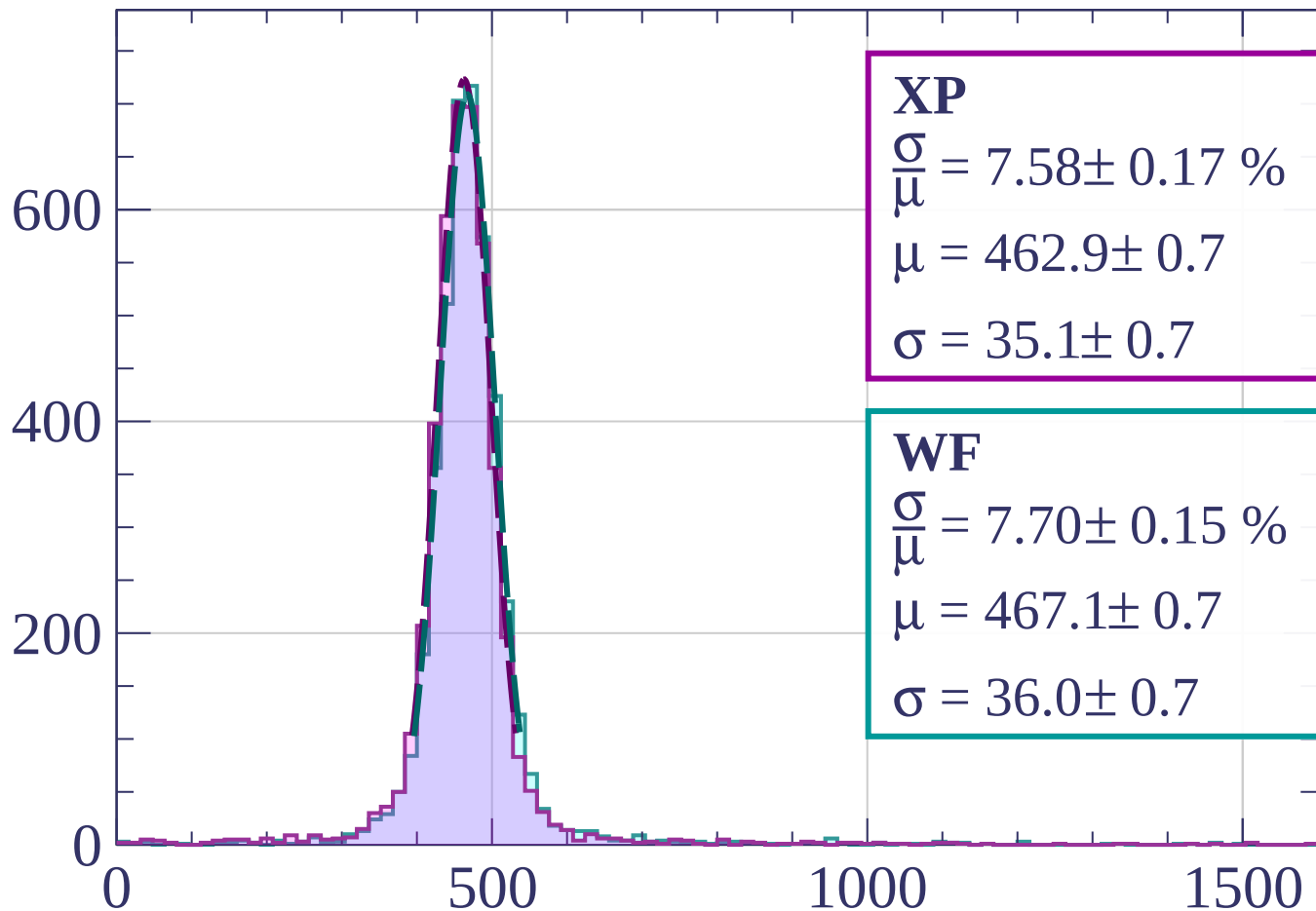
$$\mu = 464.5 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.58 \pm 0.17 \%$$

$$\mu = 462.9 \pm 0.7$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.15 \%$$

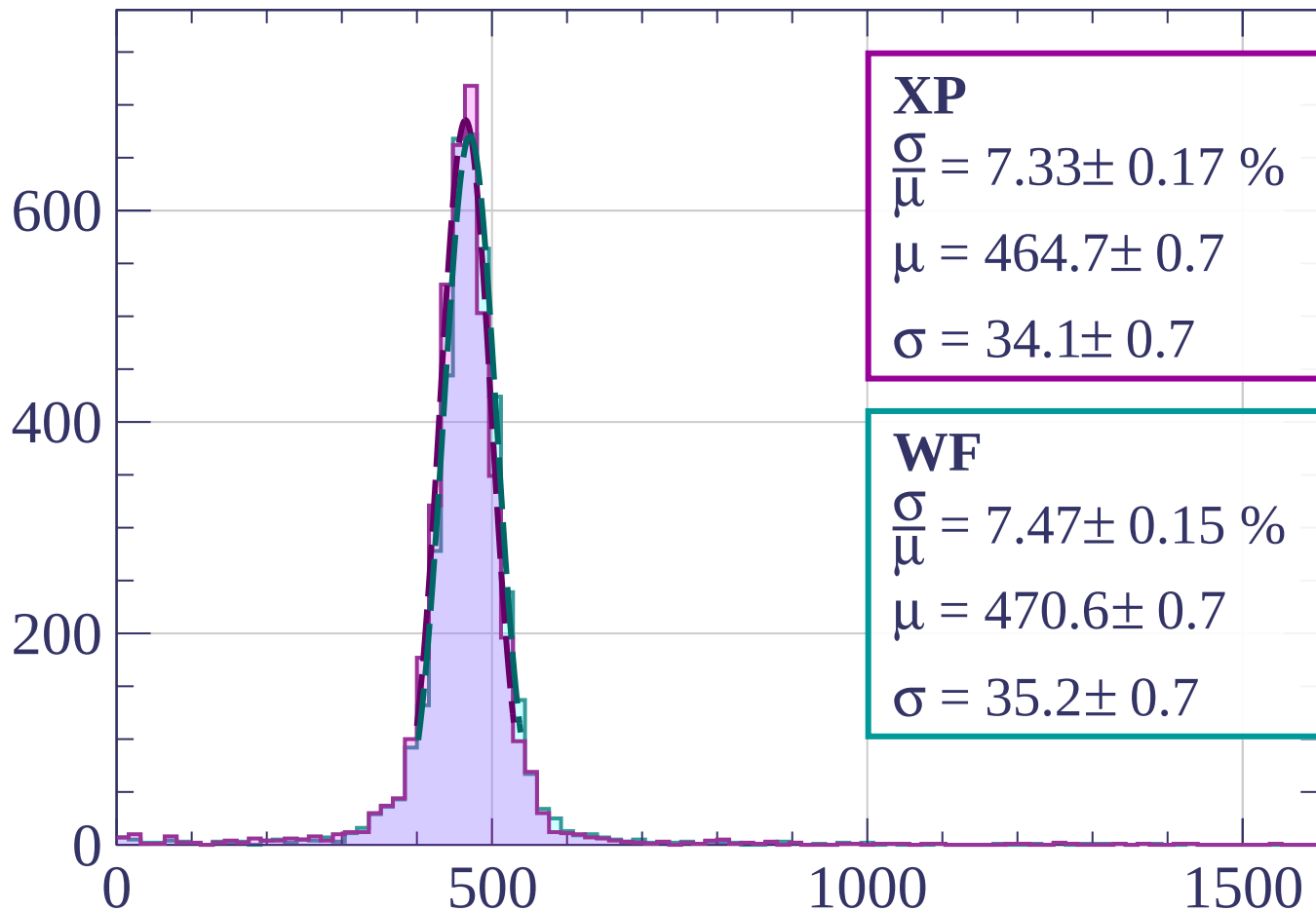
$$\mu = 467.1 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.17 \%$$

$$\mu = 464.7 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.47 \pm 0.15 \%$$

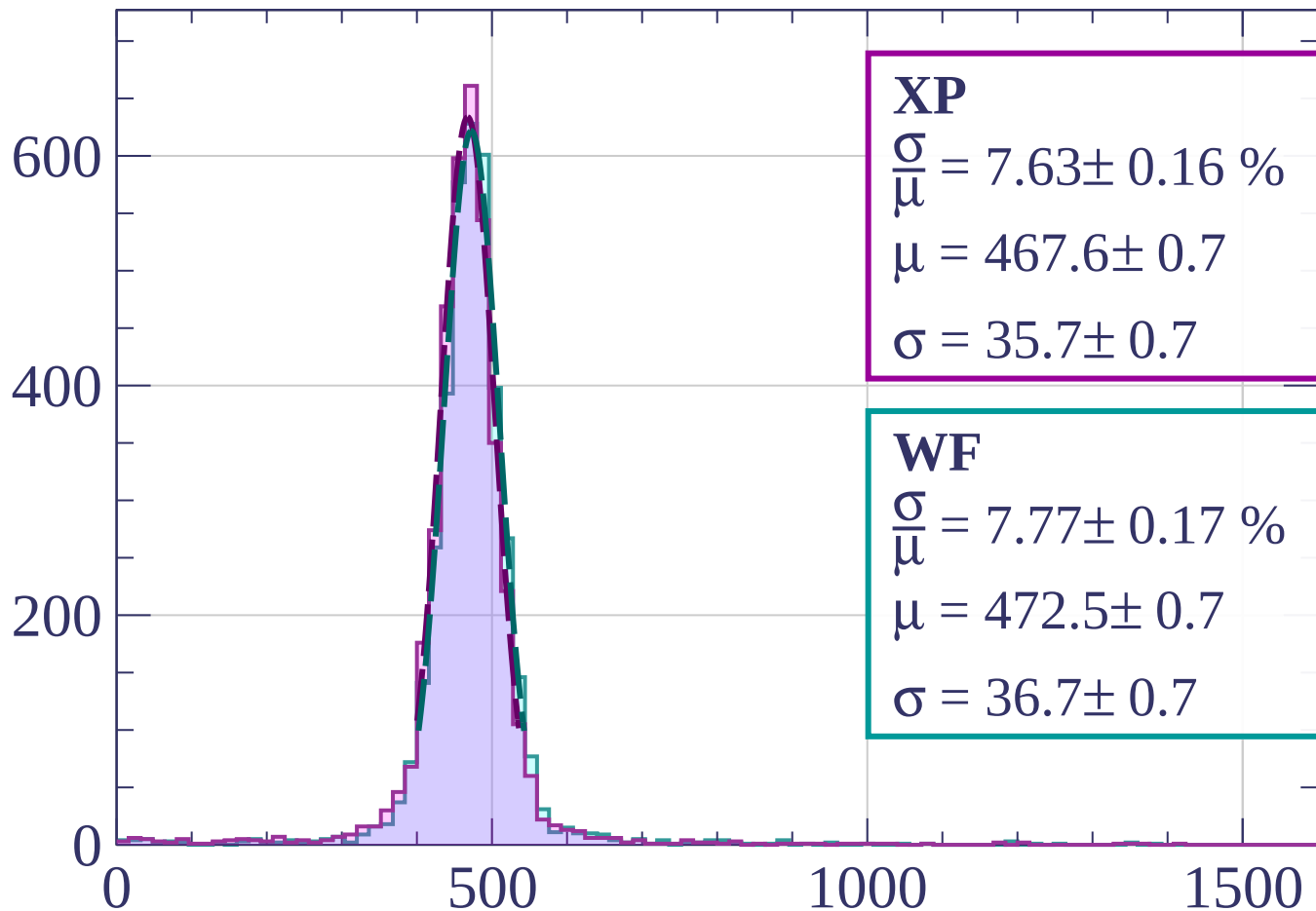
$$\mu = 470.6 \pm 0.7$$

$$\sigma = 35.2 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.63 \pm 0.16 \%$$

$$\mu = 467.6 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.17 \%$$

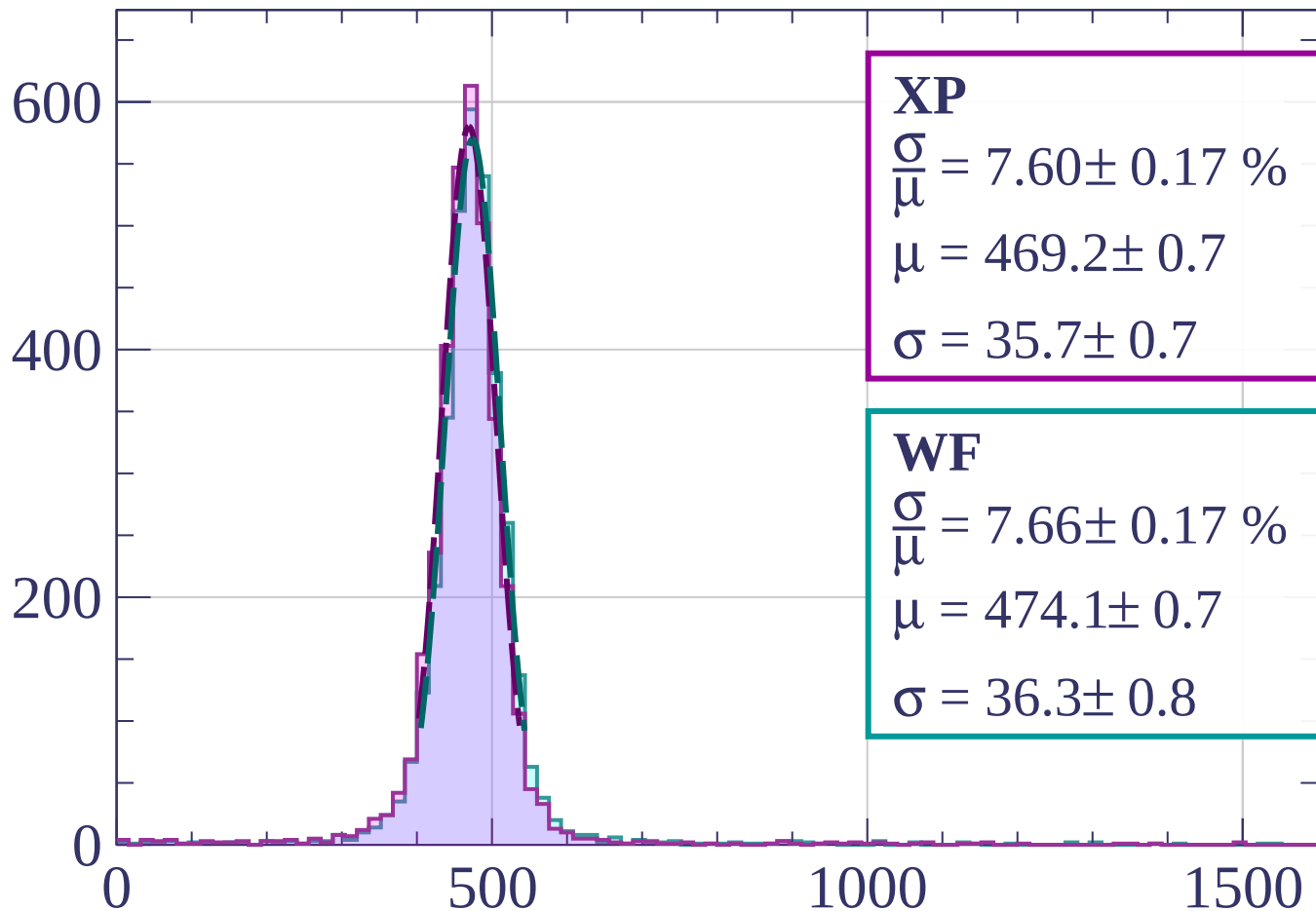
$$\mu = 472.5 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.17 \%$$

$$\mu = 469.2 \pm 0.7$$

$$\sigma = 35.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.17 \%$$

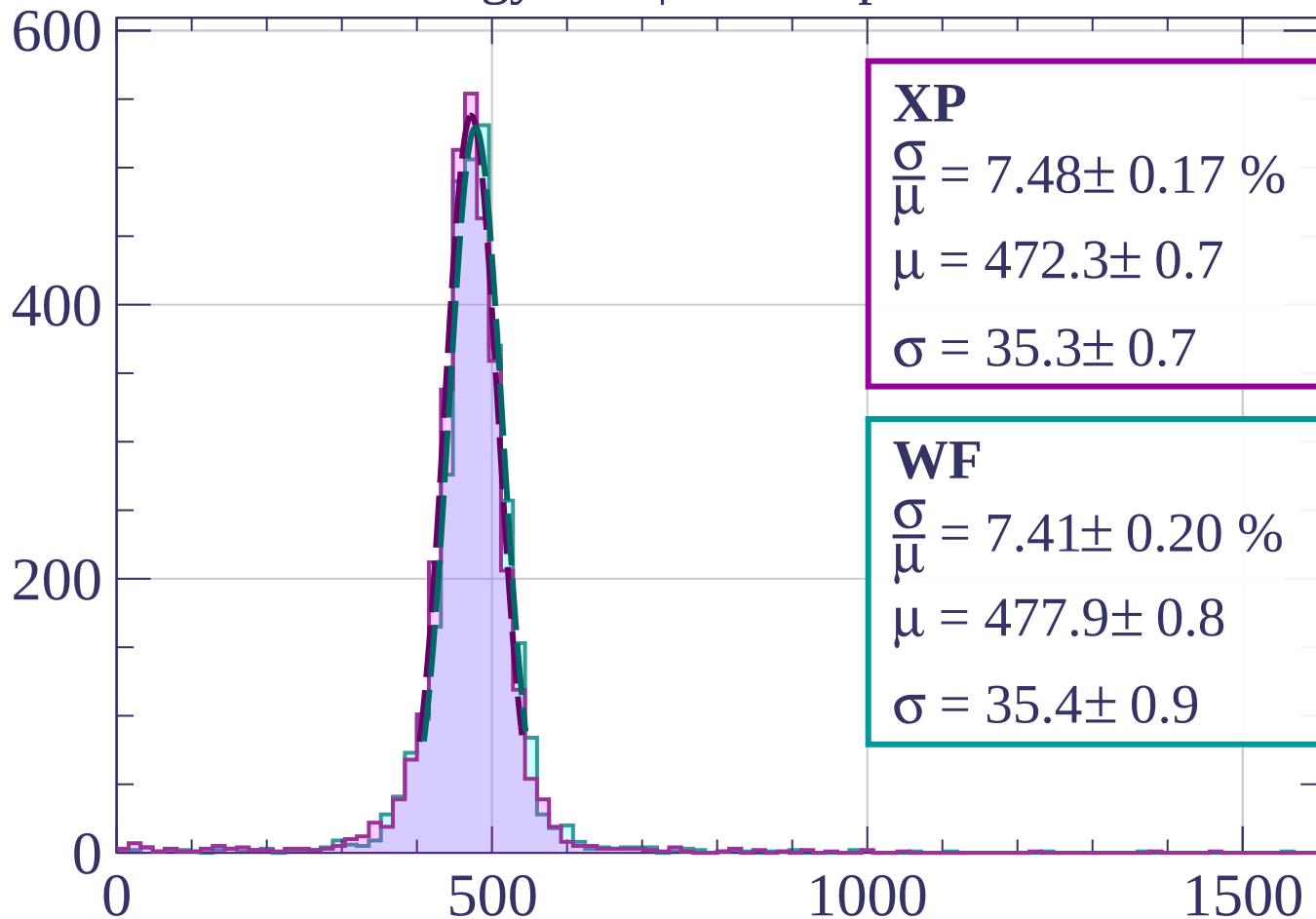
$$\mu = 474.1 \pm 0.7$$

$$\sigma = 36.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.48 \pm 0.17 \%$$

$$\mu = 472.3 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.41 \pm 0.20 \%$$

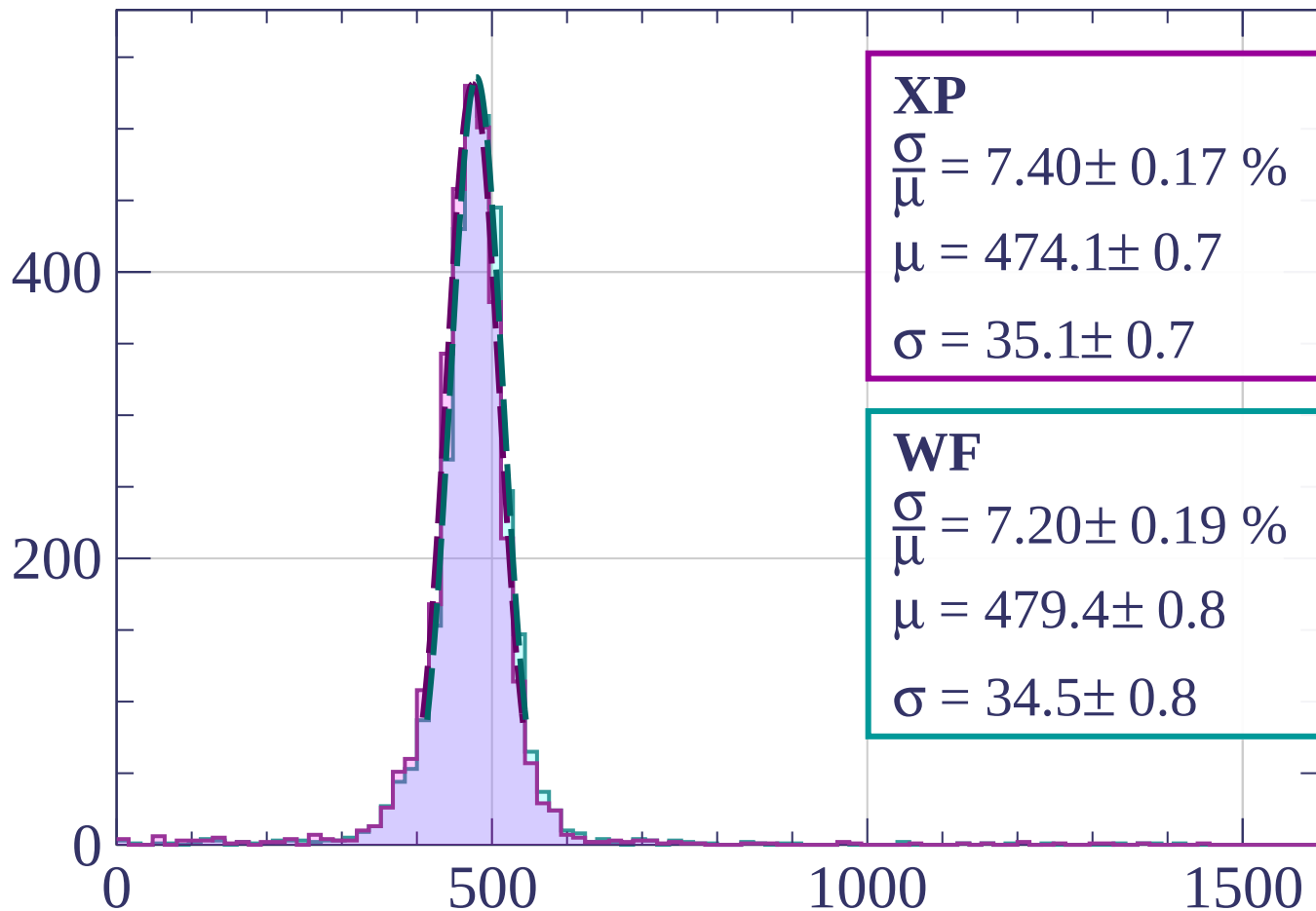
$$\mu = 477.9 \pm 0.8$$

$$\sigma = 35.4 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.17 \%$$

$$\mu = 474.1 \pm 0.7$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.20 \pm 0.19 \%$$

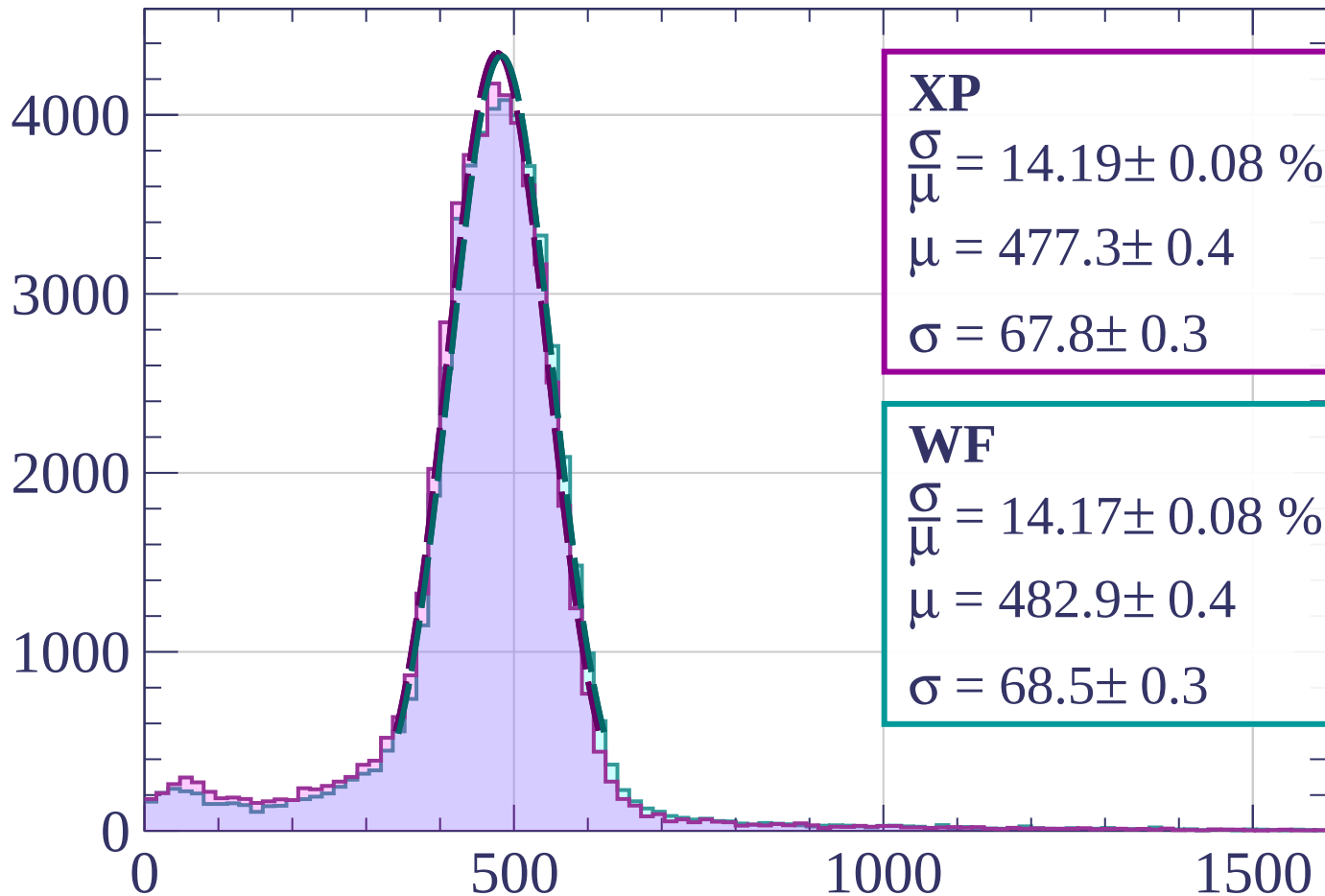
$$\mu = 479.4 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

Count



XP

$$\frac{\sigma}{\mu} = 14.19 \pm 0.08 \%$$

$$\mu = 477.3 \pm 0.4$$

$$\sigma = 67.8 \pm 0.3$$

WF

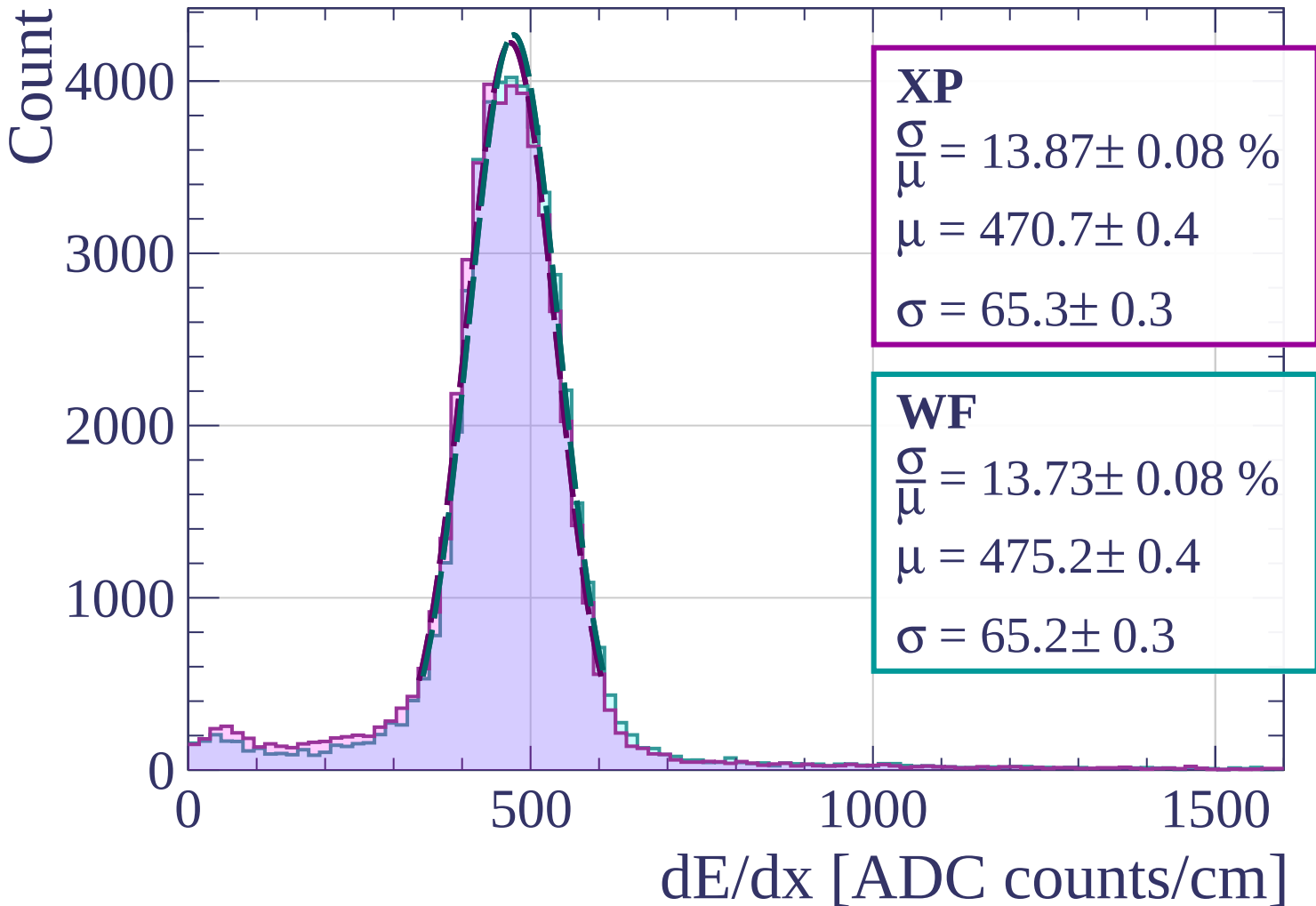
$$\frac{\sigma}{\mu} = 14.17 \pm 0.08 \%$$

$$\mu = 482.9 \pm 0.4$$

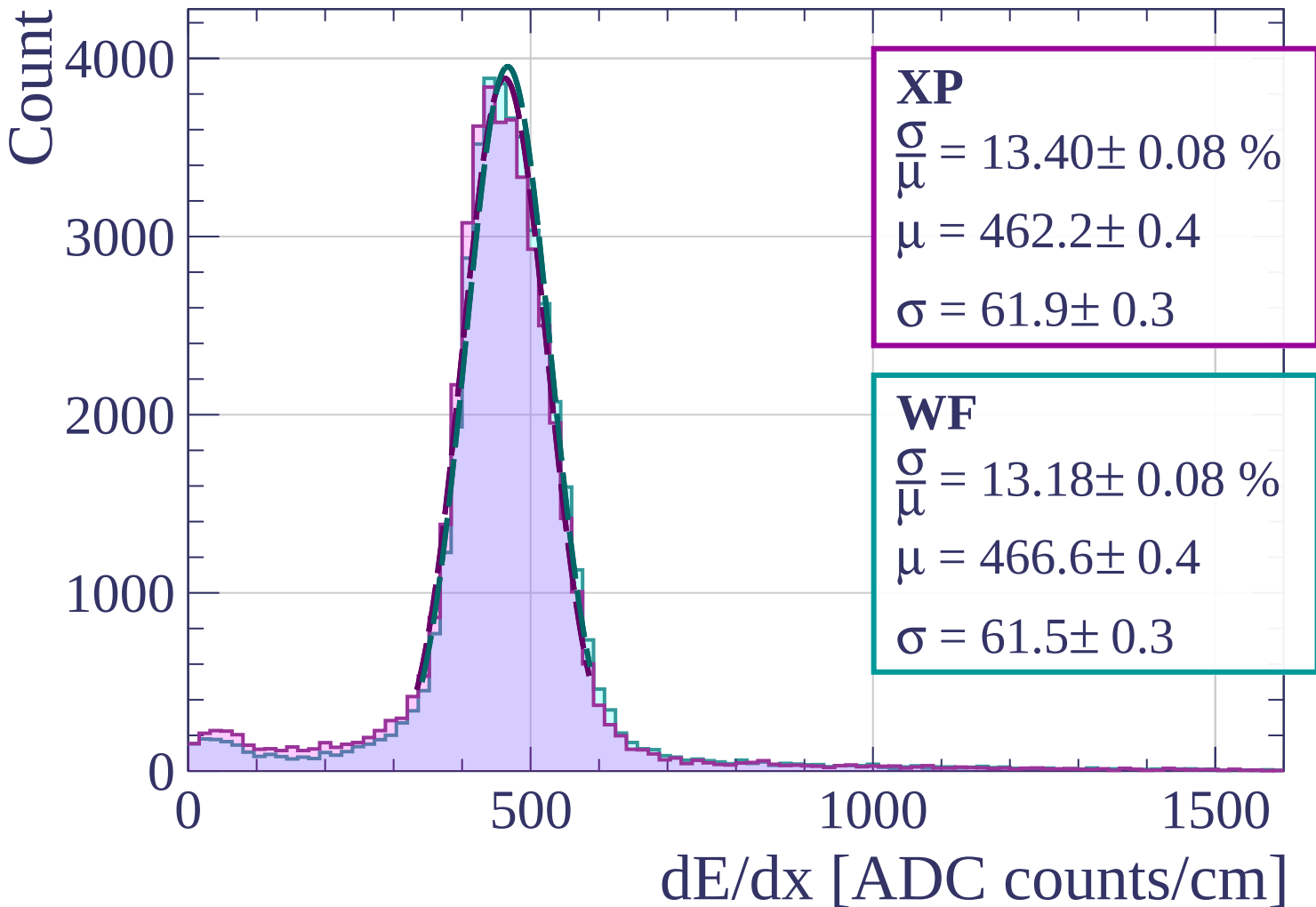
$$\sigma = 68.5 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

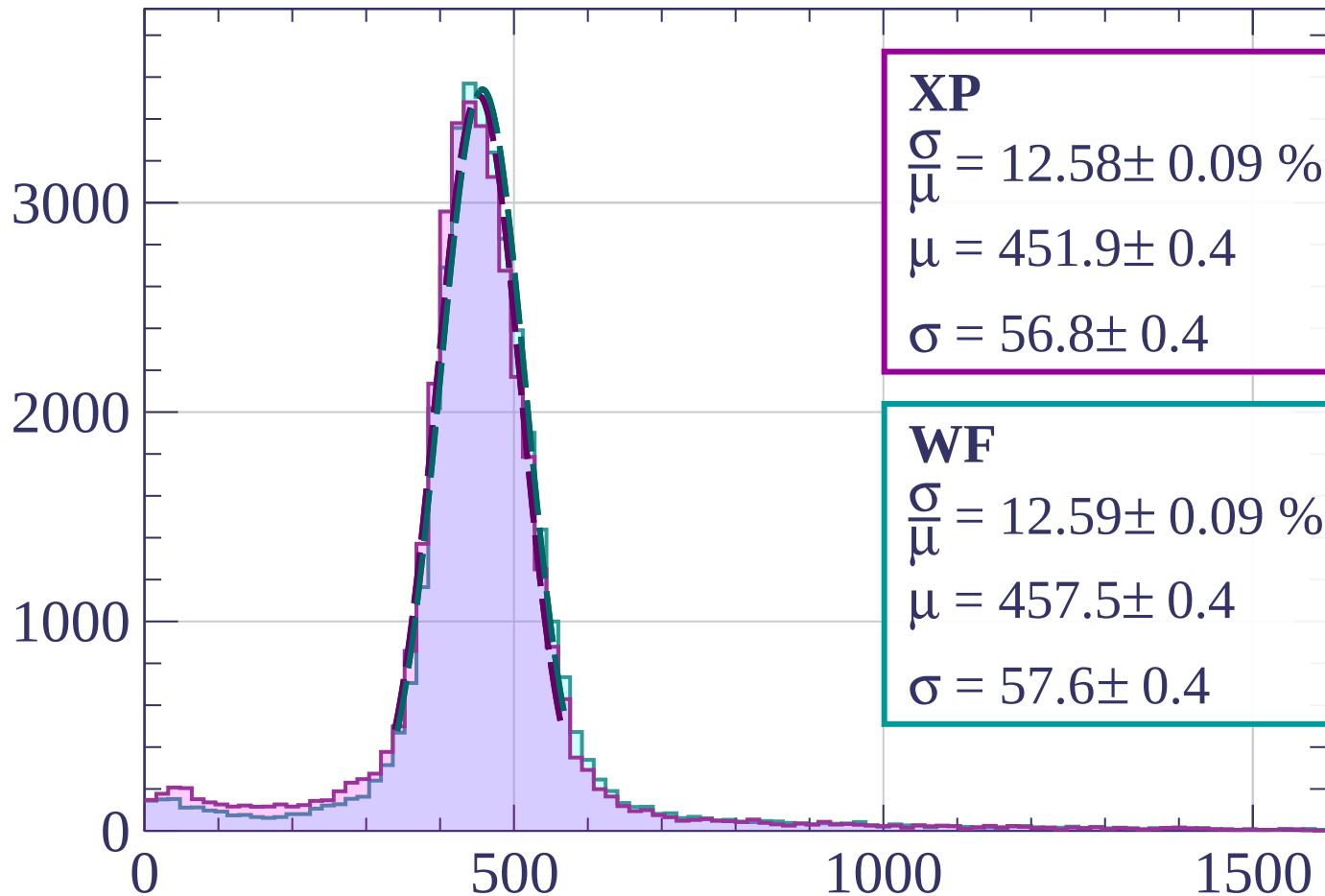


Energy loss | $6 < \phi < 9$



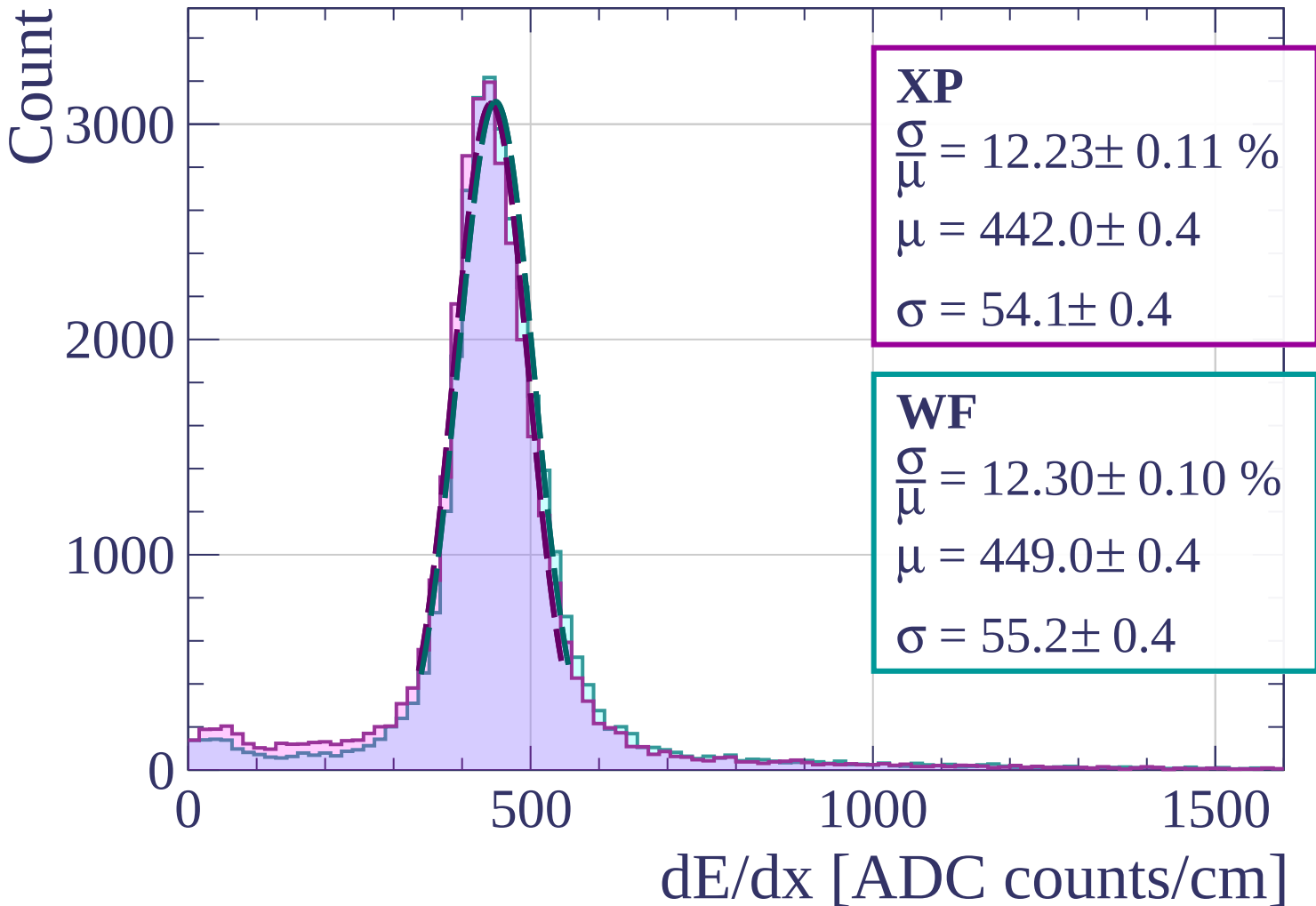
Energy loss | $9 < \phi < 12$

Count



dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$



Energy loss | $15 < \phi < 18$

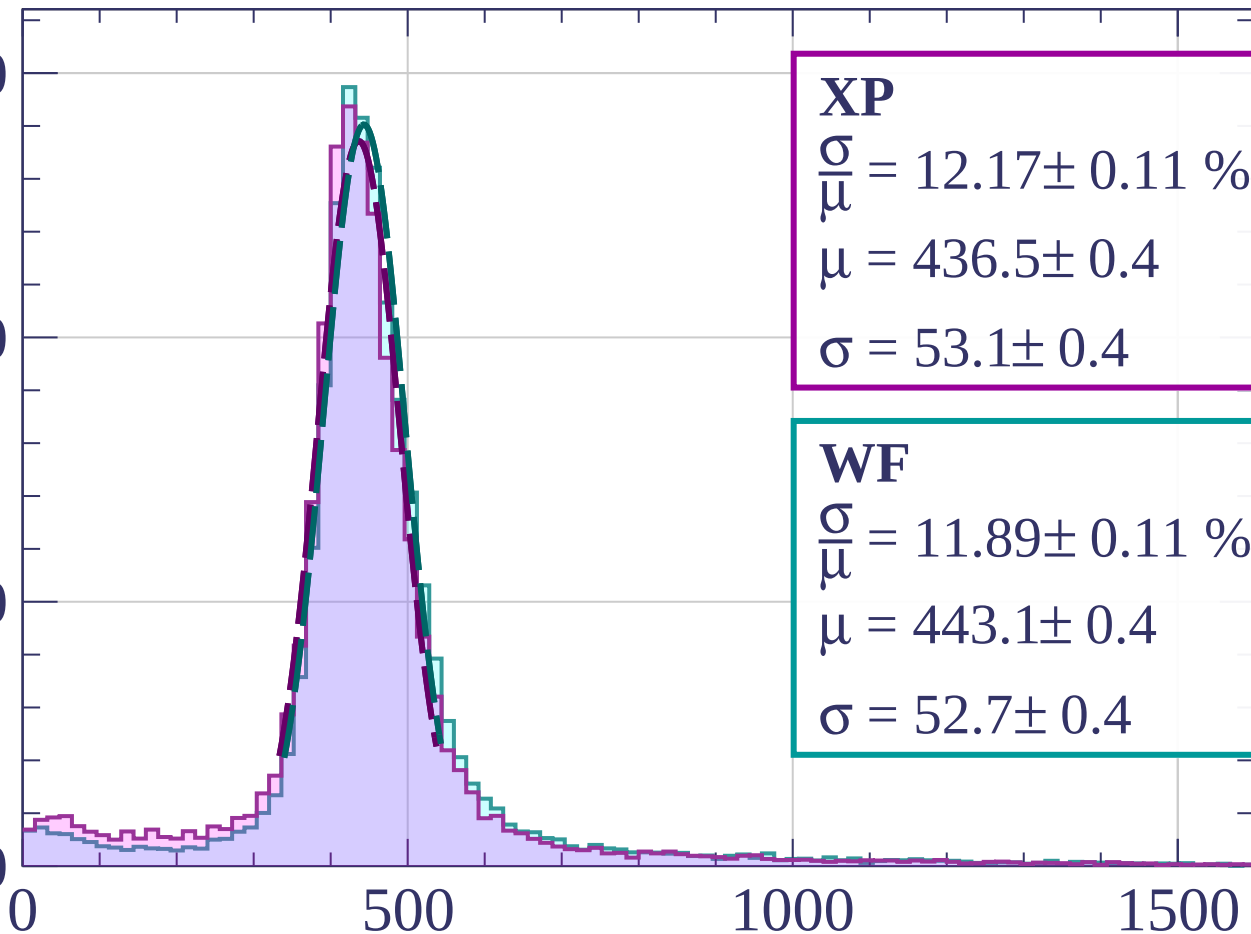
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 12.17 \pm 0.11 \%$$

$$\mu = 436.5 \pm 0.4$$

$$\sigma = 53.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 11.89 \pm 0.11 \%$$

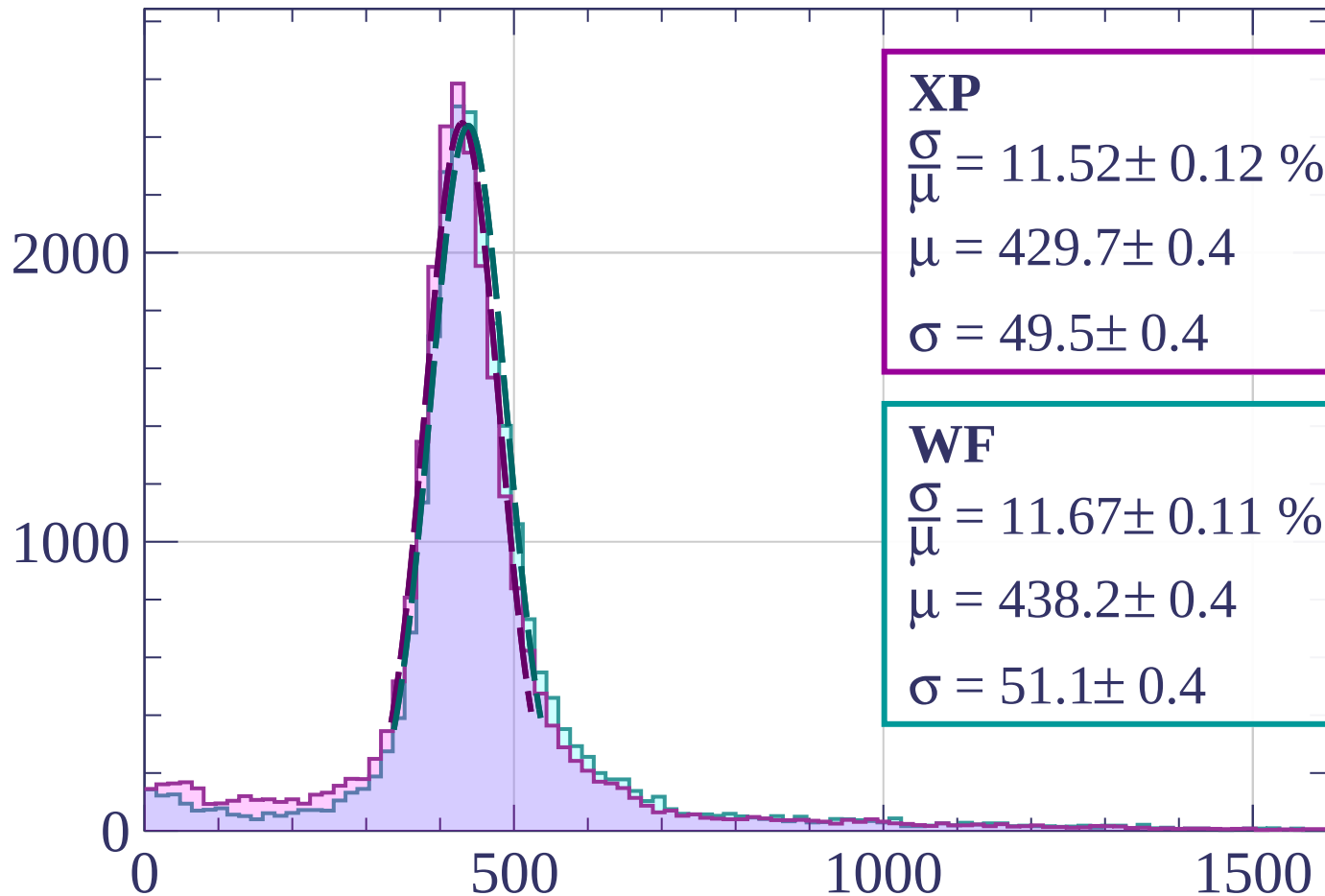
$$\mu = 443.1 \pm 0.4$$

$$\sigma = 52.7 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

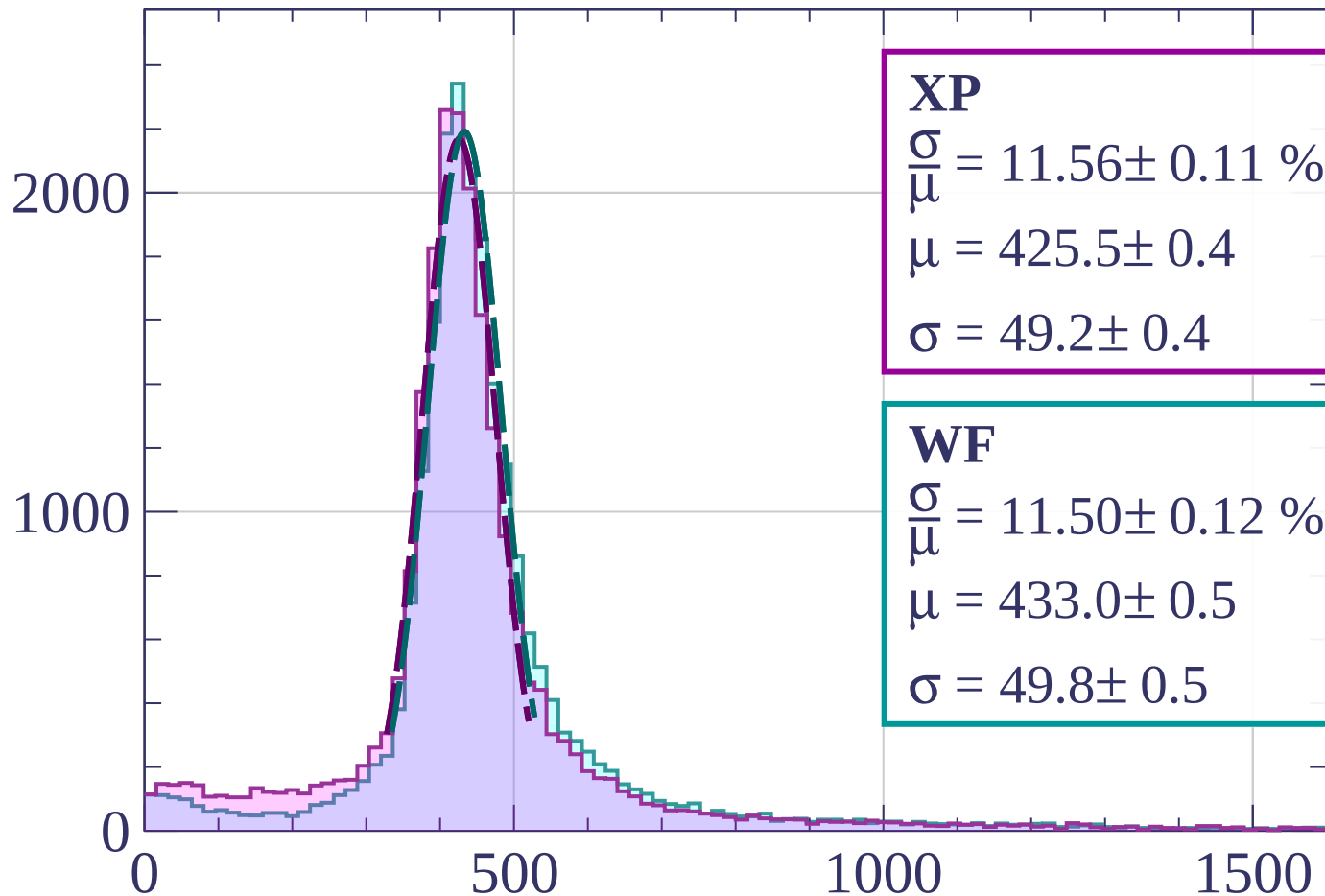
Count



dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 11.56 \pm 0.11 \%$$

$$\mu = 425.5 \pm 0.4$$

$$\sigma = 49.2 \pm 0.4$$

WF

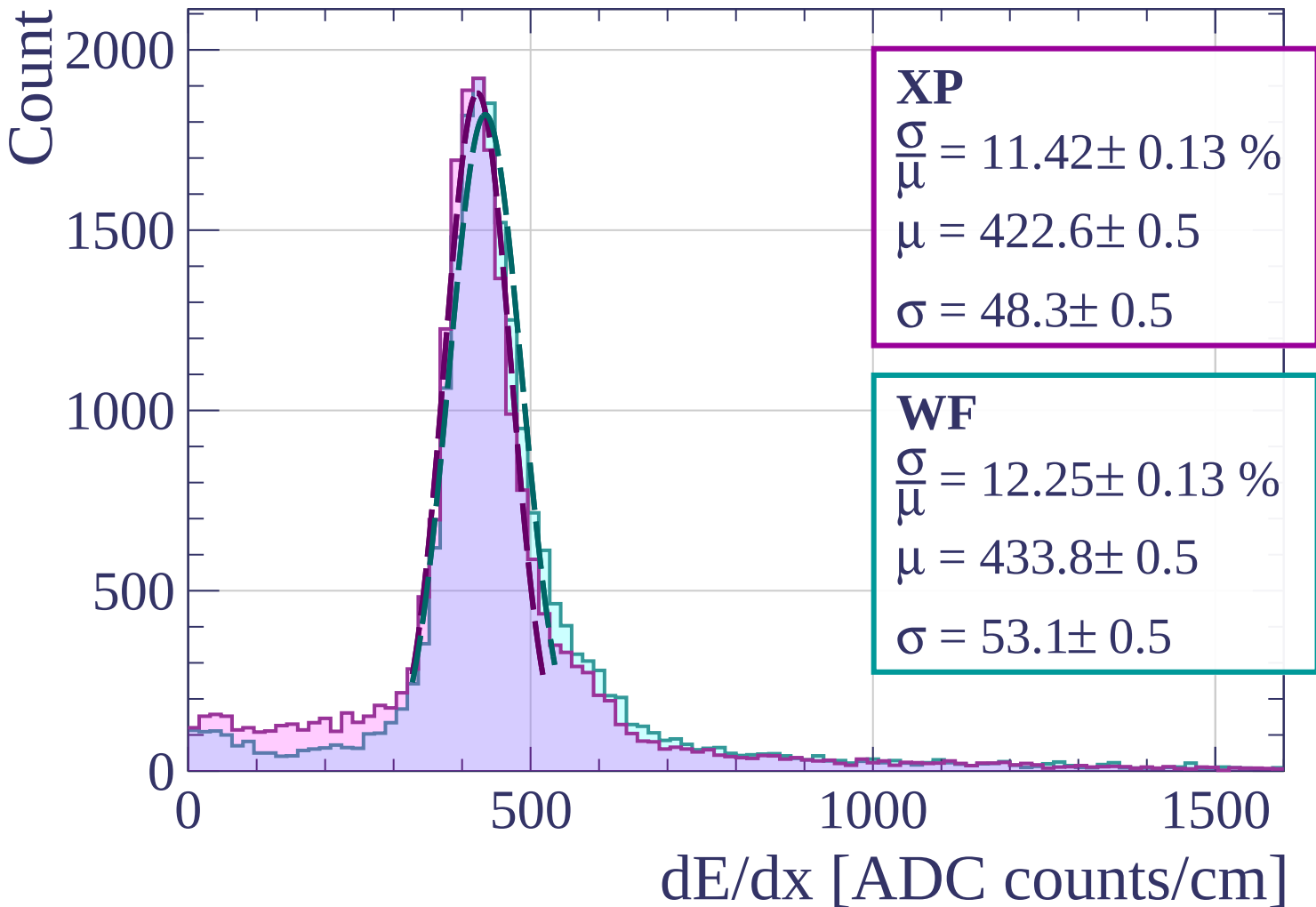
$$\frac{\sigma}{\mu} = 11.50 \pm 0.12 \%$$

$$\mu = 433.0 \pm 0.5$$

$$\sigma = 49.8 \pm 0.5$$

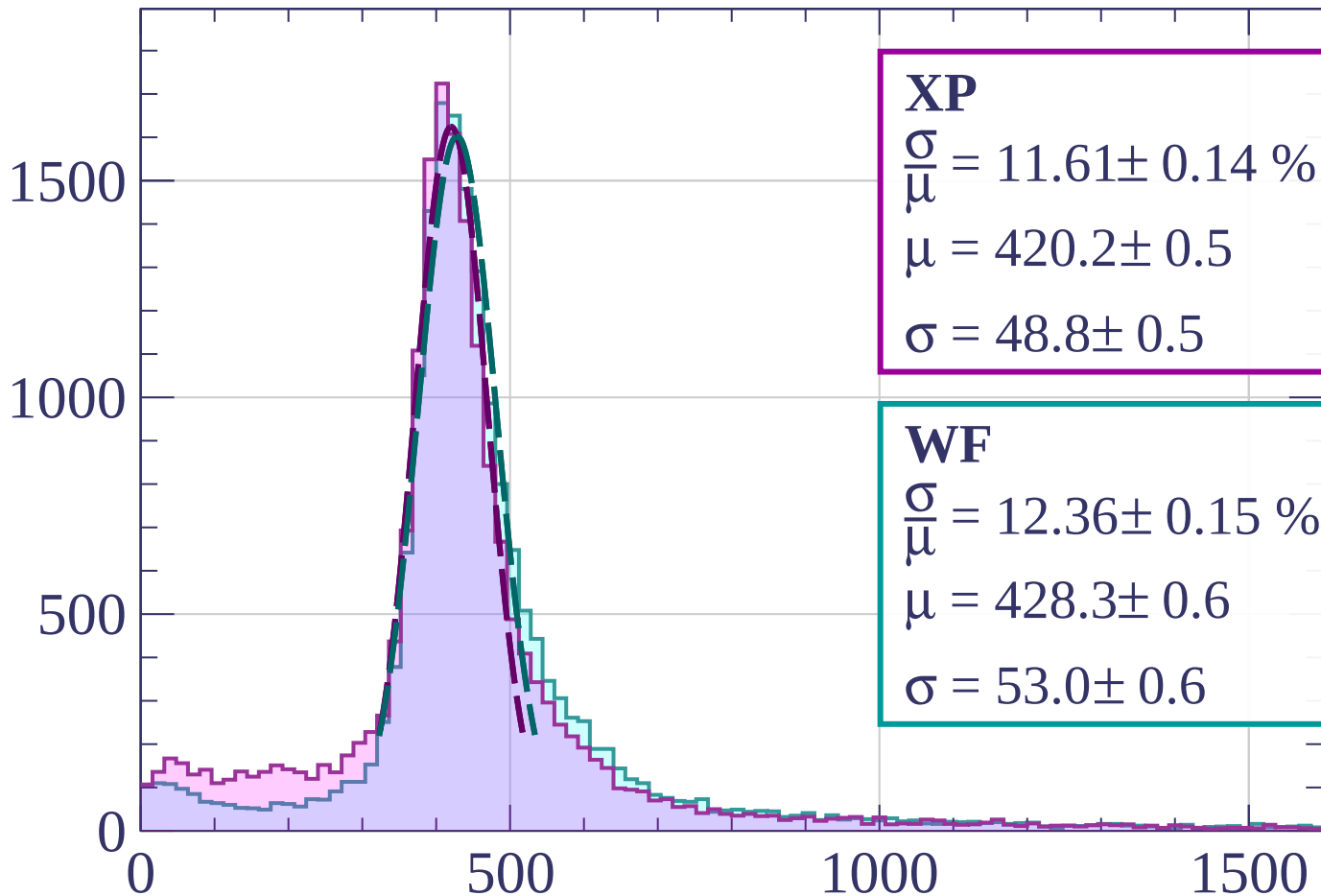
dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$



Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 11.61 \pm 0.14 \%$$

$$\mu = 420.2 \pm 0.5$$

$$\sigma = 48.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 12.36 \pm 0.15 \%$$

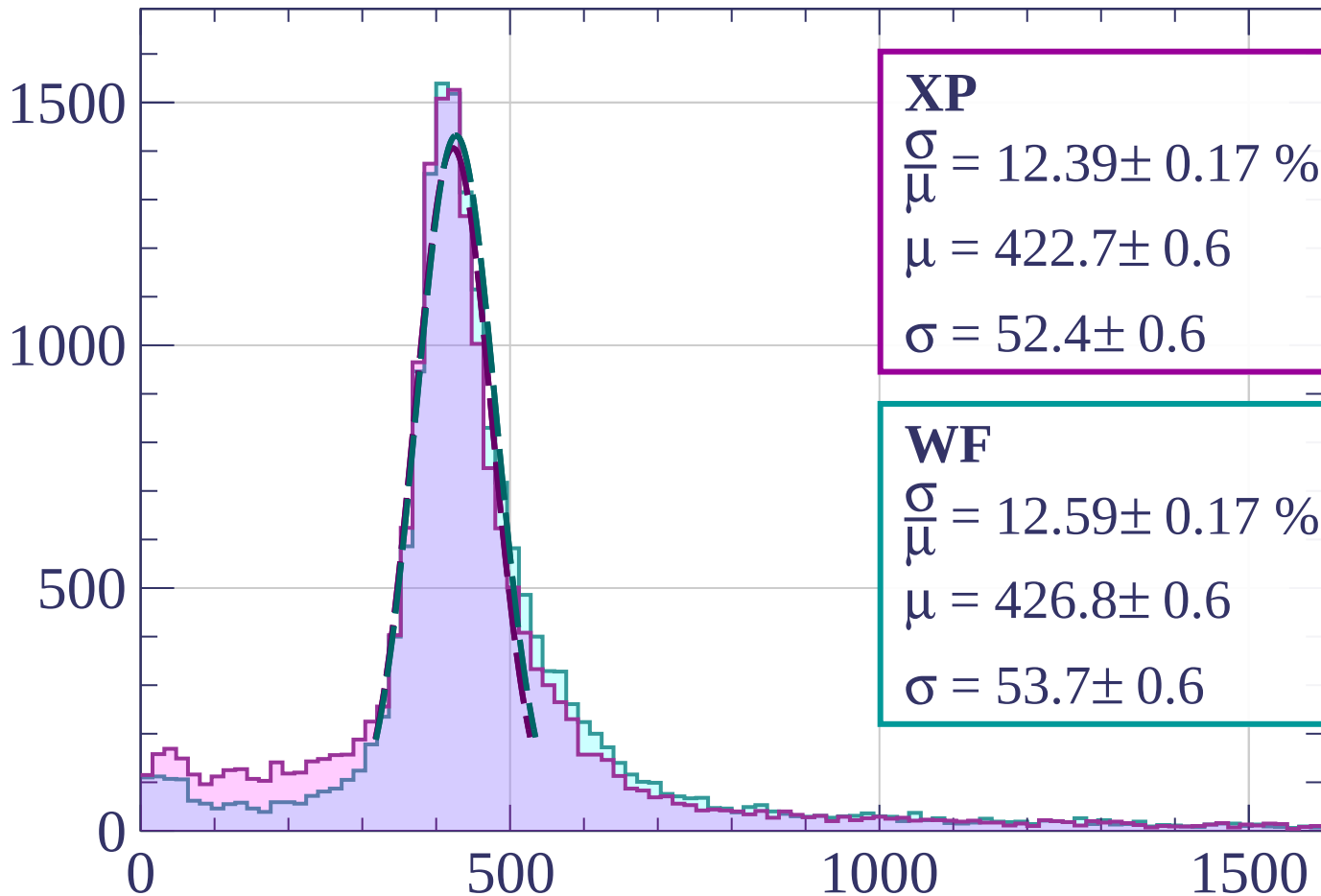
$$\mu = 428.3 \pm 0.6$$

$$\sigma = 53.0 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 12.39 \pm 0.17 \%$$

$$\mu = 422.7 \pm 0.6$$

$$\sigma = 52.4 \pm 0.6$$

WF

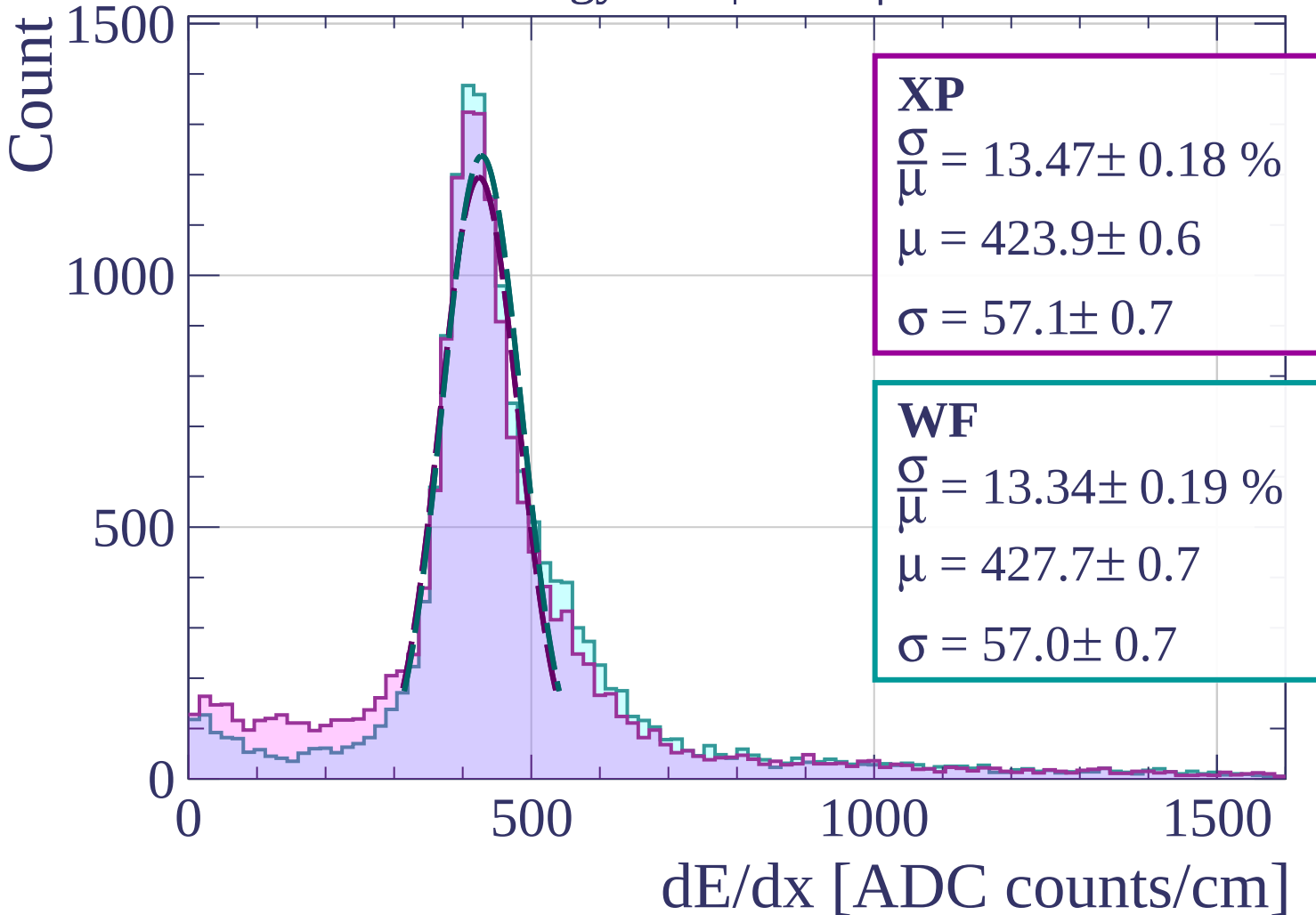
$$\frac{\sigma}{\mu} = 12.59 \pm 0.17 \%$$

$$\mu = 426.8 \pm 0.6$$

$$\sigma = 53.7 \pm 0.6$$

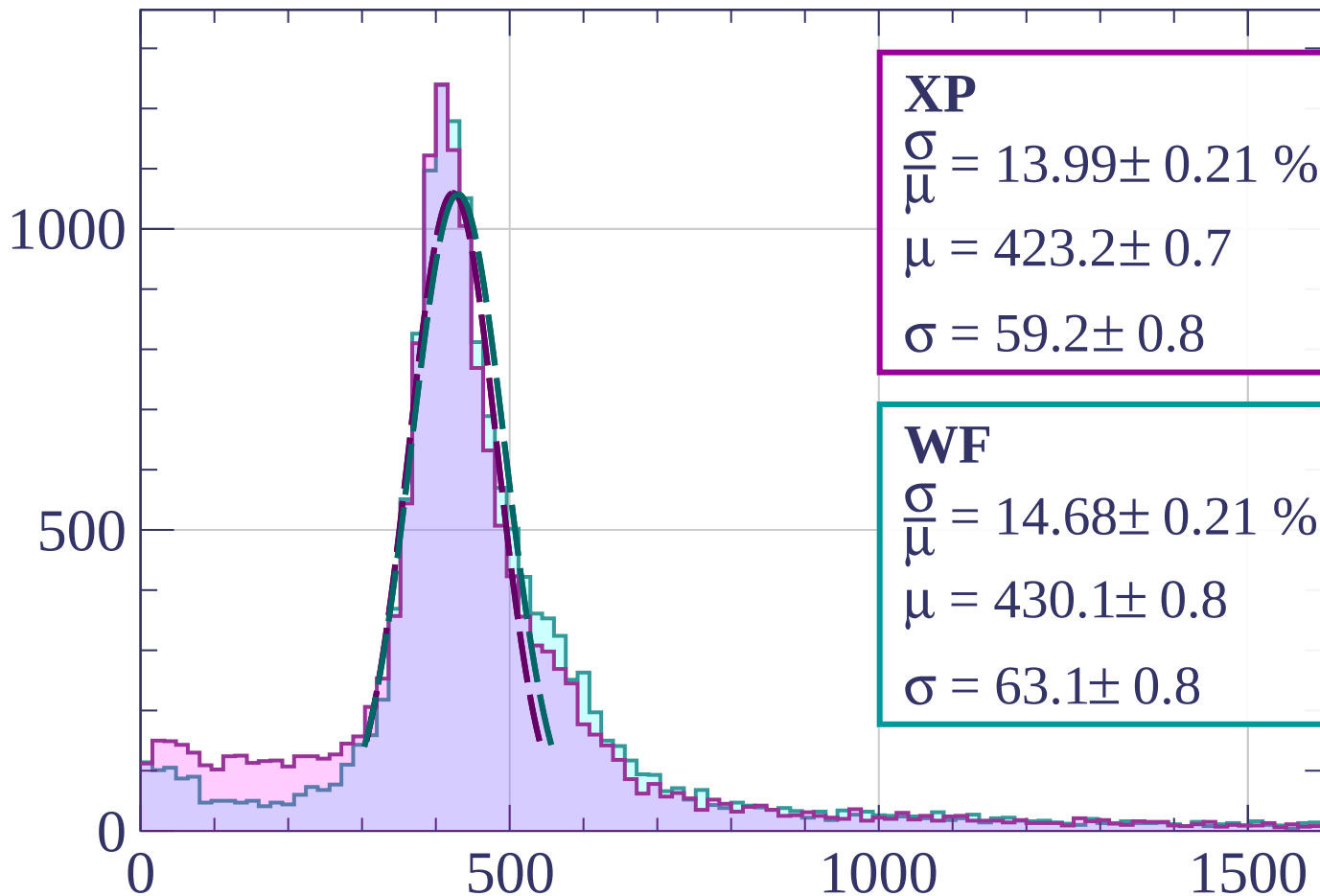
dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$



Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 13.99 \pm 0.21 \%$$

$$\mu = 423.2 \pm 0.7$$

$$\sigma = 59.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 14.68 \pm 0.21 \%$$

$$\mu = 430.1 \pm 0.8$$

$$\sigma = 63.1 \pm 0.8$$

dE/dx [ADC counts/cm]

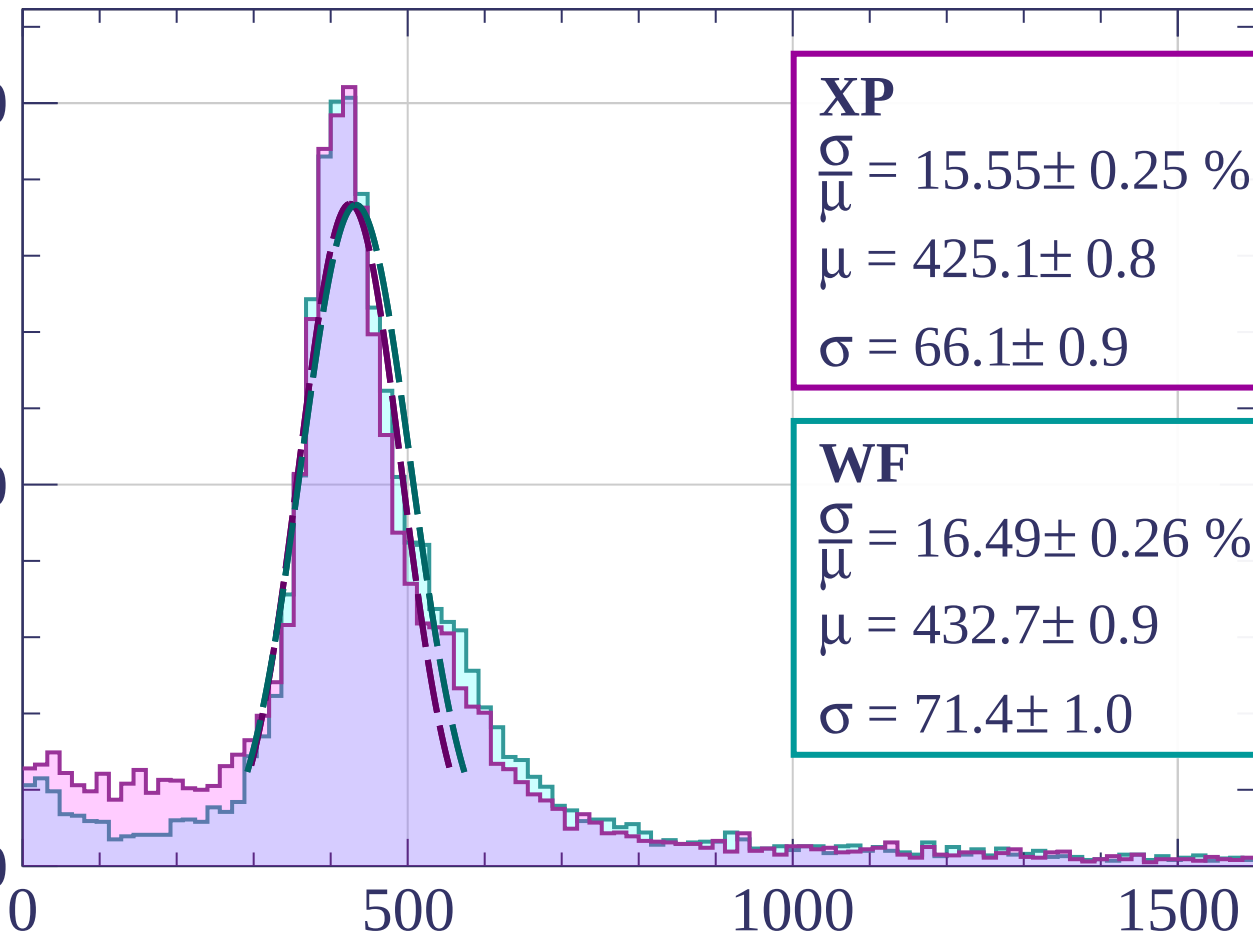
Energy loss | $39 < \phi < 42$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 15.55 \pm 0.25 \%$$

$$\mu = 425.1 \pm 0.8$$

$$\sigma = 66.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 16.49 \pm 0.26 \%$$

$$\mu = 432.7 \pm 0.9$$

$$\sigma = 71.4 \pm 1.0$$

dE/dx [ADC counts/cm]

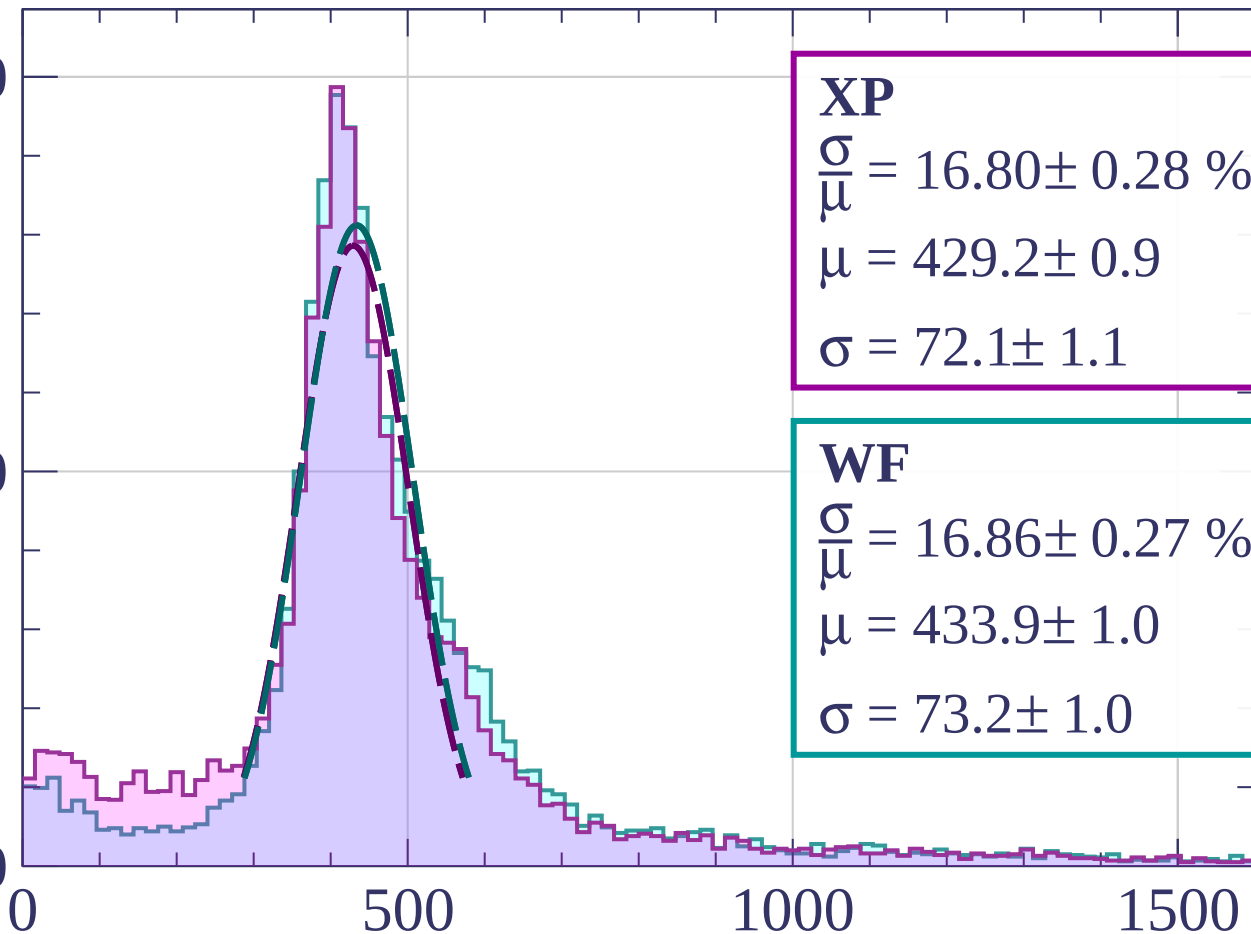
Energy loss | $42 < \phi < 45$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 16.80 \pm 0.28 \%$$

$$\mu = 429.2 \pm 0.9$$

$$\sigma = 72.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 16.86 \pm 0.27 \%$$

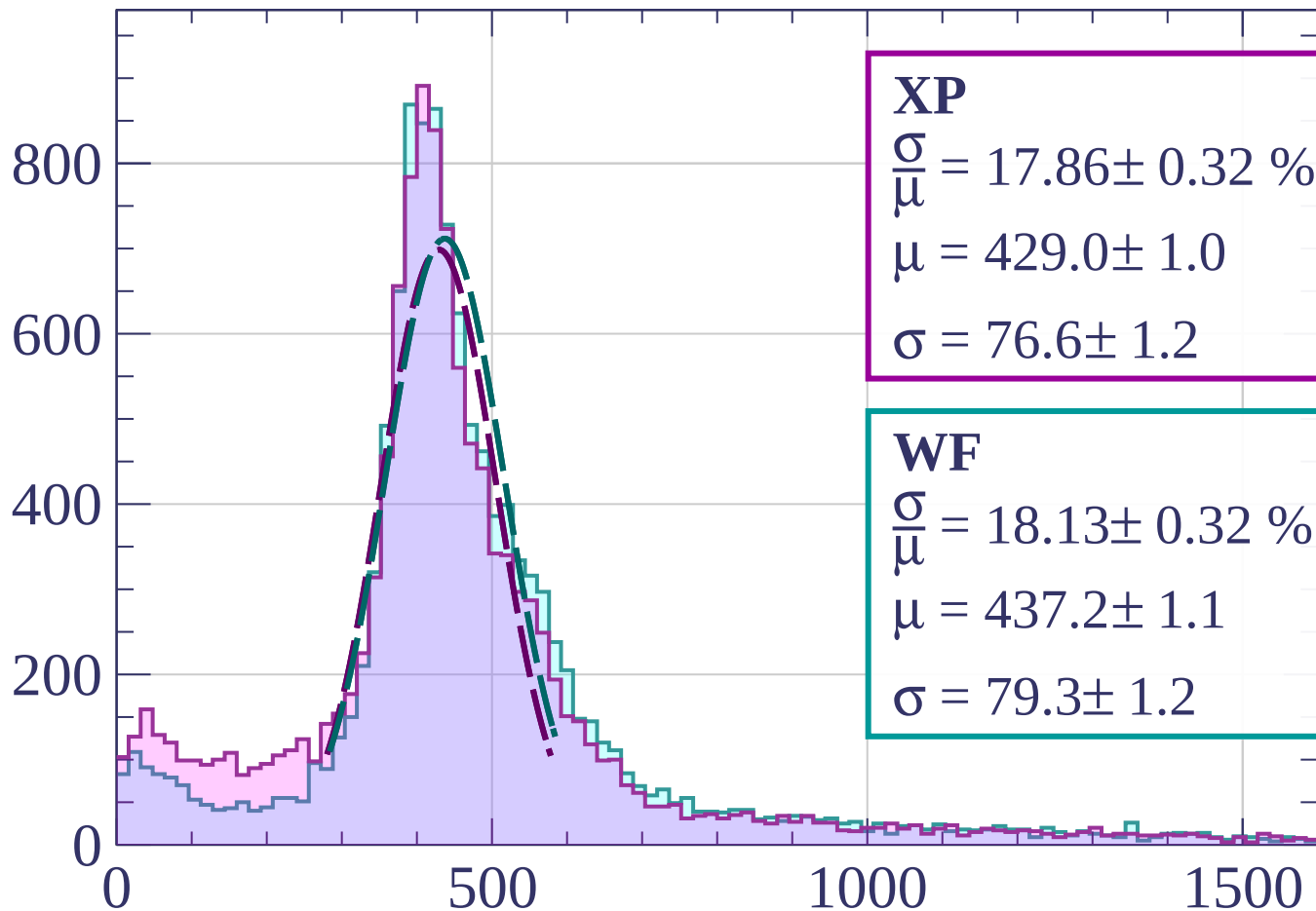
$$\mu = 433.9 \pm 1.0$$

$$\sigma = 73.2 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 17.86 \pm 0.32 \%$$

$$\mu = 429.0 \pm 1.0$$

$$\sigma = 76.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 18.13 \pm 0.32 \%$$

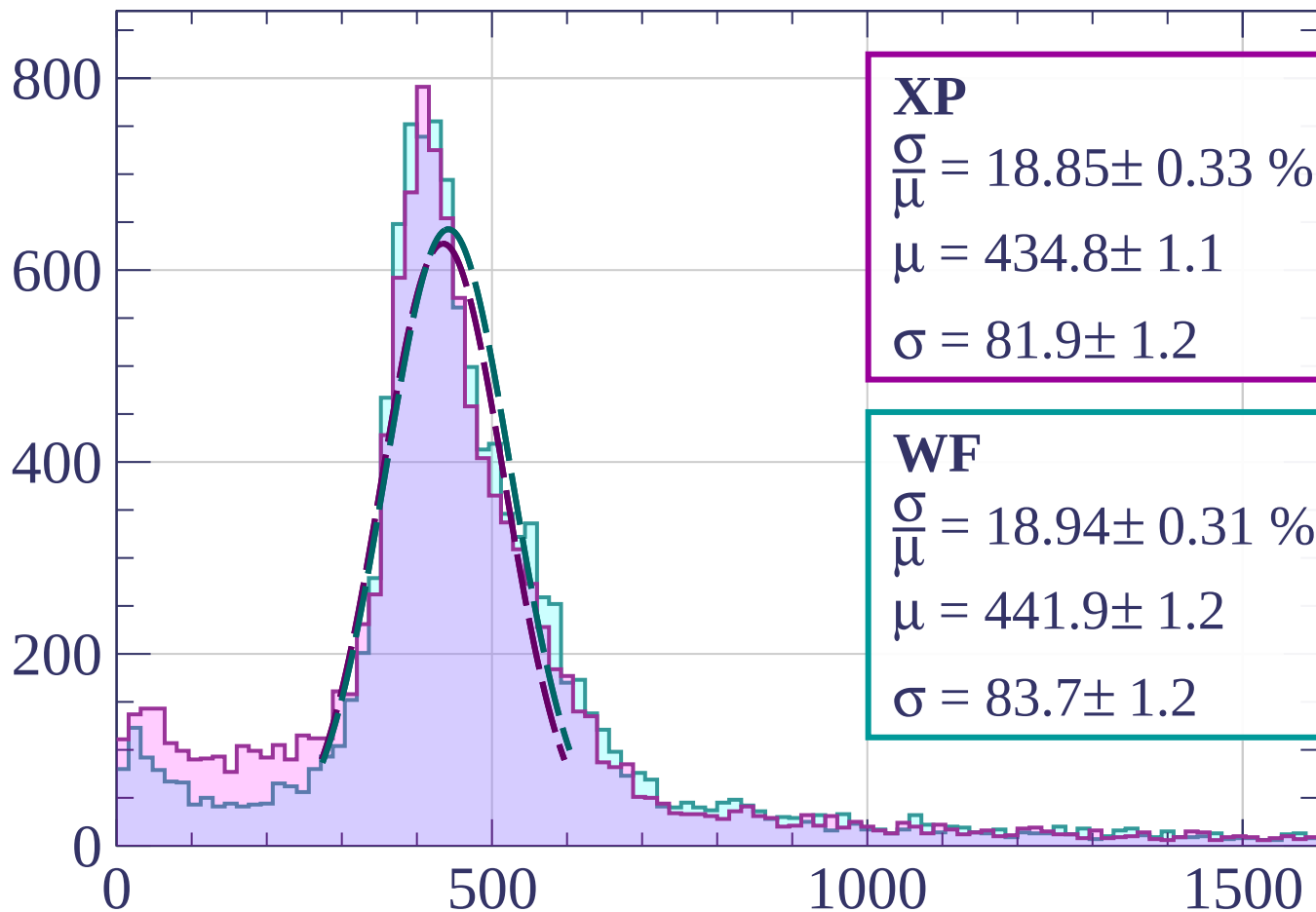
$$\mu = 437.2 \pm 1.1$$

$$\sigma = 79.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \phi < 51$

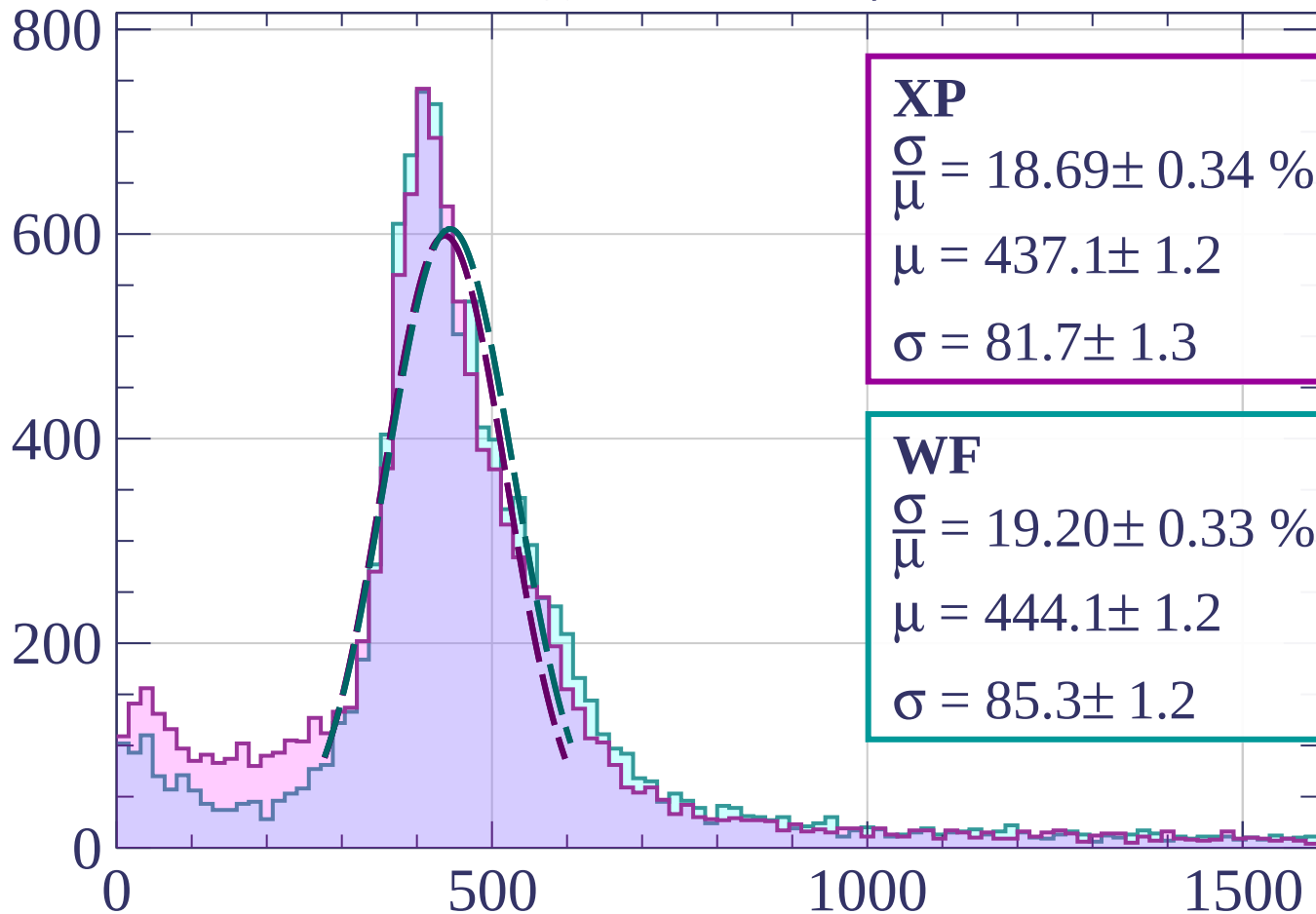
Count



dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 18.69 \pm 0.34 \%$$

$$\mu = 437.1 \pm 1.2$$

$$\sigma = 81.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 19.20 \pm 0.33 \%$$

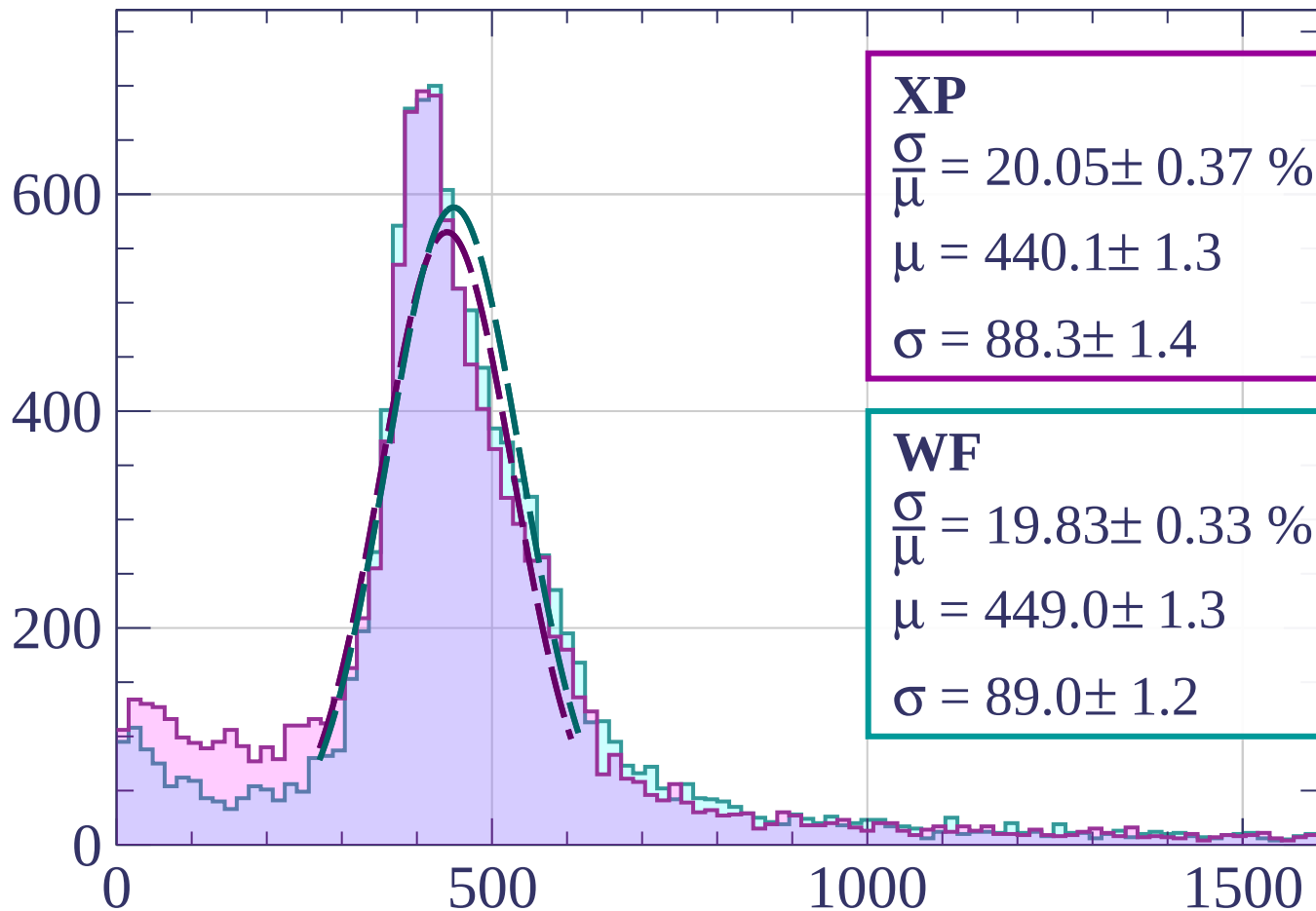
$$\mu = 444.1 \pm 1.2$$

$$\sigma = 85.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 20.05 \pm 0.37 \%$$

$$\mu = 440.1 \pm 1.3$$

$$\sigma = 88.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 19.83 \pm 0.33 \%$$

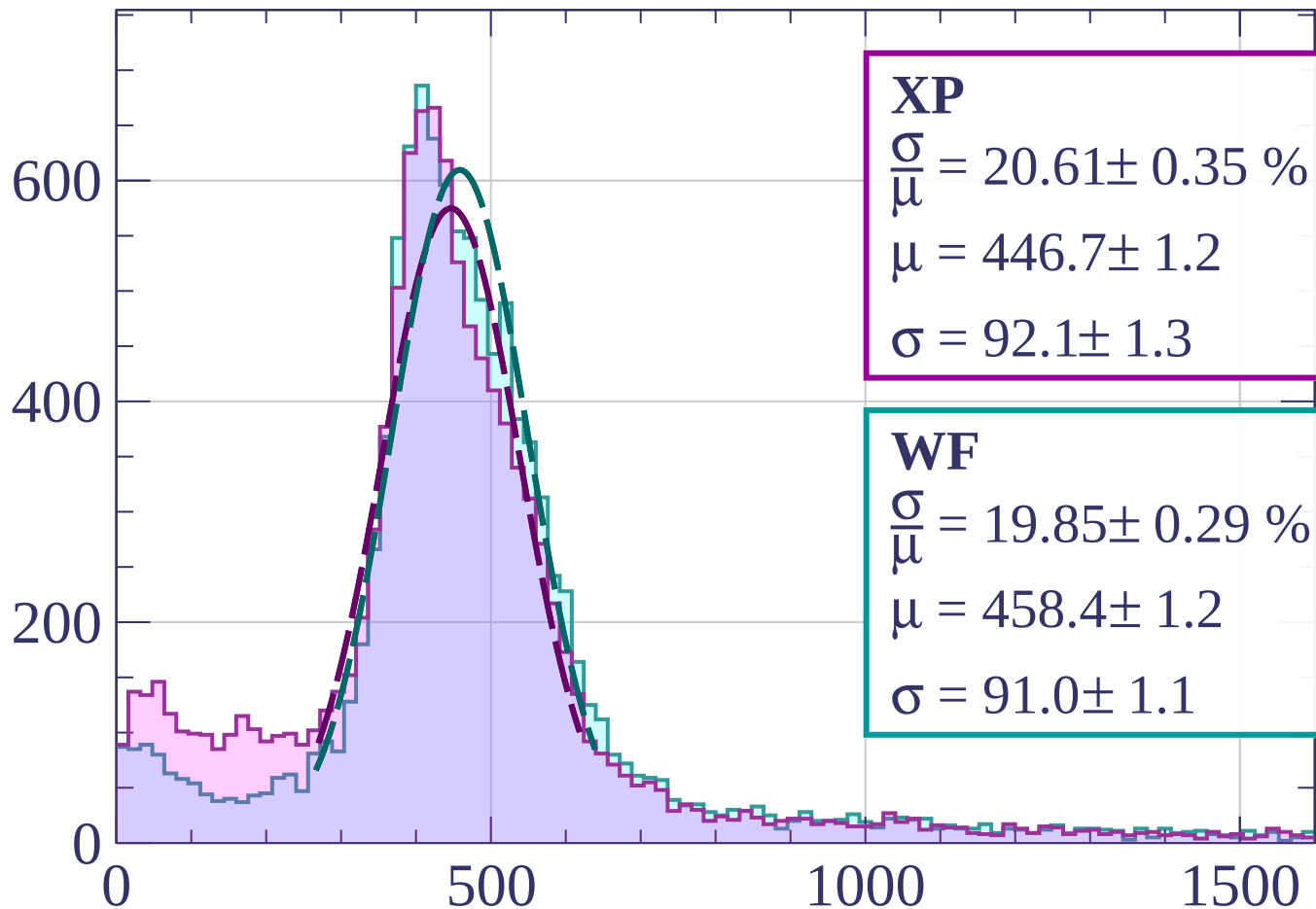
$$\mu = 449.0 \pm 1.3$$

$$\sigma = 89.0 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 20.61 \pm 0.35 \%$$

$$\mu = 446.7 \pm 1.2$$

$$\sigma = 92.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 19.85 \pm 0.29 \%$$

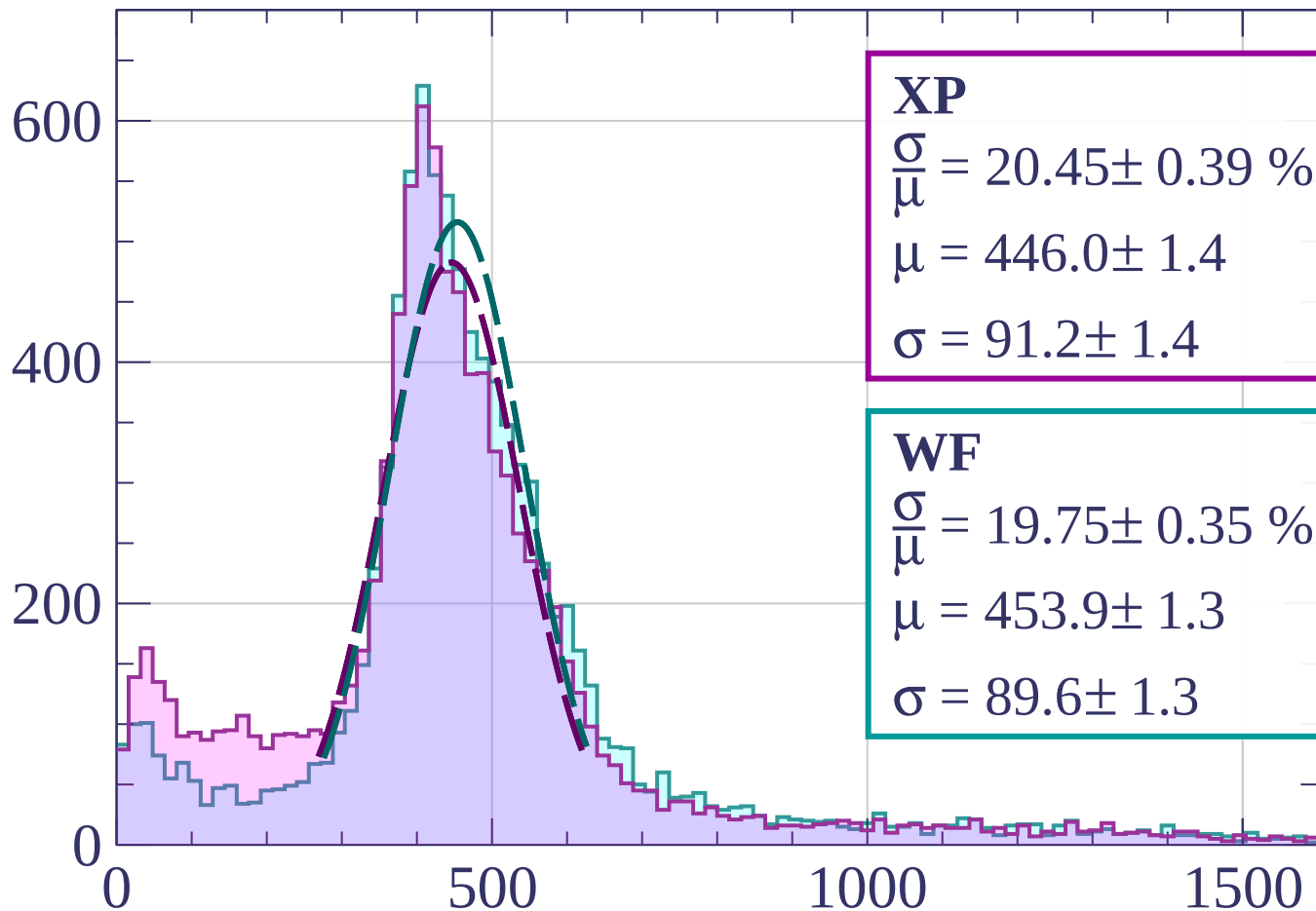
$$\mu = 458.4 \pm 1.2$$

$$\sigma = 91.0 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 20.45 \pm 0.39 \%$$

$$\mu = 446.0 \pm 1.4$$

$$\sigma = 91.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 19.75 \pm 0.35 \%$$

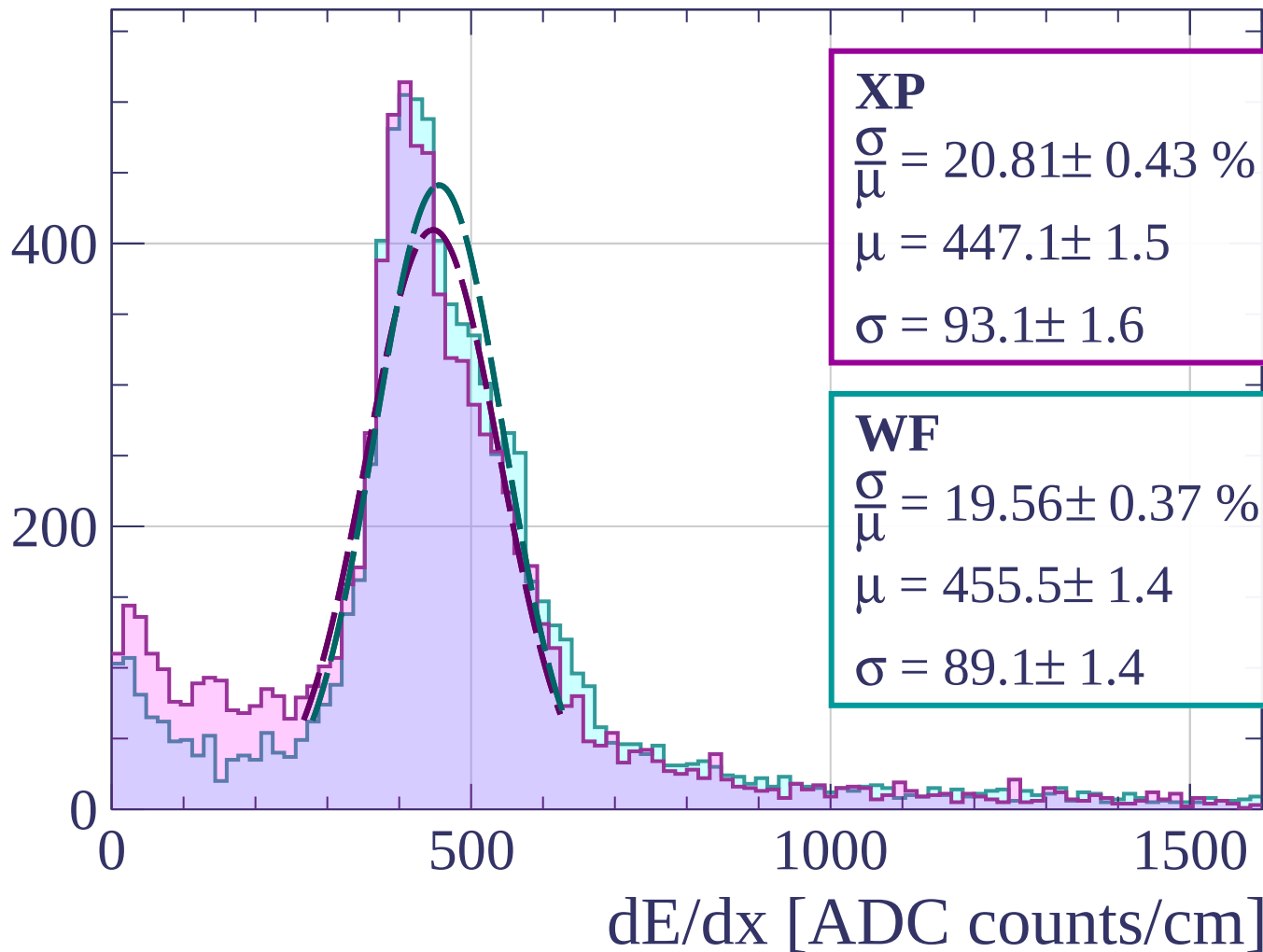
$$\mu = 453.9 \pm 1.3$$

$$\sigma = 89.6 \pm 1.3$$

dE/dx [ADC counts/cm]

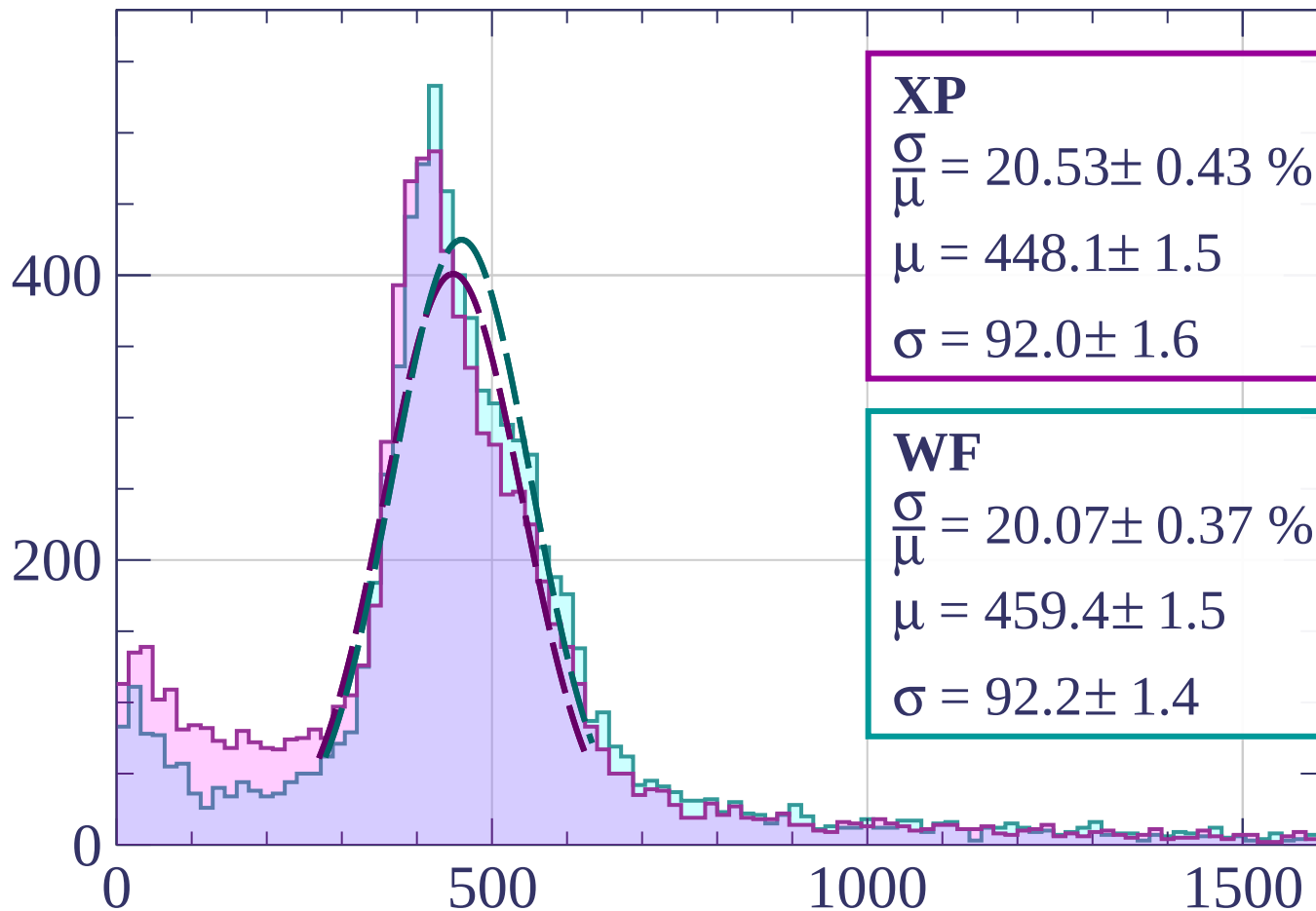
Energy loss | $63 < \phi < 66$

Count



Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 20.53 \pm 0.43 \%$$

$$\mu = 448.1 \pm 1.5$$

$$\sigma = 92.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.07 \pm 0.37 \%$$

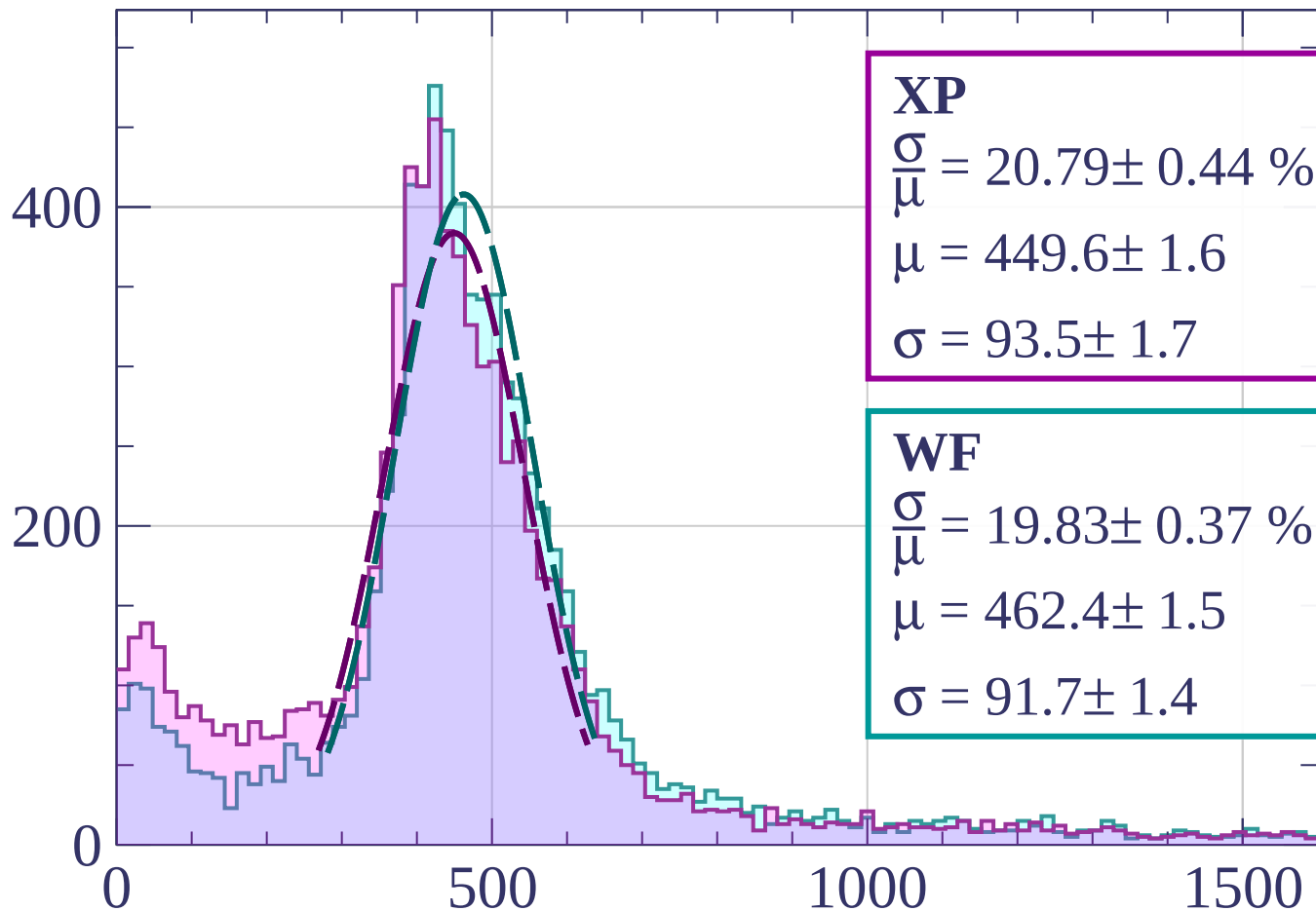
$$\mu = 459.4 \pm 1.5$$

$$\sigma = 92.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 20.79 \pm 0.44 \%$$

$$\mu = 449.6 \pm 1.6$$

$$\sigma = 93.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 19.83 \pm 0.37 \%$$

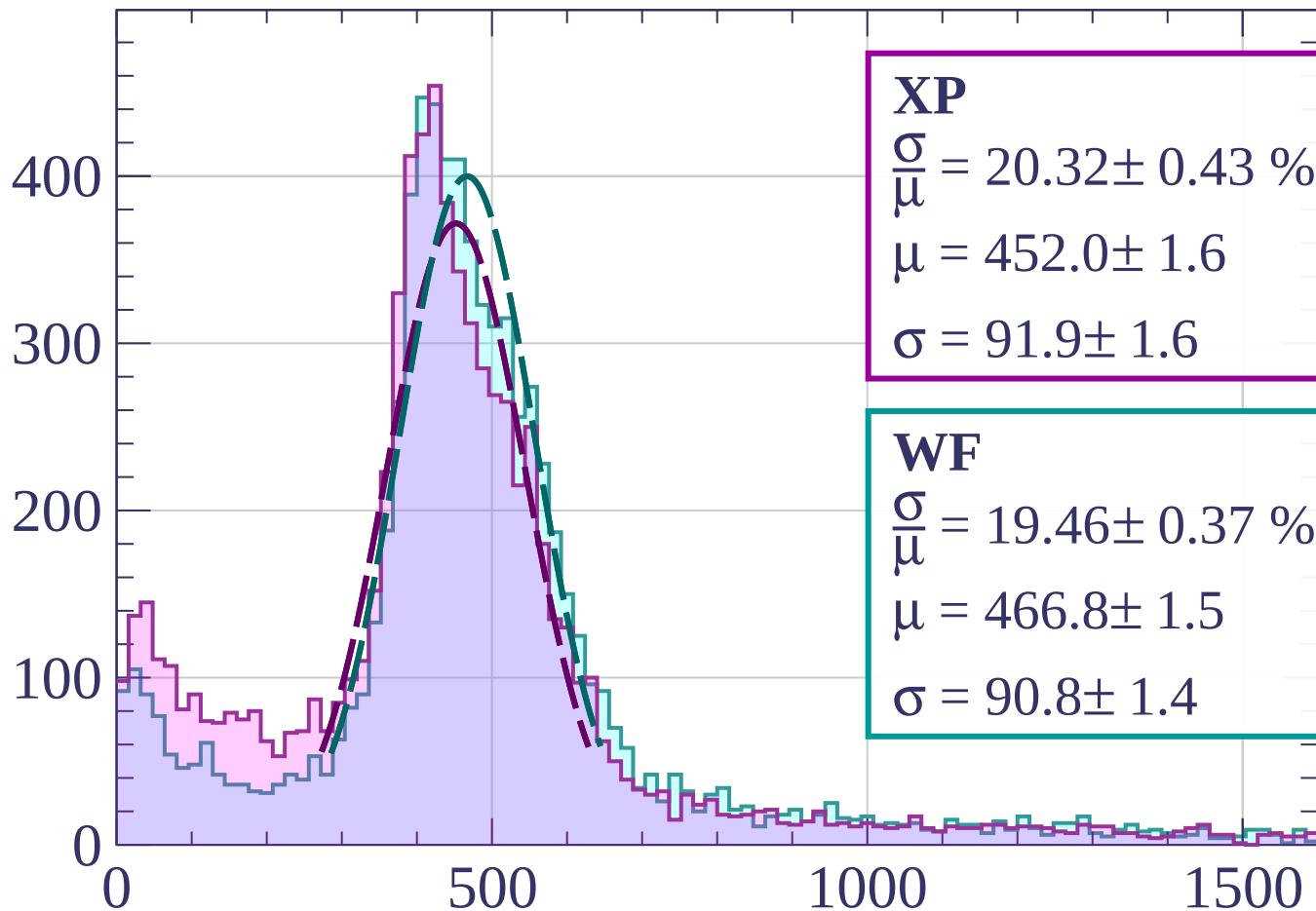
$$\mu = 462.4 \pm 1.5$$

$$\sigma = 91.7 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 20.32 \pm 0.43 \%$$

$$\mu = 452.0 \pm 1.6$$

$$\sigma = 91.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 19.46 \pm 0.37 \%$$

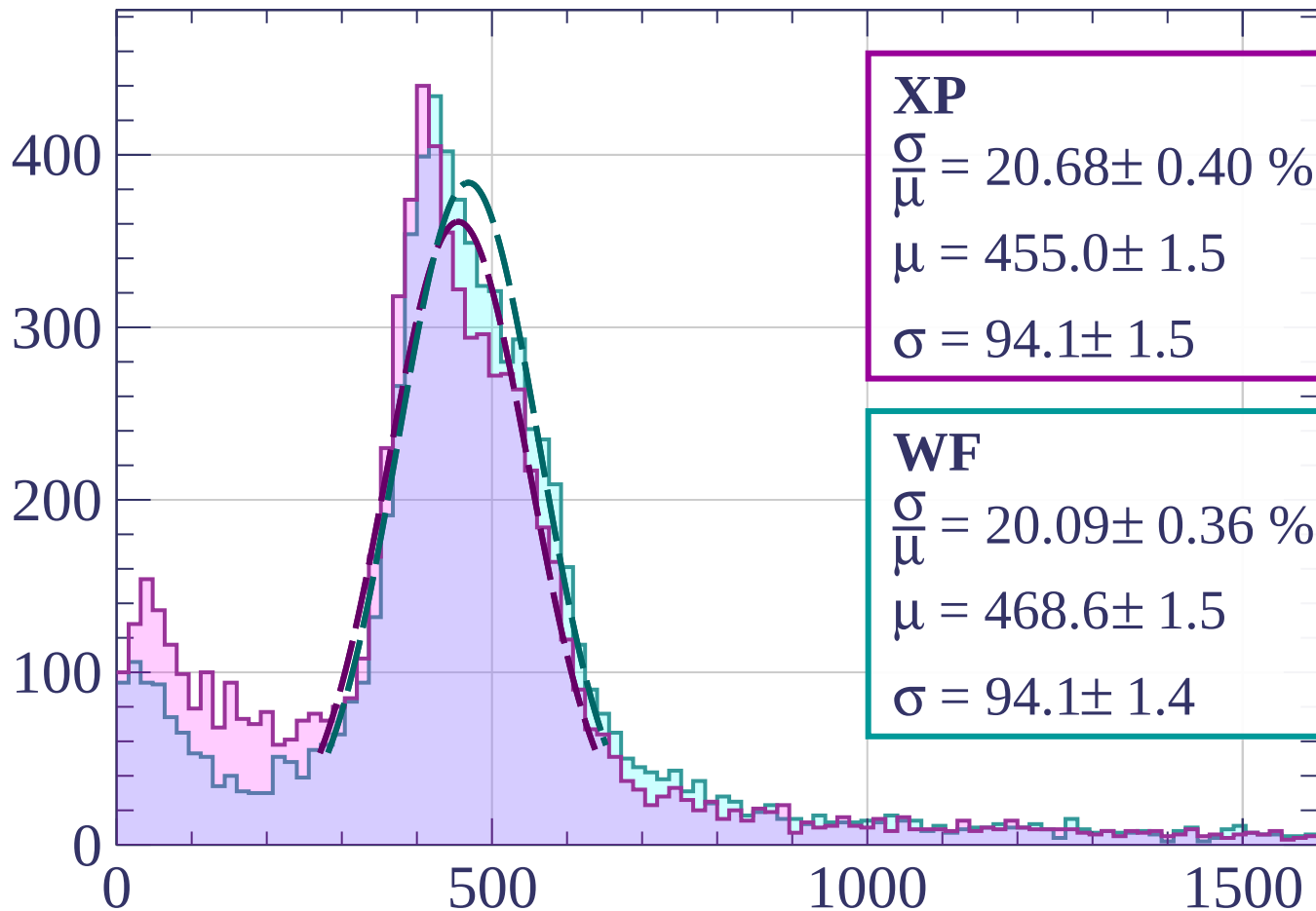
$$\mu = 466.8 \pm 1.5$$

$$\sigma = 90.8 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 20.68 \pm 0.40 \%$$

$$\mu = 455.0 \pm 1.5$$

$$\sigma = 94.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 20.09 \pm 0.36 \%$$

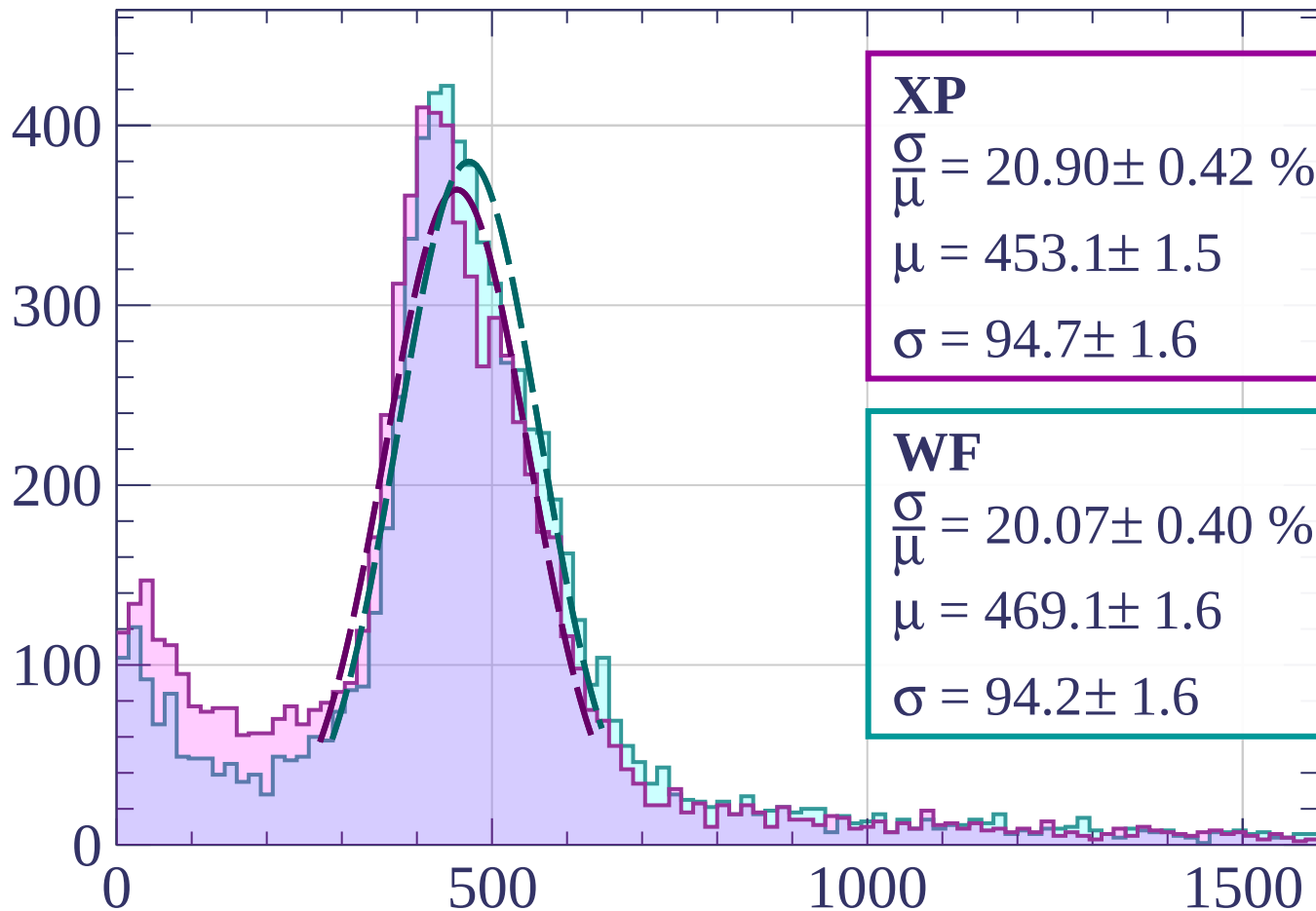
$$\mu = 468.6 \pm 1.5$$

$$\sigma = 94.1 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $78 < \phi < 81$

Count



XP

$$\frac{\sigma}{\mu} = 20.90 \pm 0.42 \%$$

$$\mu = 453.1 \pm 1.5$$

$$\sigma = 94.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.07 \pm 0.40 \%$$

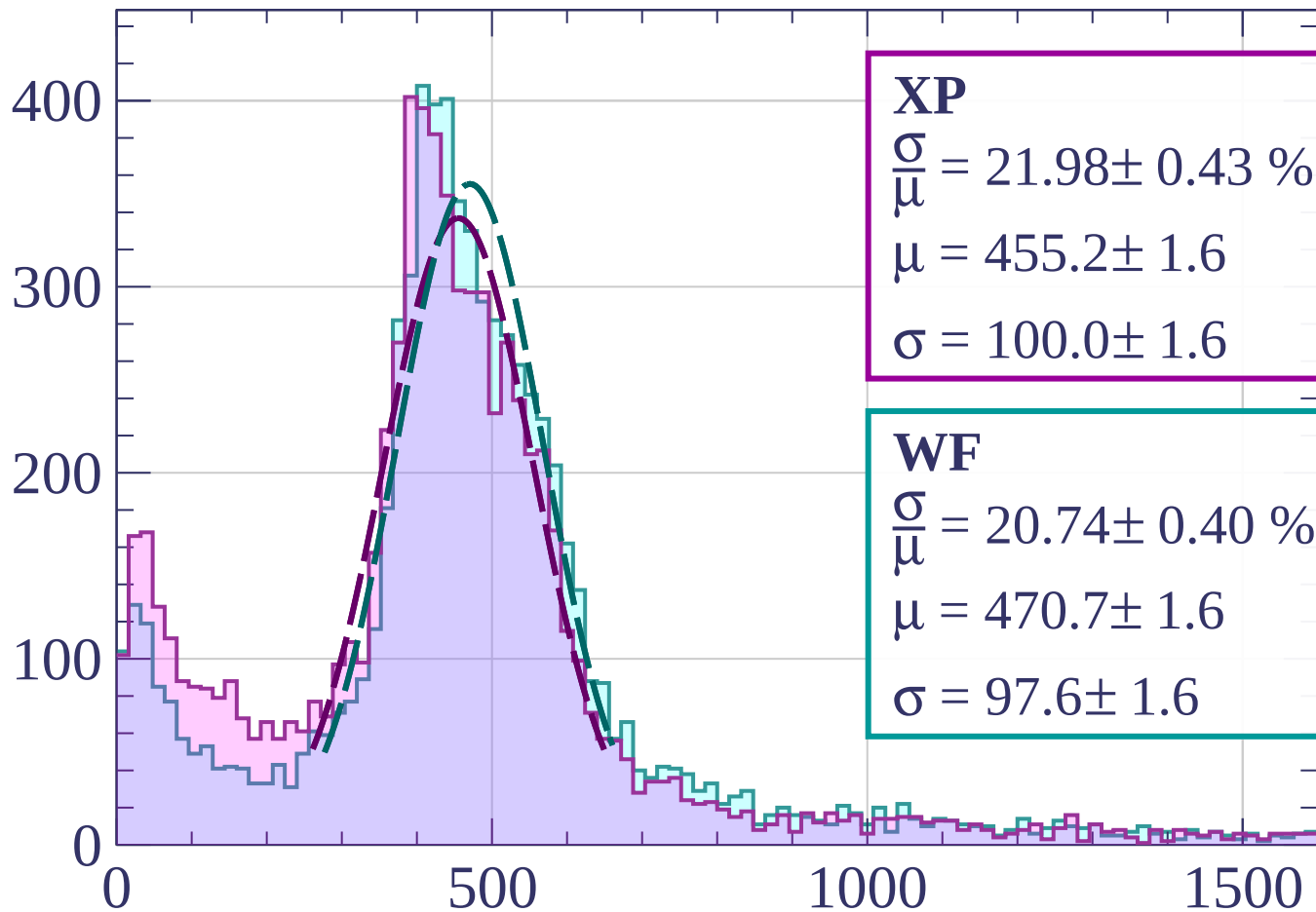
$$\mu = 469.1 \pm 1.6$$

$$\sigma = 94.2 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \phi < 84$

Count



XP

$$\frac{\sigma}{\mu} = 21.98 \pm 0.43 \%$$

$$\mu = 455.2 \pm 1.6$$

$$\sigma = 100.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 20.74 \pm 0.40 \%$$

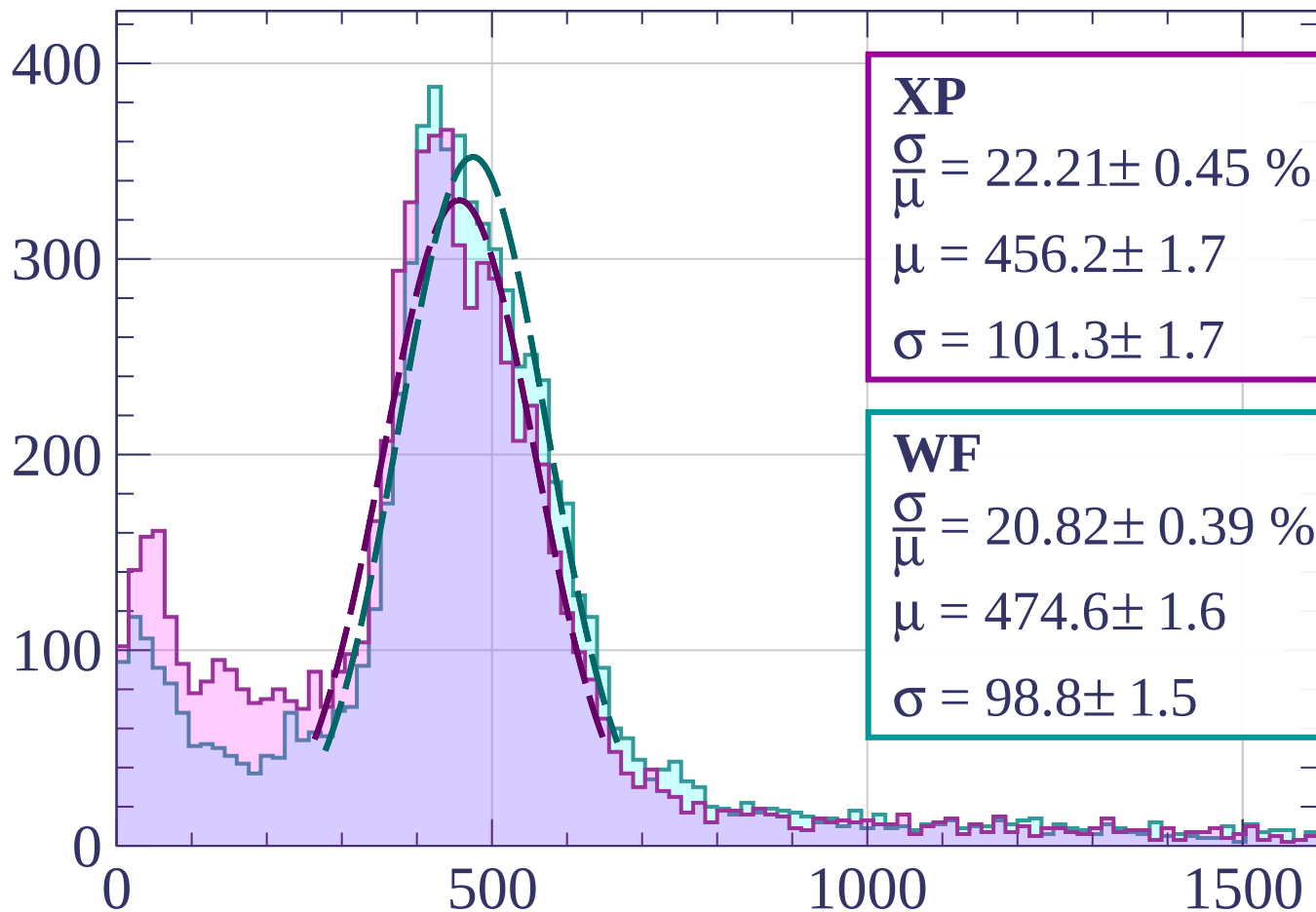
$$\mu = 470.7 \pm 1.6$$

$$\sigma = 97.6 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 22.21 \pm 0.45 \%$$

$$\mu = 456.2 \pm 1.7$$

$$\sigma = 101.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 20.82 \pm 0.39 \%$$

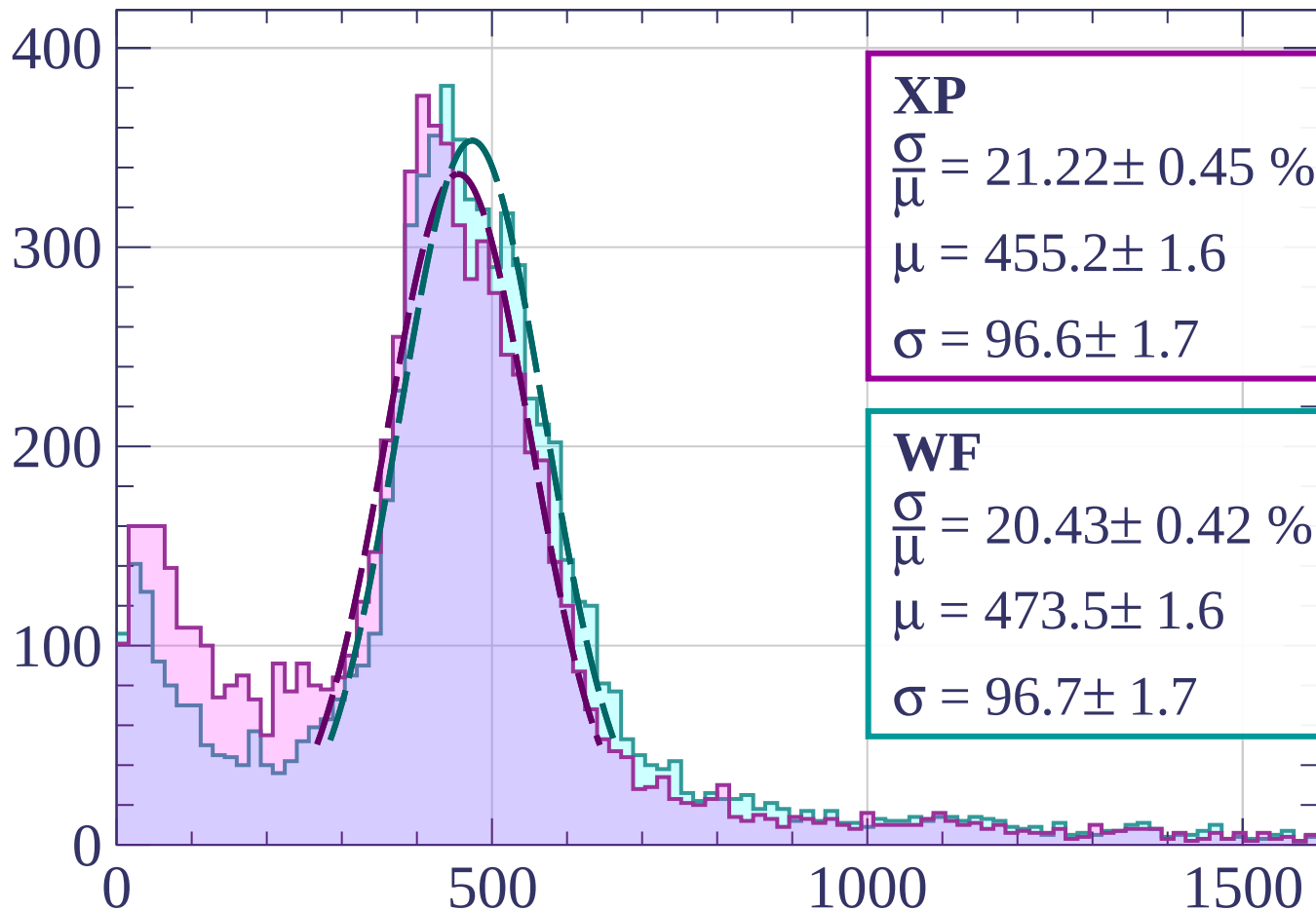
$$\mu = 474.6 \pm 1.6$$

$$\sigma = 98.8 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $87 < \phi < 90$

Count



XP

$$\frac{\sigma}{\mu} = 21.22 \pm 0.45 \%$$

$$\mu = 455.2 \pm 1.6$$

$$\sigma = 96.6 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 20.43 \pm 0.42 \%$$

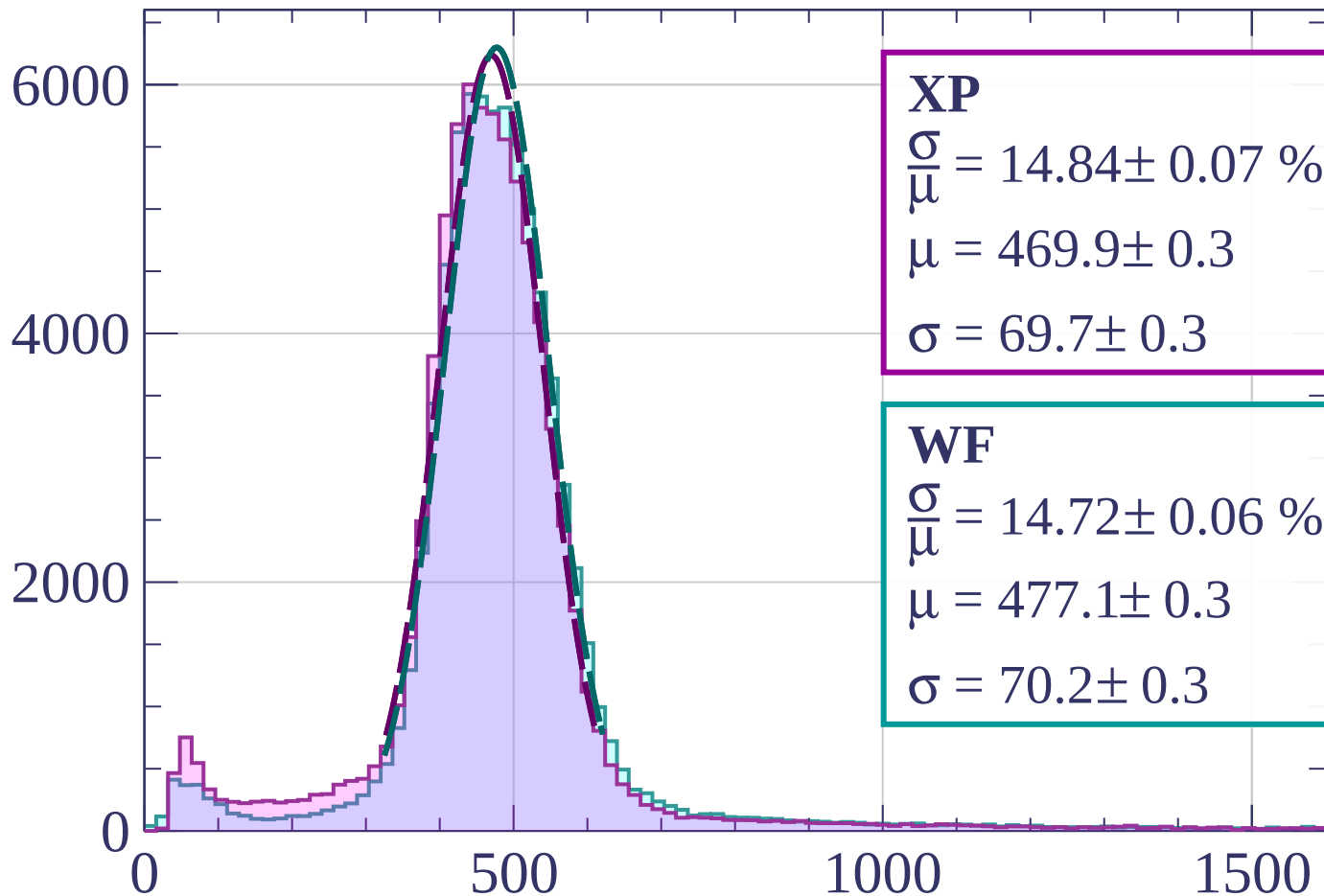
$$\mu = 473.5 \pm 1.6$$

$$\sigma = 96.7 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 14.84 \pm 0.07 \%$$

$$\mu = 469.9 \pm 0.3$$

$$\sigma = 69.7 \pm 0.3$$

WF

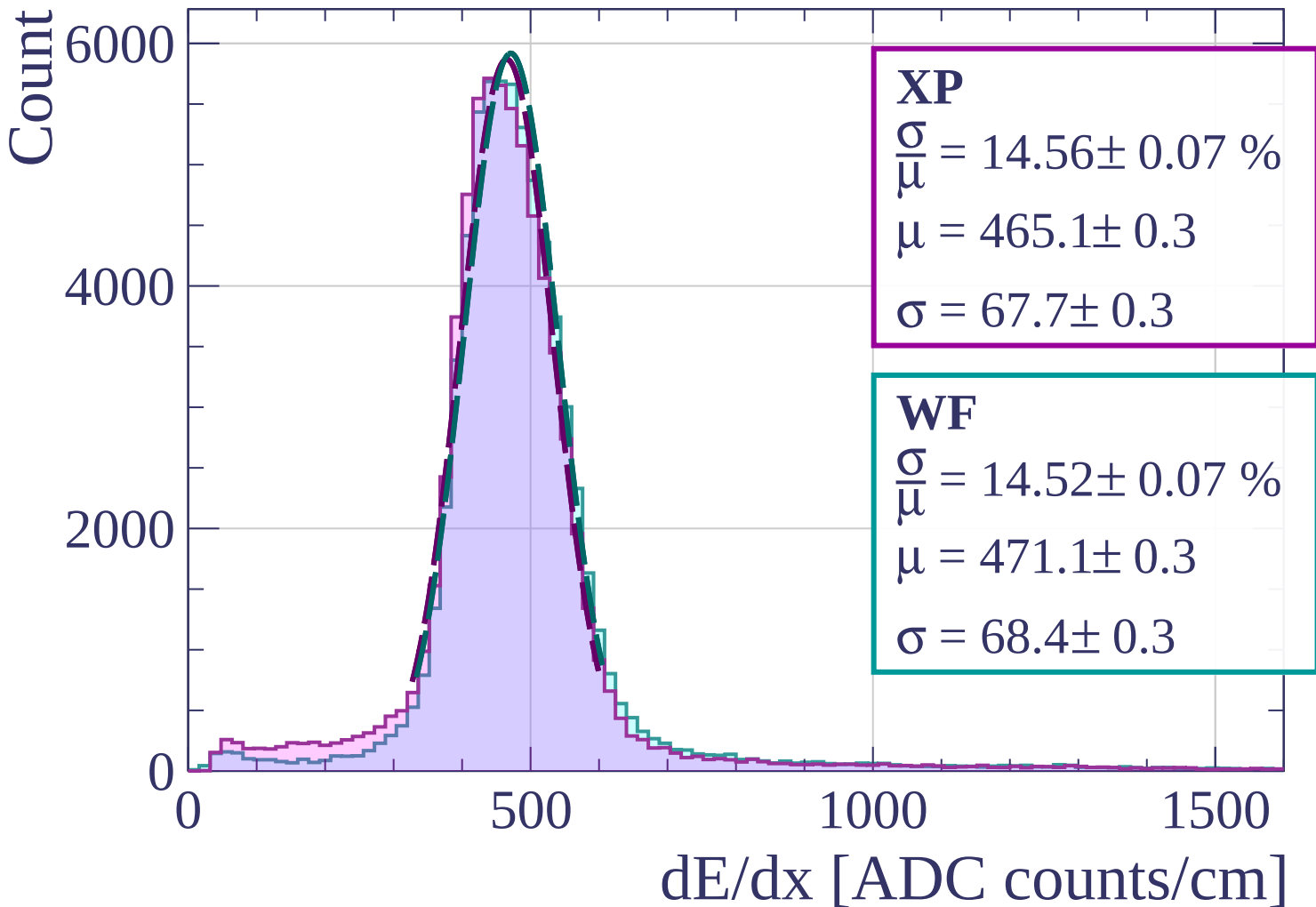
$$\frac{\sigma}{\mu} = 14.72 \pm 0.06 \%$$

$$\mu = 477.1 \pm 0.3$$

$$\sigma = 70.2 \pm 0.3$$

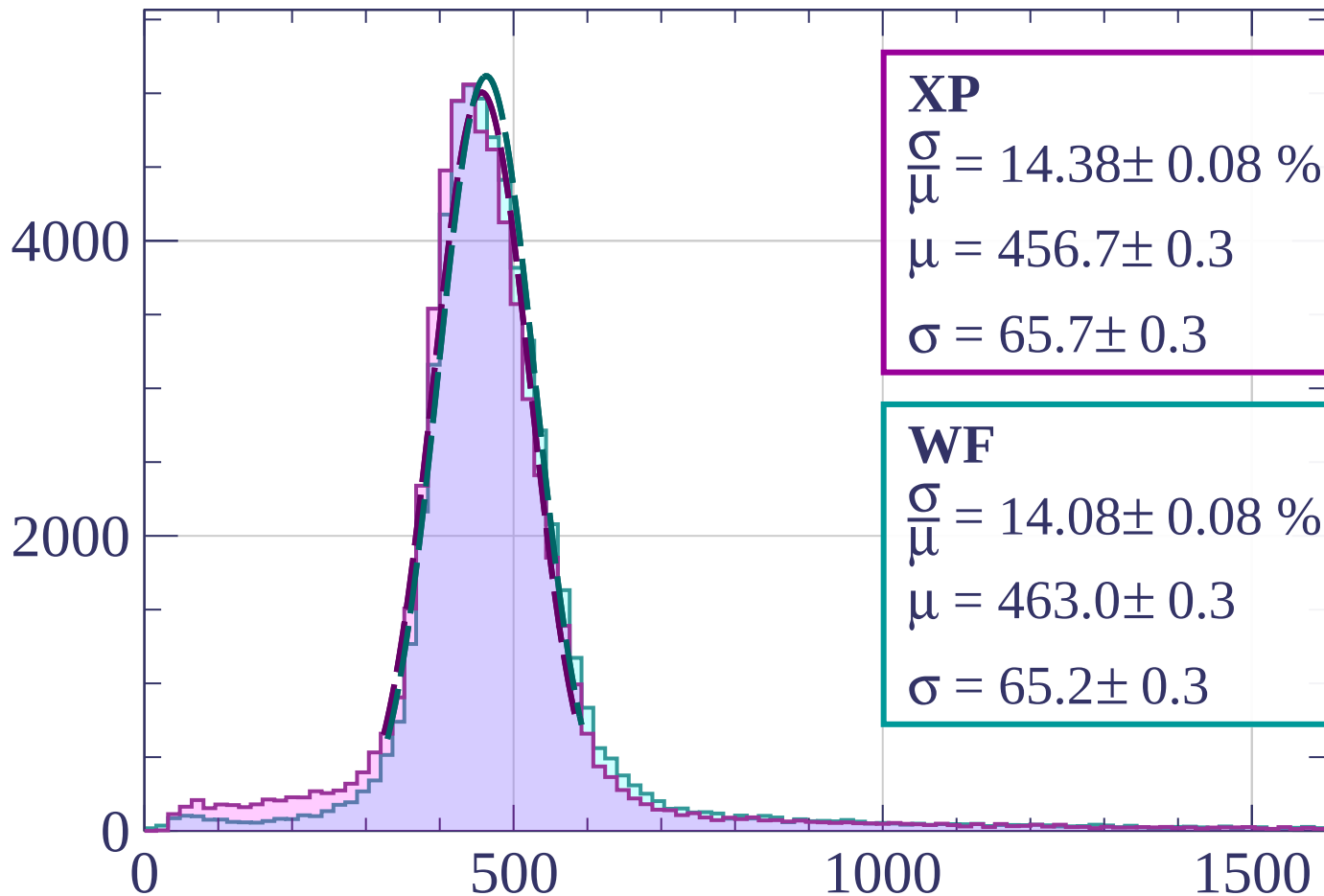
dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 14.38 \pm 0.08 \%$$

$$\mu = 456.7 \pm 0.3$$

$$\sigma = 65.7 \pm 0.3$$

WF

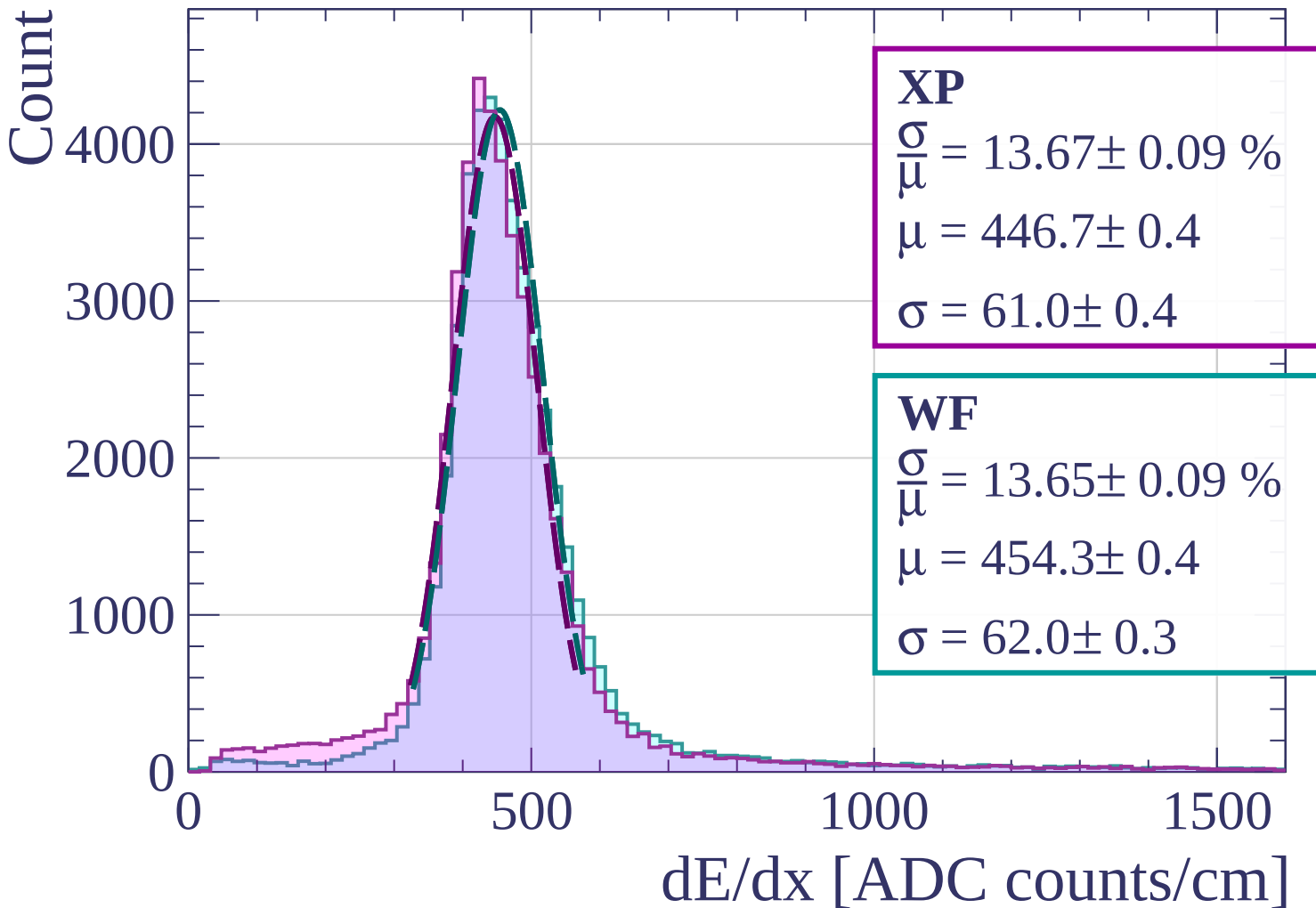
$$\frac{\sigma}{\mu} = 14.08 \pm 0.08 \%$$

$$\mu = 463.0 \pm 0.3$$

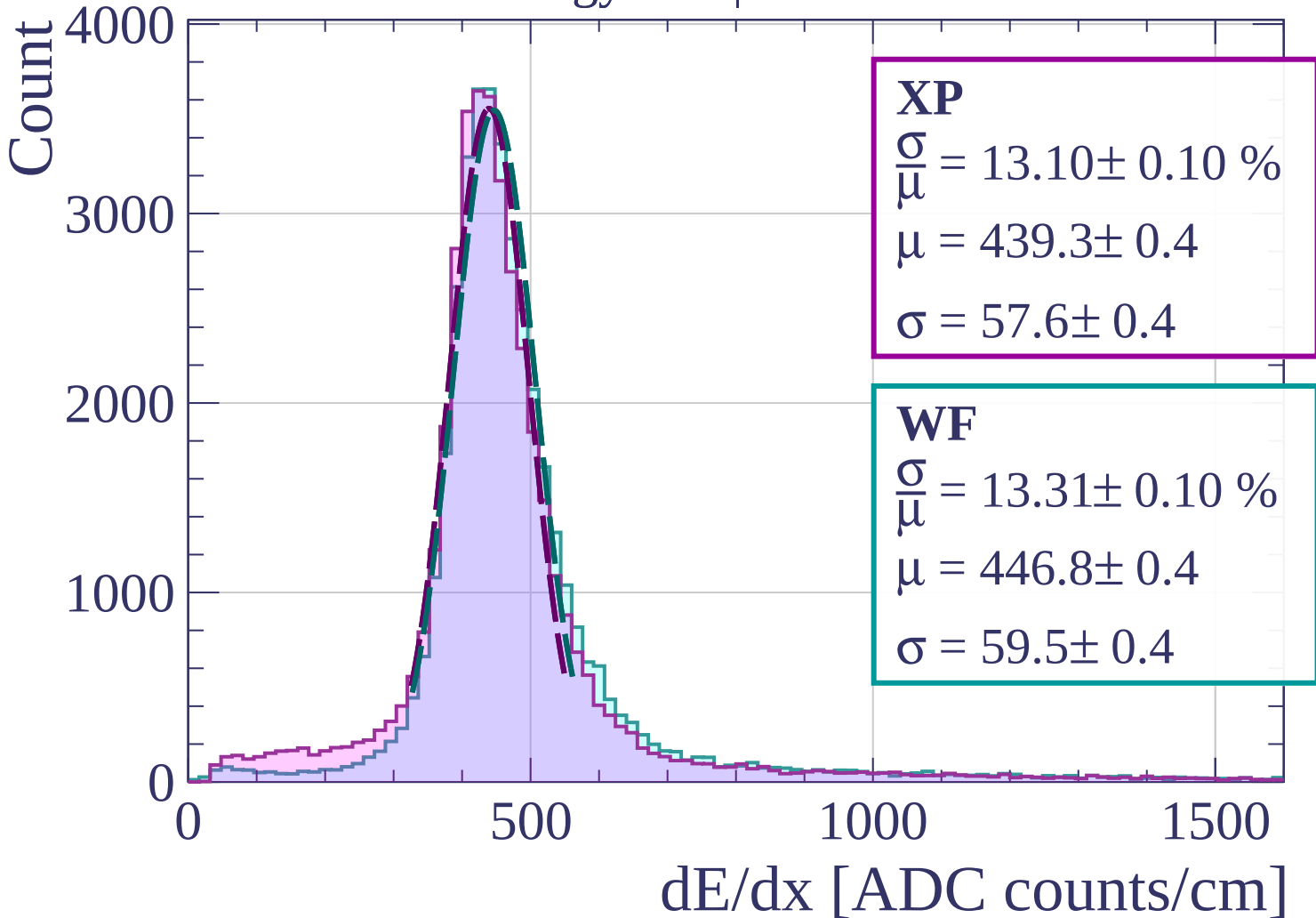
$$\sigma = 65.2 \pm 0.3$$

dE/dx [ADC counts/cm]

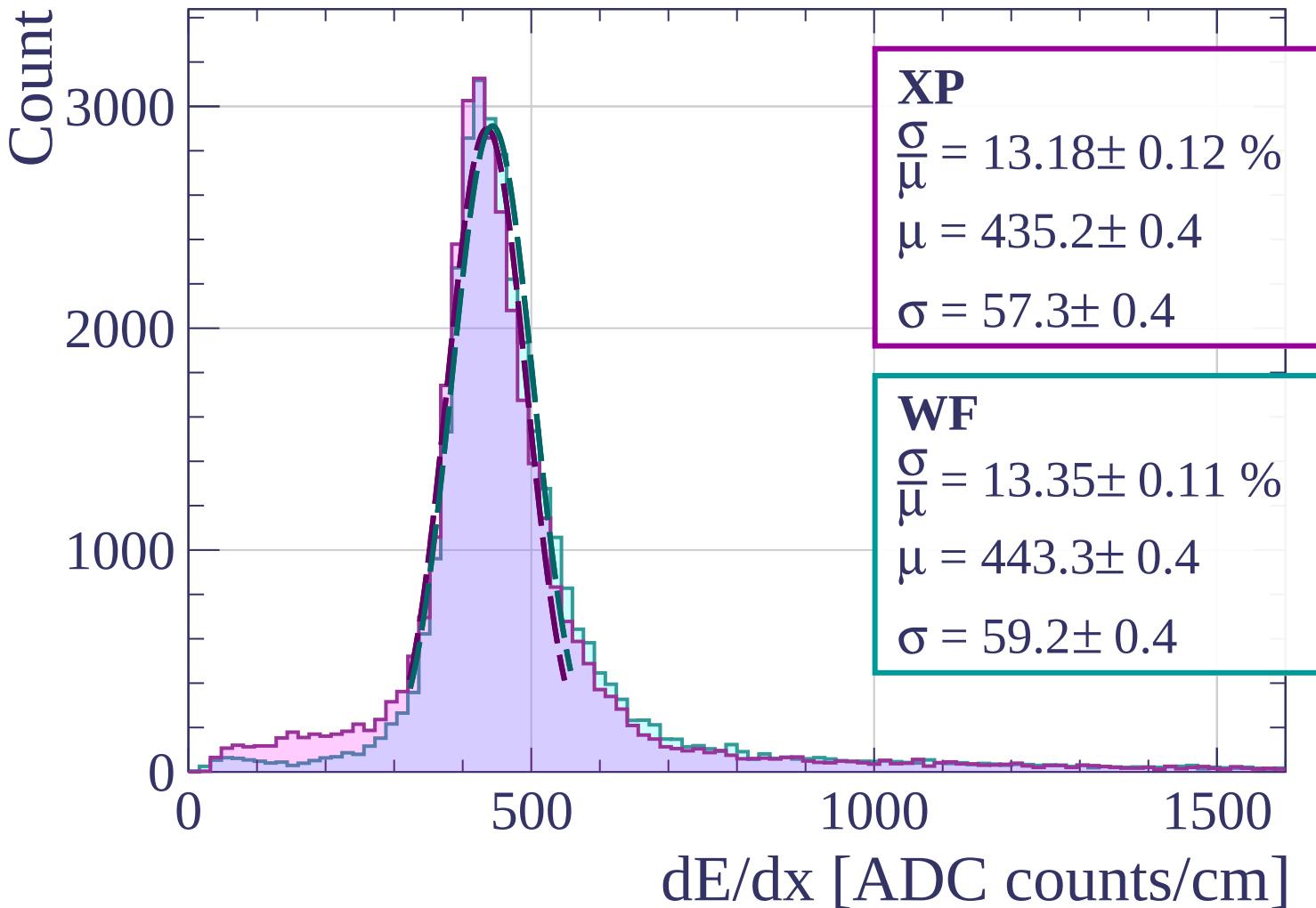
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

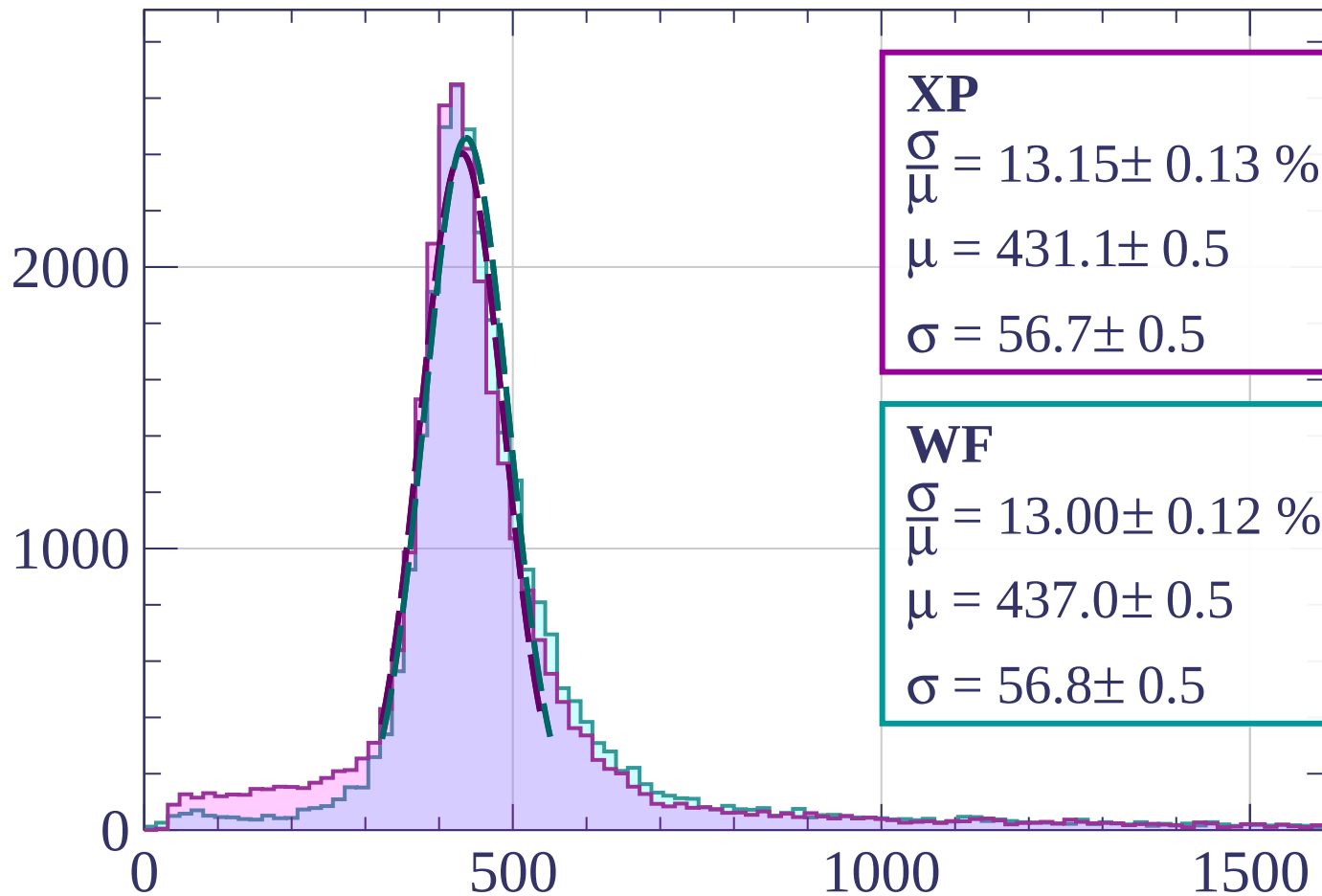


Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 13.15 \pm 0.13 \%$$

$$\mu = 431.1 \pm 0.5$$

$$\sigma = 56.7 \pm 0.5$$

WF

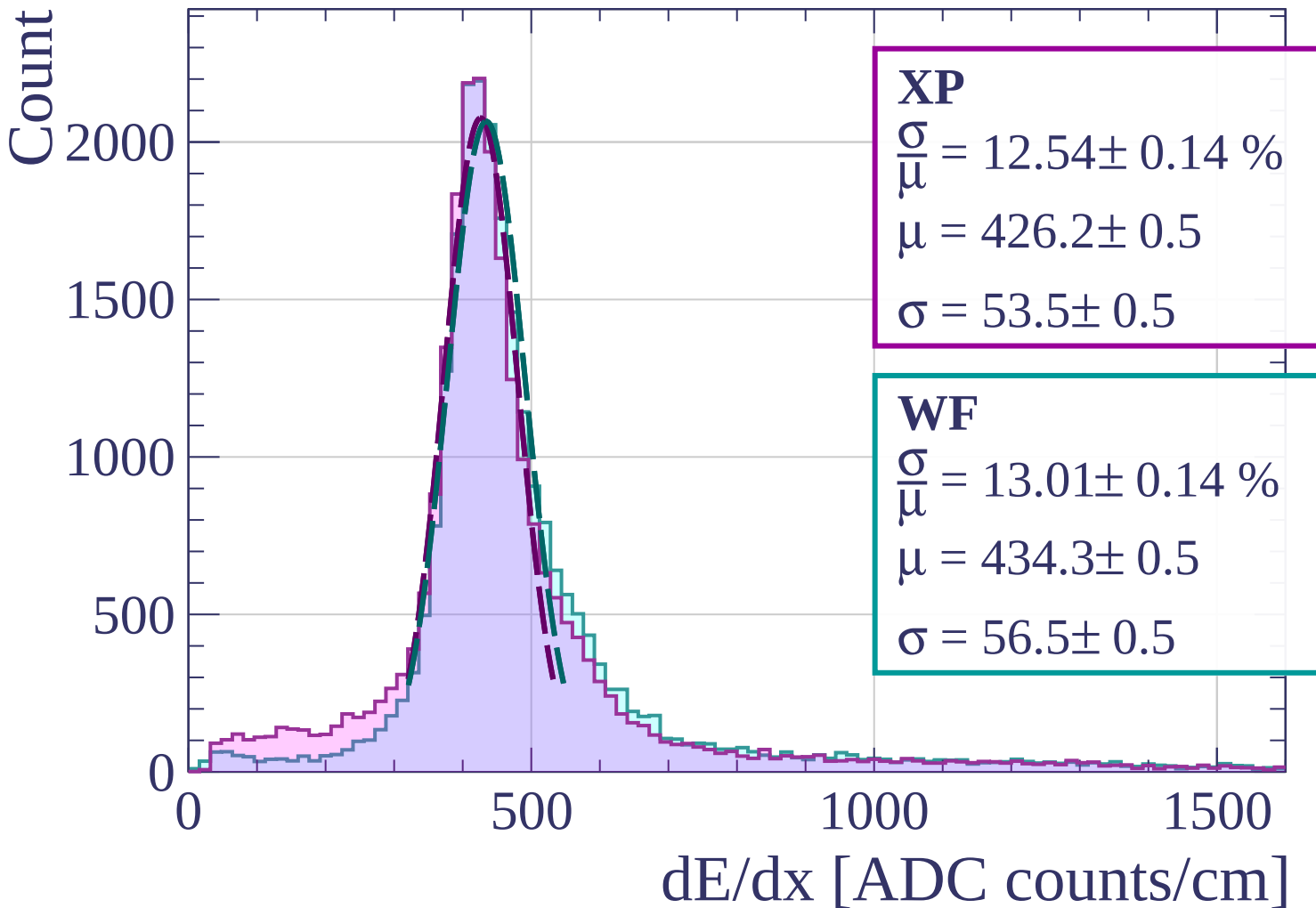
$$\frac{\sigma}{\mu} = 13.00 \pm 0.12 \%$$

$$\mu = 437.0 \pm 0.5$$

$$\sigma = 56.8 \pm 0.5$$

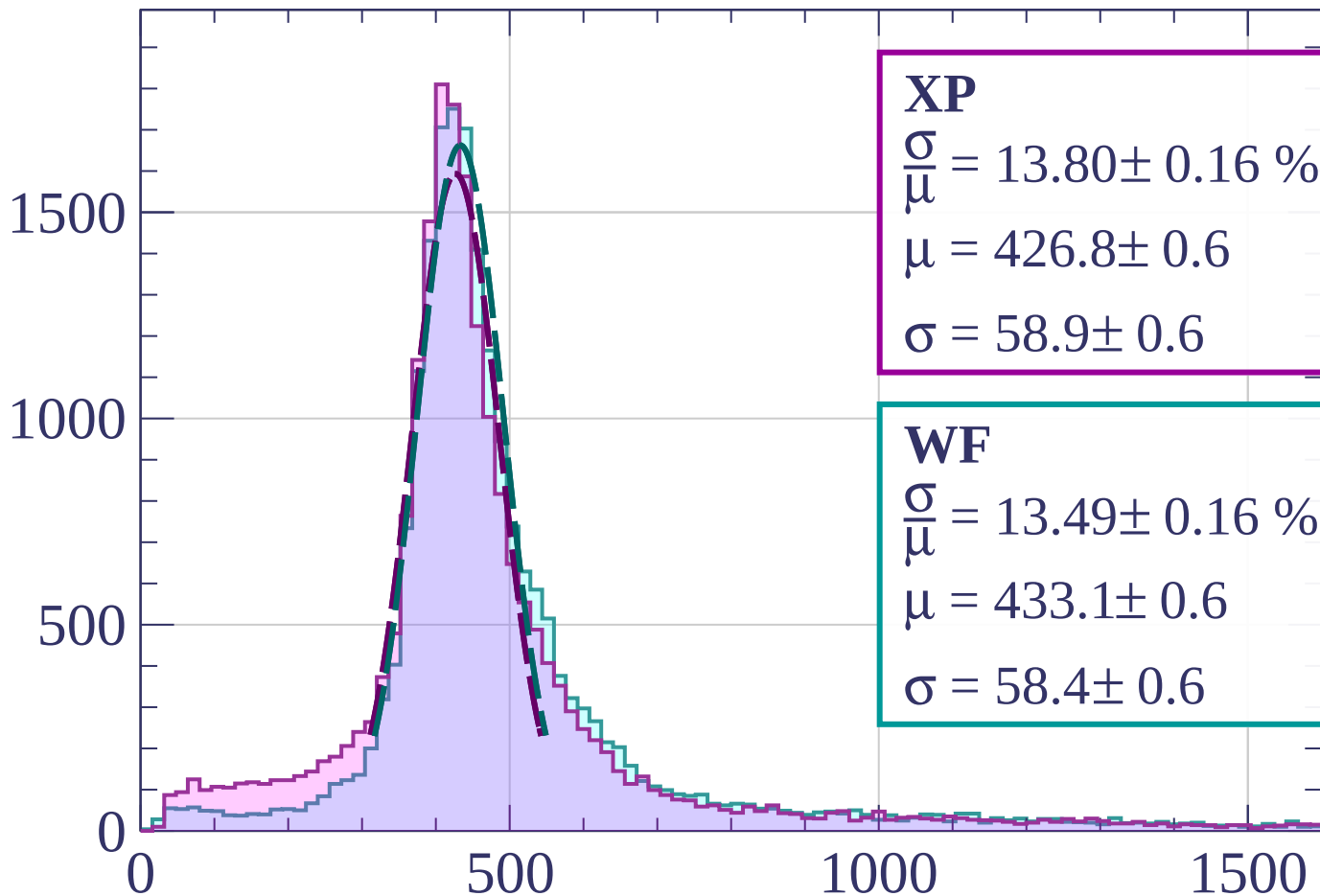
dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$



Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 13.80 \pm 0.16 \%$$

$$\mu = 426.8 \pm 0.6$$

$$\sigma = 58.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 13.49 \pm 0.16 \%$$

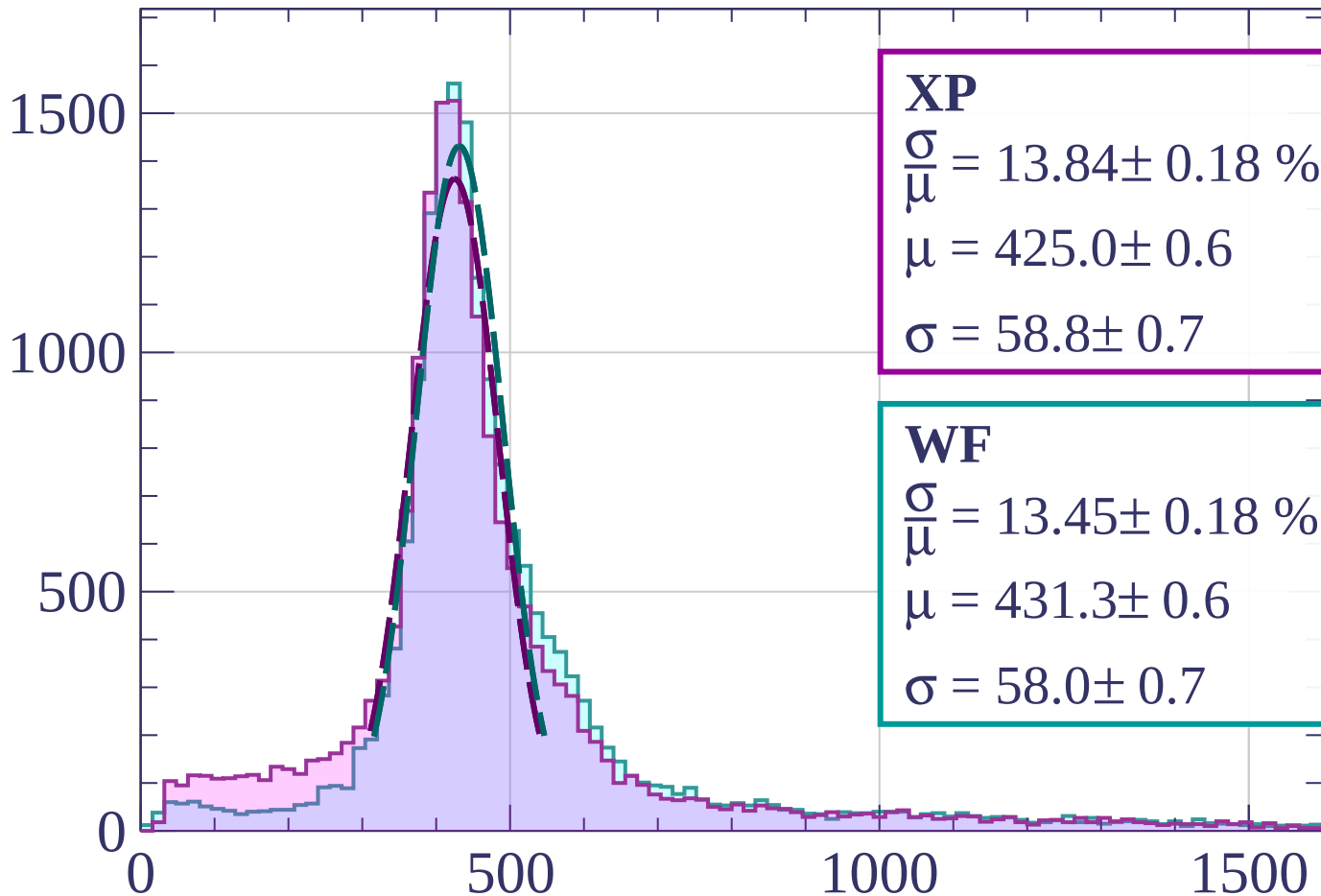
$$\mu = 433.1 \pm 0.6$$

$$\sigma = 58.4 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 13.84 \pm 0.18 \%$$

$$\mu = 425.0 \pm 0.6$$

$$\sigma = 58.8 \pm 0.7$$

WF

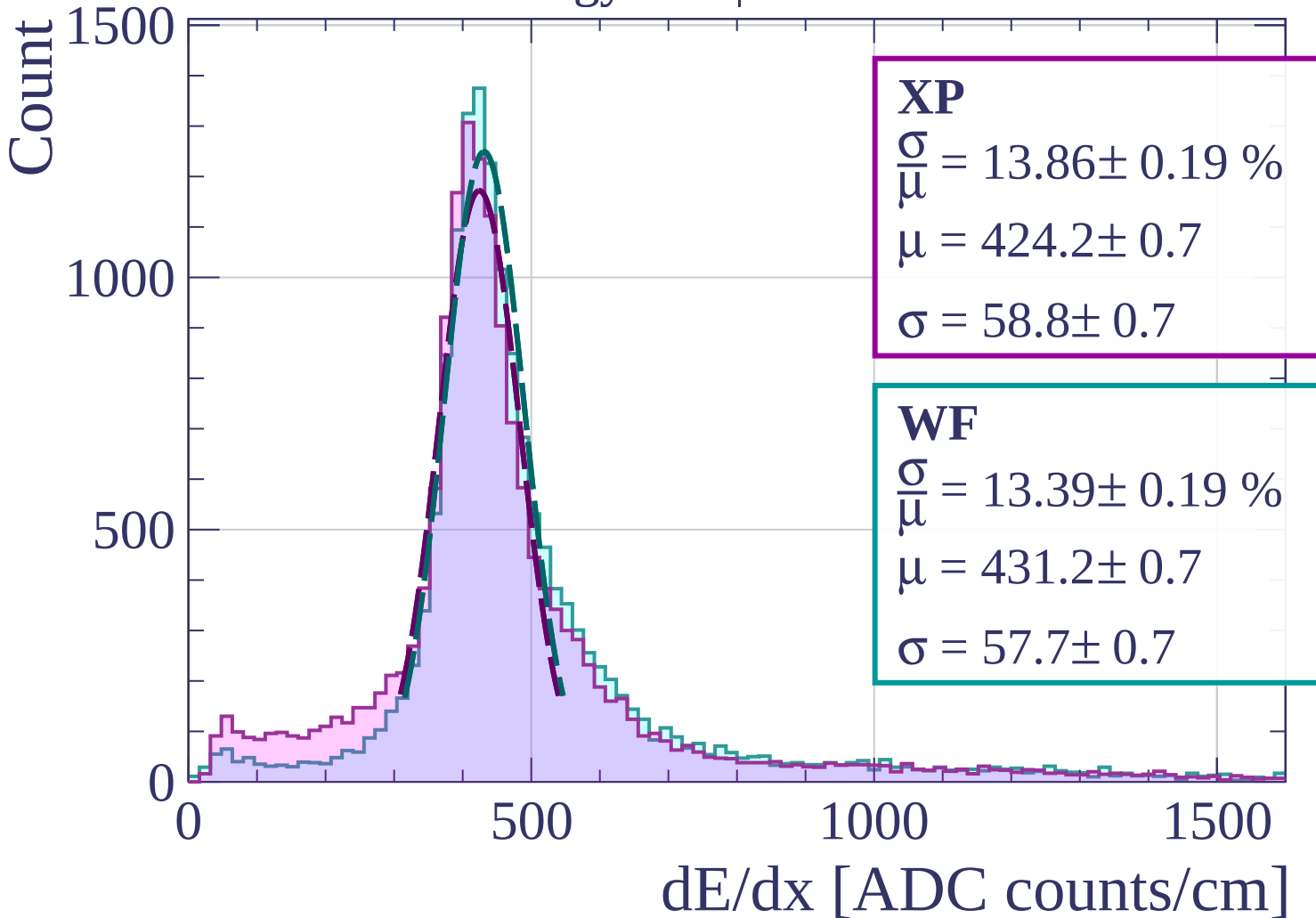
$$\frac{\sigma}{\mu} = 13.45 \pm 0.18 \%$$

$$\mu = 431.3 \pm 0.6$$

$$\sigma = 58.0 \pm 0.7$$

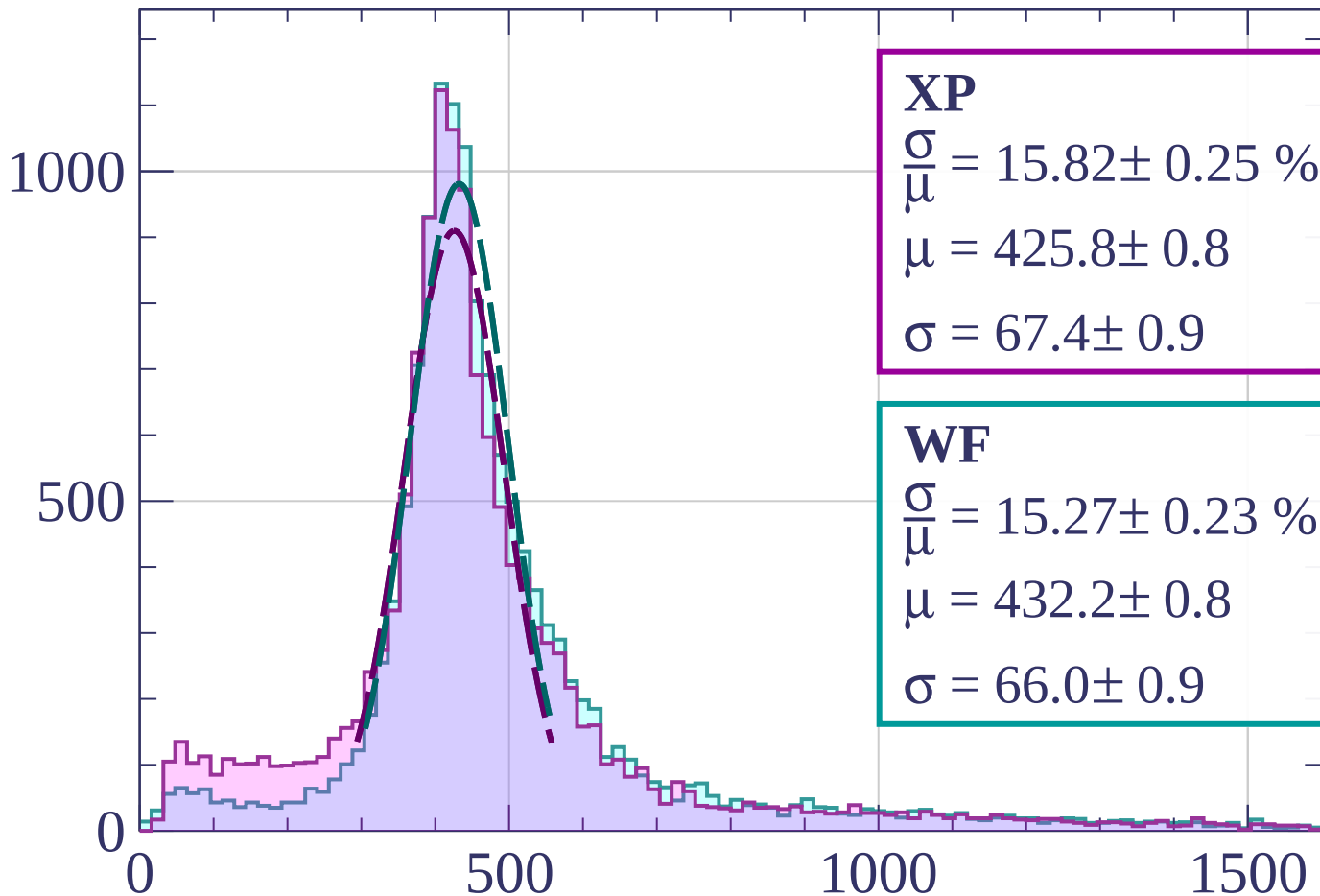
dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$



Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 15.82 \pm 0.25 \%$$

$$\mu = 425.8 \pm 0.8$$

$$\sigma = 67.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 15.27 \pm 0.23 \%$$

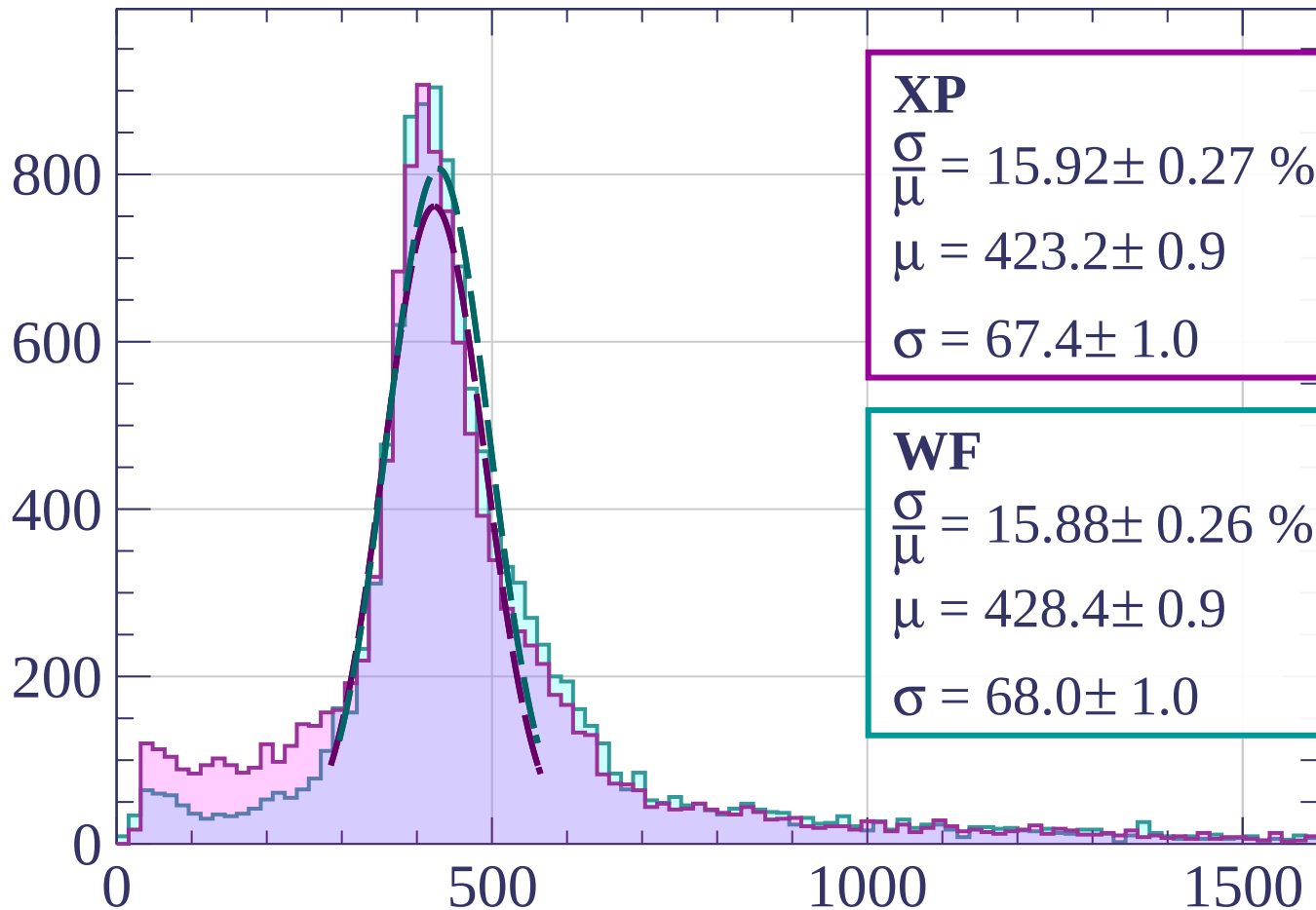
$$\mu = 432.2 \pm 0.8$$

$$\sigma = 66.0 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 15.92 \pm 0.27 \%$$

$$\mu = 423.2 \pm 0.9$$

$$\sigma = 67.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 15.88 \pm 0.26 \%$$

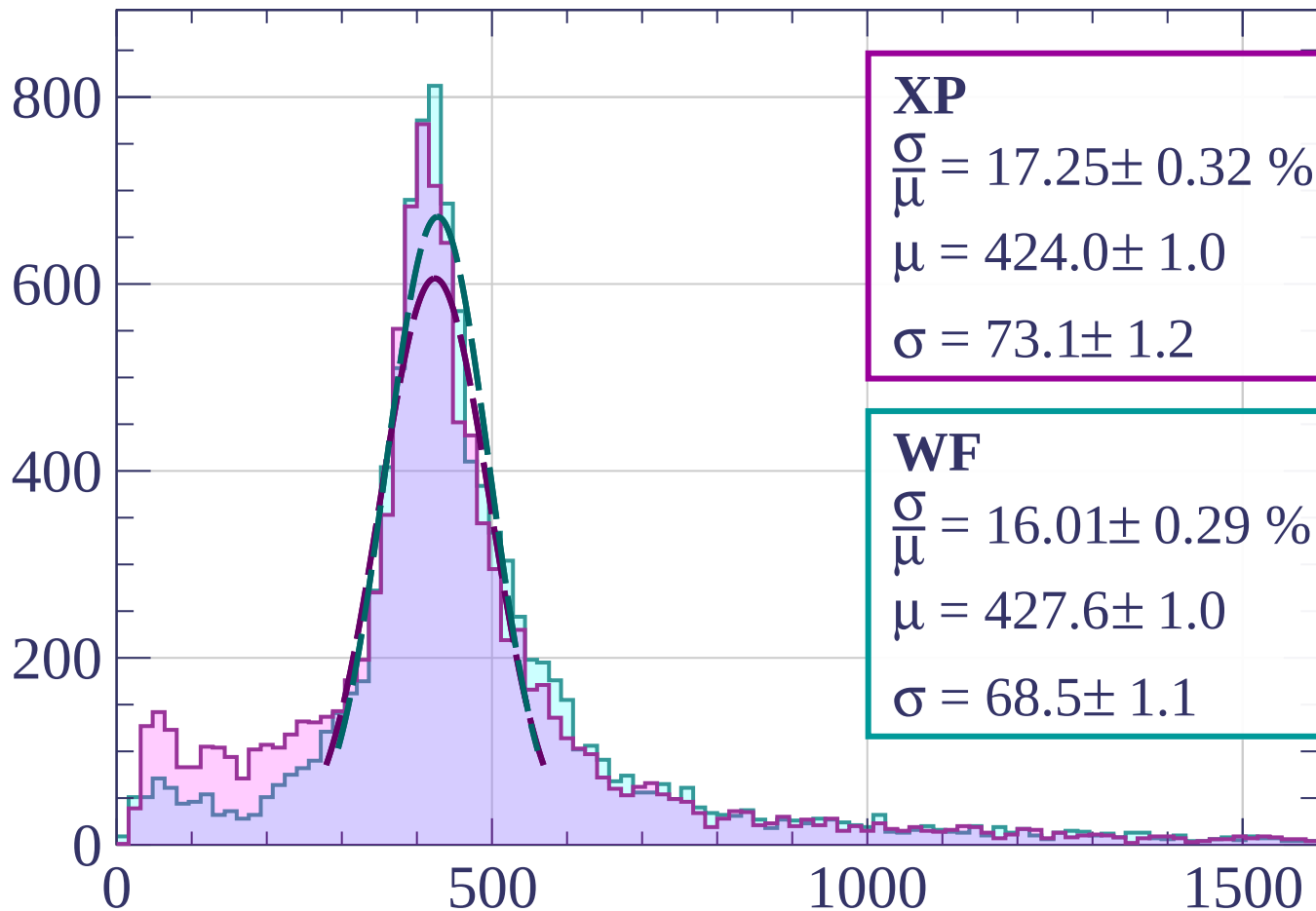
$$\mu = 428.4 \pm 0.9$$

$$\sigma = 68.0 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 17.25 \pm 0.32 \%$$

$$\mu = 424.0 \pm 1.0$$

$$\sigma = 73.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 16.01 \pm 0.29 \%$$

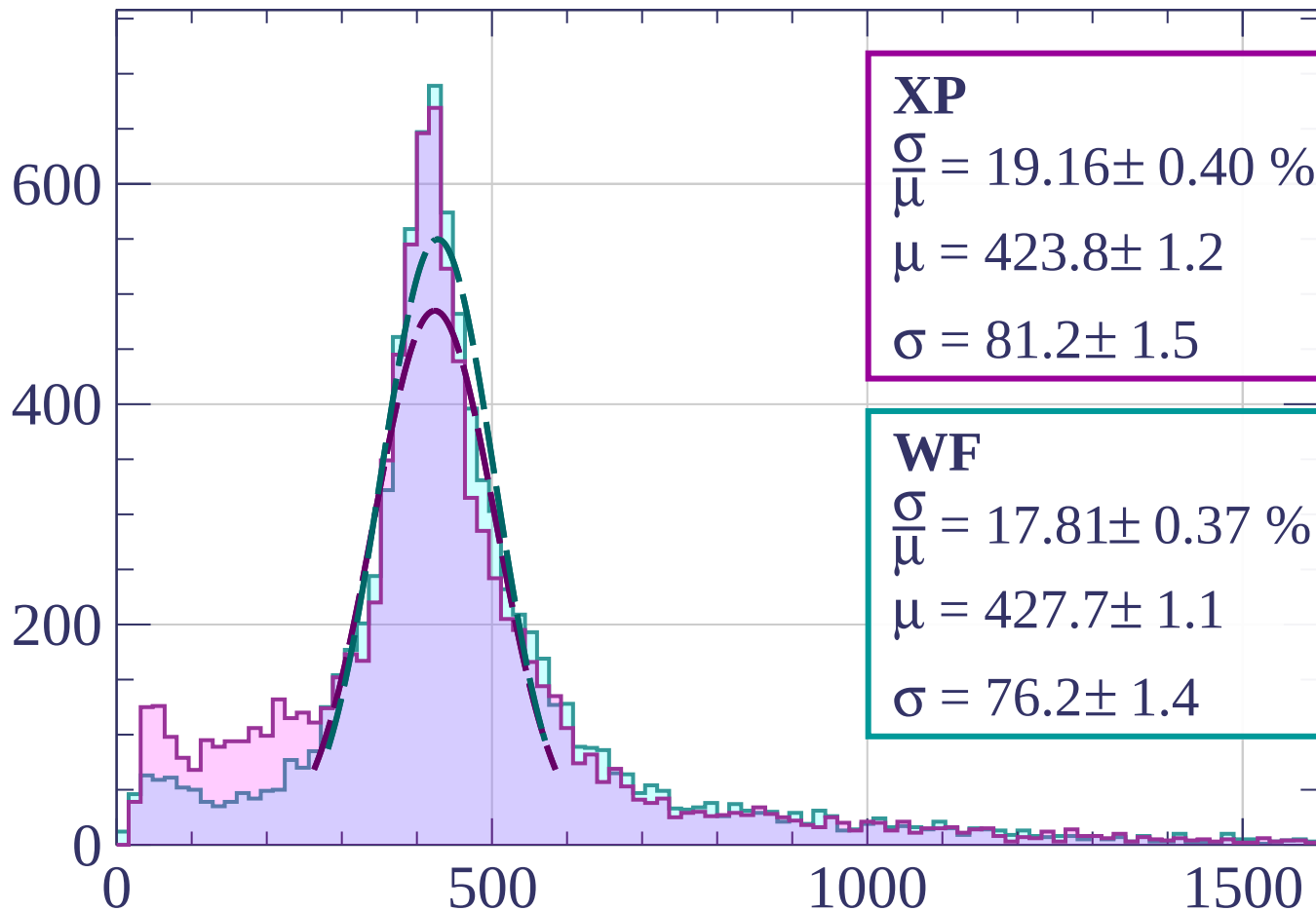
$$\mu = 427.6 \pm 1.0$$

$$\sigma = 68.5 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 19.16 \pm 0.40 \%$$

$$\mu = 423.8 \pm 1.2$$

$$\sigma = 81.2 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 17.81 \pm 0.37 \%$$

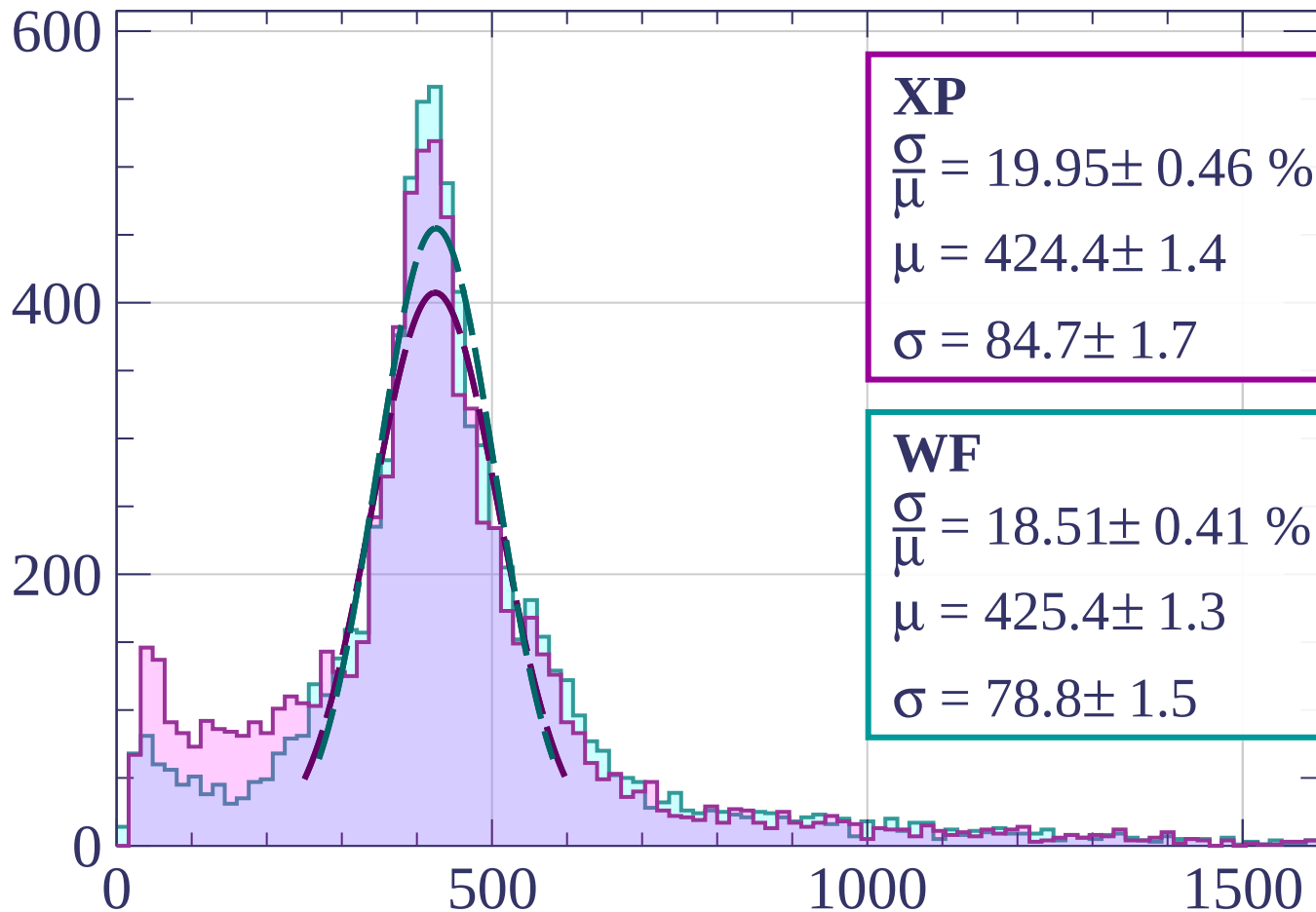
$$\mu = 427.7 \pm 1.1$$

$$\sigma = 76.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 19.95 \pm 0.46 \%$$

$$\mu = 424.4 \pm 1.4$$

$$\sigma = 84.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 18.51 \pm 0.41 \%$$

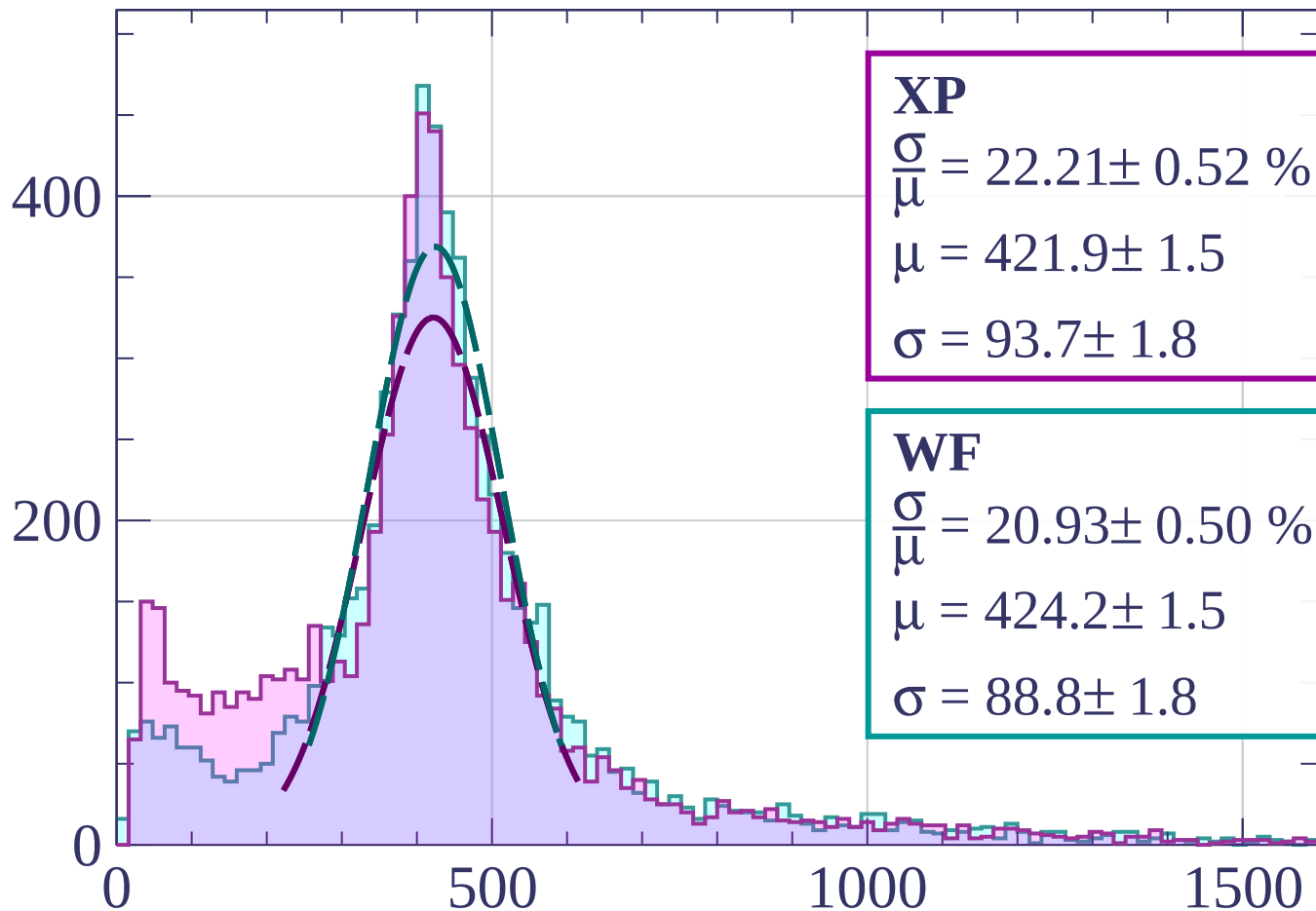
$$\mu = 425.4 \pm 1.3$$

$$\sigma = 78.8 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 22.21 \pm 0.52 \%$$

$$\mu = 421.9 \pm 1.5$$

$$\sigma = 93.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 20.93 \pm 0.50 \%$$

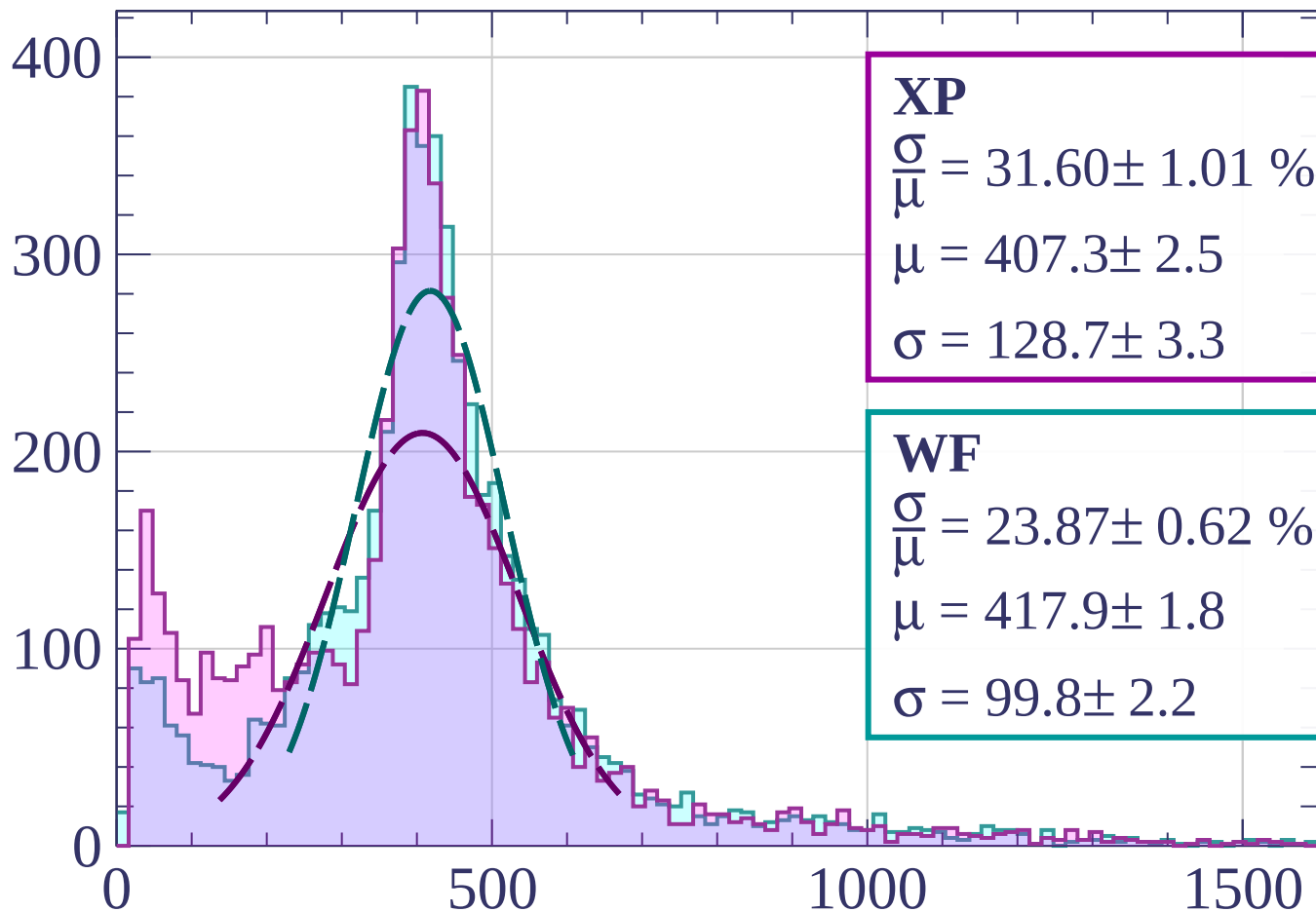
$$\mu = 424.2 \pm 1.5$$

$$\sigma = 88.8 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 31.60 \pm 1.01 \%$$

$$\mu = 407.3 \pm 2.5$$

$$\sigma = 128.7 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 23.87 \pm 0.62 \%$$

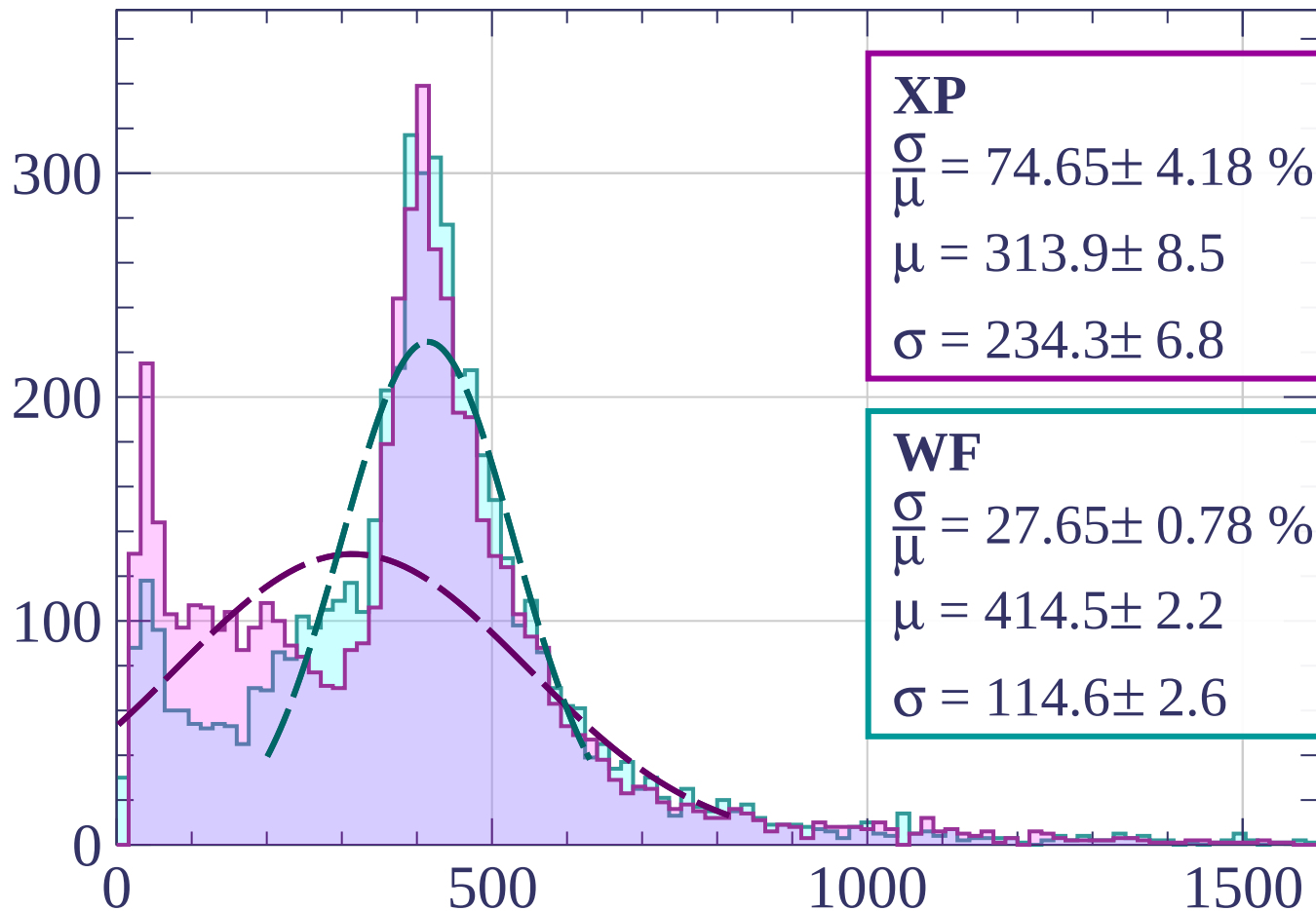
$$\mu = 417.9 \pm 1.8$$

$$\sigma = 99.8 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 74.65 \pm 4.18 \%$$

$$\mu = 313.9 \pm 8.5$$

$$\sigma = 234.3 \pm 6.8$$

WF

$$\frac{\sigma}{\mu} = 27.65 \pm 0.78 \%$$

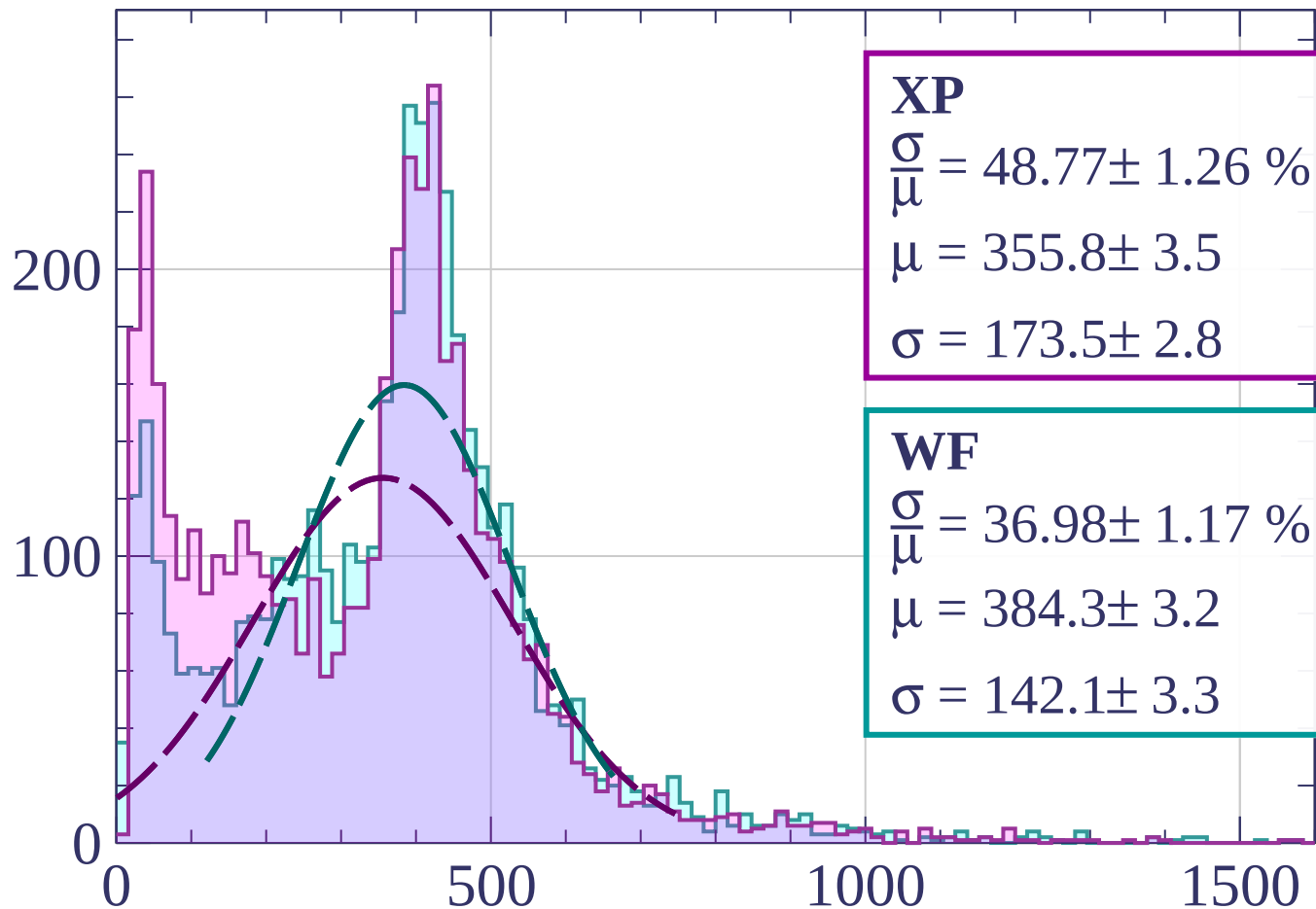
$$\mu = 414.5 \pm 2.2$$

$$\sigma = 114.6 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 48.77 \pm 1.26 \%$$

$$\mu = 355.8 \pm 3.5$$

$$\sigma = 173.5 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 36.98 \pm 1.17 \%$$

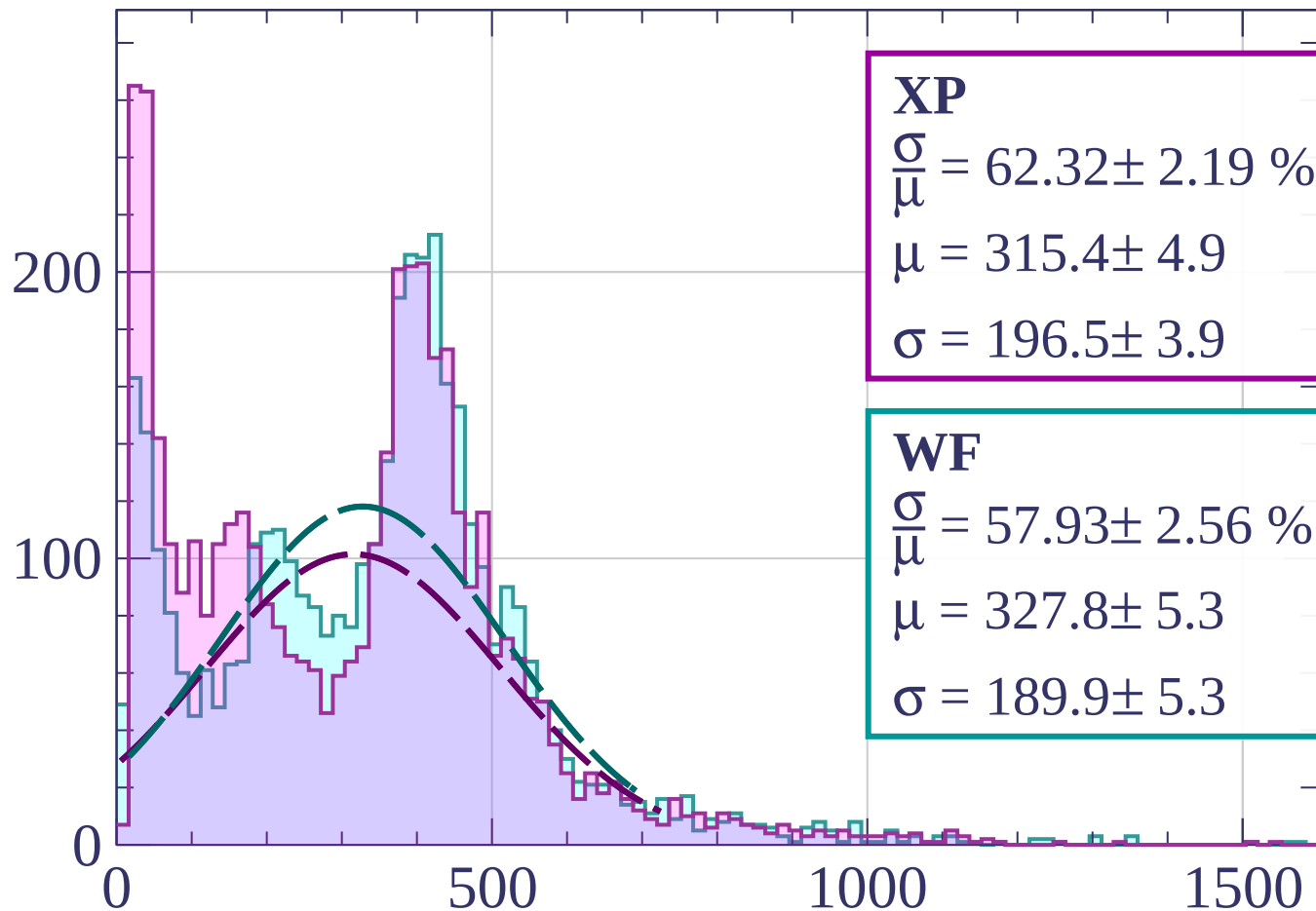
$$\mu = 384.3 \pm 3.2$$

$$\sigma = 142.1 \pm 3.3$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 62.32 \pm 2.19 \%$$

$$\mu = 315.4 \pm 4.9$$

$$\sigma = 196.5 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 57.93 \pm 2.56 \%$$

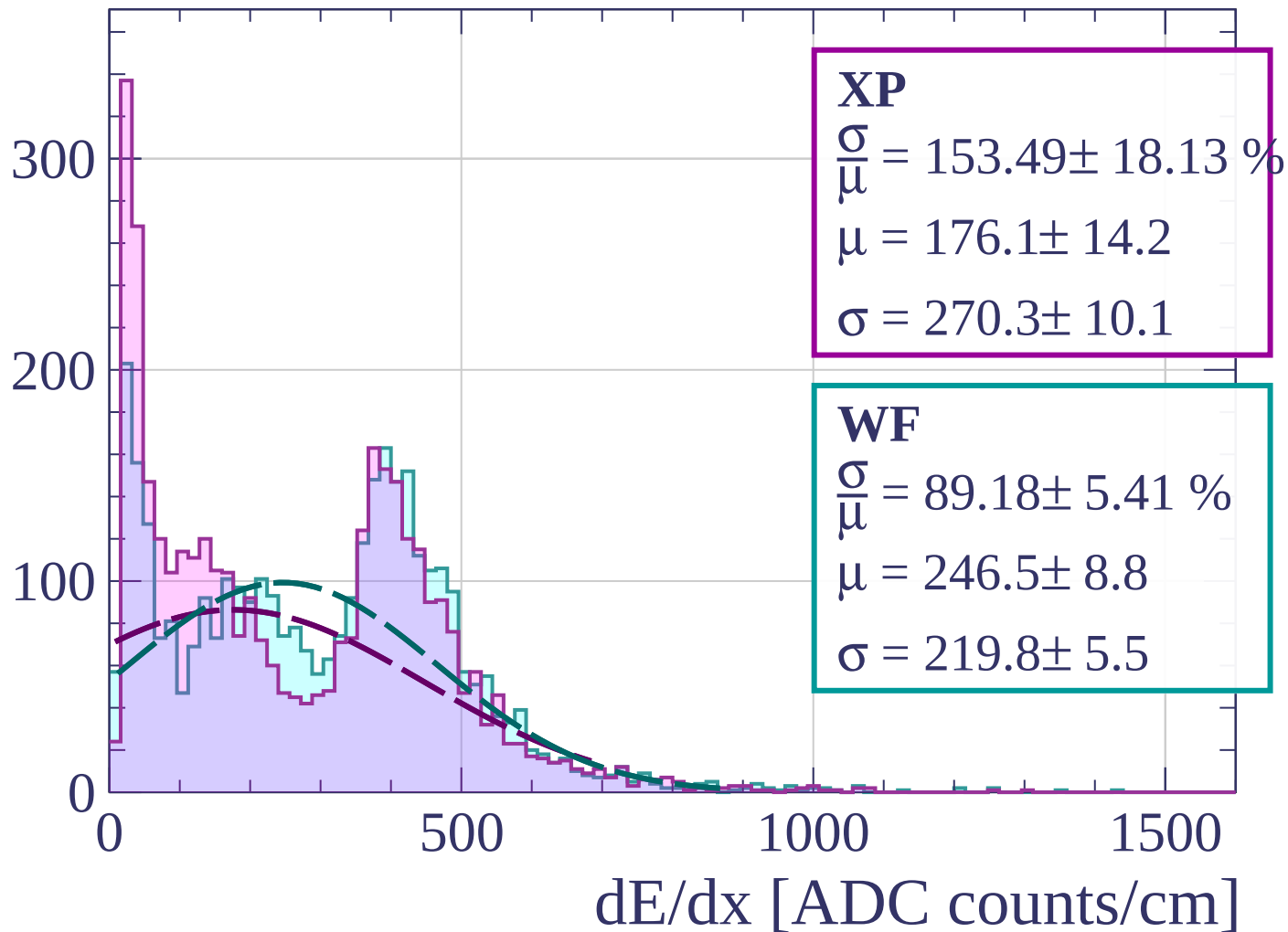
$$\mu = 327.8 \pm 5.3$$

$$\sigma = 189.9 \pm 5.3$$

dE/dx [ADC counts/cm]

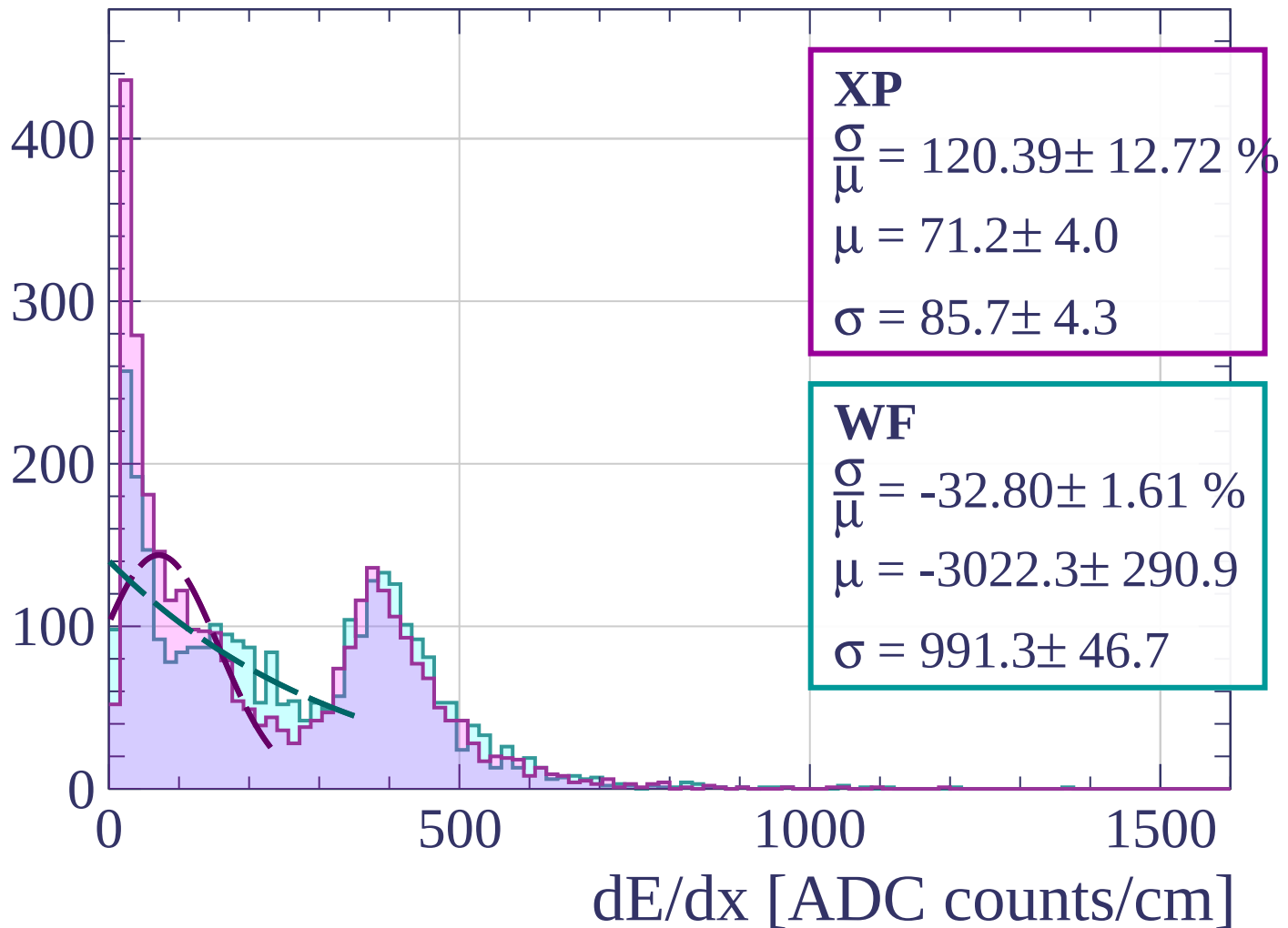
Energy loss | $63 < \theta < 66$

Count



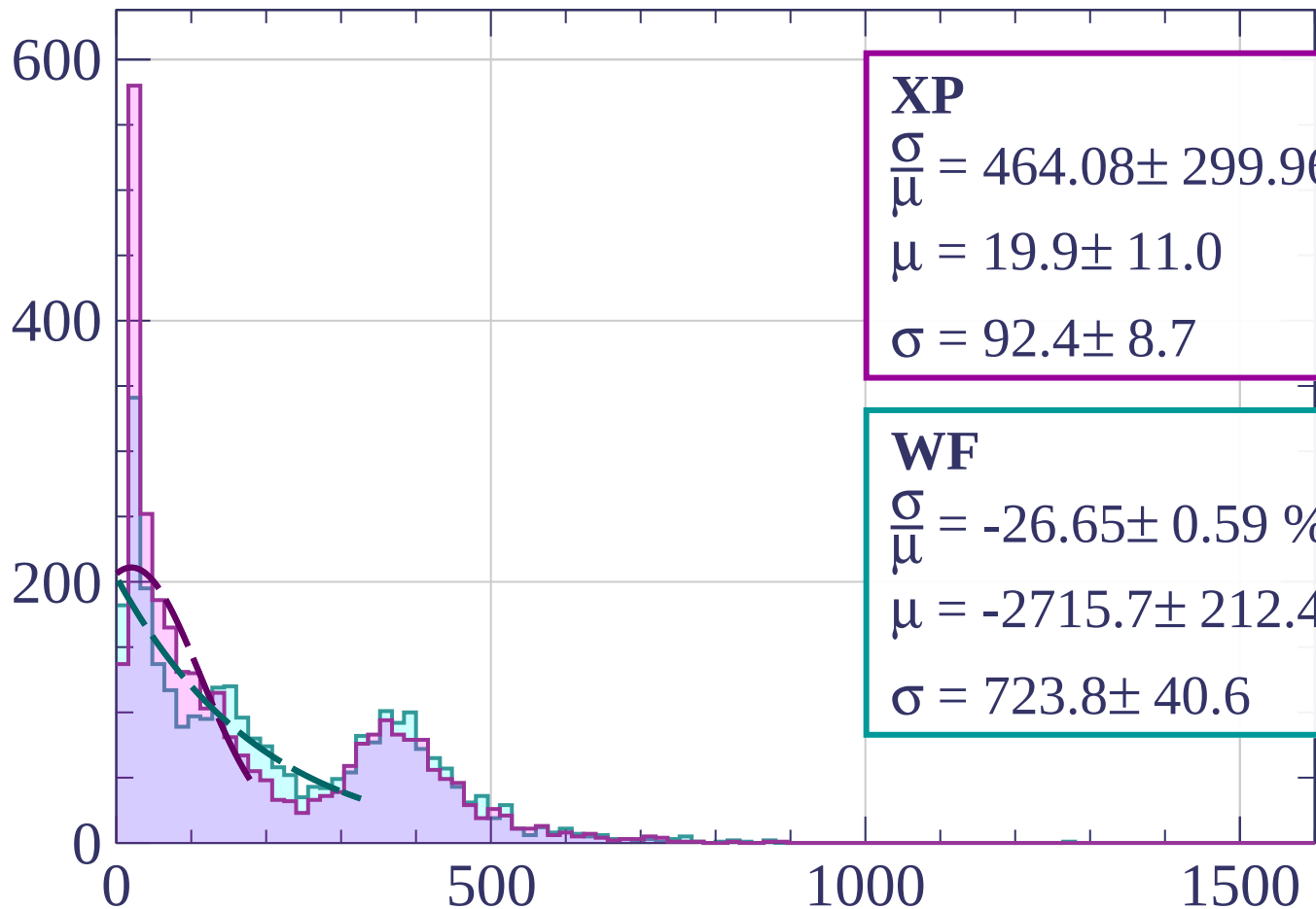
Energy loss | $66 < \theta < 69$

Count



Energy loss | $69 < \theta < 72$

Count



XP

$$\frac{\sigma}{\mu} = 464.08 \pm 299.96 \%$$

$$\mu = 19.9 \pm 11.0$$

$$\sigma = 92.4 \pm 8.7$$

WF

$$\frac{\sigma}{\mu} = -26.65 \pm 0.59 \%$$

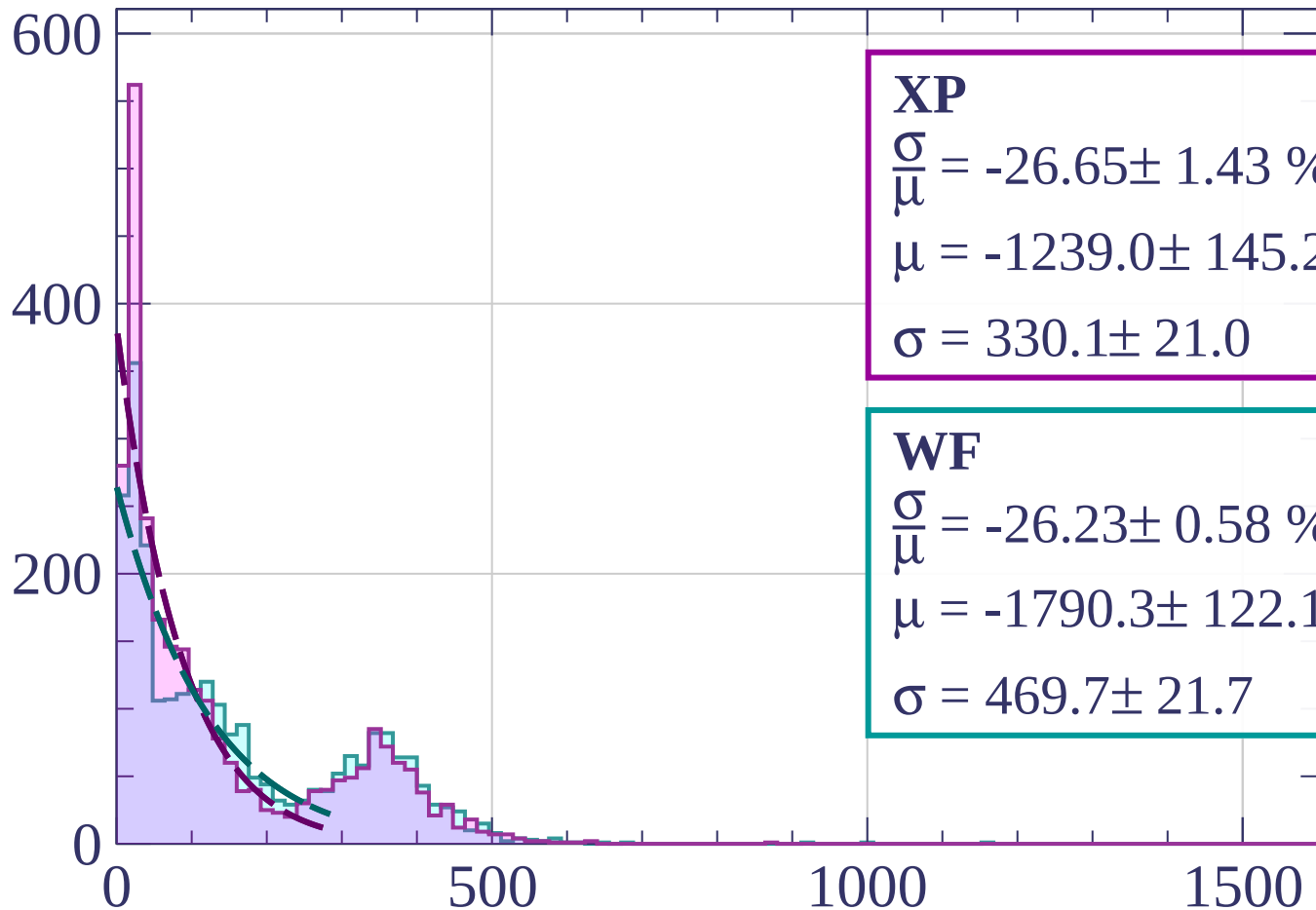
$$\mu = -2715.7 \pm 212.4$$

$$\sigma = 723.8 \pm 40.6$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \theta < 75$

Count



XP

$$\frac{\sigma}{\mu} = -26.65 \pm 1.43 \%$$

$$\mu = -1239.0 \pm 145.2$$

$$\sigma = 330.1 \pm 21.0$$

WF

$$\frac{\sigma}{\mu} = -26.23 \pm 0.58 \%$$

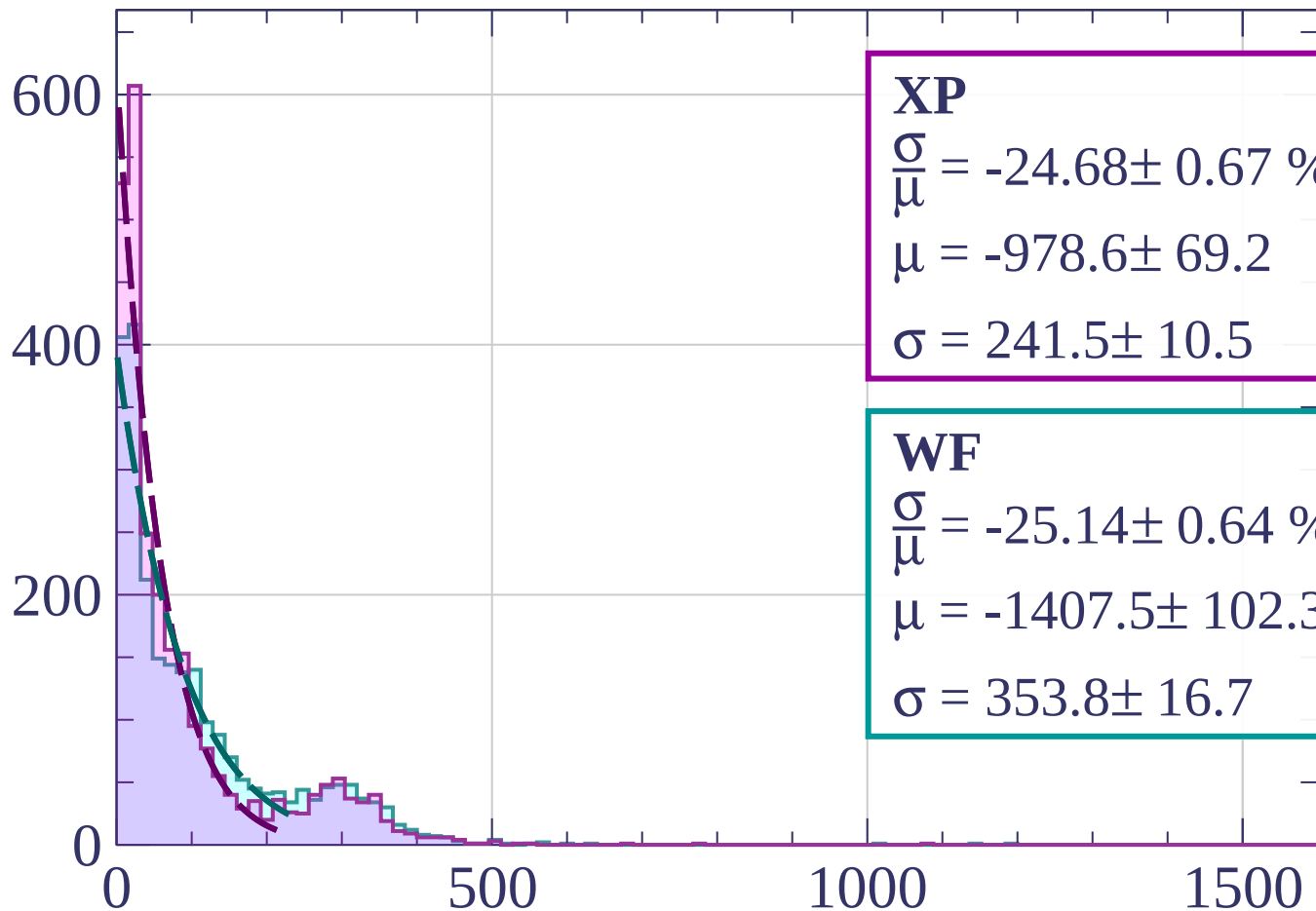
$$\mu = -1790.3 \pm 122.1$$

$$\sigma = 469.7 \pm 21.7$$

dE/dx [ADC counts/cm]

Energy loss | $75 < \theta < 78$

Count



XP

$$\frac{\sigma}{\mu} = -24.68 \pm 0.67 \%$$

$$\mu = -978.6 \pm 69.2$$

$$\sigma = 241.5 \pm 10.5$$

WF

$$\frac{\sigma}{\mu} = -25.14 \pm 0.64 \%$$

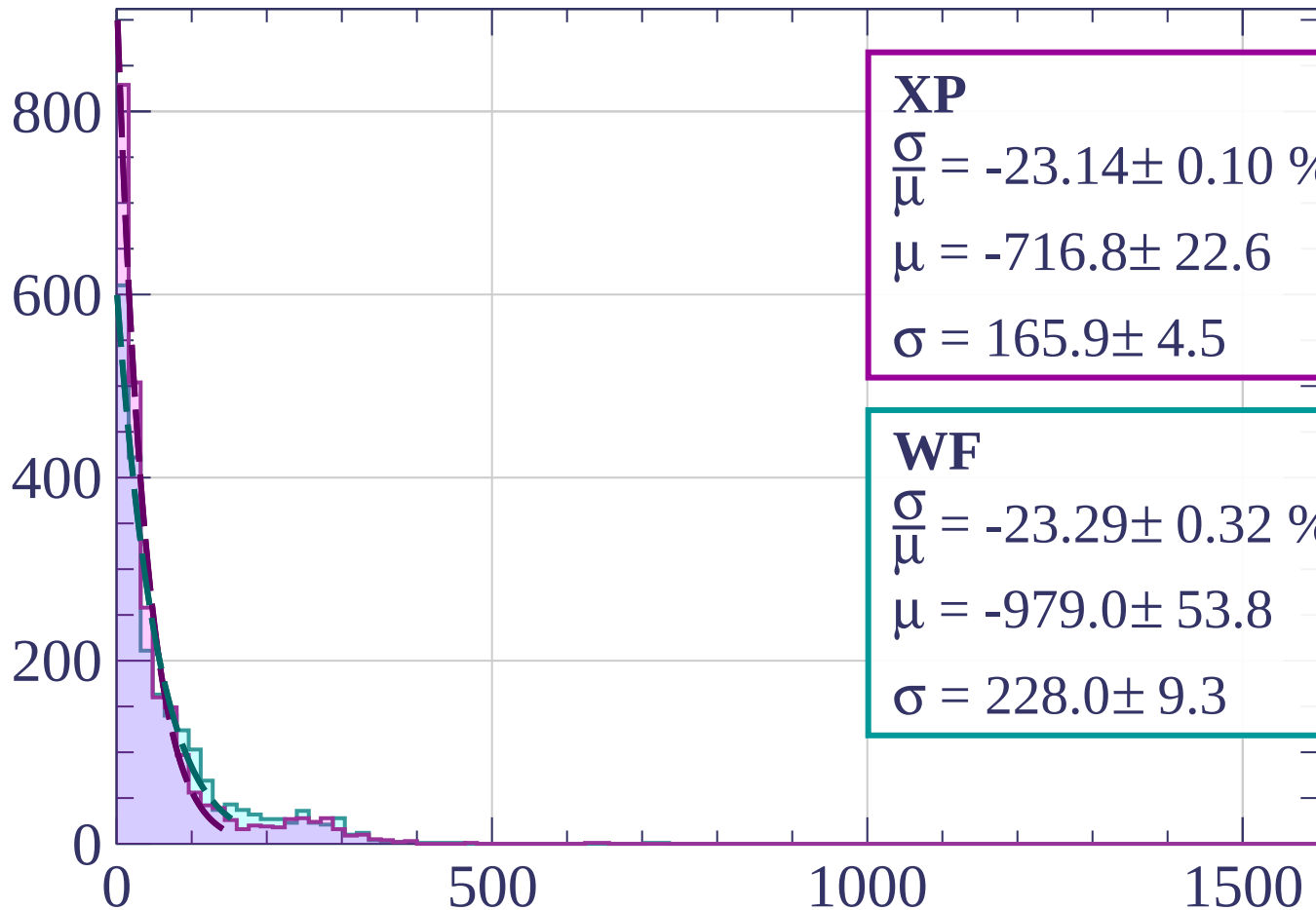
$$\mu = -1407.5 \pm 102.3$$

$$\sigma = 353.8 \pm 16.7$$

dE/dx [ADC counts/cm]

Energy loss | $78 < \theta < 81$

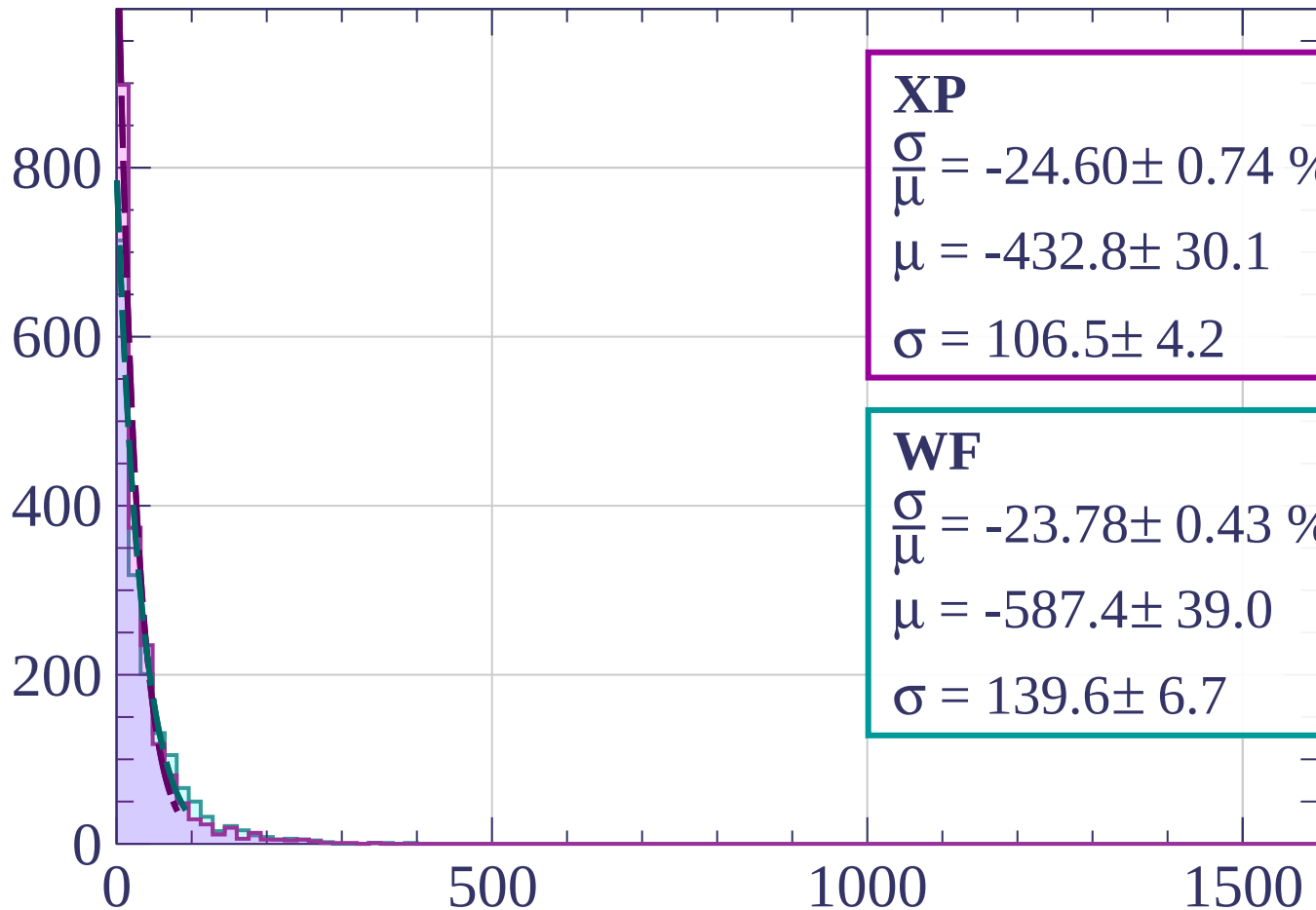
Count



dE/dx [ADC counts/cm]

Energy loss | $81 < \theta < 84$

Count



XP

$$\frac{\sigma}{\mu} = -24.60 \pm 0.74 \%$$

$$\mu = -432.8 \pm 30.1$$

$$\sigma = 106.5 \pm 4.2$$

WF

$$\frac{\sigma}{\mu} = -23.78 \pm 0.43 \%$$

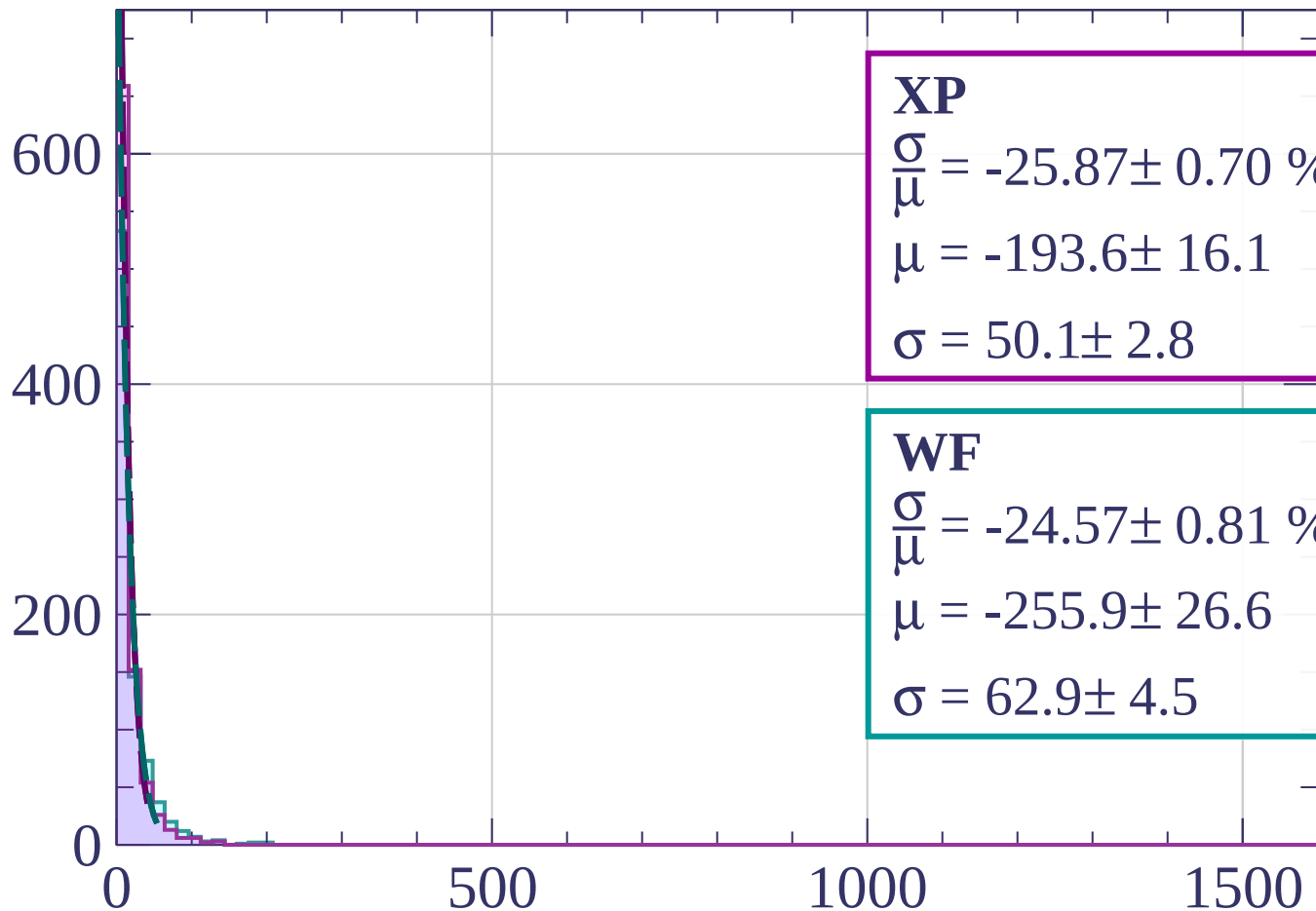
$$\mu = -587.4 \pm 39.0$$

$$\sigma = 139.6 \pm 6.7$$

dE/dx [ADC counts/cm]

Energy loss | $84 < \theta < 87$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = -25.87 \pm 0.70 \%$$

$$\mu = -193.6 \pm 16.1$$

$$\sigma = 50.1 \pm 2.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = -24.57 \pm 0.81 \%$$

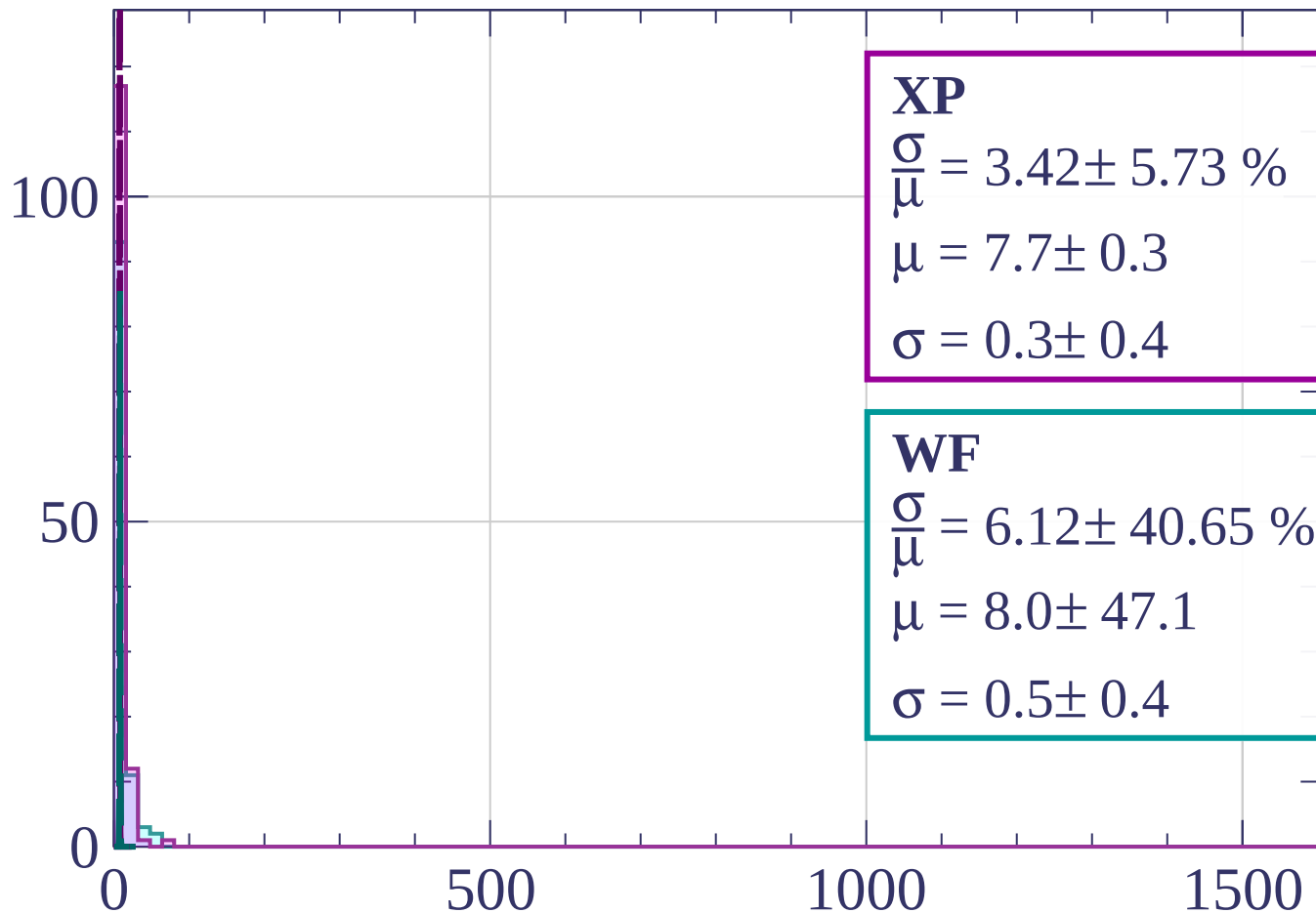
$$\mu = -255.9 \pm 26.6$$

$$\sigma = 62.9 \pm 4.5$$

dE/dx [ADC counts/cm]

Energy loss | $87 < \theta < 90$

Count



XP

$$\frac{\sigma}{\mu} = 3.42 \pm 5.73 \%$$

$$\mu = 7.7 \pm 0.3$$

$$\sigma = 0.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 6.12 \pm 40.65 \%$$

$$\mu = 8.0 \pm 47.1$$

$$\sigma = 0.5 \pm 0.4$$

dE/dx [ADC counts/cm]

